

# A First-Principles Dynamic Model of a Rubber V-Belt CVT for General Transient Simulation

With Mechanical Actuation and Traction-Limited Torque Transfer

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# Abstract

Mechanically actuated rubber V-belt continuously variable transmissions couple ratio change, belt clamping, torque transmission, and shaft dynamics through the same passive hardware. This makes their transient behaviour difficult to predict or diagnose when pulley speed, clamping force, shift motion, or operating regime must be prescribed in advance. This work develops CINDER, a first-principles transient formulation that instead treats the mechanically actuated CVT as a complete dynamic component driven by mechanical conditions at its two shafts.

The model couples primary centrifugal actuation, torque-reactive secondary-helix mechanics, pulley rotation, belt transport and geometric closure, and traction-limited belt–pulley contact. Engagement, ratio limits, free shifting, sticking, sliding, and support transitions are represented as admissible states of the same mechanical system, allowing their sequence to emerge from the evolving forces and constraints. The formulation is organized so that predicted behaviour can be traced through these internal mechanisms rather than only reproduced at the shaft level.

Internal mechanical, energy, numerical-convergence, and closure checks establish the consistency of the implemented system over the tested operating domain. The retained actuator dynamics are strongly regime dependent: primary flyweight inertia appreciably changes the engagement transient but becomes negligible during smooth shifting, while secondary-helix inertia can alter clamping and traction-transition timing even when the corresponding shift displacement changes only slightly. An independent comparison with a published transient rubber-belt model further shows that similar controlled shaft histories can coexist with substantially different clamp, shift, and slip behaviour, identifying the reduced belt representation as an important remaining fidelity limit.

Finally, coupled engine–CVT–vehicle simulations show how hardware changes propagate through clamping, ratio, traction, engine operating point, and vehicle response under launch, severe grade, repeated loading, and reverse-power conditions. The resulting framework therefore provides not only a transient prediction of CVT behaviour, but a mechanically traceable explanation of why that behaviour occurs.

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# Nomenclature and Notation

All quantities are expressed in SI units unless otherwise stated. Angles are expressed in radians. **Table N.1** establishes the notation conventions used throughout the formulation. The remaining tables are organized by the role each quantity plays in a model evaluation rather than alphabetically: fixed CVT data define the selected hardware, the current CVT condition is retained from one instant to the next, shaft-interface conditions are supplied by the systems attached to the two shafts, and instantaneous CVT quantities are recomputed from those inputs.

## Notation and Conventions

| Notation                                                        | Convention                                                                                                                                                                                          |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $j \in \{p, s\}$                                                | Generic pulley index; $p$ denotes the primary and $s$ the secondary.                                                                                                                                |
| $i \in \{u, l\}$                                                | Straight-span index; $u$ denotes the upper span and $l$ the lower span.                                                                                                                             |
| M, F                                                            | Movable-side and fixed/shaft-side secondary assemblies, respectively.                                                                                                                               |
| B                                                               | Shaft boundary; quantities carrying this subscript belong to the external system as presented at a CVT shaft interface.                                                                             |
| $\dot{(\ )}, \ddot{(\ )}$                                       | First and second derivatives with respect to time.                                                                                                                                                  |
| $(\ )', (\ )''$                                                 | First and second derivatives with respect to a quantity's established independent variable. The independent variable is stated when it is not already clear from the defining map or local context. |
| $(\ )^-, (\ )^+$                                                | Values immediately before and immediately after a finite boundary event.                                                                                                                            |
| $\Delta(\ )$                                                    | Finite change across an event, defined as outgoing minus incoming.                                                                                                                                  |
| $\mathbf{x}, \mathbf{z}, \boldsymbol{\nu}, \boldsymbol{\Gamma}$ | Vector quantities are bold. Greek vectors retain the case of their scalar components.                                                                                                               |
| $\mathbf{A}, \mathbf{J}, \mathbf{W}, \mathbf{C}$                | Matrix quantities are bold uppercase.                                                                                                                                                               |
| $[\cdot]^\top$                                                  | Transpose.                                                                                                                                                                                          |
| $\hat{\mathbf{e}}_{(\ )}$                                       | Unit vector in the direction indicated by its subscript.                                                                                                                                            |

Table N.1: Mathematical and index conventions used throughout the formulation.

## Fixed CVT Data

The quantities in **Table N.2** define the CVT selected for a simulation. They include directly specified hardware properties as well as geometry, mass properties, and configuration maps calculated once during initialization. They are not advanced as model state.

| Symbol                                 | Unit              | Meaning                                                                                                                      |
|----------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Belt and common geometry</b>        |                   |                                                                                                                              |
| $d_c$                                  | m                 | Pulley centre-to-centre distance.                                                                                            |
| $L_b$                                  | m                 | Total belt length.                                                                                                           |
| $\beta$                                | rad               | Sheave-face half-angle used for axial-radial conversion.                                                                     |
| $h_b$                                  | m                 | Radial belt height.                                                                                                          |
| $b_{\text{out}}, b_{\text{in}}$        | m                 | Belt widths at the outer and inner radial surfaces.                                                                          |
| $d_{\text{cord}}$                      | m                 | Radial depth of the cord-layer centre measured inward from the outer belt surface.                                           |
| $\rho_b$                               | kg/m <sup>3</sup> | Belt material density.                                                                                                       |
| $A_b$                                  | m <sup>2</sup>    | Belt cross-sectional area; fixed once the belt section is specified.                                                         |
| $y_{\text{cm}}$                        | m                 | Radial depth of the belt cross-section centroid measured inward from the outer surface.                                      |
| $m_b$                                  | kg                | Total belt mass used in the complete-belt transport balance.                                                                 |
| <b>Pulley geometry and travel</b>      |                   |                                                                                                                              |
| $x_{j,\text{max}}$                     | m                 | Maximum closing travel of pulley $j$ .                                                                                       |
| $x_{j,\text{ref}}, r_{j,\text{ref}}$   | m                 | Known seated axial position and corresponding outer-surface radius used to anchor the local radius map.                      |
| $r_{p,\text{min}}$                     | m                 | Minimum primary outer-surface belt radius, retained through the primary deadzone.                                            |
| $x_{p,\text{dz}}$                      | m                 | Primary position at which the movable sheave first reaches the belt.                                                         |
| $s_{\text{dz}}$                        | m                 | System-coordinate deadzone boundary; $s_{\text{dz}} \equiv x_{p,\text{dz}}$ .                                                |
| $s_{\text{max}}$                       | m                 | Upper system shift bound; $s_{\text{max}} \equiv x_{p,\text{max}}$ .                                                         |
| $r_j(x_j)$                             | m                 | Hardware-defined outer-surface radius map for pulley $j$ .                                                                   |
| $x_s(s)$                               | m                 | Secondary movable-sheave position imposed by belt-length compatibility for a given system shift coordinate.                  |
| $L_{\text{geo}}(r_p, r_s, d_c)$        | m                 | Exact geometric belt-path length associated with a proposed pair of pulley radii.                                            |
| $\mathcal{R}_L(r_p, r_s, d_c)$         | m                 | Belt-length residual $L_{\text{geo}} - L_b$ , used to determine compatible installed geometry.                               |
| $\mathcal{R}_x(x_p, x_s)$              | m                 | Belt-length residual after the local pulley radius maps are substituted; its zero defines compatible pulley positions.       |
| <b>Primary actuation and inertia</b>   |                   |                                                                                                                              |
| $m_p$                                  | kg                | Translating mass of the primary movable sheave, ramp carrier, and components moving rigidly with them.                       |
| $I_f$                                  | kg m <sup>2</sup> | Total flyweight-set mass moment of inertia about the flyweight pivot axes.                                                   |
| $m_f$                                  | kg                | Total flyweight-set mass; used explicitly in the classical point-mass reduction and flyweight construction appendix.         |
| $q_f(x_p)$                             | rad               | Flyweight pivot-angle map as a function of primary position.                                                                 |
| $J_{f,p}(x_p)$                         | kg m <sup>2</sup> | Flyweight-set mass moment of inertia about the primary shaft axis as a function of primary position.                         |
| $I_{p,\text{const}}$                   | kg m <sup>2</sup> | Constant shaft-axis inertia of primary CVT hardware excluding the configuration-dependent flyweight contribution $J_{f,p}$ . |
| $k_p$                                  | N/m               | Primary compression-spring stiffness.                                                                                        |
| $x_{p,\text{pre}}$                     | m                 | Installed primary spring compression at $x_p = 0$ .                                                                          |
| <b>Secondary actuation and inertia</b> |                   |                                                                                                                              |
| $m_s$                                  | kg                | Translating mass of the secondary movable sheave and components translating rigidly with it.                                 |

| Symbol                     | Unit             | Meaning                                                                                          |
|----------------------------|------------------|--------------------------------------------------------------------------------------------------|
| $I_{s,F}$                  | $\text{kg m}^2$  | Shaft-axis inertia of the secondary fixed/shaft-side assembly inside the CVT boundary.           |
| $I_{s,M}$                  | $\text{kg m}^2$  | Shaft-axis inertia of the secondary movable assembly.                                            |
| $r_h$                      | m                | Effective cylindrical radius used to unwrap the secondary helix.                                 |
| $\alpha_s$                 | rad              | Local unwrapped helix-track angle profile measured from the circumferential direction.           |
| $\theta_s(x_s)$            | rad              | Relative movable-sheave rotation imposed by the helix as a function of secondary axial position. |
| $k_{s,\theta}$             | $\text{N m/rad}$ | Secondary torsional spring stiffness.                                                            |
| $\theta_{s,\text{pre}}$    | rad              | Signed installed torsional preload at $x_s = 0$ .                                                |
| $k_{s,x}$                  | $\text{N/m}$     | Secondary axial spring stiffness.                                                                |
| $x_{s,\text{pre}}$         | m                | Installed secondary axial spring compression at $x_s = 0$ .                                      |
| <b>Belt–pulley contact</b> |                  |                                                                                                  |
| $\mu_s$                    | 1                | Static belt–pulley traction coefficient used to test sticking admissibility.                     |
| $\mu_k$                    | 1                | Kinetic belt–pulley traction coefficient used on sliding branches.                               |

Table N.2: Fixed CVT data and hardware-defined maps used by the model.

## Current CVT Condition

Only the continuous state and discrete mechanical regime are retained as the CVT’s time-varying internal condition from one evaluation to the next.

| Symbol               | Unit  | Meaning                                                                                                    |
|----------------------|-------|------------------------------------------------------------------------------------------------------------|
| $\mathbf{x}(t)$      | mixed | Continuous CVT state vector $[\omega_p, \omega_s, s, \dot{s}, v_b]^\top$ .                                 |
| $\omega_p, \omega_s$ | rad/s | Primary and secondary shaft angular speeds.                                                                |
| $s$                  | m     | System shift coordinate, equal to the primary movable-sheave position $x_p$ .                              |
| $\dot{s}$            | m/s   | Shift velocity.                                                                                            |
| $v_b$                | m/s   | Signed belt transport speed.                                                                               |
| $q(t)$               | –     | Discrete mechanical regime identifying the currently active structural and tangential contact constraints. |

Table N.3: Quantities retained as the current internal CVT condition.

## Shaft-Interface Conditions

The attached systems communicate with the CVT only through the mechanical conditions in [Table N.4](#). These quantities are supplied for the current evaluation and are not additional CVT state variables.

| Symbol       | Unit              | Meaning                                                                       |
|--------------|-------------------|-------------------------------------------------------------------------------|
| $\tau_{B,j}$ | N m               | Signed external torque presented to shaft $j$ by the attached system.         |
| $I_{B,j}$    | kg m <sup>2</sup> | Equivalent rotational inertia of the attached system presented at shaft $j$ . |

Table N.4: Mechanical conditions supplied at the primary and secondary shaft interfaces.

## Instantaneous CVT Quantities

The quantities in [Table N.5](#) are evaluated from the fixed CVT data, current CVT condition, shaft-interface conditions, and active regime. They are useful throughout the mechanics and results but are recomputed rather than retained as independent state. Purely local dummy variables and temporary algebraic abbreviations are defined where they are used and are not repeated here.

| Symbol                                   | Unit             | Meaning                                                                                                          |
|------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Geometry and kinematics</b>           |                  |                                                                                                                  |
| $x_p, x_s$                               | m                | Current primary and secondary movable-sheave positions; $x_p = s$ and $x_s = x_s(s)$ .                           |
| $r_j \equiv r_{j,\text{outer}}$          | m                | Current belt outer-surface radius on pulley $j$ .                                                                |
| $r_{j,\text{inner}}$                     | m                | Current belt inner-surface radius on pulley $j$ .                                                                |
| $r_{j,\text{eff}}$                       | m                | Cord-line effective radius used as the belt-torque lever arm.                                                    |
| $r_{j,\text{cm}}$                        | m                | Radius of the belt cross-section centre of mass on pulley $j$ .                                                  |
| $\ell$                                   | m                | Length of either straight belt span for the current two-pulley geometry.                                         |
| $\delta$                                 | rad              | Signed angle of the straight spans relative to the pulley-centre line.                                           |
| $\phi_p, \phi_s$                         | rad              | Primary and secondary belt wrap angles.                                                                          |
| $\theta$                                 | rad              | Local angular coordinate through a pulley wrap, increasing in the positive belt-transport direction.             |
| $\omega_{s,M}$                           | rad/s            | Absolute angular speed of the secondary movable assembly.                                                        |
| $\bar{\omega}_j$                         | rad/s            | Representative angular speed of the belt-contact surface on pulley $j$ .                                         |
| $v_{\text{rel},j}$                       | m/s              | Relative tangential belt–pulley speed at contact $j$ .                                                           |
| $a_{\text{rel},j}$                       | m/s <sup>2</sup> | Relative tangential acceleration used as the sticking residual.                                                  |
| <b>Belt contact, tension, and torque</b> |                  |                                                                                                                  |
| $N_p, N_s$                               | N                | Total integrated belt normal resultants over both sheave faces of the primary and secondary.                     |
| $\lambda_j$                              | 1                | Signed traction utilization defined by $dF_{t,j} = \lambda_j dN_j$ .                                             |
| $\lambda_{j,\text{st}}$                  | 1                | Static traction utilization candidate on a sticking wrap; admissible when $ \lambda_{j,\text{st}}  \leq \mu_s$ . |
| $\sigma_j$                               | 1                | Sliding-direction sign, with $\lambda_j = \sigma_j \mu_k$ on a sliding wrap.                                     |
| $T_j(\theta)$                            | N                | Belt tension distribution through wrap $j$ .                                                                     |
| $T_{j,\text{in}}, T_{j,\text{out}}$      | N                | Belt tensions entering and leaving wrap $j$ in the positive transport direction.                                 |
| $\tau_j$                                 | N m              | Total signed belt torque acting on pulley $j$ .                                                                  |

| Symbol                                            | Unit                     | Meaning                                                                                                                     |
|---------------------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| $H_{\text{CVT},j}$                                | $\text{kg m}^2/\text{s}$ | Angular momentum of CVT hardware inside the boundary on shaft side $j$ .                                                    |
| $H_j$                                             | $\text{kg m}^2/\text{s}$ | Total pulley-side angular momentum including the equivalent boundary inertia.                                               |
| <b>Primary and secondary actuation</b>            |                          |                                                                                                                             |
| $F_{\text{fw},x_p}$                               | N                        | Axial closing force transmitted by the primary flyweights.                                                                  |
| $F_{\text{spring},x_p}$                           | N                        | Primary spring opening force.                                                                                               |
| $F_{\text{belt},x_p}$                             | N                        | Axial belt reaction on the primary movable sheave.                                                                          |
| $\tau_{s,F}, \tau_{s,M}$                          | N m                      | Belt-torque shares applied to the fixed/shaft-side and movable secondary faces.                                             |
| $\tau_{s,\text{spring}}$                          | N m                      | Torque applied to the secondary movable assembly by the wound spring.                                                       |
| $\tau_{s,\text{helix}}$                           | N m                      | Circumferential reaction torque at the secondary pin–helix contact.                                                         |
| $F_{\text{helix},x_s}$                            | N                        | Axial force transmitted through the secondary helix.                                                                        |
| $F_{\text{spring},x_s}$                           | N                        | Secondary axial spring closing force.                                                                                       |
| $F_{\text{belt},x_s}$                             | N                        | Axial belt reaction on the secondary movable sheave.                                                                        |
| <b>Instantaneous mechanical solve</b>             |                          |                                                                                                                             |
| $\dot{\mathbf{x}}$                                | mixed                    | State derivative returned by the active-regime mechanics.                                                                   |
| $\mathcal{F}_q$                                   | –                        | Regime-dependent evolution map relating the current state and shaft-interface conditions to $\dot{\mathbf{x}}$ .            |
| $\mathbf{z}$                                      | mixed                    | Engaged mechanical unknown vector $[\dot{\omega}_p, \dot{\omega}_s, \dot{v}_b, \ddot{s}, N_p, N_s]^\top$ .                  |
| $\mathbf{A}$                                      | mixed                    | Coefficient matrix of the common engaged linear mechanical solve.                                                           |
| $\mathbf{b}$                                      | mixed                    | Known-side vector of the common engaged linear mechanical solve.                                                            |
| $\mathbf{z}_{\text{dz}}$                          | mixed                    | Free-deadzone unknown vector $[\dot{\omega}_p, \dot{\omega}_s, \dot{v}_b, \ddot{s}]^\top$ .                                 |
| $\mathbf{A}_{\text{dz}}, \mathbf{b}_{\text{dz}}$  | mixed                    | Coefficient matrix and known-side vector of the reduced free-deadzone mechanical solve.                                     |
| <b>Structural-regime quantities</b>               |                          |                                                                                                                             |
| $F_{\text{stop},0}$                               | N                        | Reaction at the fully open primary stop.                                                                                    |
| $F_{\text{stop},\text{LR}}$                       | N                        | Reaction associated with the engaged low-ratio structural seat.                                                             |
| $F_{\text{stop},\text{max}}$                      | N                        | Reaction at the upper primary closing stop.                                                                                 |
| $\ddot{s}_{\text{cf}}$                            | $\text{m}/\text{s}^2$    | Contact-free shift acceleration used to test release from an active structural stop.                                        |
| <b>Finite boundary events</b>                     |                          |                                                                                                                             |
| $\boldsymbol{\nu}$                                | mixed                    | Retained event-velocity vector $[\omega_p, \omega_s, v_b, \dot{s}]^\top$ .                                                  |
| $\mathbf{J}_{\text{eng}}, \mathbf{J}_{\text{dz}}$ | mixed                    | Maps from retained event velocities to physical body velocities in the engaged and deadzone structural arrangements.        |
| $\mathbf{J}_-, \mathbf{J}_+$                      | mixed                    | Incoming and outgoing physical-motion maps selected for a finite event.                                                     |
| $\mathbf{W}$                                      | mixed                    | Diagonal physical mass/inertia weighting matrix used in the event momentum construction.                                    |
| $\mathbf{p}$                                      | mixed                    | Physical momentum vector associated with the body-velocity inventory, $\mathbf{p} = \mathbf{W}\mathbf{J}\boldsymbol{\nu}$ . |
| $\Delta\mathbf{p}$                                | mixed                    | Physical momentum change across a finite event.                                                                             |
| $\mathbf{C}_+$                                    | mixed                    | Matrix whose rows impose the active outgoing velocity constraints.                                                          |
| $\mathbf{c}_i$                                    | mixed                    | One outgoing event-constraint row; examples include stop, pulley-stick, and deadzone-lock constraints.                      |
| $\Gamma_i, \boldsymbol{\Gamma}$                   | N s                      | Scalar constraint impulse and assembled vector of active event impulses.                                                    |
| $T^-, T^+$                                        | J                        | Total retained kinetic energy immediately before and after a finite event.                                                  |

| Symbol                                                                                 | Unit | Meaning |
|----------------------------------------------------------------------------------------|------|---------|
| Table N.5: Principal instantaneous and derived quantities used by the CVT formulation. |      |         |

# Chapter 1

## Introduction & Background

### 1.1 Problem and Motivation

Mechanically actuated rubber V-belt continuously variable transmissions (CVTs) are widely used in compact powersport drivetrains, including snowmobiles, all-terrain vehicles (ATVs), and utility terrain vehicles (UTVs) (Messick, 2018), because they can vary the speed ratio automatically while transmitting power through a relatively simple mechanical assembly. Their visible layout is straightforward: two variable-radius pulleys are connected by a belt. Their transient behaviour, however, is not. The operating state of the transmission emerges from the interaction of its internal mechanisms, the belt–pulley contacts, and the mechanical conditions under which it operates.

Several coupled effects determine that response. Engine speed influences the centrifugal actuation of the primary pulley, while transmitted torque influences the torque-reactive response of the secondary pulley. Together, these mechanisms generate the clamping forces that press the belt into the sheave grooves. Clamping force determines the traction available at the belt–pulley contacts, which in turn determines how torque is carried and whether the belt and pulleys can remain coupled without gross slip. At the same time, pulley motion changes the belt radii and therefore the transmission ratio, which changes the speed and torque relationships between the two pulleys. These effects do not occur independently; each changes the conditions under which the others act.

This coupling makes the behaviour of a mechanically actuated CVT difficult to interpret from external measurements alone. A shift curve, pulley-speed trace, or observed loss of traction may reflect the combined effect of primary actuation, secondary reaction, belt traction, and shift motion. The underlying states responsible for that behaviour are largely internal to the transmission and are not usually available directly during testing. This is especially consequential because mechanically actuated CVTs are tuned by changing physical components such as springs, flyweights, ramps, and helices; determining the effect of a change can therefore require repeated hardware changes and testing (Skinner, 2020).

Addressing this problem requires a physically interpretable dynamic framework that preserves the connection between the internal CVT mechanisms and the behaviour they produce. A useful model should do more than reproduce an output trace: it should make it possible to follow how actuation, clamping, belt traction, shift motion, and inertia combine as the transmission responds to changing operating conditions. If an undesirable response can be traced to its mechanical cause, changes to the hardware or operating conditions can be made deliberately rather than through trial and

error. The motivation is therefore not only to predict transient CVT behaviour, but to make that behaviour understandable enough to diagnose and improve.

The remainder of this chapter develops the context for that framework. [Section 1.2](#) introduces the physical layout and operating principles of the belt–sheave CVT, including ratio change, traction, and mechanical actuation. [Section 1.3](#) contains a literature review that organizes prior work by modelling emphasis and identifies the foundations available for mechanical actuation, belt contact, and transient CVT dynamics. Given this context, [Section 1.4](#) then states the aim, scope, and contributions of the present formulation, while [Section 1.5](#) outlines the development that follows.

## 1.2 Conceptual Overview of a Continuously Variable Transmission

This section introduces the operating principles of the continuously variable transmission (CVT) considered in this work. The goal is not yet to derive the model equations, but to build the mechanical intuition needed to understand them. In particular, this section explains why a CVT is useful, how a rubber V-belt CVT changes ratio, why clamping and traction matter, how the primary and secondary pulleys generate their forces, and how these mechanisms interact to determine the transmission operating state.

### 1.2.1 Why Use a Continuously Variable Transmission?

A transmission is responsible for matching the operating speed of an engine to the speed required at the vehicle wheels. In a conventional stepped gearbox, this is achieved using a fixed set of discrete gear ratios. Each gear imposes a fixed relationship between engine speed and wheel speed. As the vehicle accelerates, the engine speed rises through one gear, drops when the transmission shifts, and then rises again in the next gear. This repeated sweep through the engine speed range is a direct consequence of having only a small number of available ratios.

This matters because an internal combustion engine (ICE) does not produce the same torque and power at every speed. Instead, the engine has a useful operating region where it can produce power more effectively, as shown conceptually in [Figure 1.1](#). A stepped gearbox can pass through this region, but it cannot generally stay there while vehicle speed continues to increase. [Figure 1.2](#) contrasts this stepped behavior with the smoother operating path made possible by a continuously variable transmission.

A continuously variable transmission (CVT) addresses this limitation by allowing the transmission ratio to vary smoothly within a finite range rather than changing through a fixed set of gear steps. Since the ratio can change continuously, the engine can remain closer to a desired operating region while the vehicle accelerates. This is the main benefit of a CVT: it gives the drivetrain more freedom to match engine behavior to vehicle load.

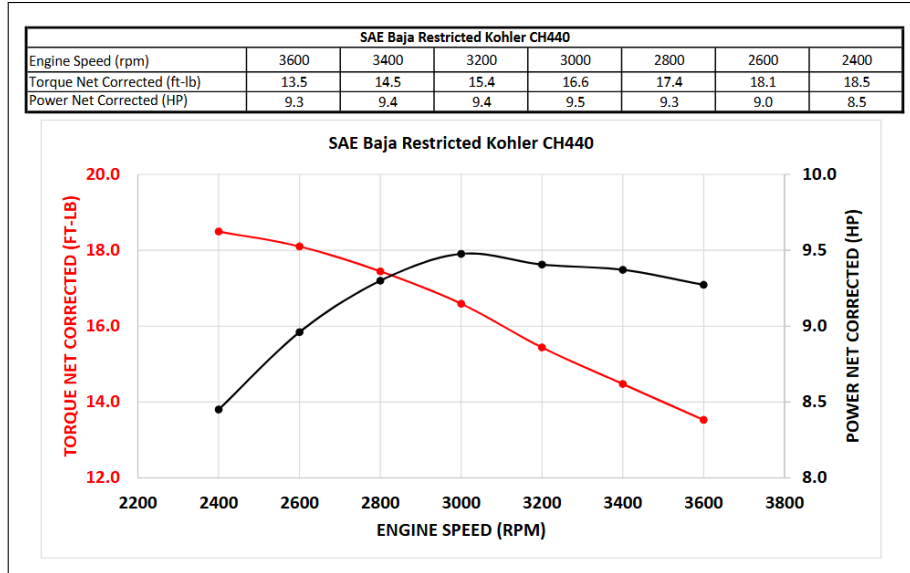


Figure 1.1: Torque and power of the Kohler CH440 engine as functions of engine speed. Both quantities vary substantially across the operating range, with peak torque occurring at a lower engine speed than peak power. (Baja SAE, 2022)

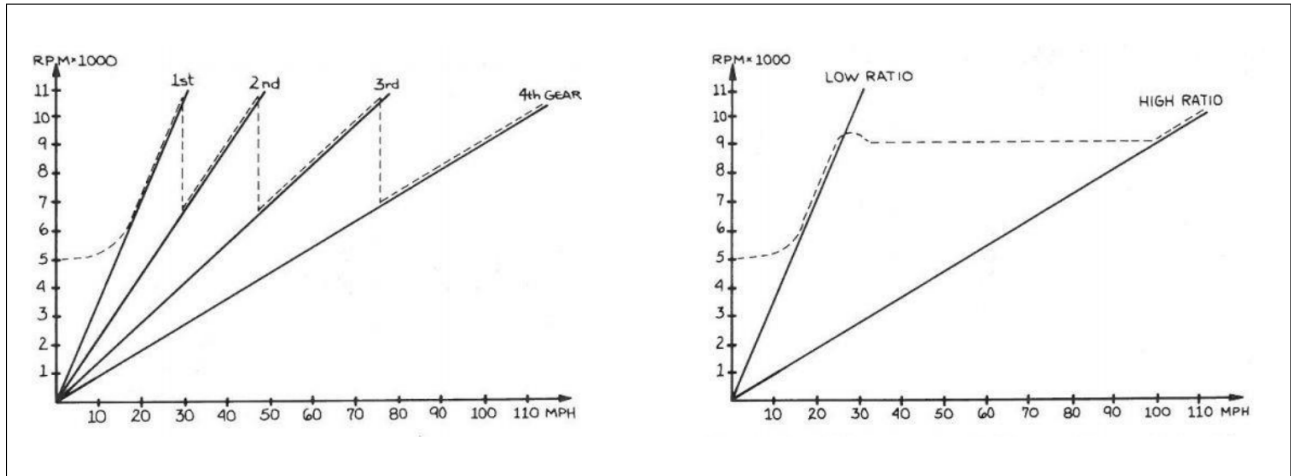


Figure 1.2: Conceptual comparison of engine-speed behavior during acceleration. Solid lines show the engine-speed–vehicle-speed relationship imposed by the available transmission ratios, while the dotted line illustrates a representative engine-speed trajectory during acceleration. With a stepped gearbox (left), the engine repeatedly sweeps through a speed range and drops in speed at each gear change. With a CVT (right), engine speed can rise to a desired operating speed and remain approximately constant while the transmission continuously changes ratio, increasing again only after the high-ratio limit is reached. (Aaen, 2007)

### 1.2.2 Physical Layout and Single-Pulley Geometry

There are several forms of continuously variable transmission, including different mechanical layouts and different ways of producing the variable speed relationship. This work focuses on a belt-sheave CVT, shown conceptually in Figure 1.3. In this arrangement, two variable-radius pulleys are

connected by a rubber V-belt. The pulley connected to the engine side is called the primary (or driving) pulley. The pulley connected to the downstream drivetrain is called the secondary (or driven) pulley. The belt transmits motion and torque between these two pulleys.

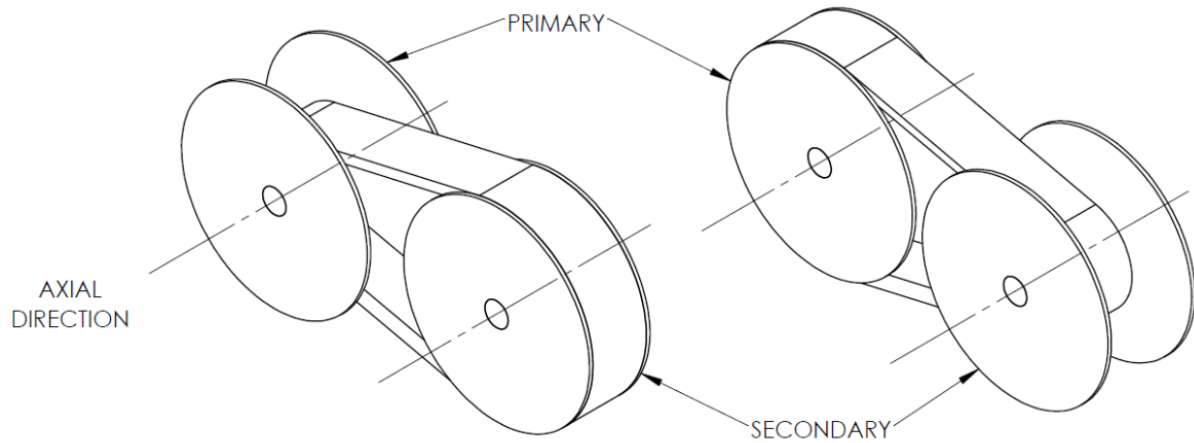


Figure 1.3: Two sample configurations of the belt-sheave CVT considered in this work. The primary pulley is connected to the engine side, the secondary pulley is connected to the downstream drivetrain, and the rubber V-belt transmits motion and torque between them. (Messick, 2018)

Before considering the full two-pulley system, it is helpful to look at the geometry of a single pulley. Both the primary and secondary pulleys share the same basic anatomy. Each pulley is made from two opposing conical sheaves that form a V-shaped groove. One sheave is fixed axially to the shaft, while the other can move along the shaft. The rubber V-belt sits between the sheave faces. This basic layout is shown in [Figure 1.4](#).

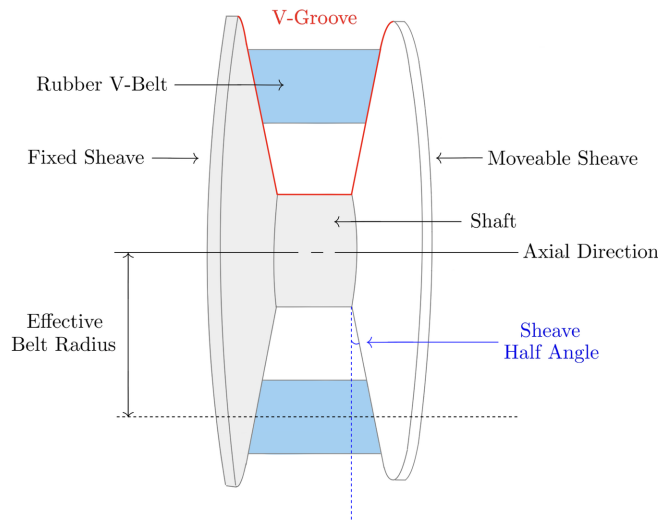


Figure 1.4: Generic anatomy of a belt-sheave CVT pulley. Each pulley consists of opposing conical sheaves that form a V-groove. Motion of the movable sheave changes the groove width and therefore changes the radius at which the belt rides.

The key geometric idea is simple: changing the axial separation between the sheaves changes the effective radius of the belt. If the movable sheave moves toward the fixed sheave, the groove becomes narrower. Since the belt has a finite width, it cannot occupy the same radial position in the narrower groove, so it is pushed outward along the conical sheave faces. The effective belt radius therefore increases.

If the movable sheave moves away from the fixed sheave, the groove becomes wider and the belt can fall deeper between the sheaves. This reduces the effective belt radius. At this level, the mechanism is just wedge geometry: axial motion of the sheave becomes radial motion of the belt. Closing the sheaves makes the belt radius larger, while opening the sheaves makes the belt radius smaller, as shown in [Figure 1.5](#).

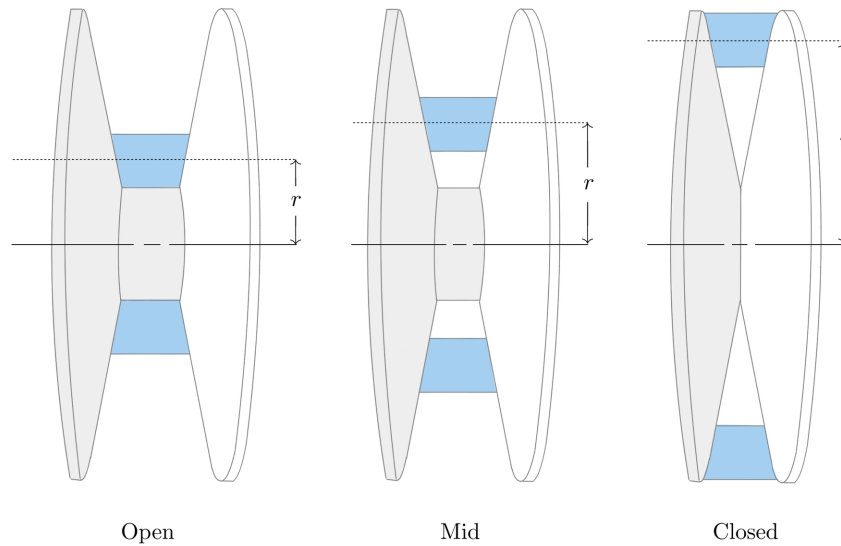


Figure 1.5: Single-pulley radius change. As the sheaves close, the V-groove narrows and the belt is pushed outward to a larger effective radius. As the sheaves open, the belt can fall deeper in the groove and the effective radius decreases.

### 1.2.3 Two-Pulley Coupling and Ratio Change

The full CVT contains two of the variable-radius pulleys described above. The important difference is that the two pulleys are connected by the same belt. Because the belt length is approximately fixed, the effective radii of the two pulleys cannot change independently.

If the belt rides outward on the primary pulley, it must ride inward on the secondary pulley. Likewise, if the primary radius decreases, the secondary radius must increase. This opposite motion is the core geometric behavior that allows the CVT ratio to change. In the physical mechanism, the actual motion of the sheaves occurs through contact forces, springs, actuator mechanisms, and the axial force balance of the pulleys.

The CVT begins launch in a low-speed, high-reduction configuration. In this state, the primary pulley has a small effective radius and the secondary pulley has a large effective radius. This behaves like a low gear in a conventional drivetrain: the engine-side pulley turns relatively quickly while the downstream side turns more slowly, amplifying torque through the drivetrain.

As the CVT upshifts, the primary sheaves close and the primary effective radius increases. At the same time, the secondary effective radius decreases. The transmission therefore moves from its launch configuration toward a higher-speed configuration. By the end of the shift, the primary radius is larger and the secondary radius is smaller. This coupled motion is shown in [Figure 1.6](#).

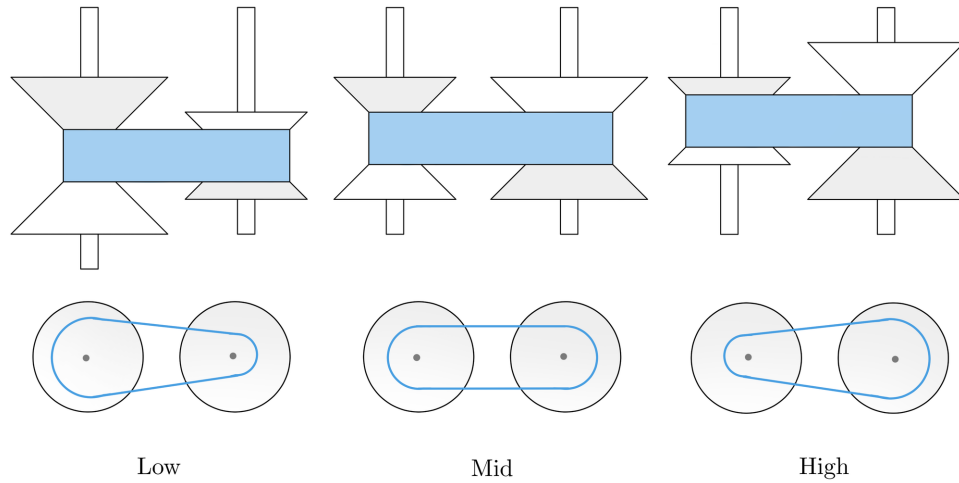


Figure 1.6: Low-, intermediate-, and high-ratio states of a belt-sheave CVT. As the belt moves outward on the primary pulley and inward on the secondary pulley, the effective radii vary in opposite directions while the belt length remains approximately constant. The lower views highlight the corresponding change in transmission ratio.

#### 1.2.4 Torque Transfer, Clamping Force, and Slip

The geometry described so far determines the speed relationship between the primary and secondary pulleys. However, a speed relationship alone does not guarantee that torque can be transmitted. Unlike a gear pair, the belt and pulley are not mechanically locked together by teeth. Torque is transmitted through frictional traction between the belt and the sheave faces.

For traction to develop, the belt must be physically squeezed between the sheave faces. [Figure 1.7](#) shows how this happens at a small portion of the belt-pulley contact. The circled region on the pulley wrap is enlarged to show the belt cross-section within the pulley V-groove. At that location, the movable sheave presses the belt toward the fixed sheave. This clamping action produces contact normal forces on the two sides of the belt, creating the contact pressure needed for frictional traction.

The same local interaction occurs continuously around the wrapped portion of the belt. At each position along the wrap, the normal forces between the belt and sheave faces allow a tangential friction force to act. Together, these small tangential traction forces transfer torque between the pulley and the belt. In practical terms, transferring more torque through the CVT requires enough clamping force to create enough available traction at the belt-pulley interface. More clamping force increases the normal contact force, which in turn increases the tangential force that friction can support.

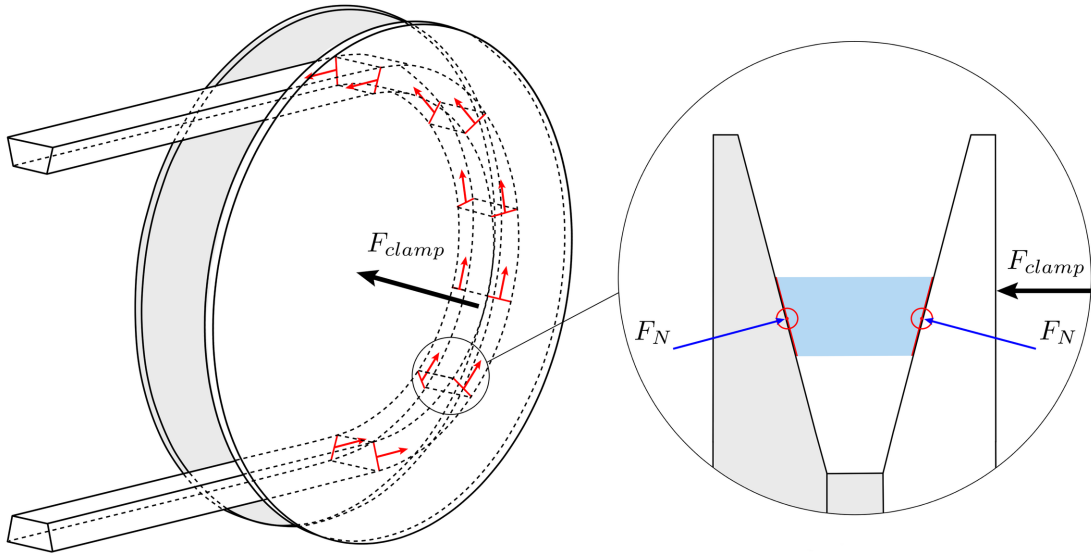


Figure 1.7: Clamping and traction in a rubber V-belt CVT. The axial clamping force  $F_{\text{clamp}}$  presses the movable sheave toward the fixed sheave. At the circled location, the enlarged cross-section shows representative local face-normal forces  $F_N$  acting on the belt. These normal forces set the frictional capacity for the local tangential traction shown around the wrap, whose distributed action transfers torque between pulley and belt.

If the available traction is not sufficient for the torque being transferred, the belt slips relative to the pulley. Slip reduces useful torque transfer and can produce heat, wear, and efficiency loss. A working CVT must therefore do two things at once: it must allow the pulley radii to change so the ratio can shift, and it must maintain enough clamping force for the belt to transmit torque without excessive slip.

This distinction is central to the rest of the model. The pulley geometry determines how the speed ratio changes. The contact mechanics determine how much torque can actually be carried through the belt–pulley contacts. A CVT is therefore not only a variable-geometry mechanism; it is also a clamping and traction problem.

### 1.2.5 Mechanical Actuation in the Present CVT

Different CVTs generate radius change and clamping force in different ways. Some systems use electronic or hydraulic actuation to command pulley motion or clamping pressure directly. The system studied here is mechanically actuated: its pulley forces arise from mechanisms inside the primary and secondary pulley assemblies rather than from an externally commanded ratio.

These mechanisms are important because they determine both how the belt moves through the ratio range and how much clamping force is available for torque transfer. The primary pulley is mainly speed-sensitive, while the secondary pulley is mainly torque-reactive.

The primary pulley uses flyweights, ramps, and a spring. As engine speed increases, the flyweights move outward under centrifugal effects. The ramp surfaces convert this outward flyweight motion

into a closing force on the primary sheaves. The spring resists this motion and helps set the speed at which the primary begins to close. As the primary closes, the belt is pushed outward, the primary effective radius increases, and the CVT shifts in the upshift direction introduced in [Figure 1.6](#). The primary actuation mechanism is shown conceptually on the left side of [Figure 1.8](#).

The secondary pulley uses a helix mechanism and spring loading. The helix can be understood like a screw-like interface: when torque is transmitted through the secondary, the angled helix surfaces convert part of that torque reaction into a clamping force between the sheaves. In this way, torque through the secondary helps create the clamping needed to carry torque through the belt. The spring provides preload and helps set the baseline clamping behavior.

The helix also means that secondary shift is not purely axial. As the secondary moves through its shift range, axial motion is coupled with relative rotation between secondary sheave components. This detail becomes important later when the secondary force model is developed, but the basic intuition is simply that the helix converts torque and relative rotation into axial clamping. The secondary actuation mechanism is shown conceptually on the right side of [Figure 1.8](#).

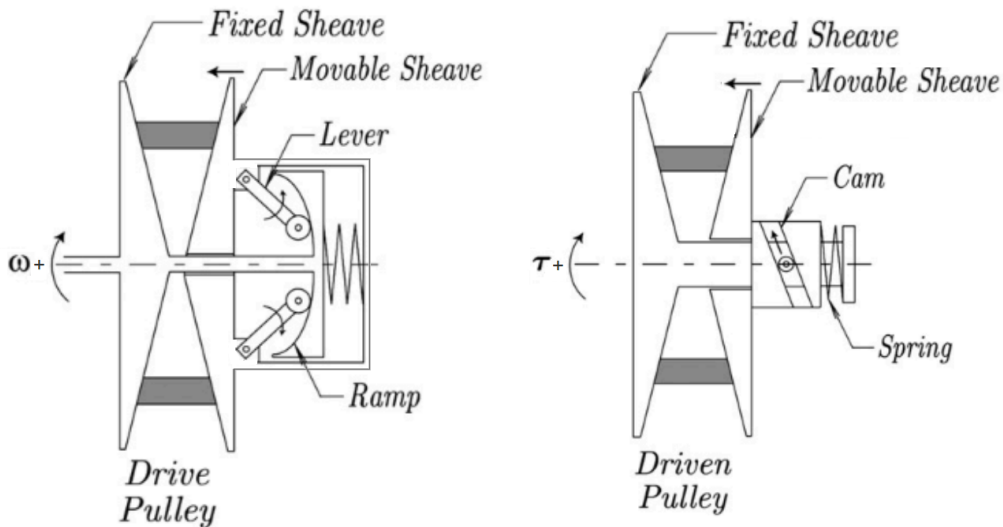


Figure 1.8: Conceptual illustration of the mechanical actuation mechanisms in the present CVT. Left: the primary (drive) pulley, where increasing rotational speed moves the flyweights outward and the ramp geometry converts that motion into a closing force on the sheaves, opposed by a spring. Right: the secondary (driven) pulley, where transmitted torque loads a cam or helix mechanism that converts part of the torque reaction into sheave clamping force, together with spring preload. In this way, the primary is mainly speed-sensitive and the secondary is mainly torque-reactive. ([Julió and Plante, 2011](#))

Although the primary and secondary mechanisms respond most strongly to different physical quantities, they act on the same belt and the same shift process. Primary speed affects centrifugal actuation, transmitted torque affects the torque-reactive secondary, the resulting clamping forces determine the available belt traction, and the belt geometry couples motion of the two pulleys. As the operating conditions change, the balance among these effects can change as well, allowing the transmission to shift in either direction or to enter a different traction state. These coupled

relationships form the physical foundation for the dynamic behaviour developed later in this work.

## 1.3 Motivation and Literature Review

CVT modelling is not a single, uniform problem. A model developed to size a variator, estimate belt efficiency, design a control strategy, or predict transient behaviour may involve the same physical components, but it will usually make different choices about which quantities are prescribed and which are solved. Some models are primarily design tools. Some focus on mechanical actuation and force balance. Some study belt forces, axial thrust, slip, or efficiency at fixed operating points. Others attempt to resolve transient shift behaviour or detailed belt–pulley contact mechanics.

The progression followed here is from design intuition, to component force balance, to mechanical actuation, to belt contact, to ratio change, and finally to general transient CVT behaviour. This ordering mirrors the way the modelling problem becomes difficult. A belt–sheave CVT can first be understood as a variable-radius transmission, then as a collection of mechanical actuators that generate axial force, then as a friction-limited belt contact problem, and finally as a coupled dynamic system in which pulley rotation, shift motion, belt behaviour, and contact state evolve together. The literature contains strong foundations for each of these pieces, but those foundations are distributed across several modelling traditions.

### 1.3.1 Framework for Classifying CVT Models

A useful way to compare CVT models is to ask what part of the system is being allowed to move, and what part is being held fixed. The same physical transmission can be treated as a geometry problem, a force-balance problem, a belt-contact problem, a shift-dynamics problem, or a coupled system-dynamics problem. Each viewpoint can be useful, but each one answers a different question. This is why the literature is best read by tracking the modelling choices that define each formulation, rather than by treating all CVT models as attempts at the same final system.

One of the first choices is how the ratio is represented. In the simplest case, the transmission is evaluated at a fixed ratio. A slightly broader approach computes the ratio from a quasi-static force balance at each operating point. Other models prescribe a ratio trajectory, or use an empirical relationship for the rate of ratio change. More detailed shift models treat ratio change as a mechanical process in its own right, involving radial belt motion, pulley motion, and contact forces (Kim et al., 2002; Sorge, 2008; Cammalleri and Sorge, 2009; Sorge and Cammalleri, 2011). These approaches are not interchangeable. A fixed-ratio model can explain forces at a known condition, while a shift model can explain how the system moves from one condition to another. Even then, a formulation that describes the mechanics of an imposed or already active shift does not necessarily determine when a shift begins, which direction it takes, or what operating condition follows it.

A second choice is how axial force is produced. Some models take pulley clamping forces as known inputs. This is natural when the goal is to study belt response under specified loading, or when the variator is externally controlled. Other models include the mechanism that generates those forces, such as centrifugal weights, springs, torque-sensitive cams, hydraulic pressure, or electronic actuation (Cammalleri, 2005; Kim et al., 2002; Skinner, 2020; Sorge and Cammalleri, 2011). The important distinction is not only the actuator type, but how completely its mechanical action is resolved. A clamping force may be prescribed directly, obtained from an instantaneous force balance or empirical relation, or produced by a mechanism whose own motion is treated dynamically. Each choice answers a different question about how the pulley force arises.

A third choice is how much of the belt–pulley contact is represented. At low fidelity, the belt may be treated through a capstan-style tension relation, often adjusted for the wedge action of the V-groove. More detailed models include radial and tangential friction, sliding angle, belt penetration, bending stiffness, local normal pressure, tension variation, slip mechanisms, and efficiency losses (Gerbert, 1996; Kim and Kim, 1989; Sorge, 1996; Kong and Parker, 2005, 2006; Chen et al., 1998; Bertini et al., 2014; Messick, 2018). At still higher fidelity, the belt may be discretized into nodes or elements so that local belt motion and contact behaviour are solved directly (Julió and Plante, 2011; Ballew, 2015). Increasing contact fidelity can make the local mechanics more realistic, but it does not by itself determine how much of the complete transmission response is being solved. Local mechanical fidelity and global dynamic closure are therefore separate modelling choices.

A fourth choice is how the rotating system and its interaction with the surroundings are treated. Many models prescribe one or both pulley speeds, or impose the speed ratio algebraically. This is often appropriate for steady contact analysis, efficiency prediction, or controlled operating-point studies. Other models solve rotational equations of motion so that pulley speeds and transmitted torques evolve dynamically (Srivastava and Haque, 2007; Ballew, 2015; Duan et al., 2016). Any physical transmission must, of course, interact with whatever machinery is attached to it. The important distinction is whether those external systems supply only the mechanical loading associated with that interaction, or whether quantities belonging to the internal CVT mechanics—such as pulley-speed histories, clamping-force histories, axial pulley motion, or ratio trajectories—must also be prescribed. This distinction determines how much of the transmission operating state is actually predicted by the model.

A further choice is the operating region over which the same formulation remains valid. Some models are intended for steady or fully engaged operation. Others describe transient shifting once the variator is already engaged, focus specifically on clutching or engagement, or move through a predefined sequence of operating phases. A broader formulation must determine which mechanical regime is active as conditions change and remain valid as the transmission enters or leaves those regimes. A model may therefore reproduce behaviour accurately within one operating phase without necessarily providing the mechanics that determine how the transmission reaches, leaves, or reverses that phase.

These questions form a useful progression for understanding the modelling problem, but the associated modelling choices are partly independent. A formulation may resolve belt contact in great detail while prescribing global pulley motion. Another may solve pulley rotation dynamically while receiving clamping force as an external input. Another may model mechanical actuation carefully while treating ratio change through quasi-static or empirical relations. None of these choices is inherently wrong; they reflect the intended modelling objective. The important point is that model fidelity in one part of the CVT does not by itself imply closure of the complete transmission response.

The following review therefore considers both what each formulation was developed to predict and what quantities or operating conditions remain prescribed. It proceeds from practical and quasi-static design models, through mechanical actuation and belt-contact mechanics, to transient rubber-belt formulations and related dynamic models in metal-belt and chain CVTs. This provides a consistent basis for identifying which parts of the CVT dynamics are already well established and where the remaining modelling gaps lie.

### 1.3.2 Practical Design and Tuning Foundations

The practical understanding of mechanically actuated rubber V-belt CVTs developed largely through machine design and tuning practice. General machine design references such as [Budynas and Nisbett \(2015\)](#) provide the background needed to treat the CVT as a real mechanical assembly involving belts, springs, clutches, shafts, friction interfaces, uncertainty, and design iteration. Not all of these effects are included in a compact model, but they provide useful context for deciding which behaviours are retained and which are idealized.

Practical tuning literature provides a more application-specific foundation. [Aaen \(2007\)](#) describes powersport CVTs using the language of engagement speed, shift speed, flyweights, primary springs, secondary springs, torque sensing, backshift, and speed variance. Its value is not that it provides a complete dynamic model, but that it builds component-level intuition from the perspective of a tuner or racer. It explains how changes to weights, springs, ramps, and helices are expected to affect behaviour, while leaving much of the coupled system response to experience and testing.

These sources are valuable because they describe the physical machine before it is reduced to equations. They also show why prediction is difficult: the same hardware change can affect engagement, shift rate, belt grip, backshift, engine speed, and vehicle acceleration at the same time. Qualitative tuning rules are therefore useful starting points, but they are not enough to determine the complete transient response of the CVT under changing operating conditions.

### 1.3.3 Design-Oriented and Quasi-Static CVT Models

Design-oriented CVT models commonly evaluate the transmission at a fixed operating condition or through a sequence of quasi-static operating points. Rather than asking how every transient state evolves in time, these models usually ask which geometry, actuator, belt, or tuning parameters should be selected to satisfy a desired operating requirement. In a fixed-operating-point analysis, quantities such as ratio and pulley speed may be specified while the corresponding forces, tensions, or capacities are calculated. In a quasi-static model, the operating condition may instead be obtained from an instantaneous force balance, while inertia and the time required to move between neighbouring equilibria are neglected. These approaches commonly relate vehicle requirements, pulley geometry, belt length, wrap angle, speed ratio, flyweight force, spring preload, helix force, belt tension, and torque capacity. Their main value is that they connect tunable hardware to expected operating behaviour.

A useful starting point is the numerical design method of [Arango and Muñoz Alzate \(2020\)](#), which begins from vehicle requirements, road specifications, and desired performance before working toward the CVT design. This is valuable because it places component selection in its proper context: the belt, pulley geometry, speed-ratio range, engine or motor selection, and mechanical centrifugal actuation are not independent choices, but responses to the requirements imposed by the vehicle and its operating route. In this sense, the work provides a design-level framework where system requirements drive CVT component choices.

Other design-oriented works focus more narrowly on particular parts of the variator design. [Cammalleri \(2005\)](#) develops a speed-torque-controlled rubber V-belt variator design using equilibrium relationships for a centrifugal driver and torque-sensitive driven pulley. [La Battaglia et al. \(2022\)](#) focuses on the kinematic design of motorcycle rubber V-belt CVTs, emphasizing the roller sliding profile of the front pulley as a critical design variable. Both works show how simplified mathematical models can be used to choose actuator or geometry parameters without requiring a complete

transient simulation of the transmission.

[Skinner \(2020\)](#) provides a practical force-balance and tuning-oriented model. The work emphasizes quantifiable tuning objectives such as engagement speed, shift-out speed, static RPM shifting, and efficient power transfer, and relates those objectives to the primary and secondary subsystems. It is useful as an applied tuning reference because it connects the variables adjusted by teams to predicted operating behaviour, while remaining primarily an operating-point and force-balance model.

The common strength of these works is that they make CVT design less arbitrary. They identify the parameters that matter and show how geometry, springs, flyweights, helices, belt forces, and vehicle requirements can be related. They also highlight a recurring tension in CVT modelling: models that are simple enough to support early design may omit important coupled behaviour, while models that resolve more detailed mechanics can become difficult to apply directly as design tools. In this sense, design-oriented and quasi-static models occupy an important middle ground. They provide usable relationships for component selection and tuning, but they do not generally determine the complete time evolution of engagement, traction, shift motion, and pulley response.

### 1.3.4 Mechanical Actuation and Clamping Force Generation

After the basic design variables are identified, the next modelling question is how the CVT generates the axial forces that clamp the belt and drive ratio change. In externally controlled CVTs, these forces may be treated as prescribed hydraulic or electronic inputs. In mechanically actuated rubber V-belt CVTs, however, the forces are produced by the hardware itself. The primary force is typically speed-sensitive, while the secondary force is torque-sensitive.

[Cammalleri \(2005\)](#) provides an important design treatment of this problem through a speed-torque-controlled rubber V-belt variator. The centrifugal driver and torque-sensitive driven pulley are treated as part of the variator design, with equilibrium relationships connecting actuator geometry, spring loading, pulley torque, and the intended operating behaviour of the CVT. This establishes a quantitative link between the mechanical hardware and the clamping forces it is designed to produce.

[Kim et al. \(2002\)](#) also study a rubber belt CVT with mechanical actuators, including both steady-state and transient characteristics. Their driver and driven thrusts are obtained from the mechanical actuator geometry and instantaneous force balances, while the transient ratio response is represented separately by an experimentally identified first-order relationship. The work therefore connects mechanical primary and secondary actuation to transient CVT behaviour, but the motion of the actuator hardware itself is not what determines the shift acceleration.

[Sorge and Cammalleri \(2011\)](#) address the secondary side in substantially greater mechanical detail through the helical shift mechanics of rubber V-belt variators. Their formulation allows the fixed and movable driven faces to rotate differently and resolves how the belt-contact state determines the torque carried by the movable face. The resulting helix response therefore depends on spring loading, helix geometry, transmitted torque, and the internal relative motion of the driven pulley rather than on a simple assumed torque split. The instantaneous pulley radius and shift speed are nevertheless supplied to the contact calculation, so the resulting actuator relation determines the forces associated with a specified shift state rather than the acceleration of the movable secondary assembly itself.

Together, these works establish much of the mechanical foundation needed to model passive

CVT actuation. They show how centrifugal primary mechanisms and torque-reactive secondaries generate configuration-dependent axial forces, and how detailed actuator geometry can materially change those forces. They also illustrate an important distinction introduced in [Section 1.3.1](#): including the actuator mechanism does not necessarily mean that the actuator is itself a dynamically propagated subsystem. In these formulations, the actuator force is generally closed configuration by configuration, while transient ratio motion is either outside the model or supplied through a separate shift description. This distinction becomes important when the actuator hardware has inertia and its motion is coupled directly to the shift and shaft dynamics.

### 1.3.5 Rubber V-Belt Contact, Slip, and Efficiency Models

Once axial force is generated by the actuation mechanisms, the belt-pulley interface determines what that force actually does. The sheaves may clamp the belt, but the resulting torque capacity, axial reaction, radial belt motion, slip, and loss all depend on how the rubber belt interacts with the pulley groove. A rubber V-belt therefore cannot be treated as an ideal kinematic constraint. It wedges into the sheaves, develops a tension distribution around the wrap, experiences radial and tangential friction, deforms under load, and dissipates energy through several mechanisms.

Classical belt theory gives the starting point: friction and wrap angle determine the maximum tension ratio that can be supported before gross sliding occurs. For V-belts, this picture is modified by the groove angle and the associated wedge effect. However, real belt slip is more complicated than a single capstan-style traction limit. [Gerbert \(1996\)](#) presents a unified treatment of belt slip in which creep, radial compliance, shear deflection, and seating/unseating effects all contribute to speed loss. The important point is not that every model must include all of these effects, but that “slip” is not one physical mechanism. A model should be clear about which slip and loss mechanisms it represents and which it leaves outside its scope.

The contact problem also determines the axial reaction felt by the sheaves. [Kim and Kim \(1989\)](#) developed a theoretical analysis of axial forces in a rubber V-belt CVT using simplified assumptions about the contact regions and belt force distribution. [Sorge \(1996\)](#) developed a compact model for axial thrust in V-belt drives, emphasizing radial penetration and the connection between belt forces and pulley thrust. These works are useful because they connect the belt contact state back to the sheave force balance: axial force is not only generated by the actuator, but also shaped by how the belt seats, slides, and transmits load in the groove.

A more detailed treatment is given by [Kong and Parker \(2005\)](#), who formulate the steady mechanics of belt-pulley systems using distributed belt equations. [Kong and Parker \(2006\)](#) extend this framework to pulley grooves and sliding friction. This line of work resolves local belt mechanics at much greater fidelity than simple force-balance models, including the distribution of forces and sliding behaviour around the contact arc.

[Messick \(2018\)](#) extends this contact-mechanics tradition for rubber V-belt CVTs by building on the Kong and Parker framework and validating axial-force and efficiency predictions experimentally. Messick’s work is valuable for its contact-region formulation, belt property measurement, operating-range analysis, and dynamometer validation. It is also clear about its modelling scope: shifting is treated as a quasi-equilibrium process rather than as a transient response-time problem. In this sense, it is a strong reference for belt force and efficiency modelling, while leaving the time evolution of the complete CVT operating state to a different class of model.

Efficiency-focused studies provide another view of the same interface. [Chen et al. \(1998\)](#) experi-

mentally investigated speed and torque losses in a rubber V-belt CVT, while [Bertini et al. \(2014\)](#) developed an analytical model for rubber V-belt CVT power losses including sliding, hysteresis, and entry/exit effects. These works show that transmission losses arise from several interacting mechanisms whose relative importance depends on the operating condition.

The contact and efficiency literature therefore provides the foundation for understanding how belt force, axial thrust, slip, and loss arise. It also reinforces the distinction between local fidelity and global dynamic closure. A high-fidelity contact model may be necessary for efficiency prediction, axial-force prediction, or detailed local belt mechanics. A lower-dimensional contact model may instead be appropriate when the modelling objective requires the belt to remain coupled transparently to actuator dynamics, pulley rotation, shift motion, and the broader transient operating state of the CVT.

### 1.3.6 Transient Rubber-Belt Models and Dynamic Closure

The contact and force-balance literature establishes how a rubber belt CVT behaves at a specified operating condition. Transient models extend this picture by asking how the transmission moves between operating conditions and which parts of that motion must be resolved in time. In the rubber-belt literature, this has been approached through models of shift mechanics, distributed belt motion, pulley rotational dynamics, and mechanically actuated vehicle response. A useful comparison among these works is therefore which parts of the CVT state are advanced dynamically and which quantities are supplied as operating conditions or inputs.

[Sorge \(2008\)](#) provides an important bridge from quasi-static analysis to transient shift mechanics. Rather than representing ratio change as a sequence of neighbouring equilibria, the formulation treats the radial migration of the belt through the pulley grooves as a time-dependent process coupled to circumferential belt motion, mass conservation, and sliding. [Cammalleri and Sorge \(2009\)](#) later develop approximate closed-form solutions for the same class of shift problem. These works establish a mechanical basis for transient ratio change, while taking the pulley loading and active shift condition as part of the problem definition.

A more detailed transient treatment is given by [Julió and Plante \(2011\)](#), who discretize the rubber belt and apply Newton’s law to the individual belt elements. Their model predicts the time response of transmission ratio under specified drive-pulley axial force, drive-pulley speed, and load torque, while resolving belt elasticity, bending, damping, and frictional contact directly. The formulation is validated for steady and shifting conditions and provides substantially greater local belt detail than reduced force-balance models. The authors also report difficulty obtaining a stabilized prediction for the low-torque start-up condition in which belt clamping itself acts as the clutch, indicating that the operating range of the formulation is narrower than the full sequence of CVT engagement and shift behaviour.

[Ballew \(2015\)](#) extends this discretized-belt model family by removing the prescribed input-pulley speed used in earlier formulations. Applied torques, pulley inertias, and belt forces instead determine the pulley angular accelerations, allowing both shaft speeds to evolve dynamically. Engine and vehicle submodels are also included so that changes in external loading can feed back through the transmission rather than being represented by a prescribed pulley-speed history.

Axial actuation in [Ballew \(2015\)](#) remains a separate input to the belt–pulley model. The thesis notes that the axial-force inputs could be supplied either by a control system or by a model of a mechanically governed actuator. In the simulations presented, a PI controller with feed-forward

gain generates the primary axial force, while the secondary axial force is held fixed. The formulation therefore extends the dynamic closure of the rotating system substantially, while leaving the source of the pulley clamping forces to a separate actuation model.

[da Silva et al. \(2023\)](#) present a reduced mechanically actuated rubber-belt CVT model coupled to a longitudinal vehicle simulation. Axial equations of motion are written for the movable sheaves, with flyweight loading on the primary and a torque-dependent secondary contribution. The operating sequence is divided into a clutching phase followed by a fully engaged phase, and the model is validated against a full-throttle acceleration. The work therefore provides an example of movable-sheave dynamics and mechanical actuation being retained in a transient application, while using a simplified, phase-specific representation of the CVT behaviour.

Taken together, these works show a clear progression in transient rubber-belt modelling. Ratio change has been treated as a mechanical process, belt motion and contact have been propagated directly, pulley rotational motion has been allowed to respond to applied torques, and movable-sheave inertia has been introduced in mechanically actuated applications. The formulations differ, however, in which parts of the transmission remain prescribed or are restricted to a particular operating phase. This makes the broader CVT literature useful for examining how other variator architectures have divided the boundary between prescribed actuation and dynamically resolved transmission behaviour.

### 1.3.7 Adjacent Metal-Belt and Chain CVT Dynamic Models

The broader CVT literature contains an extensive modelling tradition for metal push-belt and chain variators, particularly in automotive applications. These systems are not direct equivalents of dry rubber V-belt CVTs, since their load-carrying mechanisms, deformation, lubrication, and actuation systems differ. They nevertheless provide useful comparisons for transient ratio change, frictional contact, pulley dynamics, and the treatment of applied clamping forces.

[Srivastava and Haque \(2009\)](#) review the dynamics and control of belt and chain CVTs and summarize a wide range of steady and transient formulations, including slip and traction models, control-oriented shift laws, and detailed belt or chain mechanics. The review illustrates both the maturity of dynamic CVT modelling outside the rubber-belt literature and the importance of keeping the underlying transmission architecture in view when comparing models.

The Carbone–Mangialardi–Mantriota (CMM) family provides a well-developed example of mechanically derived shift dynamics for metal-belt and chain variators. [Carbone et al. \(2001, 2005\)](#) derive ratio-changing behaviour from belt or chain kinematics, mass conservation, friction, and pulley deformation, including creep-mode and slip-mode shifting. Experimental work by [Yildiz et al. \(2018\)](#) validates the resulting CMM shift dynamics under load. In these tests, the predicted quantity is the speed-ratio response to a prescribed change in clamping-force ratio, while drive-pulley angular speed, driven-side clamping force, and load torque are specified operating quantities. The formulation is therefore directed at the variator shift response under known actuation and loading.

Other metal-belt studies similarly treat transient ratio change and traction behaviour directly. [Srivastava and Haque \(2005, 2008\)](#) develop dynamic formulations that retain transient belt motion and examine the effects of friction characteristics and band-pack slip. [Bonsen et al. \(2003\)](#) study variator slip and its relationship to transmitted torque and clamping. These works extend the treatment of shift and traction beyond steady operating-point analysis while retaining the controlled clamping architecture typical of metal-belt automotive CVTs.

[Schindler et al. \(2012\)](#) develop a spatial transient multibody model of a pushbelt variator with detailed representation of the pulley sheaves, ring packages, friction, and unilateral contact. The model is used to study the internal transient response of the pushbelt system, including deformation and three-dimensional contact behaviour. In the simulations presented, the output pulley receives prescribed clamping force and load torque, while the input pulley is assigned angular-velocity and clamping-velocity histories. The detailed belt and pulley response is therefore resolved under a specified global variator excitation.

Chain CVTs provide another example of strongly coupled transient mechanics. [Duan et al. \(2016\)](#) include pulley rotational dynamics, movable-pulley axial dynamics, chain radial motion, chain and pulley deformation, and a speed-dependent friction model spanning micro-slip and macro-slip. The axial and rotational pulley dynamics are coupled directly through the chain forces. The movable-pulley equation is driven by an applied clamping force, however, so the generation of that clamping force remains outside the variator model.

These adjacent models show how transient shift mechanics, detailed contact behaviour, pulley rotation, and movable-pulley motion can be represented at substantially different levels of detail. They also illustrate a recurring modelling boundary: in hydraulically or otherwise externally actuated variators, clamping force and pulley actuation are naturally supplied as inputs to the transmission model. For a mechanically actuated rubber V-belt CVT, those same functions are produced by hardware contained within the transmission, making the treatment of the actuation mechanism itself an important part of how the model boundary is defined.

### 1.3.8 Synthesis of Prior Work

The literature does not point to a single missing ingredient in CVT modelling. Instead, several mature lines of work resolve different parts of the same machine. Practical and design references establish the component relationships used to size and tune rubber V-belt CVTs. Mechanical-actuation studies explain how centrifugal and torque-reactive hardware generate pulley forces. Contact and efficiency models resolve the belt–pulley interface at progressively greater local detail. Transient formulations add radial belt motion, ratio change, pulley rotation, movable-sheave dynamics, or distributed belt and chain motion. The resulting body of work provides strong foundations for the individual mechanisms required in a dynamic transmission model.

A recurring distinction among these formulations is the location of the model boundary. Any CVT must receive mechanical interaction from the systems attached to it, but the quantities treated as external inputs differ substantially among models. Depending on the modelling objective, pulley speed, axial force, clamping pressure, ratio motion, or actuator commands may be prescribed while the remaining transmission response is solved. These choices are appropriate for many component, control, and contact studies, but they also determine how much of the CVT itself is contained within the dynamic system.

For a mechanically actuated rubber V-belt CVT, the mechanism that generates clamping is itself part of the transmission. Its motion therefore cannot be separated completely from the response that it produces: the actuation, pulley, belt, and rotational mechanics form one coupled physical system. No formulation identified in the reviewed literature was found to close that complete mechanically actuated system dynamically from the loading applied at the primary and secondary shafts. Treating the clamping mechanism dynamically is one part of this gap, but the broader requirement is that the internal CVT response be solved as a coupled whole rather than assembled from separately prescribed or separately closed pieces.

A separate gap concerns the range of operating behaviour covered by the same formulation. Much of the literature is developed for an already engaged variator, a specified shift, or a prescribed sequence of operating phases. Individual works address clutching, upshift and backshift, ratio limits, or traction behaviour, but no formulation identified in the reviewed literature was found to remain mechanically valid across the complete operating sequence of disengagement and engagement, seated ratio limits, free shifting, and traction loss and recovery without imposing a predefined operating phase or trajectory. This is a different requirement from accurately modelling any one of those operating conditions: it requires the same physical formulation to remain valid as the active mechanical constraints change.

The remaining modelling opportunity is therefore defined by two independent forms of closure. The first is system-level closure of the complete mechanically actuated CVT, so that its internal state evolves as one coupled mechanical system in response to the loading applied at the primary and secondary shafts. The second is operating-domain closure, so that the same model remains valid as the transmission moves between its major mechanical regimes rather than being constructed around one specified phase or trajectory.

Table 1.1 summarizes selected formulations that most directly establish this progression. The table is not a ranking of model quality; each formulation reflects a different modelling objective. It instead shows which parts of the transmission are resolved within each model and which quantities remain prescribed as operating conditions or inputs.

| Formulation                                                       | Principal behaviour resolved                                                      | Actuation / clamping                                                          | Quantities supplied or separately closed                                           | Primary operating scope                                 |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------|
| <a href="#">Cammalleri (2005)</a>                                 | Mechanical variator design and equilibrium force relationships                    | Centrifugal primary and torque-sensitive secondary represented mechanically   | Operating speed, torque, and equilibrium condition                                 | Steady / quasi-static design                            |
| <a href="#">Kim et al. (2002)</a>                                 | Steady mechanical response with an empirical first-order ratio transient          | Mechanical primary and secondary thrust relations                             | Operating conditions and experimentally identified transient shift relation        | Engaged upshift and downshift response                  |
| <a href="#">Sorge (2008)</a>                                      | Transient radial belt migration, sliding, and ratio-change mechanics              | Pulley loading forms part of the specified shift problem                      | Pulley loading and active shift condition                                          | Engaged transient shift                                 |
| <a href="#">Sorge and Cammalleri (2011)</a>                       | Torque-reactive secondary helix mechanics with differential sheave rotation       | Helix and spring represented mechanically                                     | Instantaneous pulley radius and shift motion supplied to the secondary calculation | Engaged secondary shift mechanics                       |
| <a href="#">Messick (2018)</a>                                    | Distributed rubber-belt contact, axial thrust, and efficiency                     | Axial loading prescribed                                                      | Pulley speed, torque, clamping, and operating point                                | Steady / quasi-equilibrium contact and efficiency       |
| <a href="#">Julió and Plante (2011)</a>                           | Transient discretized rubber-belt motion and contact                              | Drive-pulley axial force prescribed                                           | Drive-pulley speed, axial force, and load torque                                   | Engaged steady and shifting operation                   |
| <a href="#">Ballew (2015)</a>                                     | Discretized belt dynamics with pulley rotational response to applied torques      | Primary axial force controlled; secondary axial force fixed in reported cases | Axial-force inputs; actuator mechanism remains separate                            | Engaged transient operation with dynamic shaft response |
| <a href="#">Carbone et al. (2001, 2005); Yildiz et al. (2018)</a> | Mechanically derived metal-variator ratio dynamics including creep and slip modes | Clamping-force ratio prescribed                                               | Drive speed, clamping quantities, and load torque                                  | Engaged underdrive / overdrive shifting                 |
| <a href="#">Schindler et al. (2012)</a>                           | Detailed spatial pushbelt, pulley, friction, and unilateral-contact dynamics      | Global pulley actuation prescribed                                            | Input angular and clamping velocities; output clamping force and load torque       | Detailed transient response of an engaged variator      |
| <a href="#">Duan et al. (2016)</a>                                | Coupled chain, pulley axial and rotational dynamics, deformation, and slip        | Applied clamping force drives movable-pulley motion                           | Applied clamping and external loading                                              | Engaged chain-CVT shift dynamics                        |

Table 1.1: Selected CVT formulations illustrating the progression from operating-point and component models to increasingly coupled transient descriptions. The comparison emphasizes which behaviour is resolved within each formulation and which quantities remain prescribed according to its modelling objective.

## 1.4 Aim, Scope, and Contributions

The literature review identifies two remaining modelling needs for mechanically actuated rubber V-belt CVTs: closure of the transmission as one coupled dynamic system, and a formulation that remains valid as the CVT moves between its different operating regimes. The present work addresses these needs with a first-principles transient model of the CVT itself.

### 1.4.1 Aim

The aim of this work is to develop a general transient model of a mechanically actuated dry rubber V-belt CVT whose behaviour follows from its internal mechanics and from the mechanical loading applied at the primary and secondary shafts. The model is intended to determine its own ratio motion, clamping response, pulley dynamics, belt behaviour, and traction state rather than requiring those quantities to be prescribed as an operating history. External systems may therefore be coupled to either shaft without changing the internal CVT formulation.

The same physical model is also intended to remain applicable as the transmission moves between its different operating conditions. Engagement, operation at the ratio limits, free shifting, and changes in traction state are treated as different admissible states of the same CVT rather than as separate scenario-specific models. The direction and sequence of those transitions are determined by the evolving mechanical conditions.

A further aim is physical interpretability. The derivation is organized so that predicted behaviour can be traced back to the geometry, actuation, inertial response, belt mechanics, and active constraints that produced it. The assumptions, numerical implementation, and reported simulation cases are also documented so that the formulation can be reproduced and independently examined.

### 1.4.2 Scope

The modeled system is the mechanically actuated CVT itself. It includes the primary and secondary pulley assemblies, the mechanical hardware that produces their clamping forces, and the complete rubber V-belt loop. The model is deliberately independent of any particular engine, motor, dynamometer, vehicle driveline, or other surrounding system. Instead, those systems may be coupled to the CVT through the primary and secondary shafts without changing the internal transmission formulation.

The formulation retains the rotational motion of both pulley assemblies, the axial motions required for ratio change, the motion of the retained actuation hardware, and the transport motion of the belt. The retained component masses and inertias participate in the corresponding dynamic balances, while the actuation mechanisms and belt–pulley contacts provide the forces and reactions through which these motions influence one another.

The belt and contact mechanics are intentionally represented at lower order than in distributed deformable-belt formulations. The model does not attempt to resolve the complete spatial deformation of the rubber belt, local creep and shear fields, detailed seating and unseating behaviour, full actuator tribology, flexible pulley and shaft deformation, thermal evolution, wear, or a complete transmission-loss model. The principal consequences of these and the other idealizations are stated explicitly in [Section 2.2](#). These choices define the physical resolution of the present formulation; they do not prevent more detailed component laws from being introduced in future work.

### 1.4.3 Contributions

The principal contribution of this work is a first-principles transient formulation of the mechanically actuated rubber V-belt CVT as a complete dynamic component. Its internal response is solved as one coupled mechanical system under the loading applied at the primary and secondary shafts, without requiring prescribed histories of ratio, clamping force, pulley speed, or shift direction. The same formulation is constructed to remain valid across the major operating regimes of the transmission rather than around one predefined shift or operating sequence.

Realizing this overall capability requires several specific mechanical and formulation developments. These are contributions in their own right, while also providing the physical closure needed by the complete CVT model:

1. **Dynamic primary centrifugal actuation.** The primary flyweight mechanism is treated as a moving inertial subsystem rather than as an instantaneous centrifugal-force relation. The derivation retains the effect of flyweight acceleration in the axial force balance and the configuration-dependent contribution of the flyweights to primary shaft angular momentum. During engagement and shift, flyweight acceleration changes the closing force at the same time that the changing shaft-axis inertia contributes to the rotational response of the primary.
2. **Dynamic axial-rotational coupling of the torque-reactive secondary.** The movable secondary sheave is allowed to rotate relative to the shaft as required by the helix geometry while translating axially. Its retained rotational inertia enters both the axial dynamics of the movable assembly and the angular-momentum balance of the secondary shaft. Changes in transmitted torque can alter the helix reaction, secondary clamping, and shift motion, while acceleration of the movable sheave feeds back into the secondary shaft response.
3. **Reduced transient belt, geometry, and traction closure.** The pulley geometry and fixed belt length are carried through position, velocity, and acceleration level so that the evolving pulley radii enter the dynamic balances consistently. The belt retains an independent transport motion, and the reduced wrap mechanics preserve the radial and tangential acceleration terms implied by the selected kinematics. As the radii change, the same belt mechanics determine transport acceleration, wrap loading, and the normal forces and torques that act back on the pulley and shift dynamics. They also determine whether each contact has enough static traction to remain sticking. When the required traction exceeds the static limit, that contact slides.
4. **Common mechanics across CVT operating regimes.** The formulation includes the mechanical constraints associated with primary deadzone motion, engagement, seated ratio limits, free shift, and traction-limited stick-slip operation within one regime-dependent model. Contact conditions, support reactions, and travel boundaries determine when those constraints can be retained and when the mechanics must change. If a rigid constraint is reached at finite speed, the outgoing velocities are obtained from the retained inertia and impulse-momentum balance rather than an arbitrary state reset. Starting from a physically admissible initial condition, the same mechanics can follow engagement, upshift and backshift, arrival at and release from the ratio limits, and traction loss and recovery without prescribing that sequence in advance.

Together, these developments provide the system-level and operating-domain closure identified as missing in [Section 1.3.8](#). They also preserve a direct mechanical interpretation of the model:

changes in the predicted CVT response can be traced to specific actuator, inertial, geometric, belt, or contact effects within the formulation.

## 1.5 Document Roadmap

The remainder of this work develops the model from system definition through simulation results. **Chapter 2** formalizes the CVT as a simulation problem by defining the modeled transmission, its interaction with surrounding systems, the retained assumptions and coordinates, the state variables, and the mechanical regimes through which the model evolves. **Chapter 3** then derives the governing mechanics, proceeding from pulley and belt geometry to axial actuation, pulley rotational dynamics, belt transport and contact, and finally the coupled transient model and its regime transitions.

**Chapter 4** documents the numerical implementation and begins the numerical investigation by verifying the coupled model and comparing it with an independent transient formulation. It then examines when the retained actuator and belt dynamics change the predicted behaviour before bringing these mechanisms together in full transient simulations over a common off-road course. **Chapter 5** draws together the conclusions of the study and identifies directions for extending the present formulation. The appendices collect supporting derivations, implementation details, reference inputs, and study configurations that are useful for reproducing or extending the work but would interrupt the main development.

## Chapter 2

# Model Definition and Simulation Problem

With the aim and scope established in [Section 1.4](#), the next step is to define the CVT as a precise simulation problem. This requires a clear account of what belongs to the modeled transmission, how surrounding systems interact with it mechanically, and what information describes the CVT at a given instant.

The definition proceeds from the outside inward. [Section 2.1](#) first establishes the modeled CVT and its shaft interfaces, after which [Section 2.2](#) states the physical idealizations that set the resolution of the formulation. [Section 2.3](#) then defines the coordinates and sign conventions used to describe the retained motions. With those choices fixed, [Section 2.4](#) identifies the internal state carried from one instant to the next, while [Section 2.5](#) introduces the mechanical regimes through which that state can evolve. Finally, [Section 2.6](#) brings these pieces together by distinguishing the fixed model data, current state, shaft loading, and quantities solved at an instant. Together, these definitions establish the simulation problem used by the derivation in [Chapter 3](#).

### 2.1 Modeled CVT and Shaft Interfaces

The CVT is connected to the surrounding machinery through its primary and secondary shafts. These two connections form the external mechanical interfaces of the modeled transmission. The model itself contains the primary and secondary pulley assemblies, the complete belt loop and both belt–pulley contacts, together with the primary flyweights, ramp, and spring and the secondary spring and helix. The masses and inertias retained for these components remain inside the same model boundary.

[Figure 2.1](#) shows this separation. Systems outside the CVT act on it through the two shafts, while the transmission mechanics between those interfaces are resolved by the CVT model. The surrounding machinery can therefore be changed without redefining the transmission itself.

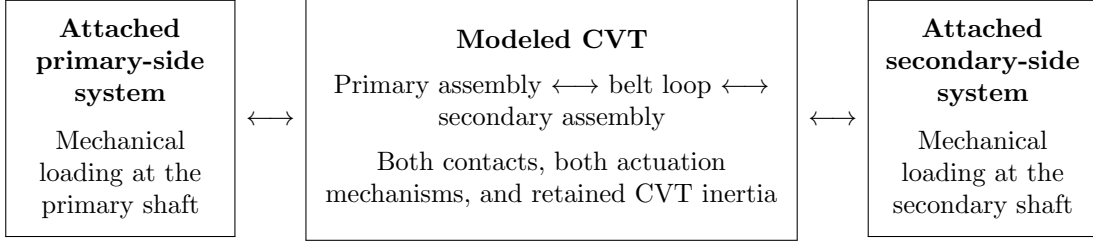


Figure 2.1: The modeled CVT and its external mechanical interfaces. Everything in the centre box is resolved as part of the transmission. An attached system acts on the CVT through the corresponding shaft, while the shaft motion remains part of the modeled response.

At either shaft, let  $j \in \{p, s\}$  identify the primary or secondary interface. The attached system contributes the mechanical loading represented by

$$\tau_{B,j}, \quad I_{B,j}.$$

Here,  $\tau_{B,j}$  is the torque exerted on the CVT through shaft  $j$ , while  $I_{B,j}$  is the equivalent inertia of the attached system as seen at that shaft. A rotor connected directly to the shaft contributes its own inertia; a wheel, drivetrain, or other mechanism connected through gearing contributes the corresponding inertia reflected to the shaft. The rotational sign convention for the shaft speed and applied torque is defined later in [Section 2.3.1](#).

These interface quantities load the CVT without prescribing its shaft motion. An attached system may determine its torque and equivalent inertia from the current shaft speed, time, and its own internal state. Those quantities then act together with the inertia and mechanics retained inside the CVT to determine the resulting shaft acceleration.

Only inertia belonging to machinery outside the CVT boundary is included in  $I_{B,j}$ . Pulley, sheave, flyweight, helix, and belt inertia retained by the CVT model are therefore excluded from the boundary inertia so that no component is counted twice. The labels *primary* and *secondary* identify the two shaft locations; they do not prescribe a direction of power flow.

## 2.2 Modeling Assumptions

The assumptions below define the physical resolution of the formulation rather than a particular operating case. Each assumption is stated together with its principal consequence for the mechanics that can and cannot be represented. The consequence summaries are not intended to enumerate every secondary effect of an idealization; they identify the main physical resolution removed by that choice so that later results can be interpreted against a clear model boundary.

### 2.2.1 Material and Mechanism Idealizations

#### Assumption 1 (Time-Invariant Material and Surface State)

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**Model statement**

The material properties and surface condition assigned to the CVT remain unchanged during a simulation. Intrinsic component dimensions are likewise not allowed to evolve through wear or thermal growth. The selected properties therefore describe the hardware throughout the simulation rather than evolving with its operating history.

---

**Consequences for model fidelity**

Temperature-dependent stiffness or friction, thermal state evolution, wear, glazing, conditioning, degradation, and other feedback from accumulated contact history are not represented. The formulation therefore predicts the mechanical response for a specified material and surface state rather than the progressive evolution of that state.

## Assumption 2 (Rigid Pulley and Actuation Hardware)

---

**Model statement**

The pulleys, shafts, actuation hardware, guides, and supporting CVT structure are represented as rigid bodies. Their prescribed geometry does not deform under the loads developed during operation.

---

**Consequences for model fidelity**

Pulley bending, shaft bending or torsion, guide and housing deflection, load-dependent groove geometry, and structural vibration generated by those compliances are outside the model. More detailed CVT formulations show that pulley deformation can alter local belt mechanics and axial loading; those effects would require compliant component models rather than changes to the rigid-body balances alone ([Carbone et al., 2005](#); [Duan et al., 2016](#)).

## Assumption 3 (Idealized Actuator Mechanics)

---

**Model statement**

The primary ramp and flyweight mechanism, secondary helix, guides, and springs follow their prescribed kinematic and force laws without additional contact friction, backlash, or local compliance.

---

**Consequences for model fidelity**

Ramp or roller friction, pin–helix friction, roller or pin spin, guide friction, backlash, helix-flank transitions, contact compliance, and unmodelled spring hysteresis or internal damping are omitted. Relaxing this assumption would add actuator friction and compliance to the same mechanism free bodies; for example, contact friction appears explicitly in more detailed centrifugal and helix treatments ([Kim et al., 2002](#); [Sorge and Cammalleri, 2011](#)).

## Assumption 4 (Fixed, Uniform Belt Cross-Section)

### Model statement

The same prescribed cross-section is assigned everywhere along the belt and is carried unchanged as the belt runs and shifts. This description fixes the cross-sectional shape and dimensions and treats mass density as uniform through the section and along the belt. The belt therefore has a constant mass per unit length, and its cross-sectional centre of mass coincides with the geometric centroid. The cord layer is retained only as a prescribed tension-carrying line at the fixed depth  $d_{\text{cord}}$ ; its distinct density and contribution to the through-section mass distribution are not resolved. As the section follows the curved modeled belt path, it does not compress, bulge, shear, warp, or otherwise change its cross-sectional shape under centrifugal or sheave loading.

### Consequences for model fidelity

Accordingly, neither spatial variations caused by cogs, ribs, joints, local reinforcement, manufacturing variation, or nonuniform wear nor load-induced changes caused by lateral and radial compression, bulging, cross-sectional shear and warping, centrifugal cross-sectional deformation, or local cog and rib deformation are represented. Concentrated cord mass and differences between the densities of the belt layers cannot shift the modeled centre of mass away from the geometric centroid. Changes in belt width, height, contact geometry, mass distribution, and internal strain energy caused by these omitted effects therefore lie outside the present formulation. Detailed belt models retain some of these spatial and compliance effects explicitly ([Gerbert, 1996](#); [Kong and Parker, 2005](#); [Messick, 2018](#)).

## 2.2.2 Belt Geometry and Kinematics

## Assumption 5 (Aligned Pulley Axes and Fixed Centre Distance)

### Model statement

The primary and secondary pulley axes remain parallel at a fixed spacing. The pulley centres and shaft directions therefore do not change as the CVT shifts.

### Consequences for model fidelity

Shaft misalignment, eccentricity, geometric runout, bearing-centre motion, and the associated changes in belt path and face loading are not represented. Removing this assumption would require the belt geometry and contact loading to respond to the additional shaft and bearing motion.

## Assumption 6 (Fixed Belt Length)

### Model statement

The modeled belt has a fixed total length. Longitudinal extension is not treated as an independent deformation of the belt, so every admissible pulley configuration must be compatible with the same installed length.

---

**Consequences for model fidelity**

Tension-dependent elongation, longitudinal strain energy, extension waves, permanent elongation, and the longitudinal component of elastic creep are not represented. Detailed traction-belt models can retain these effects through a longitudinal compliance field. In the present formulation, geometric ratio change instead follows from pulley-radius motion constrained by the fixed belt length ([Gerbert, 1996](#); [Kong and Parker, 2005](#); [Messick, 2018](#)).

## Assumption 7 (Single Belt Transport Speed)

---

**Model statement**

The circumferential motion of the complete belt is represented by one signed transport speed,  $v_b$ , shared by the whole belt loop. Belt material still moves around the loop, but the model does not retain an independent transport history for individual material points or a spatial field of local tangential belt speed.

---

**Consequences for model fidelity**

Local differences in tangential material speed that arise as the belt moves through changing pulley geometry, together with the associated momentum transport, are not resolved. Because the belt properties are uniform around the loop and no local material history is carried, the model also does not distinguish which particular material point occupies a given location. It still retains acceleration of the common transport speed and motion of the prescribed wrap geometry. Shift-mechanics and discretized belt formulations resolve the omitted local transport in greater detail ([Sorge, 2008](#); [Cammalleri and Sorge, 2009](#); [Julió and Plante, 2011](#)).

## Assumption 8 (Prescribed Planar Belt Path)

---

**Model statement**

At each instant, the belt is taken to wrap each pulley along a circular arc and to pass between the pulleys along two straight tangent spans in one common plane. The belt path is therefore set by the current pulley geometry rather than solved as a flexible dynamic curve.

While both pulleys are engaged, the belt is treated as taut and perfectly flexible. It carries load along its length through longitudinal tension, but it carries no compression, transverse shear force, or bending moment. The belt also remains in continuous contact with the sheave faces over each prescribed wrap. This engaged description is physically admissible only while the belt tension and distributed normal contact loading remain nonnegative.

---

**Consequences for model fidelity**

Slack or buckled spans within the engaged CVT, local belt lift-off, partial wrap, and contact reattachment are not represented. Bending stiffness, flexural strain energy, and bending hysteresis are also absent even though the prescribed belt path is curved. A calculated negative tension or normal loading therefore indicates that the assumed engaged path is no longer physically admissible; it does not represent a physical negative belt load.

Progressive or spiral radial migration within a wrap, seating and unseating trajectories, span sag, transverse belt vibration, and transverse inertia associated with straight-span reorientation are omitted. The same planar description also excludes the out-of-plane span skew, twist, and axial belt inertia that would arise when the two physical groove centreplanes move differently during shift. More detailed belt and shift-mechanics models resolve subsets of these path and contact effects explicitly (Sorge, 2008; Kong and Parker, 2005; Messick, 2018).

---

**Assumption 9 (Reduced Belt Motion While the Primary Is Disengaged)**

---

**Model statement**

While the primary sheaves are separated from the belt, no primary sheave normal resultant, tangential traction, or belt torque is retained. Any radial support provided at the minimum primary radius is not assigned a tangential contact law. Belt pretension and the changing shape of the slack spans are not retained as additional mechanical states.

To close the disengaged-primary mechanics without introducing a separate slack-belt description, the belt is instead taken to remain seated on the fully closed secondary and to be carried without slip at the secondary effective radius. The belt transport speed and secondary speed are therefore joined by one kinematic constraint throughout the disengaged-primary regime.

---

**Consequences for model fidelity**

The disengaged belt and secondary do not have independent transport coordinates. Slack-span motion, sag, belt whip, intermittent primary rubbing, secondary slip, and tension waves created as slack is taken up cannot be predicted. The model also cannot determine deadzone belt pretension or the transient contact loads produced by a detailed re-engagement history. Within this reduced closure, the entire belt instead moves at the speed imposed by the secondary contact.

### 2.2.3 Belt–Pulley Contact Representation

---

**Assumption 10 (Gross Tangential Coulomb Traction)**

---

**Model statement**

During continuous motion, each belt–pulley interface is represented by a dry gross stick–slip law with fixed static and kinetic friction coefficients for the selected surface state. The frictional

contact action retained by the model acts in the circumferential belt-transport direction.

---

**Consequences for model fidelity**

The formulation can distinguish torque-carrying stick from gross kinetic slip, but it does not resolve radial friction, local creep or microslip, detailed presliding behaviour, lubrication or Stribeck effects, or friction variation with pressure, temperature, conditioning, or local sliding speed. Friction associated specifically with radial migration, seating, and unseating is also outside the retained contact direction. These mechanisms are treated separately in higher-fidelity rubber-belt contact models ([Gerbert, 1996](#); [Kong and Parker, 2006](#)).

---

**Assumption 11 (Wrap-Level Contact Resolution)**

---

**Model statement**

Belt tension and normal loading may vary around a pulley wrap, but each wrap is assigned one overall traction state at an instant. Different portions of the same wrap therefore cannot independently occupy different stick-slip states. The contact is not resolved through the belt width or depth.

---

**Consequences for model fidelity**

Local adhesion and sliding subregions, spatial traction reversals, detailed pressure concentrations, edge loading, and cross-sectional contact-stress fields cannot be predicted. Resolving those effects requires a distributed or discretized contact description of the type used in higher-fidelity belt models ([Kong and Parker, 2006](#); [Messick, 2018](#); [Julió and Plante, 2011](#)).

---

**Assumption 12 (Symmetric Sheave-Face Sharing)**

---

**Model statement**

The two sheave faces at a pulley are assumed to share the total normal contact load equally. On the secondary pulley, the belt torque is also divided equally between the fixed and movable faces for the purpose of the movable-sheave balance. The primary faces rotate together with their shaft, whereas helix motion allows the secondary movable face to rotate relative to the fixed face. The face contacts are nevertheless represented by the single secondary traction state of [Assumption 11 \(Wrap-Level Contact Resolution\)](#) rather than solved separately.

---

**Consequences for model fidelity**

Face-to-face differences in normal loading, transmitted torque, and local sliding state are not resolved. The aggregate secondary contact therefore uses one representative surface speed consistent with the equal torque sharing, without making the two physical face speeds equal. Helix-induced rotation of the movable face remains in the actuation and rotational dynamics; what is omitted is the separate contact response of the two faces. Secondary sticking therefore

denotes gross sticking of the aggregate contact, rather than simultaneous pointwise adhesion to both physical faces during helix motion. Face-resolved shift models show that the two secondary face torques and sliding conditions can depart substantially from this aggregate description during shifting ([Sorge and Cammalleri, 2011](#)).

#### 2.2.4 Boundary and Transition Idealizations

##### **Assumption 13 (Instantaneous, Perfectly Inelastic Constraint Activation)**

###### **Model statement**

When a mechanical stop or a new kinematic constraint becomes active at finite relative velocity, its finite contact duration and local compliance are replaced by an instantaneous, perfectly inelastic event. This treatment applies to arrival at a mechanical travel stop, first belt engagement, and activation of the deadzone belt lock.

Positions remain continuous through the event, but velocities may change. The outgoing velocities are determined by impulse–momentum balance using all masses and inertias retained by the continuous mechanics while enforcing the newly active kinematic constraints. Applied torques, spring forces, damping forces, and other bounded forces act over no resolved time during the event and therefore contribute no finite impulse. A constraint release produces no impulse unless another constraint activates at the same instant.

###### **Consequences for model fidelity**

The event model does not resolve contact duration, local indentation, peak impact force, stress waves, rebound, chatter, or the high-frequency vibration created by compliant contact. Individual velocities are neither preserved nor set independently; incoming momentum is redistributed among the retained motions that remain compatible with the new constraints. Because the event is perfectly inelastic, kinetic energy need not be conserved. The decrease in kinetic energy is retained as the inelastic-capture loss, but the model cannot divide that loss among local belt deformation, material damping, sound, heat, or structural vibration. Predicting those details would require a finite-duration compliant-contact model. An impulsive load applied through an external shaft boundary would likewise require an explicit boundary impulse rather than the continuous torque description.

##### **Assumption 14 (Tangential Contact Through Boundary Events)**

###### **Model statement**

A belt–pulley contact that is sticking immediately before an instantaneous boundary event remains constrained against tangential slip through the event. Contacts that are sliding, together with contacts created by the event, are not assigned a new impulsive no-slip constraint. Their relative tangential motion after the event, together with the ordinary continuous traction limits, determines the stick–slip state that follows.

---

**Consequences for model fidelity**

An existing sticking contact may transmit whatever tangential constraint impulse is required to preserve no slip; that impulse is not checked against a separate impulse-level friction bound. The model therefore cannot predict impact-induced loss of an existing stick, brief impulsive slip followed by readhesion, or a transition caused by exhausting an impulse-level friction capacity.

Conversely, normal contact or kinematic capture does not automatically impose tangential sticking at a newly formed belt–pulley interface. In particular, a newly formed primary contact may retain tangential relative motion and begin its continuous motion in slip. Resolving the omitted impulsive friction behaviour would require an impulse-level traction law coupled to the normal constraint impulse.

### 2.2.5 Other Omitted Losses

---

**Assumption 15 (Omitted Parasitic Component Losses)**

---

**Model statement**

Bearing, seal, windage, and auxiliary-hardware losses are neglected unless they are introduced explicitly by a component or external boundary model. No global empirical efficiency factor is inserted to represent these omitted mechanisms.

---

**Consequences for model fidelity**

Energy loss and damping within the CVT arise only from mechanisms represented explicitly by the formulation, including gross kinetic slip and perfectly inelastic constraint-activation events. Predicted transmission efficiency may therefore be optimistic when the omitted component losses are significant, and it should not be interpreted as a complete transmission-loss prediction.

**Resulting fidelity** Taken together, these assumptions define the physical resolution used by the remainder of the formulation. The model retains rigid-body actuation, pulley rotation, sheave-shift motion, one global belt-transport motion, wrap tension and normal loading along the prescribed belt path, gross tangential stick–slip behaviour, and momentum redistribution when a new mechanical constraint activates. These are the motions and interactions resolved by the model; the assumptions above state where additional material, contact, structural, or loss mechanisms have been left outside that resolution.

## 2.3 Coordinate and Sign Conventions

This section defines the coordinates and sign conventions used throughout the formulation.

### 2.3.1 Rotational and Belt-Transport Directions

Figure 2.2 shows the two pulley assemblies face-on, with their parallel shafts extending away from the viewer. Let  $\omega_p$  and  $\omega_s$  denote the angular velocities of the primary and secondary pulley assemblies, respectively. In this view, counterclockwise is positive for both pulleys, as shown by the arrows in

the figure. Thus,  $\omega_p > 0$  and  $\omega_s > 0$  denote counterclockwise rotation, while negative values denote clockwise rotation.

Let  $v_b$  denote the signed transport speed of belt material around the loop. The positive direction is the direction shown in **Figure 2.2**: from the secondary pulley to the primary pulley along the upper span, and from the primary pulley to the secondary pulley along the lower span. Thus,  $v_b > 0$  denotes transport in this direction, while  $v_b < 0$  denotes transport in the opposite direction.

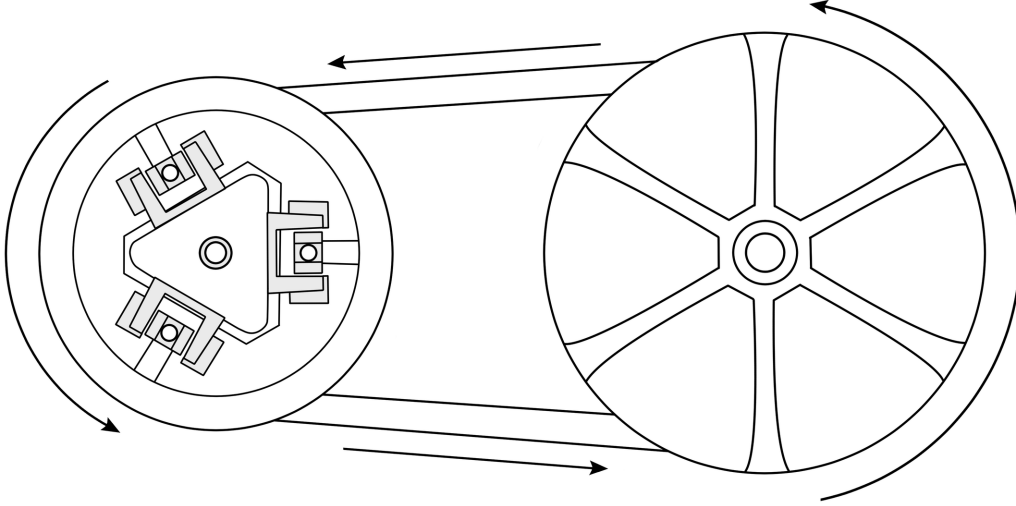


Figure 2.2: Rotational and belt-transport sign convention viewed from the outboard pulley faces, looking inward along the parallel shaft axes. Counterclockwise rotation is positive for both pulleys. With the primary pulley on the left and the secondary pulley on the right, positive belt transport runs from secondary to primary along the upper span and from primary to secondary along the lower span.

Torque follows the corresponding shaft convention. A torque acting on shaft  $j \in \{p, s\}$  is positive when it acts in the rotational sense defined by  $\omega_j > 0$ . In particular, the interface torque  $\tau_{B,j} > 0$  acts on the CVT in this positive direction. The sign is fixed by the rotational coordinate, not by whether the torque aids or opposes the shaft's instantaneous motion.

### 2.3.2 Local Axial Sheave Coordinates

Each movable sheave is assigned a local axial coordinate. Let  $x_p$  denote the position of the primary movable sheave and  $x_s$  the position of the secondary movable sheave. More generally,  $x_j$  denotes the corresponding coordinate for pulley  $j \in \{p, s\}$ .

For either pulley,  $x_j = 0$  denotes the fully open position. Increasing  $x_j$  moves the movable sheave toward the fixed sheave and closes the pulley, while decreasing  $x_j$  opens it. If  $x_{j,\max}$  denotes the available travel, then

$$x_j = 0 \quad \text{is fully open,} \quad x_j = x_{j,\max} \quad \text{is fully closed.}$$

The velocity follows the same convention:  $\dot{x}_j > 0$  denotes closing motion and  $\dot{x}_j < 0$  denotes opening motion. The fully open position is only the geometric origin of the coordinate; it does not prescribe the configuration from which a simulation must begin.

**Figure 2.3** illustrates this convention on the primary pulley. The secondary uses the same local definition. Because the two coordinates are local to their respective sheave pairs, their positive directions may point in opposite physical directions along a common global axis; in both cases, positive axial motion closes the corresponding pulley.

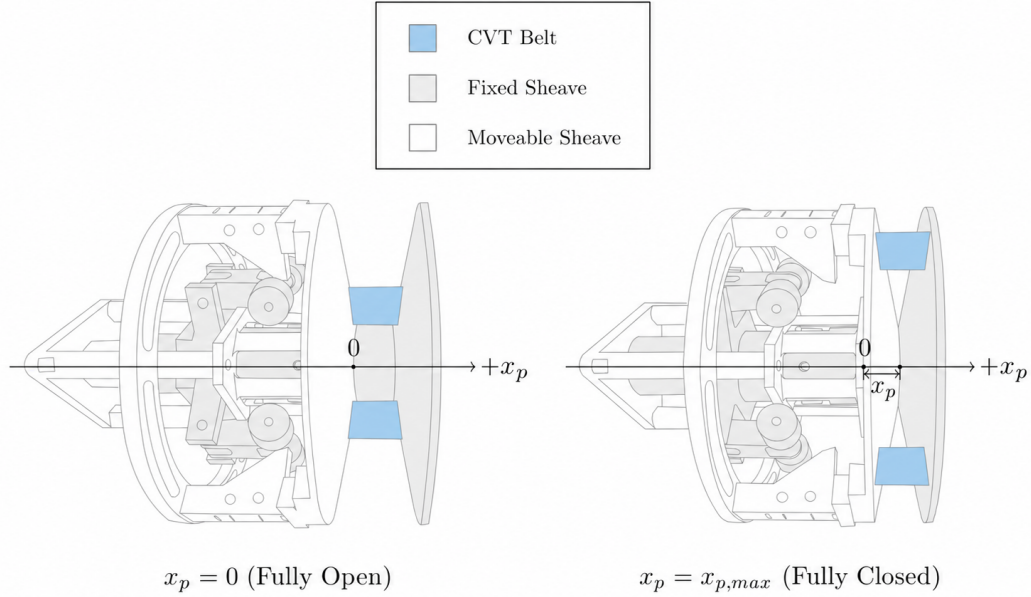


Figure 2.3: Local axial convention illustrated on the primary movable sheave.  $x_p = 0$  denotes the fully open position and increasing  $x_p$  moves the movable sheave toward the fixed sheave. The fully closed position is  $x_p = x_{p,max}$ . The secondary uses the same local convention for  $x_s$ .

### 2.3.3 Local Radial, Tangential, and Axial Directions

At either pulley, the radial and tangential directions lie in the plane of pulley rotation. At any point around the pulley, the radial unit vector  $\hat{\mathbf{e}}_r$  points outward from the pulley axis. The tangential unit vector  $\hat{\mathbf{e}}_\theta$  is perpendicular to  $\hat{\mathbf{e}}_r$  and points in the direction produced by the positive rotation defined in [Section 2.3.1](#). A radial or tangential component is positive when it points along its corresponding unit vector.

These vectors form a local polar basis rather than a fixed Cartesian pair. Their directions therefore change with the point around the pulley, while  $\hat{\mathbf{e}}_r$  remains outward and  $\hat{\mathbf{e}}_\theta$  remains tangent in the positive rotational direction. **Figure 2.4** illustrates this basis on a representative primary pulley; the direction definitions apply at either pulley. For reference, the axial direction remains the local closing direction defined in [Section 2.3.2](#); in the primary view shown here, positive axial motion points perpendicular to the page along the shaft.

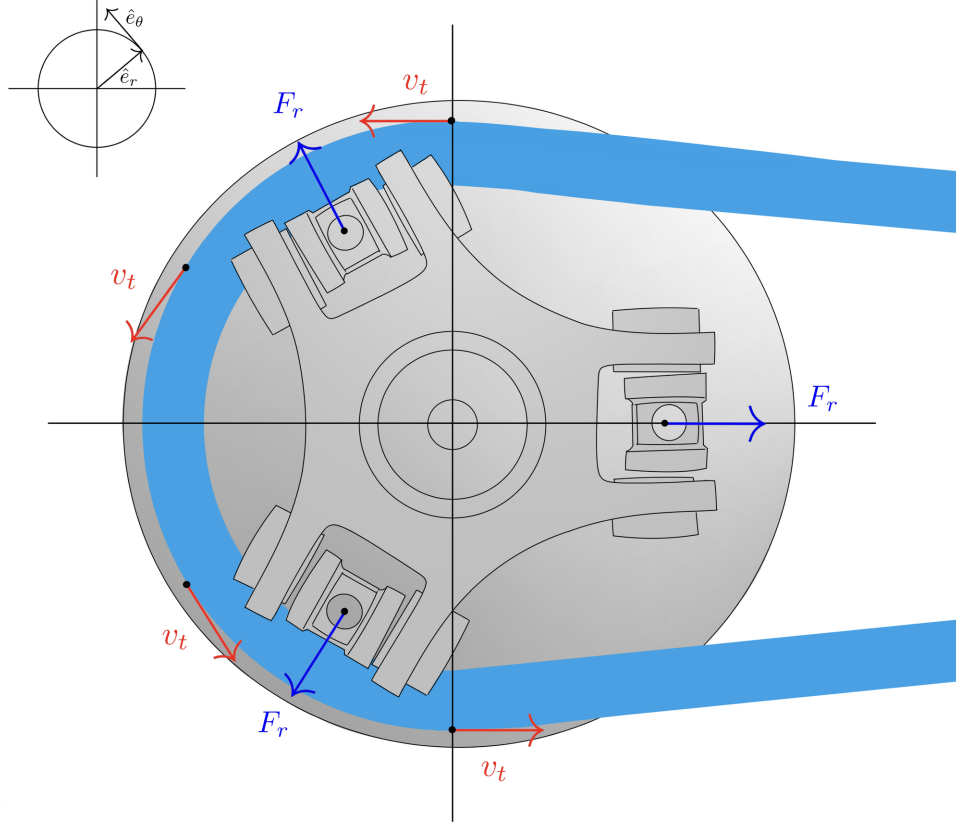


Figure 2.4: Representative primary-pulley view of the local radial and tangential directions. The inset shows the local polar basis:  $\hat{\mathbf{e}}_r$  points outward from the pulley axis, while  $\hat{\mathbf{e}}_\theta$  is tangent in the positive rotational direction defined in [Section 2.3.1](#). The basis changes orientation with the point around the pulley. All force and velocity arrows shown are illustrative.

## 2.4 CVT State Variables

To describe how the CVT evolves, five continuous quantities are carried from one instant to the next. They are collected in the state vector,

**State Vector**

**(2.4.1)**

$$\mathbf{x}(t) = \begin{bmatrix} \omega_p \\ \omega_s \\ s \\ \dot{s} \\ v_b \end{bmatrix}$$

The five entries follow the three retained motions: rotation of the two shafts, shift of the movable sheaves, and transport of the belt. For the shafts, both  $\omega_p$  and  $\omega_s$  are retained because the belt couples their motion without, in general, fixing one speed from the other. No corresponding

shaft-angle states are needed because the retained CVT mechanics do not depend on the absolute angular orientation of either pulley.

The shift motion requires both a position and a speed. Its position is represented by  $s$ , chosen to increase in the upshift direction. In the present geometry, this follows closing of the primary movable sheave. Once the belt is engaged, the fixed belt length and prescribed path of [Assumption 6 \(Fixed Belt Length\)](#) and [Assumption 8 \(Prescribed Planar Belt Path\)](#) link that primary motion to opening of the secondary. This gives the physical reason for using one shift coordinate rather than carrying  $x_p$  and  $x_s$  independently. The exact relation, including the primary deadzone before engagement, is established later in [Section 3.1.3](#). The corresponding shift speed  $\dot{s}$  must also be retained because the moving sheaves and actuation hardware carry inertia.

The remaining state variable,  $v_b$ , describes transport of belt material around the loop. It is retained independently because tangential slip allows the belt speed to differ from the surface speed of either pulley. Under [Assumption 7 \(Single Belt Transport Speed\)](#), the model carries no separate transport history for individual belt material points; together with the uniform belt properties of [Assumption 4 \(Fixed, Uniform Belt Cross-Section\)](#), this removes the need for any belt-position coordinate.

Taken together, these five entries are the continuous information carried forward in time. Their derivative makes clear what must be determined to advance that state:

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} \dot{\omega}_p \\ \dot{\omega}_s \\ \dot{s} \\ \ddot{s} \\ \dot{v}_b \end{bmatrix}. \quad (2.4.2)$$

In this derivative,  $\dot{s}$  is already known from the current state. The mechanics must supply the two shaft accelerations, the shift acceleration, and the belt-transport acceleration. Once those rates are known, the continuous state can be advanced to the next instant.

That still does not completely specify how the CVT can evolve. The same values of  $\mathbf{x}$  can occur under different contact or support conditions, for which the allowable motion need not be the same.

## 2.5 Mechanical Regimes and State Evolution

As the CVT moves, the contacts and supports that carry load do not remain fixed. A movable sheave can reach or leave a travel stop, the primary can gain or lose belt contact, and a belt-pulley interface can change between sticking and sliding. Each change alters which reactions can exist and which motions the mechanism can support. The mechanical problem can therefore change even when the hardware passes through the same positions and velocities.

A travel stop gives the simplest example. Suppose the primary movable sheave is at the end of its closing travel with  $\dot{s} = 0$ . If the other forces on the movable assembly continue to drive it closed, the stop supplies a reaction and prevents further motion into the boundary. If those forces instead tend to open the pulley, the stop reaction vanishes and the sheave can accelerate away. The same continuous state can therefore describe two different mechanical situations, distinguished by which support is active.

The additional information is the current *mechanical regime*. For the modeled CVT, the regime records which belt–pulley contacts exist, whether shift motion is free or held by a mechanical stop, and whether each existing tangential contact is sticking or sliding. Let  $q(t)$  denote the current regime. This label is not another mechanical coordinate; it is a compact description of the active mechanical arrangement.

Together,  $\mathbf{x}(t)$  and  $q(t)$  describe the internal CVT condition carried from one instant to the next. While one regime persists, the same contacts, supports, and kinematic restrictions remain active, so the state follows one continuous set of mechanical balances. Within that regime, the continuous mechanics must relate the current state and shaft loading to the state derivative. This dependence can be represented schematically as

$$\dot{\mathbf{x}} = \mathcal{F}_q(t, \mathbf{x}, \tau_{B,p}, I_{B,p}, \tau_{B,s}, I_{B,s}). \quad (2.5.1)$$

Here,  $\mathcal{F}_q$  is only a compact representation of the continuous mechanics. Its subscript records that changing the active arrangement changes which balances and constraints apply. As long as the conditions defining the regime remain satisfied,  $q$  is unchanged and the state evolves continuously.

A regime persists only while those physical conditions can be maintained. As the CVT moves, a stop may be reached or released, belt contact may form or separate, or a tangential contact may reach the boundary between sticking and sliding. Such an event changes the active mechanical arrangement. Whether the state itself changes at that instant depends on what changed physically.

If a stop releases from rest, or another contact condition changes without producing an impulse, the configuration and velocities pass through the event unchanged,

$$\mathbf{x}^+ = \mathbf{x}^-, \quad (2.5.2)$$

where the superscripts  $-$  and  $+$  denote values immediately before and after the event. The state is continuous, but its derivative can change immediately because the new regime applies a different set of balances and constraints.

A different situation occurs when a new rigid kinematic constraint becomes active while the mechanism still has finite velocity into it. The incoming motion may then be incompatible with the motion allowed after the constraint forms. Under **Assumption 13 (Instantaneous, Perfectly Inelastic Constraint Activation)**, the activation is treated as instantaneous and perfectly inelastic. The CVT does not jump to a new shift position during that instant, but its retained velocities may change to satisfy the newly active constraint. Thus,

$$s^+ = s^-, \quad \begin{bmatrix} \omega_p^+ \\ \omega_s^+ \\ \dot{s}^+ \\ v_b^+ \end{bmatrix} \text{ need not equal } \begin{bmatrix} \omega_p^- \\ \omega_s^- \\ \dot{s}^- \\ v_b^- \end{bmatrix}. \quad (2.5.3)$$

Once a compatible post-event motion has been established, continuous evolution resumes under the new regime.

The CVT motion therefore consists of continuous intervals under a fixed regime connected by discrete changes in the active mechanical arrangement. Because both forms of evolution are part

of the same problem, the model forms a *hybrid initial-value problem*. A complete initial condition specifies an initial state together with an admissible initial regime,

$$\mathbf{x}(t_0) = \mathbf{x}_0, \quad q(t_0) = q_0, \quad (2.5.4)$$

with  $\mathbf{x}_0$  satisfying the contacts and constraints represented by  $q_0$ . The regimes are not separate CVT models. They are different admissible arrangements of the same pulley assemblies, belt, actuation mechanisms, and shaft interfaces. The state vector records how that mechanism is moving; the regime records which motion is presently allowed.

## 2.6 Model Quantities and Evaluation Flow

One CVT model evaluation brings together quantities that do not all play the same role. Some define the particular CVT being modeled, some describe its current internal condition, some enter through the two shaft interfaces, and some are determined only after the current mechanics are evaluated. Keeping these roles separate makes clear what is fixed, what is carried forward, what is supplied from outside the CVT, and what must be found again at the current instant.

The fixed CVT data are the physical and geometric quantities that define the selected hardware. Some may be specified directly and others calculated once from those inputs during initialization; in either case, they remain unchanged while that simulation is advanced. [Table 2.1](#) places these data alongside the three roles that can change from one evaluation to the next.

Table 2.1: Roles of the quantities used during a CVT model evaluation.

| Quantity role                | Contents                                                                                                                                            | Use during evaluation                                                                                                                                                           | Inventory                 |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Fixed CVT data               | Physical and geometric data for the selected hardware, including values calculated once during initialization                                       | Define the selected CVT and the mechanical relations used to represent it; remain unchanged during one simulation.                                                              | <a href="#">Table N.2</a> |
| Current CVT condition        | State vector $\mathbf{x}(t)$ and mechanical regime $q(t)$                                                                                           | Carry the current internal motion and active mechanical arrangement from the preceding instant.                                                                                 | <a href="#">Table N.3</a> |
| Shaft-interface conditions   | Applied torque $\tau_{B,j}$ and equivalent inertia $I_{B,j}$ at each shaft                                                                          | Supply the current mechanical loading presented by the systems attached to the primary and secondary shafts.                                                                    | <a href="#">Table N.4</a> |
| Instantaneous CVT quantities | State derivative $\dot{\mathbf{x}}(t)$ and other mechanical quantities required for the current evaluation, including regime-admissibility measures | Are determined from the fixed CVT data, current CVT condition, and shaft-interface conditions; determine how the state can advance and whether the current regime can continue. | <a href="#">Table N.5</a> |

Taken together, these roles separate what defines an instantaneous CVT problem from what the mechanics return. The fixed data identify the hardware, the shaft-interface conditions provide its current external loading, and  $(\mathbf{x}, q)$  gives the internal condition to which that loading is applied. The instantaneous quantities are the response of that condition. One evaluation can therefore be stated compactly:

1. The current time, state  $\mathbf{x}(t)$ , and regime  $q(t)$  identify the internal CVT condition at the start of the evaluation.
2. The systems attached to the primary and secondary shafts supply the torque and equivalent inertia presented at their respective interfaces under the current shaft conditions.
3. The CVT mechanics use the fixed CVT data, current internal condition, and shaft-interface conditions to determine the instantaneous mechanical response. This provides  $\dot{\mathbf{x}}(t)$  together with the quantities needed to test whether the current regime remains admissible.
4. If the regime remains admissible, the state is advanced using  $\dot{\mathbf{x}}$  and the regime is retained. If an event changes the active mechanical arrangement, any required finite-speed transition establishes a compatible post-event state before continuous evolution resumes under the resulting regime.
5. The resulting state and regime become the current internal condition for the next evaluation, and the cycle repeats.

The information required to pose each instantaneous CVT problem is now defined. What remains is the mechanical response itself: the coupled forces, torques, and accelerations implied by the current geometry, actuation, belt motion, and contact state.

# Chapter 3

## Derivation

**Chapter 2** established the CVT as a hybrid initial-value problem: at a given instant, the current state, mechanical regime, and shaft-interface loading are known, and the model must determine the corresponding instantaneous response. This chapter derives the mechanics required to perform that evaluation. The development begins with the most general continuous configuration, in which the belt is fully engaged and the CVT is free to shift, and then returns to the mechanical boundaries and regime changes that connect successive intervals of continuous motion.

**Section 3.1** first develops the pulley and belt-loop kinematics, reducing the motion of both sheaves and both pulley radii to the shared shift coordinate. Using that kinematic structure, **Section 3.2** derives the axial actuation and movable-sheave dynamics, while **Section 3.3** develops the corresponding rotational dynamics of the two pulley assemblies. **Section 3.4** then closes the mechanics of the belt itself, relating its transport motion, tension, pulley loading, and transmitted torque. These pieces are assembled in **Section 3.5**, where the admissible belt–pulley contact state completes the instantaneous dynamics of the freely shifting engaged CVT. **Section 3.6** then extends this result to the remaining mechanical regimes, showing how the continuous equations reduce when additional constraints become active and how the state is carried consistently through transitions between them. Finally, **Section 3.7** brings these results together in the complete-system evaluation outlined in **Section 2.6**, showing how the CVT can be followed through time.

### 3.1 Pulley Geometry

The conceptual overview introduced the CVT as two variable-radius pulleys connected by a single belt. Moving the pulley sheaves changes where the belt sits on each pulley and therefore changes the transmission ratio.

Both pulleys contain a movable sheave. Let  $x_p$  denote the axial position of the movable primary sheave and  $x_s$  the axial position of the movable secondary sheave. The geometric problem is therefore twofold: how does the axial position of each movable sheave set the belt radius on its own pulley, and how does the shared belt constrain the two pulley positions to move together?

The first question can be answered from the local geometry of each pulley. The second requires following the belt around the complete loop. Once these position relationships are known, they can be differentiated to obtain the corresponding radius rates and accelerations.

The natural place to begin is the geometry of a single pulley.

### 3.1.1 Local Pulley Geometry

The first part of the geometric question can be answered on a single pulley: how does the axial position of its movable sheave determine the pulley radius? The same conical geometry appears on both pulleys, so the relationship can be derived once and then applied to the primary and secondary.

The axial direction and primary-sheave convention were introduced in [Section 2.3.2](#) and illustrated in [Figure 2.3](#). The local coordinates  $x_p$  and  $x_s$  follow the same convention on their respective pulleys.

For either pulley  $j \in \{p, s\}$ ,  $x_j = 0$  denotes the fully open configuration. Increasing  $x_j$  moves the movable sheave toward the fixed sheave and closes the pulley, while decreasing  $x_j$  opens it. The corresponding velocity follows the same sign convention:  $\dot{x}_j > 0$  denotes closing motion and  $\dot{x}_j < 0$  denotes opening motion.

The primary therefore begins at  $x_p = 0$  and moves in the positive direction as it closes. The secondary begins in a closed position, with  $x_s > 0$ , and moves toward smaller values of  $x_s$  as it opens.

Consider a belt seated between two straight sheave faces with half-angle  $\beta$ , as shown in [Figure 3.1](#). Let  $b_{\text{in}}$  denote the width of the belt at its inner surface. As assumed in [Assumption 4 \(Fixed, Uniform Belt Cross-Section\)](#), this width remains constant.

Suppose the movable sheave closes the pulley by a distance  $\Delta x_j$ . At the original belt radius, the available space between the sheaves is therefore reduced from  $b_{\text{in}}$  to

$$b_{\text{in}} - \Delta x_j.$$

The belt responds by moving outward through a radial distance  $\Delta r_j$ . At its new position, the angled sheave faces provide an additional axial distance  $d$  on each side.

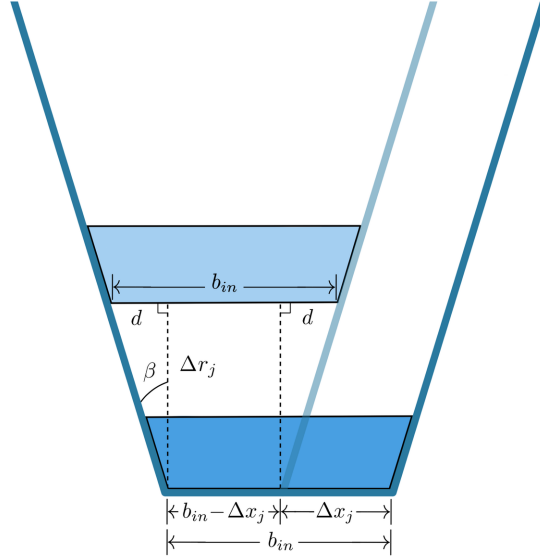


Figure 3.1: Local geometry of a belt moving outward between the sheave faces of pulley  $j$ . Closing the pulley by  $\Delta x$  reduces the available width at the original belt position to  $b_{\text{in}} - \Delta x$ . Moving outward by  $\Delta r$  provides an additional axial distance  $d$  at each sheave face.

At the new radius, the total space between the sheave faces must again equal the fixed belt width. Reading directly from [Figure 3.1](#),

$$b_{\text{in}} = 2d + (b_{\text{in}} - \Delta x_j). \quad (3.1.1)$$

The right triangle beside either sheave face relates  $d$  to the radial movement. For the sheave half-angle  $\beta$ ,

$$\tan \beta = \frac{d}{\Delta r_j}, \quad (3.1.2)$$

and therefore

$$d = \Delta r_j \tan \beta. \quad (3.1.3)$$

Substituting [Equation \(3.1.3\)](#) into [Equation \(3.1.1\)](#) gives

$$b_{\text{in}} = 2\Delta r_j \tan \beta + (b_{\text{in}} - \Delta x_j). \quad (3.1.4)$$

The constant belt-width terms cancel, leaving

$$\Delta r_j = \frac{\Delta x_j}{2 \tan \beta}. \quad (3.1.5)$$

This is the basic axial-to-radial relationship of the pulley. Closing the pulley moves the belt outward, while opening it moves the belt inward. A smaller sheave angle produces more radial movement for the same axial motion, whereas a larger sheave angle produces less.

Taking the infinitesimal limit of Equation (3.1.5) gives

$$\frac{dr_j}{dx_j} = \frac{1}{2 \tan \beta}. \quad (3.1.6)$$

Because the sheave faces are straight, this slope is constant.

**Radius convention** The derivation above follows the inner surface of the belt, where  $b_{\text{in}}$  is defined. The belt also has the finite radial height  $h_b$ , so its inner and outer surfaces occupy different radii. Let  $r_{j,\text{inner}}$  and  $r_{j,\text{outer}}$  denote those radii. They are related by

$$r_{j,\text{outer}} = r_{j,\text{inner}} + h_b. \quad (3.1.7)$$

The fixed-cross-section assumption in Assumption 4 (Fixed, Uniform Belt Cross-Section) makes  $h_b$  constant, so the axial-to-radial displacement derived above can be read from either surface. An absolute pulley radius must nevertheless identify which surface is being measured. The outer surface is used here because it provides a convenient and directly measurable reference. Throughout the geometric development,

$$r_j \equiv r_{j,\text{outer}}. \quad (3.1.8)$$

Using the outer surface gives a clear measure of where the belt is seated. It does not mean that every physical property of the belt acts at that radius.

The belt tension is carried along the cords embedded within the belt, so it acts through the centre of the cord layer. The distance from the pulley shaft to this line is therefore the lever arm through which belt tension transmits torque, making it the effective radius for gearing. This cord-line radius is commonly called the pitch radius (Gerbert, 1996).

Let  $d_{\text{cord}}$  denote the radial depth of the cord-layer centre from the outer surface, with  $0 < d_{\text{cord}} < h_b$ . Because the cord line lies this far inside the outer surface, its effective radius is

$$r_{j,\text{eff}} = r_{j,\text{outer}} - d_{\text{cord}}. \quad (3.1.9)$$

The belt mass is spread throughout the whole cross-section rather than concentrated at the outer surface or the cord line. To describe where this mass is with one radius, we use its centre of mass and denote the radius to that point by  $r_{j,\text{cm}}$ . Under the uniform-density approximation, the centre of mass is the geometric centroid of the trapezoid.

Let  $y$  measure radially inward from the outer surface, with  $y = 0$  at the outer surface and  $y = h_b$  at the inner surface, and let  $y_{\text{cm}}$  denote the depth of the centre of mass. These two depths and their corresponding radii are shown in Figure 3.2.

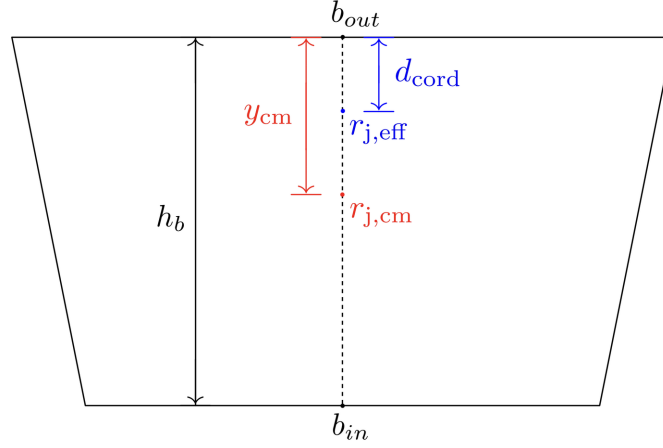


Figure 3.2: Radial reference lines within the belt cross-section. The cord-line radius  $r_{j,\text{eff}}$  is the lever arm through which belt tension transmits torque, while  $r_{j,\text{cm}}$  locates the belt's centre of mass. The depths  $d_{\text{cord}}$  and  $y_{\text{cm}}$  are measured radially inward from the outer surface.

If  $b_{\text{out}}$  and  $b_{\text{in}}$  are the outer and inner widths, respectively, the straight sides of the trapezoid give

$$b(y) = b_{\text{out}} - \frac{b_{\text{out}} - b_{\text{in}}}{h_b} y. \quad (3.1.10)$$

Because the density is uniform, a thin strip at depth  $y$  contains mass in proportion to its area  $b(y) dy$ . The centroid depth is therefore the weighted average of  $y$  over the cross-section:

$$y_{\text{cm}} = \frac{\int_0^{h_b} y b(y) dy}{\int_0^{h_b} b(y) dy}. \quad (3.1.11)$$

The denominator is the belt cross-sectional area,

$$A_b = \int_0^{h_b} b(y) dy = \frac{h_b}{2} (b_{\text{out}} + b_{\text{in}}). \quad (3.1.12)$$

Using this area in the weighted average gives

$$\begin{aligned} y_{\text{cm}} &= \frac{1}{A_b} \int_0^{h_b} y b(y) dy \\ &= h_b \frac{b_{\text{out}} + 2b_{\text{in}}}{3(b_{\text{out}} + b_{\text{in}})}. \end{aligned} \quad (3.1.13)$$

The result has the expected limiting behaviour. If  $b_{\text{out}} = b_{\text{in}}$ , the cross-section is rectangular and  $y_{\text{cm}} = h_b/2$ . For the usual trapezoid with  $b_{\text{out}} > b_{\text{in}}$ , the additional area near the outer surface

places the centroid slightly closer to that surface, so  $y_{\text{cm}} < h_b/2$ . The belt mass therefore follows the radius

$$r_{j,\text{cm}} = r_{j,\text{outer}} - y_{\text{cm}}. \quad (3.1.14)$$

The outer-surface radius map still requires one known configuration to establish its absolute placement. Let  $x_{j,\text{ref}}$  denote a known axial position at which the belt is seated, and let  $r_{j,\text{ref}}$  denote the outer radius measured at that same position. The outer radius at any other seated position is then

$$r_j(x_j) = r_{j,\text{ref}} + \frac{x_j - x_{j,\text{ref}}}{2 \tan \beta}. \quad (3.1.15)$$

The pair  $(x_{j,\text{ref}}, r_{j,\text{ref}})$  simply anchors the linear relationship to a known physical configuration.

For the secondary pulley, a convenient reference is its known closed starting position. The secondary begins with the belt seated at a large radius and opens as the CVT shifts. Since opening corresponds to decreasing  $x_s$ , [Equation \(3.1.15\)](#) produces the corresponding decrease in  $r_s$ .

The primary introduces an important geometric asymmetry: it can open farther than the width of the belt. At  $x_p = 0$ , the distance between the primary sheaves can therefore exceed  $b_{\text{in}}$ . Initial positive motion of the movable primary sheave only removes this clearance, leaving the belt at the same radius. The primary radius begins to increase only once the movable sheave reaches the belt.

This interval is the *primary deadzone*. Let  $x_{p,\text{dz}}$  denote the primary position at which the movable sheave first reaches the belt. Within the deadzone,

$$0 \leq x_p < x_{p,\text{dz}},$$

the primary coordinate changes, but the belt remains at its minimum primary radius. At  $x_p = x_{p,\text{dz}}$ , the clearance has been removed and the belt becomes seated between the sheave faces. Further closure then moves the belt outward according to the local geometry derived above.

The first-contact position provides the natural reference for the seated primary branch:

$$x_{p,\text{ref}} = x_{p,\text{dz}}, \quad r_{p,\text{ref}} = r_{p,\text{min}}.$$

The complete primary radius map is therefore

$$r_p(x_p) = \begin{cases} r_{p,\text{min}}, & 0 \leq x_p < x_{p,\text{dz}}, \\ r_{p,\text{min}} + \frac{x_p - x_{p,\text{dz}}}{2 \tan \beta}, & x_{p,\text{dz}} \leq x_p \leq x_{p,\text{max}}. \end{cases} \quad (3.1.16)$$

The linear axial-to-radial relationship in [Equation \(3.1.15\)](#) is standard in reduced CVT geometry models ([Duan et al., 2016](#); [Ballew, 2015](#)). It applies to both pulleys whenever the belt is seated, while [Equation \(3.1.16\)](#) extends the primary description through the deadzone before contact.

The local geometry now tells us the outer-surface radius associated with either movable-sheave position. The two local descriptions belong to the same closed belt, so only certain primary and secondary positions are geometrically compatible.

### 3.1.2 Complete Belt-Loop Geometry

The local pulley geometry describes how axial motion changes the belt's outer-surface radius on each pulley. The belt, however, passes around both pulleys. To connect those radii, the belt must now be followed around the complete loop.

Under **Assumption 8 (Prescribed Planar Belt Path)**, the modeled belt path consists of a circular wrap on each pulley joined by two straight tangent spans. The aligned, fixed-centre geometry of **Assumption 5 (Aligned Pulley Axes and Fixed Centre Distance)** makes the two spans equal. If each straight span has length  $l$ , and the primary and secondary wrap angles are  $\phi_p$  and  $\phi_s$ , the total geometric length of the loop is

$$L_{\text{geo}} = r_p \phi_p + r_s \phi_s + 2l. \quad (3.1.17)$$

The quantities  $l$ ,  $\phi_p$ , and  $\phi_s$  follow directly from the geometry shown in **Figure 3.3**.

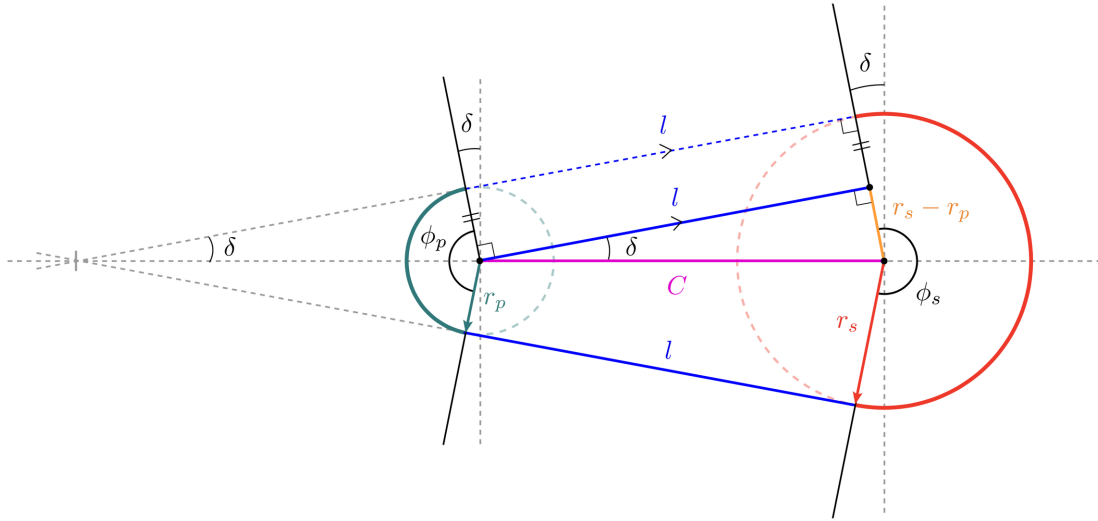


Figure 3.3: Geometry of the complete belt loop. The belt path consists of two wrapped arcs and two equal straight spans of length  $l$ . Unequal pulley radii incline the straight spans by the angle  $\delta$ , producing the primary and secondary wrap angles  $\phi_p$  and  $\phi_s$ .

Consider the right triangle in the upper span in **Figure 3.3**. The line joining the pulley centres forms its hypotenuse and has length  $d_c$ . One side lies along the straight belt span and therefore has length  $l$ .

The other side is perpendicular to the span. The primary and secondary radii drawn to their respective contact points are also perpendicular to this span and are therefore parallel. As shown

by the orange construction in [Figure 3.3](#), the difference between their lengths gives the remaining side of the triangle,  $r_s - r_p$ . The triangle therefore satisfies

$$l^2 + (r_s - r_p)^2 = d_c^2, \quad (3.1.18)$$

and the length of either straight span is

$$l = \sqrt{d_c^2 - (r_s - r_p)^2}. \quad (3.1.19)$$

This expression requires

$$|r_s - r_p| < d_c. \quad (3.1.20)$$

This condition is already enforced by the physical separation of the pulleys. The pulley circles must remain apart, requiring  $d_c > r_s + r_p$ , and  $r_s + r_p$  is necessarily greater than  $|r_s - r_p|$ .

The straight-span length accounts for only two parts of the belt path. The two wrapped portions still depend on how the straight spans meet the pulleys. If the pulley radii are equal, the spans are parallel to the line joining the pulley centres and the belt wraps halfway around each pulley. Both wrap angles are then equal to  $\pi$ .

When the radii differ, the straight spans rotate through the angle  $\delta$  shown in [Figure 3.3](#). The same right triangle gives

$$\sin \delta = \frac{r_s - r_p}{d_c}, \quad (3.1.21)$$

and therefore

$$\delta = \sin^{-1} \left( \frac{r_s - r_p}{d_c} \right). \quad (3.1.22)$$

The angle  $\delta$  is signed. For the configuration shown,  $r_s > r_p$ , so  $\delta > 0$ . The primary wrap loses an angle  $\delta$  at each end of its contact arc, while the secondary wrap gains the same amount at each end. Their wrap angles are therefore

$$\phi_p = \pi - 2\delta, \quad \phi_s = \pi + 2\delta. \quad (3.1.23)$$

If  $r_p > r_s$ , then  $\delta < 0$ , and the same equations automatically reverse which pulley has the larger wrap angle.

The complete belt length follows by adding the two wrapped arcs and the two straight spans. Substituting [Equations \(3.1.19\), \(3.1.22\) and \(3.1.23\)](#) into [Equation \(3.1.17\)](#) gives

**Exact geometric belt length****(3.1.24)**

$$\begin{aligned}
L_{\text{geo}}(r_p, r_s, d_c) = & \pi (r_p + r_s) \\
& + 2 (r_s - r_p) \sin^{-1} \left( \frac{r_s - r_p}{d_c} \right) \\
& + 2 \sqrt{d_c^2 - (r_s - r_p)^2}
\end{aligned}$$

Each term still corresponds directly to part of the belt path. The first is the combined half-circle contribution from the two pulley wraps. The second accounts for the unequal wrap angles, and the third is the combined length of the two straight spans.

For any chosen values of  $r_p$ ,  $r_s$ , and  $d_c$ , [Equation \(3.1.24\)](#) tells us how much belt is required to follow that path. The installed belt cannot change its total length as the pulleys shift. Under [Assumption 6 \(Fixed Belt Length\)](#), every possible pulley position must therefore satisfy

$$L_{\text{geo}}(r_p, r_s, d_c) = L_b, \quad (3.1.25)$$

where  $L_b$  is the fixed length of the same outer-surface path described by  $r_p$  and  $r_s$ , consistent with the convention in [Equation \(3.1.8\)](#). This is the geometric compatibility condition for the belt loop. Once any two of  $r_p$ ,  $r_s$ , and  $d_c$  are known, the condition determines the third.

For a particular CVT, the first quantity to determine is the centre distance  $d_c$ . The pulley radii change as the sheaves move, but the pulley shafts do not move with them. Once the CVT is assembled,  $d_c$  remains fixed throughout operation.

The fixed centre distance can be found from any installed pulley position that can be measured. At that position, the belt length and the two pulley radii are known. Denoting the measured radii by  $r_{p,\text{ref}}$  and  $r_{s,\text{ref}}$ , the only unknown in [Equation \(3.1.25\)](#) is  $d_c$ .

In the general unequal-radius case,  $d_c$  appears both inside the square root and inside the inverse sine. The exact relation therefore cannot be rearranged into a useful explicit expression for  $d_c$ . Instead, the centre distance is found with a one-dimensional root solve.

For a trial centre distance, the difference between the belt length required by the geometry and the belt length actually available is

$$\mathcal{R}_L(r_p, r_s, d_c) = L_{\text{geo}}(r_p, r_s, d_c) - L_b. \quad (3.1.26)$$

A positive residual means that the trial geometry requires more belt than is available. A negative residual means that it requires less. The correct centre distance is the value for which the required and available lengths match:

$$\mathcal{R}_L(r_{p,\text{ref}}, r_{s,\text{ref}}, d_c) = 0. \quad (3.1.27)$$

The root solver varies  $d_c$  until this condition is satisfied.

Once the initial root solve has determined  $d_c$ , both the centre distance and the belt length are known constants. The compatibility condition can then be used to connect the two pulley radii at every subsequent position.

Suppose the primary radius  $r_p$  is known. Substituting  $r_p$ ,  $d_c$ , and  $L_b$  into [Equation \(3.1.26\)](#) leaves the secondary radius as the only unknown. Its compatible value is found from

$$\mathcal{R}_L(r_p, r_s, d_c) = 0 \quad \text{solved for } r_s. \quad (3.1.28)$$

The resulting  $r_s$  is the secondary radius required for the same fixed belt to pass around both pulleys. The calculation also works in the opposite direction. If  $r_s$  is known instead, the compatible primary radius follows from

$$\mathcal{R}_L(r_p, r_s, d_c) = 0 \quad \text{solved for } r_p. \quad (3.1.29)$$

Thus, once  $d_c$  has been established, either pulley radius can be used to determine the other. The two radii remain local geometric quantities, but they cannot vary independently while belonging to the same fixed-length belt loop.

The arc-and-tangent construction used above is standard in belt-drive geometry. [Budynas and Nisbett \(2015\)](#) presents the same underlying construction for relating pulley radii, straight spans, wrap angles, and total belt length. The exact geometry is retained here because it provides the compatibility relation between the two shifting pulleys.

The local relations  $r_p(x_p)$  and  $r_s(x_s)$  now describe how each pulley produces its radius, while [Equation \(3.1.25\)](#) describes which pairs of radii can belong to the same belt loop. Combining these results gives the relationship between the two local axial coordinates.

### 3.1.3 From Local Pulley Coordinates to the Shift Coordinate

The preceding geometry gives a local relation for each pulley:  $x_p$  determines  $r_p(x_p)$ , and  $x_s$  determines  $r_s(x_s)$ . A complete CVT configuration, however, cannot be formed by choosing these positions independently. Both radii must belong to the same belt loop of length  $L_b$ . The remaining problem is therefore to identify the pairs  $(x_p, x_s)$  that satisfy this shared geometry.

For any proposed pair, the local pulley relations first give the two radii. Substituting those radii into the belt-length residual gives

$$\mathcal{R}_x(x_p, x_s) \equiv \mathcal{R}_L(r_p(x_p), r_s(x_s), d_c) = 0. \quad (3.1.30)$$

Every pair that makes [Equation \(3.1.30\)](#) equal to zero is a possible position of the two pulleys. Together, these pairs form the compatibility curve.

To reveal the shape of this curve, either local coordinate could be chosen as a starting point. Take  $x_p$  simply as an arbitrary choice. For each value of  $x_p$ , the primary relation gives  $r_p(x_p)$ . The shaft spacing fixes  $d_c$ , and  $L_b$  remains fixed by [Assumption 6 \(Fixed Belt Length\)](#), so the belt-length residual can then be solved for the required secondary radius  $r_s$ . The local secondary relation finally gives the corresponding position  $x_s$ :

$$x_p \longrightarrow r_p(x_p) \longrightarrow r_s \longrightarrow x_s.$$

Repeating these steps across the primary travel produces the representative compatibility curve in [Figure 3.4](#).

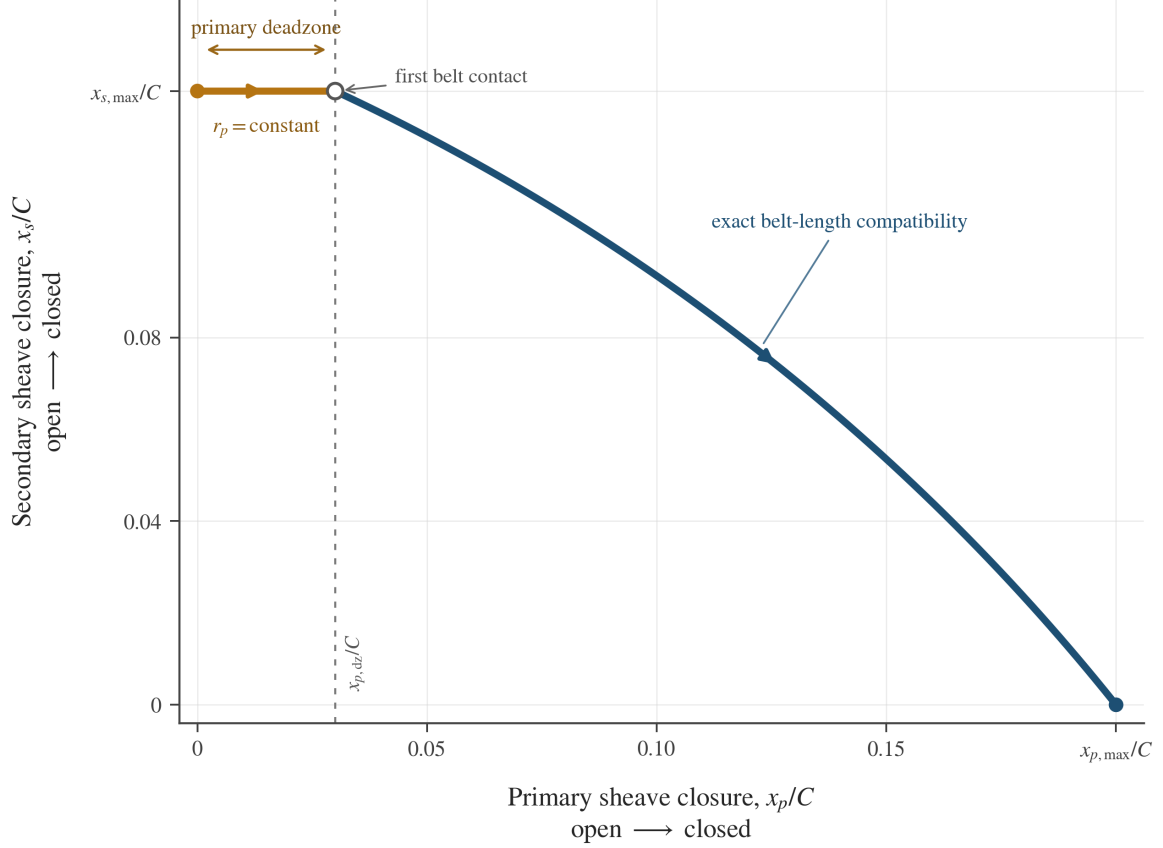


Figure 3.4: Representative compatibility curve between the local primary and secondary axial coordinates for an admissible belt–pulley geometry. Both coordinates are normalized by the same centre distance  $d_c$ , so their relative travel and the changing slope of the engaged branch are shown on a common scale. The dimensions are illustrative and do not represent a particular CVT.

The horizontal portion is created by the primary deadzone. While  $0 \leq x_p < x_{p,dz}$ , the primary sheave closes only the clearance to the belt. The primary radius remains fixed, so the compatible secondary radius and position remain fixed as well.

Once the primary reaches the belt, further closure increases  $r_p$ . The fixed belt loop then requires  $r_s$  to decrease, which means that the secondary opens and  $x_s$  decreases. The two sheaves do not move by equal amounts, and the resulting branch is not linear. Changing the radii alters both wrap lengths and both straight spans, so the compatible secondary displacement changes throughout the shift.

The curve also reveals which local coordinate can represent the complete geometry. A vertical line at any  $x_p$  meets the curve once, giving one compatible  $x_s$ . At the deadzone height, however, a

horizontal line meets every  $x_p$  between 0 and  $x_{p,dz}$ . Thus  $x_s$  is a unique function of  $x_p$  over the full travel, whereas  $x_p$  is not a unique function of  $x_s$ .

The primary coordinate is therefore adopted as the system shift coordinate. This is the same  $s$  included in the state vector of [Section 2.4](#):

|                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <span><b>System shift coordinate</b></span> <span><b>(3.1.31)</b></span> </div> <hr style="border: 0.5px solid black; margin: 5px 0;"/> $s \equiv x_p, \quad x_p(s) = s, \quad x_s = x_s(s)$ |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The secondary mapping is then

$$x_s(s) = \begin{cases} x_{s,\max}, & 0 \leq s < x_{p,dz}, \\ \text{root}_{0 \leq x_s \leq x_{s,\max}} [\mathcal{R}_x(s, x_s) = 0], & x_{p,dz} \leq s \leq x_{p,\max}. \end{cases} \quad (3.1.32)$$

The first branch holds the secondary fixed while the primary closes its clearance; the second returns the unique compatible position after belt contact. Both give  $x_{s,\max}$  at  $s = x_{p,dz}$ , so the mapping is continuous. The shift coordinate therefore determines both movable-sheave positions and, through the local pulley relations, both pulley radii. The position geometry is now complete. What remains is to determine how these positions and radii move when  $s$  changes with time.

### 3.1.4 Pulley-Radius Rates

The shift coordinate now determines the positions of both movable sheaves and the corresponding pulley radii. Its rate form asks the same geometric question over a small motion: if the primary closes by an amount  $ds$ , how far must the secondary open, and how quickly do the two pulley radii change?

We first find how each local coordinate and pulley radius changes per unit  $s$ . Throughout this subsection, a prime denotes differentiation with respect to  $s$ . For either pulley  $j \in \{p, s\}$ , the corresponding time rates are then

$$\dot{x}_j = x'_j(s)\dot{s}, \quad \dot{r}_j = r'_j(s)\dot{s}. \quad (3.1.33)$$

**Primary radius rate** The primary side follows immediately from the definition  $x_p(s) = s$ :

$$x'_p(s) = 1, \quad \dot{x}_p = \dot{s}. \quad (3.1.34)$$

The corresponding radius map is constant through the primary deadzone and linear after the movable sheave reaches the belt. Differentiating [Equation \(3.1.16\)](#) therefore gives

$$r'_p(s) = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ \frac{1}{2 \tan \beta}, & x_{p,dz} < s \leq x_{p,\max}. \end{cases} \quad (3.1.35)$$

Multiplying this slope by  $\dot{s}$  gives

|                                                                                                                                   |                 |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Primary Radius Rate</b>                                                                                                        | <b>(3.1.36)</b> |
| $\dot{r}_p = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ \frac{\dot{s}}{2 \tan \beta}, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$ |                 |

Thus, the primary sheave can move while crossing the deadzone even though the primary belt radius does not. After contact, the sheave angle fixes the radial change produced by each unit of primary closure.

**Secondary coordinate and radius rates** The secondary requires one additional step. Its local pulley relation tells us how  $r_s$  changes when  $x_s$  changes, but it does not tell us how far the secondary must move. That motion is set by the fixed belt loop.

At every compatible position,

$$L_{\text{geo}}(r_p(s), r_s(s), d_c) = L_b. \quad (3.1.37)$$

The centre distance  $d_c$  is fixed by the pulley shafts, while  $L_b$  is constant under **Assumption 6 (Fixed Belt Length)**. A small change in  $s$  therefore cannot change the geometric length of the belt path:

$$\frac{dL_{\text{geo}}}{ds} = 0. \quad (3.1.38)$$

Both radii change during this motion. The change caused by  $r_p$  is the sensitivity of the belt length to  $r_p$ , multiplied by the actual rate  $r'_p$ . The secondary contributes in the same way. Applying the chain rule to **Equation (3.1.37)** gives

$$\frac{\partial L_{\text{geo}}}{\partial r_p} r'_p + \frac{\partial L_{\text{geo}}}{\partial r_s} r'_s = 0. \quad (3.1.39)$$

The two partial derivatives have a direct purpose: each tells us how much additional belt path would be required if one pulley radius increased while the other radius remained fixed. The actual pulley motion must make these two changes cancel. We therefore calculate the two sensitivities next.

The span angle and straight-span length entering the belt-length relation were derived in **Equations (3.1.19)** and **(3.1.22)**. Writing them together in the form needed for differentiation,

$$\delta = \sin^{-1} \left( \frac{r_s - r_p}{d_c} \right), \quad l = \sqrt{d_c^2 - (r_s - r_p)^2}, \quad (3.1.40)$$

Here,  $\delta$  sets the wrap angles through **Equation (3.1.23)**, while  $l$  is the length of either straight span. Using these quantities, the exact path length from **Equation (3.1.24)** is

$$L_{\text{geo}} = \pi(r_p + r_s) + 2(r_s - r_p)\delta + 2l. \quad (3.1.41)$$

First, hold  $r_s$  and  $d_c$  fixed and increase  $r_p$ . The radius difference  $r_s - r_p$  then decreases at the same rate:

$$\frac{\partial(r_s - r_p)}{\partial r_p} = -1. \quad (3.1.42)$$

Using this result in the definitions of  $\delta$  and  $l$  gives

$$\begin{aligned} \frac{\partial \delta}{\partial r_p} &= \frac{1}{\sqrt{1 - \left(\frac{r_s - r_p}{d_c}\right)^2}} \left(-\frac{1}{d_c}\right) \\ &= -\frac{1}{\sqrt{d_c^2 - (r_s - r_p)^2}} = -\frac{1}{l}, \end{aligned} \quad (3.1.43)$$

$$\begin{aligned} \frac{\partial l}{\partial r_p} &= \frac{1}{2\sqrt{d_c^2 - (r_s - r_p)^2}} [2(r_s - r_p)] \\ &= \frac{r_s - r_p}{l}. \end{aligned} \quad (3.1.44)$$

We can now differentiate the complete path length in [Equation \(3.1.41\)](#). The first term differentiates directly, the wrap correction requires the product rule, and the final term contains the changing straight spans:

$$\begin{aligned} \frac{\partial L_{\text{geo}}}{\partial r_p} &= \pi + 2 \left[ (-1)\delta + (r_s - r_p) \left(-\frac{1}{l}\right) \right] + 2 \left( \frac{r_s - r_p}{l} \right) \\ &= \pi - 2\delta \\ &= \phi_p. \end{aligned} \quad (3.1.45)$$

The two terms containing  $(r_s - r_p)/l$  cancel. The remaining sensitivity is the primary wrap angle. The secondary calculation follows the same path, but increasing  $r_s$  increases the radius difference:

$$\frac{\partial(r_s - r_p)}{\partial r_s} = 1. \quad (3.1.46)$$

The corresponding derivatives of  $\delta$  and  $l$  are

$$\begin{aligned}\frac{\partial \delta}{\partial r_s} &= \frac{1}{\sqrt{1 - \left(\frac{r_s - r_p}{d_c}\right)^2}} \left(\frac{1}{d_c}\right) \\ &= \frac{1}{l},\end{aligned}\tag{3.1.47}$$

$$\begin{aligned}\frac{\partial l}{\partial r_s} &= \frac{1}{2\sqrt{d_c^2 - (r_s - r_p)^2}} [-2(r_s - r_p)] \\ &= -\frac{r_s - r_p}{l}.\end{aligned}\tag{3.1.48}$$

Differentiating the full path length with these signs gives

$$\begin{aligned}\frac{\partial L_{\text{geo}}}{\partial r_s} &= \pi + 2 \left[ (1)\delta + (r_s - r_p) \left(\frac{1}{l}\right) \right] + 2 \left( -\frac{r_s - r_p}{l} \right) \\ &= \pi + 2\delta \\ &= \phi_s.\end{aligned}\tag{3.1.49}$$

Again, the terms involving the changing straight-span length cancel. The sensitivity to the secondary radius is the secondary wrap angle. Together,

$$\frac{\partial L_{\text{geo}}}{\partial r_p} = \phi_p, \quad \frac{\partial L_{\text{geo}}}{\partial r_s} = \phi_s.\tag{3.1.50}$$

This result also has a simple geometric interpretation. If one pulley radius increases by a small amount while the other is held fixed, the required belt path increases by that radial change multiplied by the angle over which the belt wraps the pulley. The remaining changes in the tangent geometry cancel one another.

Returning to the fixed-length condition [Equation \(3.1.39\)](#), the two radial changes must therefore satisfy

$$\phi_p r'_p + \phi_s r'_s = 0.\tag{3.1.51}$$

After primary contact, the local pulley relation gives the same axial-to-radial slope on both pulleys. Since  $x'_p = 1$ ,

$$r'_p = \frac{1}{2 \tan \beta}, \quad r'_s = \frac{1}{2 \tan \beta} x'_s.\tag{3.1.52}$$

Substituting these two slopes into [Equation \(3.1.51\)](#) gives

$$\phi_p \left( \frac{1}{2 \tan \beta} \right) + \phi_s \left( \frac{1}{2 \tan \beta} x'_s \right) = 0.\tag{3.1.53}$$

The common local pulley slope cancels, leaving

$$\phi_p + \phi_s x'_s = 0. \quad (3.1.54)$$

Solving for the slope of the secondary mapping gives

|                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div><b>Secondary Coordinate Slope</b></div> <div><b>(3.1.55)</b></div> </div> <hr style="border: 0.5px solid black; margin: 5px 0;"/> $x'_s(s) = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s}, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$ |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Within the deadzone, the primary radius does not change, so neither the compatible secondary radius nor  $x_s$  changes. After contact,  $x'_s < 0$ : primary closure requires the secondary to open. Its magnitude is not generally one and changes with the two wrap angles, which is why the engaged branch in [Figure 3.4](#) is curved rather than straight.

The time rate of the secondary coordinate is now obtained by multiplying this slope by  $\dot{s}$ :

$$\dot{x}_s = x'_s(s)\dot{s} = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s}\dot{s}, & x_{p,dz} < s \leq x_{p,max}. \end{cases} \quad (3.1.56)$$

Applying the local axial-to-radial slope once more gives the secondary radius rate:

|                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div><b>Secondary Radius Rate</b></div> <div><b>(3.1.57)</b></div> </div> <hr style="border: 0.5px solid black; margin: 5px 0;"/> $\dot{r}_s = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s} \frac{\dot{s}}{2 \tan \beta}, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$ |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The rates above were written for  $r_p$  and  $r_s$ , which denote the outer-surface radii. The inner surface, cord line, and mass centre remain at fixed depths within the belt cross-section. When the belt moves radially on either pulley, all four reference radii therefore change by exactly the same amount. For either pulley  $j \in \{p, s\}$ ,

$$r'_j = r'_{j,inner} = r'_{j,eff} = r'_{j,cm}, \quad \dot{r}_j = \dot{r}_{j,inner} = \dot{r}_{j,eff} = \dot{r}_{j,cm}. \quad (3.1.58)$$

The choice of reference changes the absolute radius, but not its rate. The two pulley-radius rates are now determined by  $s$  and  $\dot{s}$ . However, the secondary multiplier  $-\phi_p/\phi_s$  changes as the radii and wrap angles change. Even if  $\dot{s}$  is constant, the secondary rate is not. Its acceleration must therefore include the changing slope of the compatibility curve.

### 3.1.5 Pulley-Radius Accelerations

The pulley-radius rates in [Equations \(3.1.36\)](#) and [\(3.1.57\)](#) were obtained by multiplying the slope of each geometric mapping by the shift velocity  $\dot{s}$ . Their accelerations follow by asking how those rates change. On the primary, the mapping slope is constant within each branch. On the secondary, the slope in [Equation \(3.1.55\)](#) changes as the pulley radii and wrap angles change. The secondary acceleration must therefore account for both the imposed shift acceleration and the changing geometric slope.

**Primary radius acceleration** The primary coordinate requires no additional geometry. The system-coordinate definition in [Equation \(3.1.31\)](#) gives  $x_p(s) = s$ , so

$$x'_p(s) = 1, \quad x''_p(s) = 0, \quad (3.1.59)$$

and therefore

$$\dot{x}_p = \dot{s}, \quad \ddot{x}_p = \ddot{s}. \quad (3.1.60)$$

The general shift-to-time relation in [Equation \(3.1.33\)](#) gives the primary radius rate in the form

$$\dot{r}_p = r'_p(s)\dot{s}. \quad (3.1.61)$$

Differentiating this product with respect to time gives

$$\begin{aligned} \ddot{r}_p &= \frac{d}{dt} [r'_p(s)\dot{s}] \\ &= \frac{dr'_p}{dt}\dot{s} + r'_p(s)\ddot{s}. \end{aligned} \quad (3.1.62)$$

Because  $r'_p$  is itself a function of  $s$ ,

$$\frac{dr'_p}{dt} = r''_p(s)\dot{s}. \quad (3.1.63)$$

Substituting [Equation \(3.1.63\)](#) into [Equation \(3.1.62\)](#) gives

$$\ddot{r}_p = r''_p(s)\dot{s}^2 + r'_p(s)\ddot{s}. \quad (3.1.64)$$

Within either smooth branch of the primary radius map,  $r'_p(s)$  is constant. Hence

$$r''_p(s) = 0. \quad (3.1.65)$$

Using the primary slopes from [Equation \(3.1.35\)](#), the primary radius acceleration is

### Primary Radius Acceleration

(3.1.66)

$$\ddot{r}_p = \begin{cases} 0, & 0 \leq s < x_{p,\text{dz}}, \\ \frac{\ddot{s}}{2 \tan \beta}, & x_{p,\text{dz}} < s \leq x_{p,\text{max}}. \end{cases}$$

The result is direct: within the deadzone the primary radius remains fixed, while after contact the shift acceleration is converted into radial acceleration by the same constant sheave slope found in Equation (3.1.35).

**Secondary coordinate curvature** The secondary mapping is not linear after contact. Its first derivative was found from the fixed-length rate balance in Equation (3.1.54):

$$\phi_p + \phi_s x'_s = 0.$$

The balance in Equation (3.1.54) contains the information needed for the second derivative. We will differentiate it once more with respect to  $s$ , which first requires the rates at which the two wrap angles change with  $s$ .

The required geometric quantities were defined in Equations (3.1.22) and (3.1.23). Written together, they are

$$\delta = \sin^{-1} \left( \frac{r_s - r_p}{d_c} \right), \quad \phi_p = \pi - 2\delta, \quad \phi_s = \pi + 2\delta. \quad (3.1.67)$$

As  $s$  changes, both  $r_p$  and  $r_s$  change, so the radius difference inside  $\delta$  changes at the rate

$$\frac{d}{ds}(r_s - r_p) = r'_s - r'_p. \quad (3.1.68)$$

Applying the chain rule to  $\delta$ , and using the straight-span length from Equation (3.1.19) in the final line, gives

$$\begin{aligned} \delta' &= \frac{1}{\sqrt{1 - \left( \frac{r_s - r_p}{d_c} \right)^2}} \left( \frac{r'_s - r'_p}{d_c} \right) \\ &= \frac{r'_s - r'_p}{\sqrt{d_c^2 - (r_s - r_p)^2}} \\ &= \frac{r'_s - r'_p}{l}. \end{aligned} \quad (3.1.69)$$

After primary contact, the local pulley slopes are given by Equation (3.1.52):

$$r'_p = \frac{1}{2 \tan \beta}, \quad r'_s = \frac{x'_s}{2 \tan \beta}.$$

Substituting these slopes into Equation (3.1.69) gives

$$\begin{aligned} \delta' &= \frac{\frac{x'_s}{2 \tan \beta} - \frac{1}{2 \tan \beta}}{l} \\ &= \frac{x'_s - 1}{2l \tan \beta}. \end{aligned} \quad (3.1.70)$$

The wrap-angle derivatives now follow directly from Equation (3.1.67):

$$\phi'_p = -2\delta', \quad \phi'_s = 2\delta'. \quad (3.1.71)$$

We can now differentiate the secondary slope balance in Equation (3.1.54). The primary term contributes  $\phi'_p$ , while the secondary term requires the product rule because both  $\phi_s$  and  $x'_s$  change with  $s$ :

$$\begin{aligned} 0 &= \frac{d}{ds} (\phi_p + \phi_s x'_s) \\ &= \phi'_p + \phi'_s x'_s + \phi_s x''_s. \end{aligned} \quad (3.1.72)$$

Substituting the wrap-angle derivatives from Equation (3.1.71) gives

$$-2\delta' + 2\delta' x'_s + \phi_s x''_s = 0. \quad (3.1.73)$$

Solving first for  $x''_s$ ,

$$\begin{aligned} \phi_s x''_s &= 2\delta'(1 - x'_s), \\ x''_s &= \frac{2\delta'(1 - x'_s)}{\phi_s}. \end{aligned} \quad (3.1.74)$$

Using the expression for  $\delta'$  from Equation (3.1.70),

$$\begin{aligned} x''_s &= \frac{2}{\phi_s} \left( \frac{x'_s - 1}{2l \tan \beta} \right) (1 - x'_s) \\ &= -\frac{(1 - x'_s)^2}{\phi_s l \tan \beta}. \end{aligned} \quad (3.1.75)$$

On the engaged branch, the first-derivative result in Equation (3.1.55) gives  $x'_s = -\phi_p/\phi_s$ . The remaining factor can therefore be simplified:

$$\begin{aligned} 1 - x'_s &= 1 + \frac{\phi_p}{\phi_s} \\ &= \frac{\phi_s + \phi_p}{\phi_s} \\ &= \frac{2\pi}{\phi_s}, \end{aligned} \tag{3.1.76}$$

where  $\phi_p + \phi_s = 2\pi$  follows from Equation (3.1.23). Substituting this result into Equation (3.1.75) gives the curvature of the secondary coordinate mapping:

|                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div><b>Secondary Coordinate Curvature</b></div> <div><b>(3.1.77)</b></div> </div> <hr style="border: 0.5px solid black; margin: 10px 0;"/> $x''_s(s) = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{4\pi^2}{\phi_s^3 l \tan \beta}, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$ |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The deadzone branch in Equation (3.1.32) is horizontal, so both  $x'_s$  and  $x''_s$  are zero there. On the engaged branch,  $x''_s < 0$ : as the primary closes, the secondary mapping becomes progressively steeper in the opening direction.

**Secondary axial and radial accelerations** The curvature in Equation (3.1.77) describes how the slope of the compatibility curve changes with position. To convert it into acceleration, begin with the secondary axial velocity from Equation (3.1.56):

$$\dot{x}_s = x'_s(s)\dot{s}.$$

Differentiating with respect to time gives

$$\begin{aligned} \ddot{x}_s &= \frac{d}{dt} [x'_s(s)\dot{s}] \\ &= \frac{dx'_s}{dt} \dot{s} + x'_s(s)\ddot{s}. \end{aligned} \tag{3.1.78}$$

Because  $x'_s$  changes only through  $s$ ,

$$\frac{dx'_s}{dt} = x''_s(s)\dot{s}. \tag{3.1.79}$$

The secondary axial acceleration is therefore

$$\ddot{x}_s = x'_s(s)\ddot{s} + x''_s(s)\dot{s}^2. \quad (3.1.80)$$

Substituting the slope from Equation (3.1.55) and the curvature from Equation (3.1.77) gives

$$\text{Secondary Axial Acceleration} \quad (3.1.81)$$

$$\ddot{x}_s = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s}\ddot{s} - \frac{4\pi^2}{\phi_s^3 l \tan \beta} \dot{s}^2, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$$

The local radius derivative in Equation (3.1.6) is the constant  $1/(2 \tan \beta)$ . Applying it to the secondary gives

$$\ddot{r}_s = \frac{1}{2 \tan \beta} \ddot{x}_s. \quad (3.1.82)$$

Substituting Equation (3.1.81) into Equation (3.1.82) gives

$$\text{Secondary Radius Acceleration} \quad (3.1.83)$$

$$\ddot{r}_s = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s} \frac{\ddot{s}}{2 \tan \beta} - \frac{2\pi^2}{\phi_s^3 l \tan^2 \beta} \dot{s}^2, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$$

The two terms have different physical origins. The first comes from the imposed shift acceleration, converted through the coordinate slope in Equation (3.1.55) and the local pulley slope in Equation (3.1.6). The second comes from the curvature in Equation (3.1.77): even when  $\ddot{s} = 0$ , moving through the engaged branch changes  $-\phi_p/\phi_s$ , so the secondary velocity continues to change.

As with the radius rates, the fixed offsets between the four radial references give, for either pulley  $j \in \{p, s\}$ ,

$$\ddot{r}_j = \ddot{r}_{j,inner} = \ddot{r}_{j,eff} = \ddot{r}_{j,cm}. \quad (3.1.84)$$

#### Remark (Deadzone Boundary)

The idealized maps in Equations (3.1.16) and (3.1.32) change slope when the primary first reaches the belt at  $s = x_{p,dz}$ . Their first derivatives therefore change discontinuously, and their ordinary second derivatives are not defined at that single point. The acceleration relations in Equations (3.1.66) and (3.1.83) apply within the deadzone and engaged branches, while first contact is treated as the transition between them.

### 3.1.6 Geometric Closure and Model Scope

Specifying the shift state now determines the complete modeled pulley geometry. The definition in Equation (3.1.31) sets the primary position directly, while Equation (3.1.32) gives the unique compatible secondary position. The local pulley maps in Equations (3.1.15) and (3.1.16) convert those two positions into the outer-surface radii used to describe the belt path. Finally, the rate and acceleration relations in Equations (3.1.36), (3.1.57), (3.1.66) and (3.1.83) determine how those radii move. Thus, for any admissible shift state,

$$(s, \dot{s}, \ddot{s}) \quad \longmapsto \quad (r_p, \dot{r}_p, \ddot{r}_p, r_s, \dot{r}_s, \ddot{r}_s).$$

Here  $r_p$  and  $r_s$  retain the outer-surface convention used throughout the geometric development. The corresponding inner-surface, effective, and centre-of-mass radii follow directly from their fixed positions within the belt cross-section. Nothing else has to be solved for.

These results follow from the geometry derived above under the idealizations stated in Section 2.2. The belt cross-section and total length remain fixed under Assumption 4 (Fixed, Uniform Belt Cross-Section) and Assumption 6 (Fixed Belt Length), while Assumption 8 (Prescribed Planar Belt Path) constrains the belt to circular wraps and straight tangent spans. Together with the aligned pulley axes and fixed centre distance of Assumption 5 (Aligned Pulley Axes and Fixed Centre Distance), these conditions leave one modeled belt path for each admissible value of  $s$ .

**Relation to prior transient models** These choices also define the level of detail represented by the present geometry. It determines the complete belt path that is globally compatible with the imposed shift position, but it does not track the trajectory of each belt element through the pulley wraps.

Duan et al. (2016) begin from closely related open-belt arc-and-tangent geometry and the same wedge relationship between movable-sheave motion and nominal radius motion. Their chain-CVT formulation retains additional internal deformation. Chain extension and its rate contribute to the span tension, and pulley deformation allows the local radius followed by a chain element to differ from the nominal pulley radius. The present formulation instead fixes the belt length through Assumption 6 (Fixed Belt Length) and prescribes the arc-and-tangent belt path through Assumption 8 (Prescribed Planar Belt Path). Consequently, elastic path-length change and deformation-driven departure from that prescribed path are not represented in this geometric closure.

Rubber-belt transient models resolve the same shifting process at several levels of detail. Analytical shift-mechanics models, such as those of Sorge (2008) and Cammalleri and Sorge (2009), describe local radial belt migration through a pulley wrap and couple it to circumferential transport, sliding, and the mechanics of the contact region. Discretized rubber-belt models, such as those of Julió and Plante (2011) and Ballew (2015), instead obtain the evolving belt geometry from the motion and contact state of individual belt elements. These approaches can resolve local deformation, tension redistribution, and contact migration in greater detail, but require additional spatial or nodal states and the contact laws that govern them.

The distinction is therefore one of model resolution. The present formulation uses the global compatibility condition in Equation (3.1.25) to find the representative pulley-radius pair and its derivatives directly from  $s$ ,  $\dot{s}$ , and  $\ddot{s}$ . The cited transient models additionally ask how the belt or chain deforms and migrates within that path. Those internal motions lie outside the present

geometric layer; within the assumptions stated above, the global pulley geometry and its motion are fully determined.

This closure determines which sheave and radius motions are compatible with the modeled belt path, but it does not determine which compatible acceleration occurs. In particular, every acceleration relation above still contains the shift acceleration  $\ddot{s}$ . Its value must be consistent with the forces and inertias acting on the two movable pulley assemblies.

## 3.2 Axial Actuation and Sheave Balances

The preceding section established how the two movable sheaves must move together. It did not determine the motion itself. That requires the forces acting on each movable assembly and the inertia that those forces must accelerate.

The primary and secondary remain separate mechanisms even though the belt geometry couples their positions. Each has its own actuator, spring, moving hardware, and belt reaction, so each must satisfy its own axial equation of motion. We therefore complete the primary balance in its local coordinate  $x_p$ , then do the same for the secondary in  $x_s$ . Once each mechanism has been interpreted and every local force has been derived, the geometric maps developed above in [Section 3.1](#) express both balances through the shared shift coordinate  $s$ . This final step shows how the two local free bodies act on one shift motion.

### 3.2.1 Primary Movable-Sheave Balance

The primary is easiest to understand by separating rotation about the shaft from motion along it. The entire primary rotates with the shaft at  $\omega_p$ , but its parts do not all move axially. The fixed sheave remains at one axial position. Opposite it, the movable sheave, ramp carrier, and the hardware attached rigidly to them slide together along the shaft. Motion in the closing direction narrows the pulley groove and clamps the belt; motion in the opening direction widens the groove.

The flyweights provide the closing drive. They are attached to the fixed sheave and carried around the shaft with it, but each can also pivot about its attachment to the fixed sheave. As the primary rotates, centrifugal loading tends to swing each flyweight radially outward until it presses against a ramp carried by the movable sheave. Because the ramp surface is inclined, its contact reaction has an axial component that drives the movable sheave in the closing direction. The primary spring pushes the movable sheave in the opening direction, and the belt pushes back through its normal reaction on the movable face.

This leaves two distinct moving bodies in the model. The movable sheave, ramp carrier, and attached hardware form one translating assembly with axial coordinate  $x_p$ . The flyweights rotate with the primary and pivot separately, but do not translate rigidly with  $x_p$ ; they deliver their closing action to the movable sheave through the ramp contacts. During a shift, the net axial force must accelerate the translating assembly, while the flyweights require their own rotational balance. We therefore begin by isolating the translating assembly, then determine the force transmitted to it by the flyweights. This mechanism and the resulting axial free body are shown together in [Figure 3.5](#).

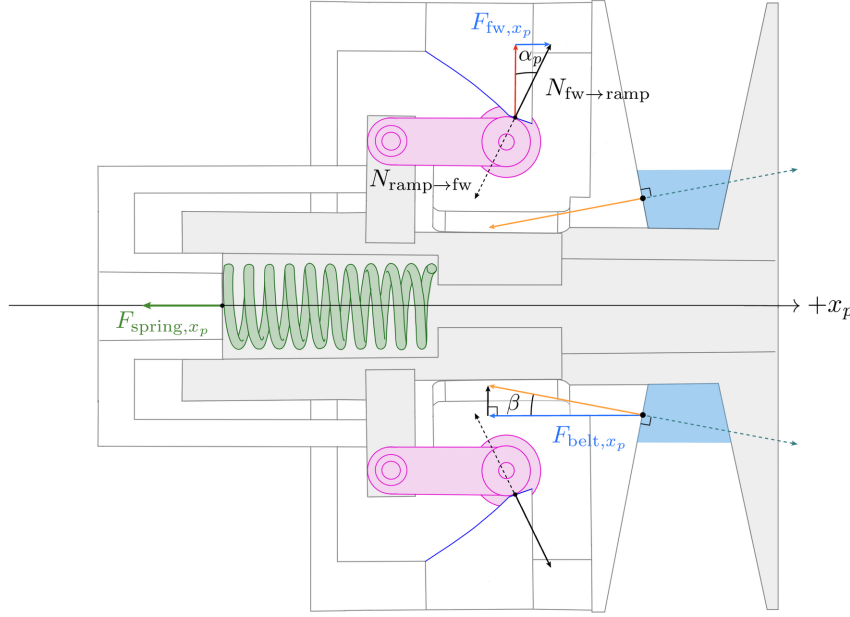


Figure 3.5: Axial free-body diagram of the primary movable assembly (unshaded). The symmetric sides of the cross-section are used to show the flyweight–ramp and belt–sheave force resolutions separately. Dotted arrows show the equal-and-opposite forces on the flyweight and belt, with  $F_{\text{clamp}}$  indicating the movable sheave’s squeeze on the belt.

As established in [Section 3.1](#), positive  $x_p$  closes the primary. Let  $m_p$  be the total mass of the movable sheave, ramp carrier, and every other component that translates rigidly with them. The three axial forces shown in [Figure 3.5](#) are:

- $F_{\text{fw}, x_p}$ , the closing force delivered to the ramp carrier by the flyweight contacts;
- $F_{\text{spring}, x_p}$ , the opening force exerted by the primary spring; and
- $F_{\text{belt}, x_p}$ , the opening force exerted by the belt on the movable face.

Newton’s second law along the shaft is therefore

$$m_p \ddot{x}_p = F_{\text{fw}, x_p} - F_{\text{spring}, x_p} - F_{\text{belt}, x_p}. \quad (3.2.1)$$

This free body tells us exactly what remains to be found. The spring force will follow from its compression, and the belt force from the normal reaction on the conical face. The flyweight force is harder. It is the axial result of a contact on a body that pivots rather than translating rigidly with the sheave. We therefore determine  $F_{\text{fw}, x_p}$  first.

### Force Transmitted by the Flyweights

**Flyweight free body** [Figure 3.6](#) isolates one flyweight in the radial–axial plane fixed to the primary. The flyweight is pinned at  $P$  and pivots about the circumferential axis through that point. Its pose is described by  $q_f$ , measured from  $+x_p$  to the body-fixed reference line shown in the figure.

At  $q_f = 0$ , this line is aligned with  $+x_p$ ; at  $q_f = \pi/2$ , it points radially outward. Increasing  $q_f$  therefore describes the outward-pivoting direction.

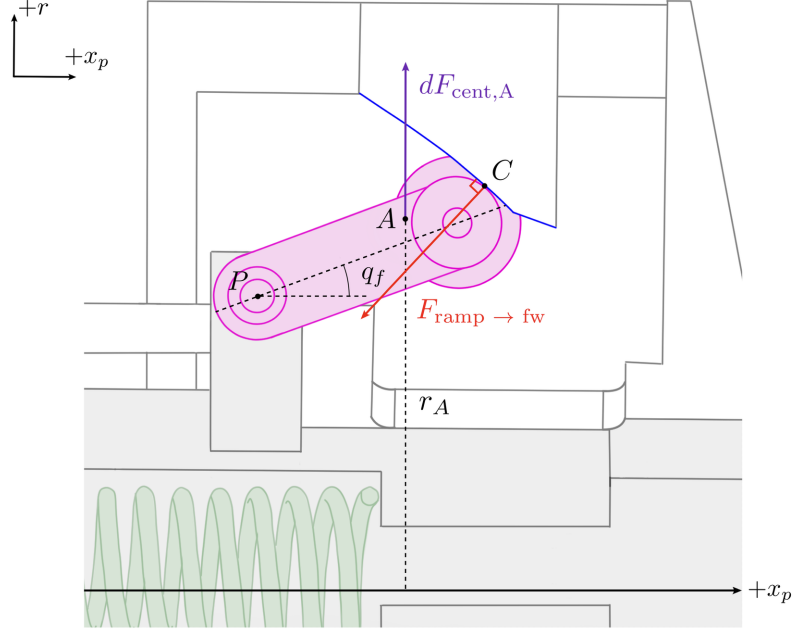


Figure 3.6: Free body of one primary flyweight in the frame rotating with the primary. Centrifugal loading acts throughout the flyweight mass; the marked element  $A$  identifies one infinitesimal mass  $dm$  at shaft radius  $r_A$  and its outward load. The ramp-contact normal at  $C$  and the pivot reaction at  $P$  complete the force system.

As the primary rotates at  $\omega_p$ , the flyweight experiences centrifugal loading throughout its mass. Each infinitesimal element is loaded radially outward according to its distance from the primary shaft. The marked element  $A$  is one example: for mass  $dm$  at shaft radius  $r_A$ , its outward load is

$$dF_{\text{cent},A} = \omega_p^2 r_A dm. \quad (3.2.2)$$

Every other element is loaded in the same way at its own shaft radius. Together, these forces form the distributed centrifugal loading of the flyweight.

Because the flyweight's relative motion is confined to the radial-axial plane, the Euler and Coriolis inertial forces are circumferential and have no moment about the circumferential pivot axis. The  $q_f$  balance therefore depends on  $\omega_p$ , but not on  $\dot{\omega}_p$ .

Taking moments about  $P$  separates the roles of these loads. The pivot reaction passes through  $P$  and therefore contributes no moment. The radial lines of action of the centrifugal loads generally do not pass through  $P$ , so together they produce a moment in the direction of increasing  $q_f$ . The ramp normal produces the opposing moment.

The figure shows one flyweight, whereas the primary contains a symmetric set whose members follow the same  $q_f$ . Let  $\tau_{f,\text{cent}}$  and  $\tau_{f,\text{contact}}$  denote the respective magnitudes of the centrifugal and contact moments summed over this complete set. Since their shared coordinate  $q_f$  is a rotation about the flyweight pivots, let  $I_f$  denote the sum of each flyweight's mass moment of inertia about its own pivot axis. With positive moment defined in the direction of increasing  $q_f$ , rotational Newton's second law gives

$$\tau_{f,\text{cent}} - \tau_{f,\text{contact}} = I_f \ddot{q}_f. \quad (3.2.3)$$

**Mechanism geometry** The moment balance is written in the flyweight coordinate  $q_f$ , whereas the force required by the movable-sheave balance must be expressed through the axial coordinate  $x_p$ . The ramp geometry connects the two: each sheave position fixes the corresponding flyweight pose.

The centrifugal moment in Equation (3.2.3) comes from the distributed loading shown in the figure, so its value depends on how the complete flyweight mass is distributed about the primary shaft. The marked element  $A$  contributes  $r_A^2 dm$  to the mass moment of inertia about that shaft. Adding the corresponding contributions from every element of every flyweight gives the complete shaft-axis inertia  $J_{f,p}$ . Since  $x_p$  fixes the flyweight pose, it fixes every element's shaft radius and therefore fixes  $J_{f,p}$ . This shaft-axis inertia should not be confused with the pivot-axis inertia  $I_f$  in Equation (3.2.3), which is constant for rigid flyweights.

The mechanism therefore supplies two configuration maps:

$$q_f = q_f(x_p), \quad J_{f,p} = J_{f,p}(x_p). \quad (3.2.4)$$

Together with the constant  $I_f$ , these maps abstract the physical geometry of the flyweight mechanism into the quantities required by the general force balance.

#### Remark (Constructing the flyweight maps)

The mechanism-specific construction is not carried out here. Appendix A.1 works through two common flyweight arrangements, showing how their geometry and mass distribution produce  $q_f(x_p)$ ,  $J_{f,p}(x_p)$ , and  $I_f$ .

With these mechanism properties specified, the remaining task is to express the two moments and  $\ddot{q}_f$  in Equation (3.2.3) through the primary motion. We begin with the moments.

Both could be found by resolving the forces and constructing their moment arms for the specific flyweight geometry. A more convenient approach is equivalent virtual work. Hold the mechanism at its current position and imagine a small compatible motion. During this motion, a moment  $\tau_f$  acting through  $dq_f$  performs the work

$$\tau_f dq_f = \delta W_f. \quad (3.2.5)$$

We can therefore find each moment by calculating the work of its underlying load during the same motion.

For the centrifugal loading, at fixed instantaneous angular speed, the work associated with a change  $dJ$  in shaft-axis inertia is  $\frac{1}{2}\omega^2 dJ$ . Applying this relation to the flyweights gives

$$\tau_{f,\text{cent}} dq_f = \delta W_{f,\text{cent}} = \frac{1}{2} \omega_p^2 dJ_{f,p}. \quad (3.2.6)$$

For the contact moment, the corresponding work is the force–displacement work  $F d\ell$ . The movable sheave moves axially through  $dx_p$ , so only the axial component of the ramp-contact force performs work. This component is  $F_{\text{fw},x_p}$ , the closing force introduced by the movable-sheave free body, and its work is therefore  $F_{\text{fw},x_p} dx_p$ . Under **Assumption 3 (Idealized Actuator Mechanics)**, the ramp contact is treated as an ideal frictionless constraint, so no work is dissipated at the contact. The work of the contact moment therefore appears entirely as axial work on the movable assembly. Using the positive magnitudes defined above,

$$\tau_{f,\text{contact}} dq_f = \delta W_{f,\text{contact}} = F_{\text{fw},x_p} dx_p. \quad (3.2.7)$$

**Flyweight acceleration** The two moment terms are now written in forms that can be returned to the rotational balance. Its remaining term is the flyweight angular acceleration. Ramp contact makes this acceleration a consequence of the sheave motion rather than an independent state. Since the contact geometry already gives  $q_f$  as a function of  $x_p$ , its angular velocity is

$$\dot{q}_f = \frac{dq_f}{dx_p} \dot{x}_p, \quad (3.2.8)$$

and one more derivative gives

$$\ddot{q}_f = \frac{dq_f}{dx_p} \ddot{x}_p + \frac{d^2 q_f}{dx_p^2} \dot{x}_p^2. \quad (3.2.9)$$

The second term appears when the amount of flyweight rotation required per unit sheave travel changes along the ramp. Even at constant  $\dot{x}_p$ , the flyweight then changes angular speed as it moves.

**Axial closing force** There is now an expression for every term in the rotational balance. Multiplying **Equation (3.2.3)** by  $dq_f$  and substituting **Equations (3.2.6)**, **(3.2.7)** and **(3.2.9)** brings them together,

$$\frac{1}{2} \omega_p^2 dJ_{f,p} - F_{\text{fw},x_p} dx_p = I_f \left( \frac{dq_f}{dx_p} \ddot{x}_p + \frac{d^2 q_f}{dx_p^2} \dot{x}_p^2 \right) dq_f. \quad (3.2.10)$$

Rearranging for the force transmitted to the movable sheave gives

$$\boxed{F_{\text{fw},x_p} = \frac{1}{2} \omega_p^2 \frac{dJ_{f,p}}{dx_p} - I_f \left( \frac{dq_f}{dx_p} \right)^2 \ddot{x}_p - I_f \frac{dq_f}{dx_p} \frac{d^2 q_f}{dx_p^2} \dot{x}_p^2.} \quad (3.2.11)$$

The result now has the form required by the movable-sheave free body. The first term is the centrifugal drive and therefore uses the shaft-axis property  $J_{f,p}$ . The final two terms arise from

acceleration about the flyweight pivots and therefore use the pivot-axis property  $I_f$ : the first reflects  $\ddot{x}_p$  through the motion ratio, while the second appears when that motion ratio changes with position.

#### Remark (Admissible flyweight maps)

The completed force law uses  $q_f''(x_p)$  and  $J_{f,p}'(x_p)$ , while the selected coordinate directions require primary closure to correspond to outward flyweight rotation. Over the admissible shift range, the supplied mechanism maps must therefore satisfy

$$q_f \in C^2, \quad J_{f,p} \in C^1, \quad 0 < \frac{dq_f}{dx_p} < \infty.$$

These are requirements on the fitted or constructed maps, not additional equations of motion.

The one-sided flyweight contact can transmit compression but not tension, so the force predicted by [Equation \(3.2.11\)](#) must satisfy  $F_{fw,x_p} \geq 0$ .

### Spring and Belt Forces

Returning to the axial force balance, the two remaining forces in [Equation \(3.2.1\)](#) act directly on the translating assembly.

**Primary spring** Let  $k_p$  be the spring stiffness and  $x_{p,\text{pre}}$  its installed compression when the primary is open at  $x_p = 0$ . Closing the sheave by  $x_p$  adds the same amount of compression, so the spring opening force is

$$F_{\text{spring},x_p} = k_p (x_{p,\text{pre}} + x_p). \quad (3.2.12)$$

**Belt reaction** Let  $N_p$  be the total integrated normal load over both primary sheave faces. This load is coupled to the belt and the secondary balance, so it remains an unknown at this stage. Under [Assumption 12 \(Symmetric Sheave-Face Sharing\)](#), the movable face carries one half of this total load. Its normal is inclined by the sheave half-angle  $\beta$  from the axial direction, so the opening force on the movable face is

$$F_{\text{belt},x_p} = \frac{N_p}{2} \cos \beta. \quad (3.2.13)$$

### Completed Primary Balance

Each force in the original free body now has a mechanism-level expression. Substitute [Equations \(3.2.11\)](#) to [\(3.2.13\)](#) into [Equation \(3.2.1\)](#) and collect the terms multiplying  $\ddot{x}_p$ :

$$\left[ m_p + I_f \left( \frac{dq_f}{dx_p} \right)^2 \right] \ddot{x}_p = \frac{1}{2} \omega_p^2 \frac{dJ_{f,p}}{dx_p} - I_f \frac{dq_f}{dx_p} \frac{d^2 q_f}{dx_p^2} \dot{x}_p^2 - k_p (x_{p,\text{pre}} + x_p) - \frac{N_p}{2} \cos \beta. \quad (3.2.14)$$

The flyweights now appear through the same two axes introduced by their free body. Rotation about the primary shaft, described by  $J_{f,p}$ , provides the speed-dependent closing drive. Rotation about the flyweight pivots, described by  $I_f$ , adds to the inertia that the axial motion must accelerate. If the motion ratio  $dq_f/dx_p$  changes along the ramp, the flyweights also accelerate even when  $\dot{x}_p$  is constant, producing the term proportional to  $\dot{x}_p^2$ . The spring and belt forces act directly on the movable assembly and oppose closure exactly as shown in the first free body.

This is the primary axial equation of motion. It states how the flyweight action is divided among spring compression, belt loading, and acceleration of the translating and pivoting hardware.

### Point-Mass Limit and Relation to Existing Models

The general force law becomes the familiar classical actuator model when the distributed flyweight mass is collapsed to its centre of mass and the flyweight's pivot dynamics are neglected. Let  $m_f$  be the total mass of the symmetric flyweight set and let  $r_f(x_p)$  be the common instantaneous centre-of-mass radius. Under the point-mass assumption, the shaft-axis inertia reduces to

$$J_{f,p} = m_f r_f^2. \quad (3.2.15)$$

Substituting this point-mass inertia into the centrifugal term of [Equation \(3.2.11\)](#), while omitting the pivot-acceleration terms, gives

$$F_{fw,x_p}^{pm} = m_f r_f \omega_p^2 \frac{dr_f}{dx_p}. \quad (3.2.16)$$

This is the familiar point-mass result. The factor  $m_f r_f \omega_p^2$  is the total centrifugal loading, while  $dr_f/dx_p$  is the local radial-to-axial motion ratio that converts that loading into sheave force. The path need not be straight, and its motion ratio need not be constant. A sliding mass on a straight ramp is one simple case; for a pivoted flyweight,  $r_f(x_p)$  instead follows from the flyweight pose and its mass geometry.

If the movable sheave is also in axial equilibrium, the point-mass actuator force balances the opening forces from the spring and belt:

$$m_f r_f \omega_p^2 \frac{dr_f}{dx_p} = k_p (x_{p,pre} + x_p) + \frac{N_p}{2} \cos \beta. \quad (3.2.17)$$

This reduction appears directly in the quasi-static model of [Messick \(2018\)](#). Their three wedge-shaped flyweights slide along linear ramps without changing orientation. Measured CAD geometry supplies the centre-of-mass radius and the radial-to-axial displacement ratio, giving a force of the same form as [Equation \(3.2.16\)](#). Friction is neglected, and the resulting primary force is supplied to the belt model as a boundary condition; shifting is treated as a sequence of equilibrium configurations rather than a time-dependent response. [Arango and Muñoz Alzate \(2020\)](#) use the same instantaneous force conversion to construct both sliding-weight ramps and pivoted flyweight-cam profiles. Their contact geometry is designed so that the available centrifugal force supplies the required clamping force through the local mechanical advantage.

Other treatments retain more of the contact geometry while keeping the same instantaneous-equilibrium closure. [Kim et al. \(2002\)](#) resolve the two contacts of a centrifugal roller and include

contact friction when converting roller centrifugal force into driver thrust. The current roller radius and contact angles vary with ratio, but no roller acceleration enters the thrust equation. Skinner (2020) resolve a flyweight, its supporting link, and the ramp reactions in a complete free body, including the centrifugal loading of both the flyweight and the link. In the present notation, those separate mass contributions could be collected in  $J_{f,p}(x_p)$ . Skinner’s derivation therefore retains a substantially richer equilibrium force path than a single fixed ramp factor, but still closes the flyweight system by force equilibrium at each configuration.

Despite their different levels of geometric detail, none of these actuator relations includes the acceleration of the flyweight mechanism. Messick does not model shift time, Skinner solves algebraically for the speed required at each configuration, and Kim places the time dependence in a separate experimentally identified first-order ratio law, using different fitted coefficients for upshift and downshift. Conversely, the transient belt model of Ballew (2015) imposes primary axial force through a controller rather than resolving the mechanical actuator. Across these examples, transient shift motion is therefore obtained without returning the flyweights’ own acceleration to the primary axial balance.

The distinction of the present formulation is therefore not that it can describe a curved ramp or a detailed flyweight shape; the equilibrium models above can already retain substantial geometric detail. Here, the complete mass distribution enters through  $J_{f,p}(x_p)$ , so the centrifugal drive is not restricted to a centre-of-mass point model. More importantly, the flyweight moment balance is not closed at instantaneous equilibrium. Mapping  $I_f\ddot{q}_f$  into  $x_p$  produces both the reflected pivot inertia and the changing-motion-ratio term in Equation (3.2.14). Consequently, the flyweight drive, spring force, belt reaction, translating mass, and pivoting mass all determine the same shift acceleration rather than being divided between an equilibrium actuator law and a separate shift-response model. No equivalent explicit reflection of the flyweights’ pivoting inertia was identified in the reviewed mechanically actuated rubber-belt CVT formulations.

This extension does not retain every effect represented in the earlier models. Structural deformation is excluded by Assumption 2 (Rigid Pulley and Actuation Hardware), while Assumption 3 (Idealized Actuator Mechanics) treats the ramp contact as an ideal constraint rather than resolving its tribology or local compliance. The roller-contact friction retained by Kim et al. (2002), for example, therefore lies outside the present force law; roller spin and contact compliance are likewise not resolved. The added fidelity is in the coupled mechanism dynamics, not in frictional or compliant contact detail.

### 3.2.2 Secondary Movable-Sheave Balance

While the primary closes mainly in response to rotational speed, the secondary is torque reactive: torque transmitted through the pulley can also increase the axial force clamping the belt.

To see how, begin with the two secondary sheaves rotationally disconnected. The fixed sheave rotates with the secondary shaft. The movable sheave can slide along that shaft and rotate relative to the fixed sheave. Because the belt contacts the two faces separately, it applies torque to each body. With no connection between them, the two bodies can accelerate differently and rotate relative to one another.

Now place a pin on the movable sheave in a straight slot running along the shaft. The movable sheave remains free to slide, but the pin prevents relative rotation. Whenever the two sheaves tend to rotate differently, the pin presses against the side of the slot and a normal reaction develops. Because the slot is axial, this normal acts entirely in the circumferential direction. It transfers

torque between the sheaves but produces no axial force.

A helix is the same slot tilted around the shaft. The contact normal remains perpendicular to the slot surface, but now has both circumferential and axial components. The circumferential component continues to couple the rotational motion of the sheaves, while the axial component acts on the movable sheave. Relative rotation is therefore allowed only when accompanied by axial travel along the helix.

These three cases and their contact reactions are compared in [Figure 3.7](#).

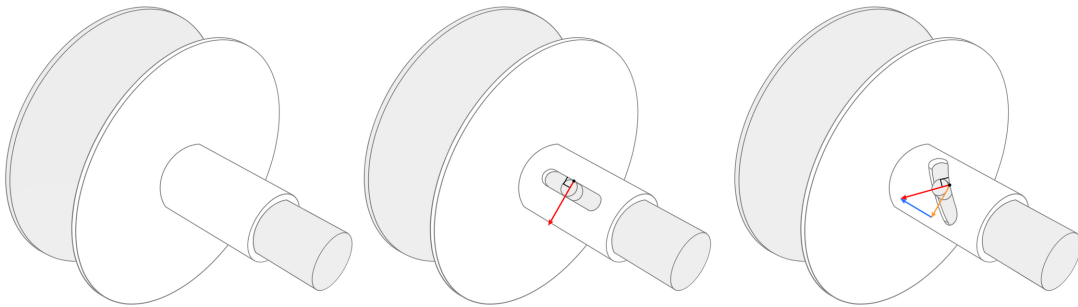


Figure 3.7: How the helix creates torque-reactive axial loading. With no connection, the two sheaves may rotate independently. A straight axial slot couples their rotation through a purely circumferential reaction; inclining the slot adds an axial component to that same reaction, coupling relative rotation to axial travel.

If the axial component of the helix reaction is balanced by the other axial loads, the movable sheave remains at its current position. Its relative angle then remains fixed, and the two sheaves rotate together with the shaft. If the axial loads do not balance, their resultant accelerates the movable sheave along the helix, and it rotates by the amount required by the slot geometry.

In the actual secondary, the belt and spring load this mechanism. The belt applies torque to the two sheave faces separately. The secondary spring is installed both compressed and twisted: its torsional preload tends to rotate the movable sheave relative to the fixed sheave, while its axial compression pushes the movable sheave toward closure. The torsional action loads the helix; the axial action enters the movable-sheave balance directly.

The axial free body gives the target for the derivation.

### Axial Free Body

[Figure 3.8](#) isolates the movable assembly in the axial direction.

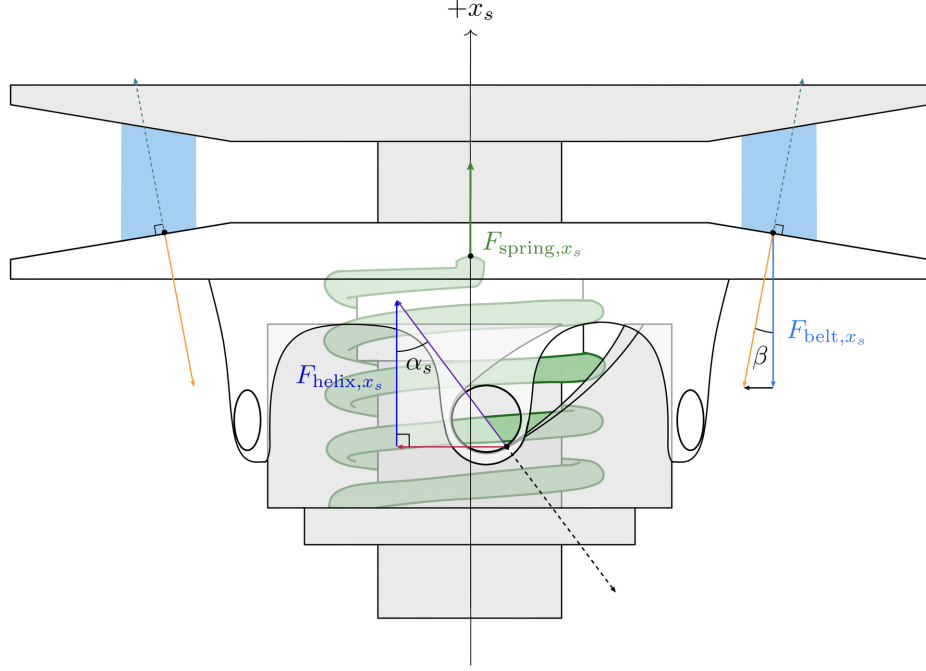


Figure 3.8: Axial free body of the complete movable secondary assembly. The axial spring force and belt reaction act directly on the translating assembly, while the secondary’s torque-reactive behaviour enters through the pin–helix force  $F_{\text{helix},x_s}$ .

Let  $m_s$  be the mass of the movable sheave and every component that translates rigidly with it. Three axial loads act on this assembly. The pin–helix contact contributes the signed force  $F_{\text{helix},x_s}$ . The compressed spring pushes the movable sheave toward the fixed sheave in  $+x_s$ . The belt pushes the two sheave faces apart, so its reaction on the movable face acts in  $-x_s$ , just as on the primary. Newton’s second law therefore gives

$$m_s \ddot{x}_s = F_{\text{helix},x_s} + F_{\text{spring},x_s} - F_{\text{belt},x_s}. \quad (3.2.18)$$

This free body makes the remaining task clear. The direct spring and belt forces follow from compression and contact geometry. The helix force does not: the tilted slot has established that an axial reaction exists, but not its magnitude or sign. Because the same contact reaction must also carry the torque that constrains relative rotation, its rotational effect is resolved first. This sets the order of the derivation:

1. use the rotational balance to find the torque carried by the helix;
2. use the helix geometry to convert that torque into axial force; and
3. add this force to the direct spring and belt loads in the axial balance.

### Rotational Loading of the Movable Sheave

As established in [Section 3.1](#),  $x_s$  is measured from the open secondary toward the fixed sheave, so  $+x_s$  closes the pulley. Let  $\theta_s$  measure the rotation of the movable sheave relative to the fixed

sheave in the same positive rotational sense already assigned to  $\omega_s$  (see [Section 2.3.1](#)). Only the conventional torque-reactive helix orientation is considered: as the secondary closes in  $+x_s$ , the movable sheave turns relative to the fixed sheave in  $+\theta_s$ .

With these directions set, the first question is what loads the helix circumferentially. The belt applies torque to the movable face, and the twisted spring applies torque between the two sheaves. The pin–helix contact supplies the torque that constrains the resulting relative motion to follow the slot.

Our goal is to find the torque reacted at the pin–helix contact. To do so, isolate the movable assembly and apply Newton’s second law for rotation. Use the following quantities:

- $I_{s,M}$ : rotational inertia of the movable sheave and every component that rotates rigidly with it;
- $\omega_{s,M}$ : absolute angular velocity of that assembly;
- $\tau_{s,M}$ : signed belt torque applied to the movable face;
- $\tau_{s,\text{spring}}$ : signed spring torque applied to the movable assembly; and
- $\tau_{s,\text{helix}}$ : the contact torque at the pin–helix connection, produced by the circumferential component of the contact normal. A positive  $\tau_{s,\text{helix}}$  acts on the movable assembly in  $-\theta_s$ .

The figures show one representative pin and track. The quantities listed above and later in this section are the totals over the complete set of loaded pin–track contacts, so no additional track multiplier is required.

The rotational balance is therefore

$$\tau_{s,M} + \tau_{s,\text{spring}} - \tau_{s,\text{helix}} = I_{s,M}\dot{\omega}_{s,M}. \quad (3.2.19)$$

Solving for the helix torque gives

$$\tau_{s,\text{helix}} = \tau_{s,M} + \tau_{s,\text{spring}} - I_{s,M}\dot{\omega}_{s,M}. \quad (3.2.20)$$

[Equation \(3.2.20\)](#) gives the contact torque we want, but its right-hand side still needs three relations:

- the share of belt torque applied to the movable face,  $\tau_{s,M}$ ;
- the torque supplied by the wound spring,  $\tau_{s,\text{spring}}$ ; and
- the absolute angular acceleration of the movable assembly,  $\dot{\omega}_{s,M}$ .

The belt term follows immediately from the face-torque assumption.

**Movable-face belt torque** Let  $\tau_s$  be the total signed torque exerted by the belt on the secondary, so  $\tau_s = \tau_{s,F} + \tau_{s,M}$ . Under [Assumption 12 \(Symmetric Sheave-Face Sharing\)](#), the two faces receive equal torque shares. Only the movable-face share appears in its free body:

$$\tau_{s,M} = \frac{\tau_s}{2}. \quad (3.2.21)$$

This equal split is a reduced-contact assumption. During shift, the movable face rotates slightly faster or slower than the fixed face. The two belt–face contacts can therefore have different sliding conditions, so their torques need not divide equally.

### Remark (Why the face-torque split can matter)

Sorge and Cammalleri (2011) resolve the two contacts separately and show that the face torques can depart substantially from an equal split; distributed tangential forces on the two faces can even act in opposite directions over parts of the wrap. Since  $\tau_{s,M}$  enters Equation (3.2.20) directly, this is also a potential error in the torque reacted by the helix.

The remaining terms both depend on the helix kinematics:  $\dot{\omega}_{s,M}$  depends on how the movable sheave rotates as it shifts, and the spring torque depends on how far that motion unwinds the spring. We therefore derive the relation between  $x_s$  and  $\theta_s$  next.

### Motion Imposed by the Helix

The helix geometry tells us how axial shift changes the relative angle of the movable sheave. Unwrap one track from its effective cylindrical radius  $r_h$ , as seen in Figure 3.9. A small relative rotation  $d\theta_s$  moves the pin by  $r_h d\theta_s$  circumferentially while the sheave moves by  $dx_s$  axially. Let  $\alpha_s$  be the local angle of the unwrapped track, measured from the positive circumferential direction toward the positive axial direction. For the conventional helix orientation considered here,

$$0 < \alpha_s < \frac{\pi}{2}.$$

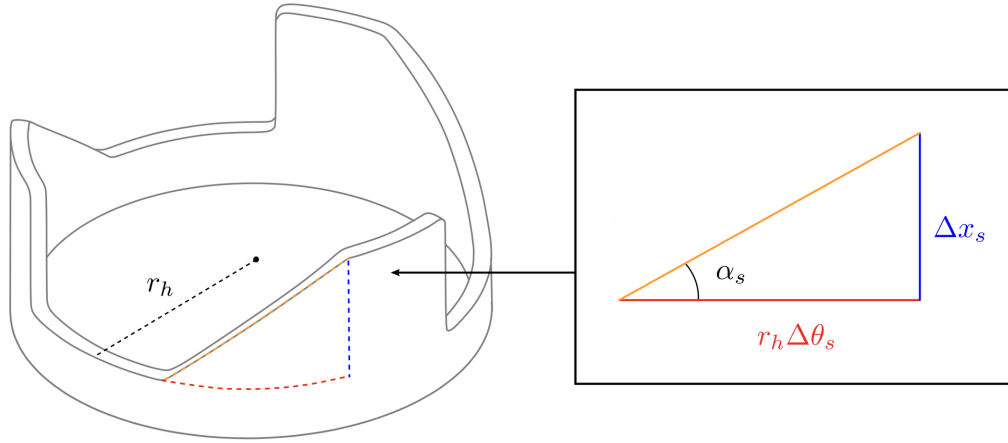


Figure 3.9: Unwrapped secondary-helix geometry. The same track angle relates relative rotation to axial travel and reaction torque to axial force.

Taking the infinitesimal limit of the triangle in Figure 3.9,

$$\tan \alpha_s = \frac{dx_s}{r_h d\theta_s}, \quad \frac{d\theta_s}{dx_s} = \frac{1}{r_h \tan \alpha_s}. \quad (3.2.22)$$

Thus,  $d\theta_s/dx_s$  is the relative rotation required per unit axial travel. It is positive for the conventional helix orientation assumed here. Returning to the straight axial slot from the opening example,  $\alpha_s \rightarrow \pi/2$  and  $d\theta_s/dx_s \rightarrow 0$ , as expected.

Taking  $\theta_s(0) = 0$  at the open secondary, integrate the local motion ratio to obtain

$$\theta_s(x_s) = \int_0^{x_s} \frac{1}{r_h \tan \alpha_s(\xi)} d\xi. \quad (3.2.23)$$

**Equation (3.2.23)** defines the helix as a function  $\theta_s(x_s)$ : for each axial position, it gives the relative angle enforced by the track. The function need not come from a constant helix angle. Any supplied function may be used provided it is single-valued and strictly increasing over the admissible shift range.

**Rotation of the movable sheave** We can now use this function to determine the two remaining quantities in the rotational balance. First consider the absolute motion of the movable sheave. The fixed sheave rotates with the secondary shaft at  $\omega_s$ , while the movable sheave also has the relative rotation  $\theta_s$ :

$$\omega_{s,M} = \omega_s + \dot{\theta}_s = \omega_s + \frac{d\theta_s}{dx_s} \dot{x}_s. \quad (3.2.24)$$

Because  $\theta_s$  was defined in the same positive rotational sense as  $\omega_s$ , the relative speed enters with the plus sign shown here. Differentiating gives the angular acceleration required by the rotational balance:

$$\dot{\omega}_{s,M} = \dot{\omega}_s + \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2. \quad (3.2.25)$$

The shaft acceleration  $\dot{\omega}_s$  remains unknown here because  $\omega_s$  is a state of the coupled model. It is solved together with the axial motion rather than by the helix relation alone. The term proportional to  $\ddot{x}_s$  is the angular acceleration caused by axial acceleration. The term proportional to  $\dot{x}_s^2$  appears only when the motion ratio varies with position.

**Torsional Spring Loading** The same helix function tells us how much the spring has unwound. Let  $k_{s,\theta}$  be the spring's torsional stiffness and  $\theta_{s,\text{pre}}$  its signed twist at  $x_s = 0$ . Increasing  $\theta_s$  reduces this twist, leaving  $\theta_{s,\text{pre}} - \theta_s(x_s)$ . The resulting torque on the movable assembly is

$$\tau_{s,\text{spring}} = k_{s,\theta} [\theta_{s,\text{pre}} - \theta_s(x_s)]. \quad (3.2.26)$$

For the installed winding assumed here, this quantity is positive while the spring remains preloaded. The opposite winding direction can be represented by a signed  $\theta_{s,\text{pre}}$ .

We now have relations for all three terms on the right-hand side of **Equation (3.2.20)**. Substituting them gives the pin-helix contact torque:

$$\begin{aligned}\tau_{s,\text{helix}} = & \frac{\tau_s}{2} + k_{s,\theta} [\theta_{s,\text{pre}} - \theta_s(x_s)] \\ & - I_{s,M} \left[ \dot{\omega}_s + \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2 \right].\end{aligned}\tag{3.2.27}$$

This completes the rotational part of the derivation. The next step is to use the pin–helix contact geometry to find the axial force associated with this torque.

### Axial Component of the Helix Reaction

The torque above is the moment produced by the circumferential component of the contact normal. Let  $F_{\text{helix},x_s}$  denote the signed axial force produced by the helix reaction, positive in  $+x_s$ .

Under **Assumption 3 (Idealized Actuator Mechanics)**, the pin–helix contact is treated as an ideal frictionless constraint. The same compatible-work argument used for the primary therefore converts the rotational contact load into its axial effect without contact dissipation. Consider the motion of the movable sheave relative to the helix carrier. The contact normal does no net work in this relative motion. Its axial component contributes  $F_{\text{helix},x_s} dx_s$ , while its circumferential component contributes  $-\tau_{s,\text{helix}} d\theta_s$ . Their sum is zero, so

$$F_{\text{helix},x_s} dx_s = \tau_{s,\text{helix}} d\theta_s.\tag{3.2.28}$$

Using the motion ratio gives

$$F_{\text{helix},x_s} = \tau_{s,\text{helix}} \frac{d\theta_s}{dx_s}.\tag{3.2.29}$$

The motion ratio  $d\theta_s/dx_s$  is therefore the mathematical origin of the axial effect: it converts the contact torque into axial force. In the straight-slot limit used in the opening example, this ratio is zero, so the slot can react torque while the axial conversion vanishes.

Substituting **Equation (3.2.27)** gives the helix force in terms of the model variables:

$$\boxed{F_{\text{helix},x_s} = \frac{d\theta_s}{dx_s} \left\{ \frac{\tau_s}{2} + k_{s,\theta} [\theta_{s,\text{pre}} - \theta_s(x_s)] - I_{s,M} \left[ \dot{\omega}_s + \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2 \right] \right\}}.\tag{3.2.30}$$

This gives  $F_{\text{helix},x_s}$ . The belt and wound spring provide the driving torques inside the braces; part of that torque may be required to accelerate the movable assembly, and only the remainder is converted into axial thrust.

#### Remark (Admissible helix maps)

The completed force uses the first two derivatives of  $\theta_s(x_s)$ . Over the admissible shift range, the supplied helix map must therefore be single-valued and twice continuously differentiable, with a finite, nonzero motion ratio:

$$\theta_s \in C^2, \quad 0 < \frac{d\theta_s}{dx_s} < \infty.$$

A manufactured multi-angle helix may still be represented, but nominal segments must be joined smoothly rather than by a mathematical corner.

The difficult force in the preliminary axial balance is now known. The two remaining forces act directly on the translating assembly.

#### Direct Spring and Belt Forces

**Direct axial spring force** The spring force follows directly from its compression. Let  $k_{s,x}$  be the axial stiffness and  $x_{s,\text{pre}}$  the compression at  $x_s = 0$ . Closing the secondary by  $x_s$  lets the spring extend by the same amount, so its remaining compression is  $x_{s,\text{pre}} - x_s$ . Hence,

$$F_{\text{spring},x_s} = k_{s,x} (x_{s,\text{pre}} - x_s). \quad (3.2.31)$$

This force and the torque in [Equation \(3.2.26\)](#) come from the same installed spring. Following the reduced decomposition of [Kim et al. \(2002\)](#), the model represents its axial compression and torsional twist with separate linear laws. This use of prescribed, uncoupled spring laws is part of [Assumption 3 \(Idealized Actuator Mechanics\)](#); compression–twist coupling, hysteresis, and end-contact friction are not retained.

**Belt opening reaction** The belt opening force follows the same projection already used on the primary. Let  $N_s$  be the integrated normal load over both secondary faces. Under [Assumption 12 \(Symmetric Sheave-Face Sharing\)](#), the movable face carries one half of this total load, giving

$$F_{\text{belt},x_s} = \frac{N_s}{2} \cos \beta. \quad (3.2.32)$$

As on the primary,  $N_s$  is not determined by this local free body. It remains a coupled unknown supplied by the belt-contact and remaining system equations.

#### Completed Local Secondary Balance

We now have a relation for every term in the axial free body. Substitute them into [Equation \(3.2.18\)](#) and collect the terms multiplying  $\ddot{x}_s$ :

$$\begin{aligned}
\left[ m_s + I_{s,M} \left( \frac{d\theta_s}{dx_s} \right)^2 \right] \ddot{x}_s = & \left[ \frac{\tau_s}{2} + k_{s,\theta} (\theta_{s,\text{pre}} - \theta_s) - I_{s,M} \dot{\omega}_s \right] \frac{d\theta_s}{dx_s} \\
& - I_{s,M} \frac{d\theta_s}{dx_s} \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2 \\
& + k_{s,x} (x_{s,\text{pre}} - x_s) - \frac{N_s}{2} \cos \beta.
\end{aligned} \tag{3.2.33}$$

The completed equation separates into four physical pieces:

- $m_s + I_{s,M}(d\theta_s/dx_s)^2$  is the local axial inertia. Just as the flyweight pivot inertia appeared in the primary balance, the helix makes the secondary's relative-rotational inertia part of the load that axial motion must accelerate;
- the first bracket contains the belt and torsional-spring torques, less the portion required to accelerate the movable assembly with the secondary shaft;
- the term proportional to  $\dot{x}_s^2$  accounts for a helix motion ratio that varies with position; and
- the last two terms are the direct spring closing force and belt opening force.

This is the axial equation of motion we set out to obtain. If its right-hand side remains zero and the sheave is initially stationary,  $x_s$  and  $\theta_s$  remain fixed while the two sheaves rotate together. A nonzero right-hand side accelerates the sheave axially, and  $\theta_s(x_s)$  fixes the accompanying relative rotation.

### Classical Limit and Relation to Existing Models

As on the primary, the full dynamic balance should recover the established actuator equilibrium when the shift mechanism is stationary. Consider a fixed secondary configuration at constant shaft speed,  $\dot{x}_s = \ddot{x}_s = \dot{\omega}_s = 0$ . Under the equal face-torque split adopted here, the helix force reduces to

$$F_{\text{helix},x_s} = \left[ \frac{\tau_s}{2} + k_{s,\theta} (\theta_{s,\text{pre}} - \theta_s) \right] \frac{d\theta_s}{dx_s}. \tag{3.2.34}$$

Adding the direct spring and belt loads gives the corresponding equilibrium condition for the complete movable assembly:

$$\left[ \frac{\tau_s}{2} + k_{s,\theta} (\theta_{s,\text{pre}} - \theta_s) \right] \frac{d\theta_s}{dx_s} + k_{s,x} (x_{s,\text{pre}} - x_s) = \frac{N_s}{2} \cos \beta. \tag{3.2.35}$$

These equations expose the classical mechanism directly. The axial spring supplies one closing force, while the belt torque and torsional-spring torque are converted into a second closing force by the helix motion ratio. The factor  $1/2$  is not a property of the helix itself; it follows from the equal face-torque closure used here. A face-resolved contact model would instead use the torque  $\tau_{s,M}$  actually applied to the movable face.

The important comparison is how this equilibrium relation is used. In the present formulation, [Equations \(3.2.34\) and \(3.2.35\)](#) are stationary limits of a dynamic balance. In much of the mechanically actuated CVT literature, an equivalent algebraic relation remains the actuator closure while the transmission is shifting.

The formulation of Kim et al. (2002) is a direct example. Their driven thrust is the sum of a spring force and a torque-cam force obtained from instantaneous equilibrium. The cam term retains the applied torque, torsional-spring load, cam geometry, and cam friction. Suppressing that friction and adopting the equal face-torque split used here gives the same spring-plus-torque-cam structure as Equation (3.2.35). Their shift transient is not obtained from the movable secondary’s equation of motion, but from a separately identified first-order ratio law. Kim et al. therefore retain contact friction that is absent here, but do not use the secondary assembly’s inertia to determine its shift acceleration.

The design-oriented sources examined here apply the same configuration-by-configuration logic at different levels of detail. Messick (2018) calculates the secondary thrust required by a steady-state belt solution and uses it as a target for helix design, explicitly excluding the shift response time. Skinner (2020) calculates the compression-spring, torsional-spring, and torque-feedback contributions to secondary clamping algebraically. Arango and Muñoz Alzate (2020) construct the torque-sensing cam profile from the transmitted torque, spring force, required axial force, and axial displacement over the operating range. These treatments are useful for sizing and tuning the mechanism, but they do not determine its motion from a movable-sheave equation of motion.

Among the reviewed sources, Sorge and Cammalleri (2011) provide the most detailed treatment of the belt–face–helix interaction during shifting. Their distributed contact model allows the fixed and movable faces to rotate at different speeds and determines the torque  $M_{N,s}$  carried by the movable face. The resulting driven-side actuator relation combines the spring load with a helix contribution having the same torque-times-motion-ratio structure as Equation (3.2.34). It therefore replaces the assumed  $\tau_s/2$  with a face-resolved torque and can capture substantial departures from an equal split.

That contact calculation is shift-aware, but the secondary actuator itself is still closed quasi-statically. Sorge and Cammalleri prescribe the instantaneous radius and the dimensionless shift speed  $\rho = \dot{r}/(\omega_f r)$ , then solve for the corresponding belt-contact state. A nonzero  $\rho$  changes the relative face speeds, sliding directions, contact tractions, and face-torque split, but it is not produced by the forces on the movable secondary. Because no axial or relative-rotational equation of motion is written for the movable half-pulley, its translational inertia and helix-reflected rotational inertia are absent by construction.

The models therefore retain complementary parts of the physics. Kim et al. retain cam friction, while Sorge and Cammalleri resolve the two belt–face contacts and their unequal torques. The present reduced model retains neither capability. It instead carries the movable assembly’s rotational balance through the general constraint  $\theta_s(x_s)$  and solves its free axial motion. This introduces the shaft-acceleration coupling, the reflected inertia  $I_{s,M}(d\theta_s/dx_s)^2$ , and the velocity-squared term generated when the helix motion ratio varies with position. Among the reviewed mechanically actuated rubber-belt formulations, no equivalent explicit projection of the movable side’s rotational inertia through a mechanical secondary helix was identified.

This distinction is deliberately narrow. The rigid-hardware, ideal-actuator, and symmetric-face idealizations are stated explicitly in Assumption 2 (Rigid Pulley and Actuation Hardware), Assumption 3 (Idealized Actuator Mechanics), and Assumption 12 (Symmetric Sheave-Face Sharing). They leave the compliant and frictional actuator details, and the face-resolved load redistribution of the models above, outside the present balance. The comparison is therefore not a claim of greater local contact fidelity; it isolates the added treatment of the movable assembly’s inertia and helix-constrained dynamics.

### 3.2.3 Coupling Through the Shared Shift Coordinate

The primary and secondary balances have been derived in their natural local coordinates,  $x_p$  and  $x_s$ . The fixed-length belt geometry makes those motions compatible. Choosing the primary displacement as the shared shift coordinate leaves the primary balance unchanged and supplies the secondary motion through the map  $x_s = x_s(s)$ .

#### Local-to-Shared Kinematics

Because the shared coordinate is the primary displacement,

$$x_p = s, \quad \dot{x}_p = \dot{s}, \quad \ddot{x}_p = \ddot{s}. \quad (3.2.36)$$

The secondary displacement follows from the fixed-length belt map. Its first two time derivatives are

$$x_s = x_s(s), \quad \dot{x}_s = \frac{dx_s}{ds} \dot{s}, \quad \ddot{x}_s = \frac{dx_s}{ds} \ddot{s} + \frac{d^2x_s}{ds^2} \dot{s}^2. \quad (3.2.37)$$

Positive  $x_p$  and  $x_s$  both denote sheave closing. Increasing  $s$  closes the primary but opens the secondary, so  $dx_s/ds < 0$ . If the belt geometry is nonlinear, this motion ratio changes through the shift and  $d^2x_s/ds^2 \neq 0$ . Both derivatives are determined by the belt-geometry relation established earlier.

#### Two Conditions on One Shift Motion

Substituting Equation (3.2.36) into Equation (3.2.14) gives the primary condition directly:

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div> <b>Primary Condition on the Shared Shift Motion</b> </div> <div> <b>(3.2.38)</b> </div> </div> <hr style="margin: 10px 0;"/> $  \begin{aligned}  \left[ m_p + I_f \left( \frac{dq_f}{ds} \right)^2 \right] \ddot{s} = & \frac{1}{2} \omega_p^2 \frac{dJ_{f,p}}{ds} \\  & - I_f \frac{dq_f}{ds} \frac{d^2q_f}{ds^2} \dot{s}^2 \\  & - k_p (x_{p,\text{pre}} + s) - \frac{N_p}{2} \cos \beta.  \end{aligned}  $ |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Because  $x_p = s$ , this is simply the local primary balance written using the shared coordinate.

The secondary condition follows just as directly. Substituting Equation (3.2.37) into Equation (3.2.33) and collecting terms gives

$$\begin{aligned}
 & \left[ m_s + I_{s,M} \left( \frac{d\theta_s}{dx_s} \right)^2 \right] \frac{dx_s}{ds} \ddot{s} \\
 &= \frac{d\theta_s}{dx_s} \left[ \frac{\tau_s}{2} + k_{s,\theta} (\theta_{s,\text{pre}} - \theta_s) - I_{s,M} \dot{\omega}_s \right] \\
 & \quad + k_{s,x} (x_{s,\text{pre}} - x_s) - \frac{N_s}{2} \cos \beta \\
 & \quad - \left\{ \left[ m_s + I_{s,M} \left( \frac{d\theta_s}{dx_s} \right)^2 \right] \frac{d^2 x_s}{ds^2} \right. \\
 & \quad \left. + I_{s,M} \frac{d\theta_s}{dx_s} \frac{d^2 \theta_s}{dx_s^2} \left( \frac{dx_s}{ds} \right)^2 \right\} \dot{s}^2.
 \end{aligned}$$

Here  $x_s$ ,  $\theta_s$ , and their local derivatives are evaluated at  $x_s = x_s(s)$ . Since  $dx_s/ds < 0$ , a force that closes the secondary drives the shift toward decreasing  $s$ , thereby opening the primary. Conversely, the secondary belt force acts in the opening direction and drives the shift toward increasing  $s$ . The kinematic map therefore carries the inverse coupling between the two pulleys directly into the acceleration balance.

The two terms proportional to  $\dot{s}^2$  have different geometric origins. The term containing  $d^2 x_s/ds^2$  appears because the belt geometry can require a changing amount of secondary travel per unit primary travel. The term containing  $d^2 \theta_s/dx_s^2$  is inherited from the local secondary balance and appears when the helix motion ratio varies with position; substituting  $\dot{x}_s = (dx_s/ds)\dot{s}$  supplies its squared local-to-shared motion ratio.

The two balances are not added into one axial free body. They remain separate conditions on the same shift acceleration: the same  $\ddot{s}$  must satisfy both movable assemblies. The belt reactions  $N_p$  and  $N_s$ , together with the coupled shaft and contact quantities, allow both conditions to hold. Those unknowns are supplied by the rotational, belt, and contact relations developed later.

### Relation to Existing Dynamic Mappings

Transient belt and pulley models provide broad precedent for dynamic ratio prediction. [Julió and Plante \(2011\)](#) discretize the rubber belt and integrate its motion, while [Ballew \(2015\)](#) allow both pulley speeds to evolve from the applied torques and inertias. In that model family, however, pulley axial force is prescribed and the radius follows from belt compression; a movable-sheave equation  $m_j \ddot{x}_j = \sum F_{j,x}$  is not integrated.

Movable-sheave dynamics and shared-coordinate reductions also predate the present model. [Duan et al. \(2016\)](#) retain movable-pulley mass in a chain CVT and couple the pulley motions through chain deformation and tension. Most directly, [da Silva et al. \(2023\)](#) write axial equations for both movable sheaves of a mechanically actuated rubber-belt CVT, impose a linear relation between their displacements, and reduce the two translations to one shift equation. Thus transient shifting, axial sheave inertia, and a shared shift coordinate are not individually new.

The distinction here lies in how these ingredients are combined. The fixed-length belt geometry supplies the nonlinear map  $x_s(s)$ , so both  $dx_s/ds$  and  $d^2 x_s/ds^2$  remain in the secondary acceleration.

At the same time, the flyweight pivot inertia and the movable secondary's relative-rotational inertia remain tied to their own mechanism maps,  $q_f(x_p)$  and  $\theta_s(x_s)$ . The result is a pair of local free-body balances expressed as simultaneous conditions on one shift acceleration.

This reduction also defines the tradeoff. It does not retain the spatial belt deformation fields of [Julió and Plante \(2011\)](#) and [Ballew \(2015\)](#), or the prescribed-shift, face-resolved torque distribution of [Sorge and Cammalleri \(2011\)](#). It instead keeps the actuator and movable-sheave inertias explicit while closing the nominal belt geometry without spatial discretization. Among the reviewed mechanically actuated rubber-belt formulations, no equivalent combination of these features was identified.

### 3.2.4 What the Axial Balances Establish

The primary and secondary now have complete axial equations of motion in their natural coordinates. On the primary, shaft speed produces a closing action because the flyweight centres of mass move outward as the movable sheave closes. Part of that action accelerates the flyweights about their pivots before the remainder is transmitted to the movable assembly. On the secondary, the belt and torsional-spring torques act through the helix. Part of their action changes the rotation of the movable sheave relative to the shaft-side assembly before the remainder appears as axial force.

Although the two actuators are physically different, their constrained motions produce the same underlying structure. The map  $q_f(x_p)$  reflects the flyweights' pivot inertia into the primary axial coordinate, while the helix map  $\theta_s(x_s)$  reflects the movable-sheave rotational inertia into the secondary coordinate. In each case, the local motion ratio contributes an effective axial inertia, and variation of that ratio with position produces a velocity-squared term. The primary closing action is additionally set by  $dr_f/dx_p$ , while the secondary balance couples to the shaft acceleration because its movable sheave rotates relative to an assembly that is itself rotating.

The shared belt geometry adds a second layer of constrained motion. The primary map  $x_p(s) = s$  carries its local balance directly into the shift coordinate, whereas the secondary map  $x_s(s)$  reflects its local motion into the same  $\ddot{s}$  and contributes its own velocity-squared curvature term. Both free-body relations must therefore be satisfied by one shift motion.

They do not, however, determine that motion in isolation. At this stage, the two axial equations contain the five unknown quantities

$$\ddot{s}, \quad N_p, \quad N_s, \quad \tau_s, \quad \dot{\omega}_s.$$

The belt normal loads, transmitted secondary torque, and secondary shaft acceleration follow only when the rotational and belt-contact mechanics are added. The two balances derived here therefore form the axial part of one simultaneous dynamic solution for the engaged CVT.

This completes the axial force description. Attention now turns from translation along the pulley shafts to rotation about them.

## 3.3 Rotational Dynamics of the Pulley Assemblies

The preceding section established how forces along the pulley shafts move the sheaves, but its equations still contain quantities belonging to rotation about those shafts. The primary flyweight force depends on the primary speed  $\omega_p$ , while the secondary balance contains the transmitted torque  $\tau_s$  and the secondary acceleration  $\dot{\omega}_s$ . The axial equations can use these quantities, but cannot supply them.

The primary pulley is carried around the primary shaft at  $\omega_p$ , while the secondary rotates about its own shaft at  $\omega_s$ . These are separate rotations, so their speeds need not change together. The mechanisms inside each pulley also move relative to these underlying shaft rotations. The primary flyweights pivot as the primary rotates, while the secondary movable sheave translates and rotates relative to its shaft through the helix. These internal motions are already tied to the shift coordinate, but they also change how angular momentum is distributed within each pulley as the CVT shifts.

The remaining question is therefore what governs the evolution of  $\omega_p$  and  $\omega_s$ , and consequently how shifting affects that evolution.

### 3.3.1 Following the Torque

To determine how  $\omega_p$  and  $\omega_s$  evolve, each pulley side requires a rotational equation of motion. That equation must account for the angular momentum carried about the corresponding shaft and the torques capable of changing it. The first step is therefore to identify the rotating systems represented by the two shaft speeds.

The primary-side system contains the primary shaft and the complete primary pulley assembly. The movable assembly and flyweights remain part of this system even while they move relative to the fixed sheave, since they continue to carry angular momentum about the primary axis. The secondary-side system is defined in the same way: it contains the secondary shaft and the complete secondary pulley assembly, including the helix-guided motion of the movable sheave. The belt passes between these two systems and is not included in either one.

The resulting division is shown in [Figure 3.10](#).

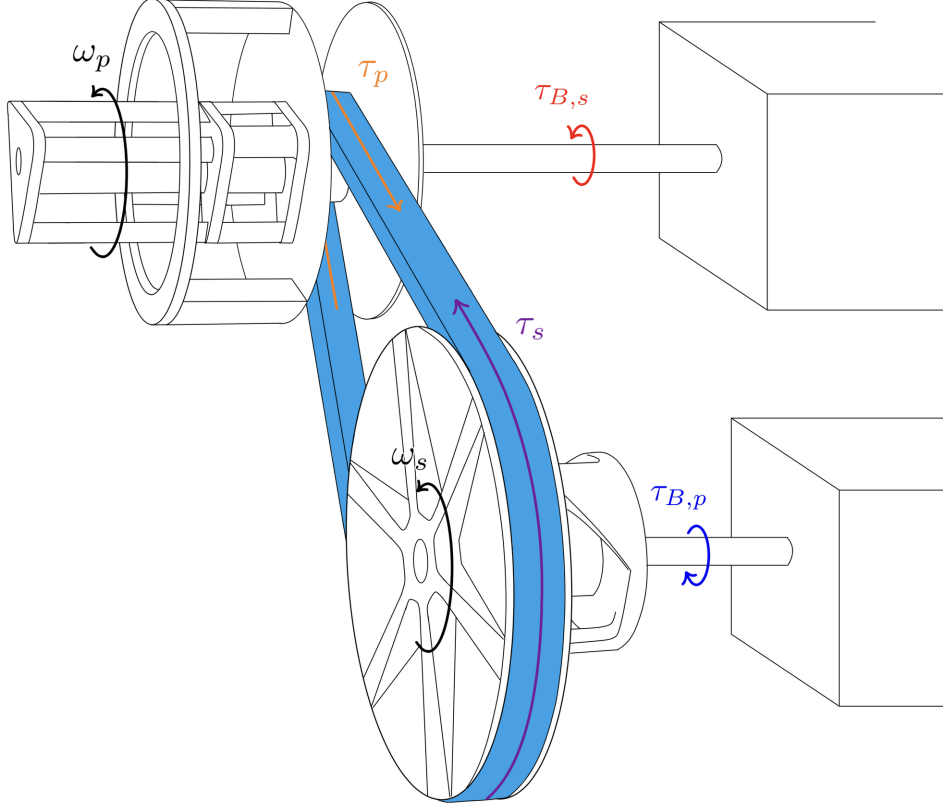


Figure 3.10: Rotating systems and torque path of the engaged CVT. The shaft-side boundaries represent the torque and equivalent inertia presented by the attached systems. The belt remains outside the two pulley-side systems and interacts with them through the contact torques  $\tau_p$  and  $\tau_s$ . The arrows show the signed directions associated with forward power transfer.

Begin with the primary. When its speed changes, the primary shaft must accelerate not only the pulley itself, but also the rotating hardware attached beyond the CVT boundary. As established in [Section 2.1](#), that external hardware is represented at the shaft by the equivalent inertia  $I_{B,p}$ , while  $\tau_{B,p}$  is the torque it applies to the primary side. The remaining external interaction is the belt, which exerts the torque  $\tau_p$  on the primary pulley.

The secondary follows the same construction. Its attached hardware contributes the equivalent inertia  $I_{B,s}$  and applies the boundary torque  $\tau_{B,s}$ , while the belt exerts  $\tau_s$  on the secondary pulley.

The signs follow directly from the convention established in [Section 2.3.1](#). During forward power transfer, the primary boundary drives the primary pulley and the belt resists it. The belt then drives the secondary pulley against the resisting secondary boundary. Thus,

$$\tau_{B,p} > 0, \quad \tau_p < 0, \quad \tau_s > 0, \quad \tau_{B,s} < 0.$$

Although the primary and secondary contain different mechanisms, their rotational balances now have the same outer structure. Let  $j \in \{p, s\}$  denote the pulley side under consideration.

The angular momentum on side  $j$  has two contributions. The fixed equivalent boundary inertia rotates at  $\omega_j$ , so it carries angular momentum  $I_{B,j}\omega_j$ . Let  $H_{\text{CVT},j}$  denote the angular momentum carried by the corresponding shaft and pulley assembly, including the internal motions of its components. The total is therefore

$$H_j = I_{B,j}\omega_j + H_{\text{CVT},j}. \quad (3.3.1)$$

Bearing, seal, windage, and other unmodeled resisting torques are omitted under **Assumption 15 (Omitted Parasitic Component Losses)**. Within the resulting pulley-side free body, the applied boundary torque and belt torque are therefore the only external torques retained. Newton's second law for rotation gives

$$\tau_{B,j} + \tau_j = \frac{dH_j}{dt}.$$

Substituting **Equation (3.3.1)** writes this as

$$\tau_{B,j} + \tau_j = \frac{d}{dt}(I_{B,j}\omega_j + H_{\text{CVT},j}).$$

The equivalent boundary inertia is fixed, so differentiating yields the common pulley-side equation of motion:

|                                                                                                     |                |
|-----------------------------------------------------------------------------------------------------|----------------|
| <b>Common Pulley-Side Rotational Balance</b>                                                        | <b>(3.3.2)</b> |
| $\tau_{B,j} + \tau_j = I_{B,j}\dot{\omega}_j + \frac{dH_{\text{CVT},j}}{dt}, \quad j \in \{p, s\}.$ |                |

The way shifting changes the angular momentum carried within a pulley is contained in  $H_{\text{CVT},j}$ . On the primary, the flyweights redistribute mass relative to the shaft as they pivot. On the secondary, the movable sheave rotates relative to the shaft-side assembly as it follows the helix. The two pulley sides therefore share the balance in **Equation (3.3.2)**, but require separate expressions for their internal angular momentum. Beginning with the primary, each assembly can now be examined in turn.

### 3.3.2 Primary Pulley Assembly

The primary is the simpler of the two pulley assemblies because nearly all of its hardware shares the shaft speed  $\omega_p$ . The shaft, sheaves, ramp carrier, and other coaxial components may translate during a shift, but that translation is along the shaft and therefore does not change their distance from its axis. Their combined shaft-axis inertia can consequently be collected into one constant quantity,  $I_{p,\text{const}}$ .

The flyweights are different. They are carried around the primary shaft at  $\omega_p$ , but their mass moves radially as they pivot. The same motion was already used in **Section 3.2.1** to determine the centrifugal action that closes the primary. There, the flyweight geometry was summarized by the shaft-axis inertia map  $J_{f,p}(x_p)$ . Since  $x_p = s$ , that map is now  $J_{f,p}(s)$ .

It is useful to recall what this inertia represents. The quantity  $J_{f,p}$  measures the flyweight mass distribution about the *primary shaft*. It is therefore distinct from the constant pivot-axis inertia  $I_f$  that appeared in the flyweight moment balance of Equation (3.2.3). The two quantities describe the same flyweights about different axes and enter different equations of motion.

At a given shift position, the angular momentum carried inside the primary CVT boundary is therefore

$$H_{\text{CVT},p} = [I_{p,\text{const}} + J_{f,p}(s)] \omega_p. \quad (3.3.3)$$

If the shift position were fixed, differentiating this expression would simply multiply the current inertia by  $\dot{\omega}_p$ . During a shift, however,  $J_{f,p}$  changes as well. Because its only configuration variable is  $s$ , differentiating Equation (3.3.3) gives

$$\frac{dH_{\text{CVT},p}}{dt} = [I_{p,\text{const}} + J_{f,p}(s)] \dot{\omega}_p + \frac{dJ_{f,p}}{ds} \dot{s} \omega_p. \quad (3.3.4)$$

The two terms describe different ways of changing the same angular momentum. The first accelerates the primary with the flyweights held at their current configuration. The second changes the angular momentum because the flyweight mass itself moves toward or away from the shaft while already rotating.

Substituting this result into the common pulley-side balance from Equation (3.3.2) gives the primary equation of motion:

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div style="flex-grow: 1;"> <p><b>Primary Pulley Rotational Balance</b></p> <hr style="border: 0.5px solid black;"/> <math display="block">\begin{aligned} \tau_{B,p} + \tau_p &amp;= [I_{B,p} + I_{p,\text{const}} + J_{f,p}(s)] \dot{\omega}_p \\ &amp;\quad + \frac{dJ_{f,p}}{ds} \dot{s} \omega_p. \end{aligned}</math> </div> <div style="text-align: right; width: 15%;"> <p><b>(3.3.5)</b></p> </div> </div> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The connection to the axial flyweight balance is now explicit. In Equation (3.2.38), the derivative  $dJ_{f,p}/ds$  determined how shaft rotation produces a centrifugal closing action. Here, the same derivative determines how that closing motion changes the shaft-axis angular momentum. The two appearances are not separate flyweight corrections: they are the axial and rotational consequences of the same radial redistribution of flyweight mass.

For the conventional primary mechanism, outward flyweight motion increases  $J_{f,p}$ . During an upshift,  $\dot{s} > 0$ , so the rotating primary must supply the angular momentum required by that outward motion in addition to accelerating the shaft. During a backshift the redistribution reverses, and the flyweights return angular momentum to the shaft-side motion.

If the shift is held fixed,  $\dot{s} = 0$ , and the flyweight mass distribution no longer changes. The additional term vanishes and the familiar fixed-inertia form is recovered:

$$\tau_{B,p} + \tau_p = [I_{B,p} + I_{p,\text{const}} + J_{f,p}(s)] \dot{\omega}_p, \quad \dot{s} = 0. \quad (3.3.6)$$

### 3.3.3 Secondary Pulley Assembly

The secondary differs from the primary in one important way. During a shift, the complete secondary remains one mechanical assembly, but not every part of that assembly has the same angular speed.

The secondary shaft, fixed sheave, and hardware locked to them rotate at  $\omega_s$ . Let their combined shaft-axis inertia inside the CVT boundary be  $I_{s,F}$ . The movable sheave and the components rigidly attached to it have inertia  $I_{s,M}$ . As established while deriving the secondary axial balance, the helix makes this movable assembly rotate relative to the shaft-side assembly as it translates. Its absolute angular speed is therefore

$$\omega_{s,M} = \omega_s + \frac{d\theta_s}{dx_s} \dot{x}_s, \quad (3.3.7)$$

which is the relation already obtained in Equation (3.2.24). The angular momentum carried by the secondary must therefore retain the two angular speeds separately:

$$H_{\text{CVT},s} = I_{s,F}\omega_s + I_{s,M}\omega_{s,M}. \quad (3.3.8)$$

The free body here is larger than the movable-sheave free body used in Section 3.2.2. There, only the movable assembly was isolated, so the spring and helix torques crossed the boundary and had to appear explicitly. Here, the shaft-side and movable-side hardware are enclosed together. The torque exerted by the helix on the movable side is accompanied by an equal-and-opposite torque on the shaft-side assembly, and both actions lie inside the same free body. The torsional spring produces the same kind of internal pair. These torques therefore cancel from the moment balance of the complete secondary. Their mechanical effect has not disappeared: the helix still allows the movable side to rotate at  $\omega_{s,M}$  rather than  $\omega_s$ , and that difference remains in Equation (3.3.8).

The common pulley-side balance in Equation (3.3.2) requires the time rate of change of the angular momentum inside this boundary. Because Equation (3.3.8) contains two angular velocities, differentiating it requires the corresponding two angular accelerations:

$$\frac{dH_{\text{CVT},s}}{dt} = I_{s,F}\dot{\omega}_s + I_{s,M}\dot{\omega}_{s,M}. \quad (3.3.9)$$

The first acceleration,  $\dot{\omega}_s$ , is the shaft acceleration governed by the present rotational balance. The second is not another independent degree of freedom. The helix already tied the movable-side rotation to the axial sheave motion in Section 3.2.2. In particular, differentiating the helix kinematics there gave Equation (3.2.25):

$$\dot{\omega}_{s,M} = \dot{\omega}_s + \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2. \quad (3.3.10)$$

The rotational balance has therefore led back to a kinematic relation already established by the axial mechanism. Substituting Equation (3.3.10) into Equation (3.3.9) gives

$$\begin{aligned} \frac{dH_{\text{CVT},s}}{dt} = & (I_{s,F} + I_{s,M}) \dot{\omega}_s \\ & + I_{s,M} \left[ \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2 \right]. \end{aligned} \quad (3.3.11)$$

Substituting this rate into the common pulley-side rotational balance in Equation (3.3.2) produces the local secondary equation of motion:

|                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div><b>Secondary Pulley Rotational Balance</b></div> <div><b>(3.3.12)</b></div> </div> <hr style="border: 0.5px solid black; margin: 10px 0;"/> $\begin{aligned} \tau_{B,s} + \tau_s = & (I_{B,s} + I_{s,F} + I_{s,M}) \dot{\omega}_s \\ & + I_{s,M} \left[ \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2 \right]. \end{aligned}$ |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The first term accelerates the rotation shared by the complete secondary. The bracketed terms change only the movable side's rotation relative to the shaft. They are the rotational counterpart of the helix-reflected inertia already encountered in the secondary axial balance.

The complete CVT uses the shared shift coordinate rather than  $x_s$  as an independent coordinate. The required mapping was already established in Equation (3.2.37). Substituting

$$\dot{x}_s = \frac{dx_s}{ds} \dot{s}, \quad \ddot{x}_s = \frac{dx_s}{ds} \ddot{s} + \frac{d^2x_s}{ds^2} \dot{s}^2$$

into Equation (3.3.12) gives the same rotational balance in the coordinate used by the complete model:

$$\begin{aligned} \tau_{B,s} + \tau_s = & (I_{B,s} + I_{s,F} + I_{s,M}) \dot{\omega}_s \\ & + I_{s,M} \frac{d\theta_s}{dx_s} \frac{dx_s}{ds} \ddot{s} \\ & + I_{s,M} \left[ \frac{d\theta_s}{dx_s} \frac{d^2x_s}{ds^2} + \frac{d^2\theta_s}{dx_s^2} \left( \frac{dx_s}{ds} \right)^2 \right] \dot{s}^2. \end{aligned} \quad (3.3.13)$$

All local helix quantities in this expression are evaluated at  $x_s = x_s(s)$ . Its signs follow directly from the kinematics already chosen. For the conventional helix,  $d\theta_s/dx_s > 0$ , while the belt geometry gives  $dx_s/ds < 0$  over the engaged shift range. Hence

$$\frac{d\theta_s}{ds} = \frac{d\theta_s}{dx_s} \frac{dx_s}{ds} < 0.$$

During an upshift,  $\dot{s} > 0$ , so  $\dot{\theta}_s < 0$ : the secondary opens and the movable side rotates more slowly than the shaft-side assembly. The sign of the corresponding angular-momentum term is therefore carried by the mechanism maps rather than assigned separately in the rotational balance.

If the shift is held fixed,  $\dot{s} = \ddot{s} = 0$ . The helix then produces no relative motion, and the fixed and movable sides rotate together. The balance reduces to

$$\tau_{B,s} + \tau_s = (I_{B,s} + I_{s,F} + I_{s,M}) \dot{\omega}_s. \quad (3.3.14)$$

### 3.3.4 What the Rotational Balances Establish

The primary and secondary now have complete shaft-axis equations of motion. Both remain direct statements of  $\sum \tau = dH/dt$ . The additional terms appear because the hardware that actuates the pulleys continues to move while the pulley assemblies rotate, so shifting changes the angular momentum stored inside the assemblies themselves.

On the primary, this connection is carried by the same flyweight inertia map  $J_{f,p}(s)$  that appeared in the axial derivation. There,  $dJ_{f,p}/ds$  contributes to the centrifugal closing action because the flyweights move outward as the primary closes. Here,  $J_{f,p}(s)$  and its derivative determine how much shaft-axis angular momentum those same flyweights carry as their radial mass distribution changes. The axial and rotational balances are therefore two views of the same flyweight motion: one asks how rotation drives the movable sheave, while the other asks how that moving flyweight mass changes the rotational response of the pulley.

The secondary shows the same reciprocity through  $I_{s,M}$  and the helix map  $\theta_s(x_s)$ . In the axial balance, the helix reflects the movable assembly's rotational inertia into the sheave motion, so part of the available torque is required to change the movable side's rotation before the remainder appears as axial force. In the rotational balance, that same relative motion means the movable side carries angular momentum at a speed different from the shaft-side assembly, so the axial shift acceleration enters the shaft-axis balance. Again, the two equations are describing one constrained piece of hardware from different free bodies.

Mechanical actuation therefore creates a direct coupling between the axial and rotational dynamics inside each pulley assembly. This is distinct from the coupling carried by the belt reaction. Belt contact can connect the two kinds of balance because its normal action loads the sheaves axially while its tangential action transmits torque about the shafts. The present terms add a second path: the flyweight and helix mechanisms that generate the clamping response also carry inertia, so their own motion appears directly in the rotational equations. The actuator is therefore part of both balances, rather than acting only as a source of clamping force whose influence reaches shaft rotation through the belt.

This distinction gives a precise comparison with earlier transient models. In the rubber-belt formulation of [Ballew \(2015\)](#), belt-contact torque changes pulley speed through a fixed lumped pulley inertia, while pulley axial force is supplied to the belt model. Axial and rotational behaviour can still interact through the belt solution, but motion of the clamping hardware does not alter the pulley angular momentum itself. [Duan et al. \(2016\)](#) provide a stronger adjacent precedent in a chain CVT: rotational dynamics, movable-pulley axial dynamics, chain motion, and contact forces are solved as a coupled system. Their shaft balance nevertheless retains a fixed pulley inertia, while the movable-pulley mass appears in a separate axial equation. The coupling is therefore carried through the chain, contact forces, and changing radius rather than through actuator motion entering both balances.

A mechanically actuated rubber-belt comparison is provided by [da Silva et al. \(2023\)](#), who retain axial equations for both movable sheaves and reduce their translations to a shared shift coordinate. Their rotational treatment uses a fixed primary-pulley inertia. This establishes direct precedent for mechanical clamping, movable-sheave inertia, and coupled shift motion, but not for the additional

shaft-axis angular-momentum terms produced by the moving actuator hardware.

The helical model of [Sorge and Cammalleri \(2011\)](#) provides a complementary secondary comparison. Their formulation resolves the relative motion of the driven sheaves and the generally unequal belt torques acting on the two faces in substantially greater local detail. That work therefore captures secondary helix and face-contact mechanics that are reduced here. The distinction in the present balance is different: the helix-constrained movable assembly is retained as part of the complete secondary angular momentum, so the same relative motion that appears in its axial dynamics also appears in the shaft-axis equation.

Taken together, these comparisons show that axial-rotational coupling itself is not the new ingredient. Existing formulations can produce that coupling through belt or chain reactions, changing radii, and shared shift motion. Among the reviewed reduced-order formulations for mechanically actuated rubber-belt CVTs, however, no equivalent pair of pulley balances was identified in which the moving clamping hardware itself provides this additional inertial coupling on both sides: the primary flyweights change their shaft-axis mass distribution as they generate closing action, while the secondary movable assembly changes its shaft-axis angular momentum through the same helix motion that produces its axial response.

The result is a single mechanical picture of each pulley rather than separate axial and rotational descriptions joined only through transmitted load. The same mechanism maps that determine how the pulley clamps also determine how its internal motion changes the shaft dynamics. When those actuator motions cease, the additional terms vanish and the familiar fixed-configuration inertia balances are recovered.

### 3.4 Belt Transport and Pulley Contact Mechanics

The preceding sections established the axial and rotational equations of motion for the two pulley assemblies, but both stop at the same remaining mechanical body: the belt. The axial balances still contain the belt normal loads  $N_p$  and  $N_s$ , while the rotational balances contain the belt torques  $\tau_p$  and  $\tau_s$ . The belt transport speed  $v_b$  is also retained as a state, but its acceleration has not yet been established.

So far, each of these quantities has appeared only through the action of the belt on one of the pulley assemblies. Newton's third law gives the same interactions from the opposite side. The force and torque exerted by the belt on a pulley are accompanied by equal-and-opposite actions on the belt itself. The remaining mechanics can therefore be viewed from the belt rather than from either pulley.

This change of free body is important because the belt is not two independent contacts. It is one closed body connecting both pulley assemblies. Whatever forces and torques are exchanged at the primary must coexist with those at the secondary and with the motion of the belt between them. The belt mechanics must therefore be self-consistent around the complete loop.

#### 3.4.1 Complete-Belt Transport

The belt transport speed provides the natural starting point. The belt speed  $v_b$  can change only if something pushes or pulls the belt along its path. Those tangential actions come from the primary and secondary pulley contacts. The belt's transport acceleration must therefore come from the two pulley wraps.

At either wrap, the mechanism is simple. The pulley contact has both normal and tangential components. Normal contact acts perpendicular to the local belt path, so it cannot push or pull the belt in the direction that defines  $v_b$ . Frictional traction, by contrast, is spread over the contact and acts tangent to the belt at every point. Traction acting in the direction of belt travel does positive work on the belt, while traction acting against belt travel does negative work and removes energy from it. Moreover, when the complete closed belt is considered, the tensions between adjacent belt sections are internal and cancel. The primary and secondary tractions are therefore the only external actions that can change the belt's transport speed.

Each small traction force also acts at a radius from the shaft and therefore produces a small moment on the pulley. Adding these moments around the wrap gives the contact torque  $\tau_j$  already used in the pulley rotational balance. Thus,  $\tau_j$  is simply the accumulated turning effect of the same tangential contact forces that can change  $v_b$ .

Newton's third law places equal-and-opposite traction forces on the belt. Their total moment is therefore  $-\tau_j$ , and at the effective contact radius  $r_{j,\text{eff}}$ , they are equivalent to the net tangential pull  $-\tau_j/r_{j,\text{eff}}$ . If the pull at one wrap supplies exactly the power removed by the pull at the other, the belt simply carries that power between the pulleys. If the two actions do not balance, the resulting energy excess or deficit must be taken up by the belt.

The belt assumptions leave that imbalance only one place to go. The fixed length of **Assumption 6 (Fixed Belt Length)** excludes longitudinal extension and the strain energy it would store. The single transport coordinate of **Assumption 7 (Single Belt Transport Speed)** gives every belt region the same speed  $v_b$ , so one region cannot speed up independently of the rest. The belt's only retained longitudinal energy is therefore  $\frac{1}{2}m_b v_b^2$ . Net power into the belt increases  $v_b$ ; net power out decreases it.

The energy argument now explains why an imbalance between the pulley actions must change  $v_b$ . Newton's second law determines the corresponding signed acceleration directly, where the whole belt loop is treated as a single body. Place one boundary around the entire loop. From the force accounting above, the only external tangential actions on this free body are the two equivalent pulley pulls, acting on the complete belt mass  $m_b$ . The complete-belt transport balance is therefore

#### Complete-Belt Transport Balance

(3.4.1)

$$\begin{aligned}\sum F_{b,t}^{\text{ext}} &= -\frac{\tau_p}{r_{p,\text{eff}}} - \frac{\tau_s}{r_{s,\text{eff}}} \\ &= \frac{d}{dt}(m_b v_b) = m_b \dot{v}_b.\end{aligned}$$

This balance gives the basic longitudinal motion of the belt. If the two pulley contacts exert a net force in the positive belt direction, the belt speeds up; if they exert a net force in the opposite direction, it slows down. If the two actions balance, the belt can carry power between the pulleys without changing  $v_b$ .

#### Interpreting the Transport Balance

As far as speeding up or slowing down the complete belt is concerned, this result is sufficient. The quantity  $v_b$ , however, describes only how quickly belt material travels along its prescribed path. Consider one marked piece of belt. It approaches a pulley along one straight span and leaves along

the other. Its direction of travel has changed. The complete-belt balance says how its speed changes, but contains nothing that explains this change in direction. Left to itself, the marked piece would continue straight past the pulley. Its motion around the wrap is therefore possible only because forces act on it throughout that turn.

The clearest way to separate these forces from the transport motion is to remove the motion altogether. Consider first the familiar case of a rope wrapped around a classical pulley. Pull equally on the two ends of the rope. If both pulls are increased together, the tension throughout the rope rises and the pulley becomes more heavily loaded, even though no tension difference has been introduced. The taut rope is redirected around the pulley, and in doing so it exerts a resultant load on it. The pulley supplies the corresponding equal-and-opposite contact reaction.

The same basic effect occurs when the CVT belt is wrapped around a sheave. Suppose the belt and pulley are stationary and the entry and exit tensions are both 500 N. Their difference is zero, so there is no transmitted torque or longitudinal acceleration, yet the tensioned belt still loads the pulley through the wrap. Raising both the entry and exit tensions to 1000 N changes neither the tension difference nor the belt's motion, but the belt now loads the pulley more strongly and the corresponding normal contact increases. The geometry and motion have not changed; the pulley loading has.

The loading described above is carried through the belt–sheave contact around the wrap. In the model, the resultant normal contact loads on the primary and secondary pulleys are denoted by  $N_p$  and  $N_s$ . Their axial components already appear in the primary and secondary movable-sheave balances; see [Equations \(3.2.14\)](#) and [\(3.2.33\)](#). The static example therefore has an important consequence: even with no transmitted torque or belt acceleration, changing the belt tension changes  $N_p$  and  $N_s$ , and can therefore change the shift balance. Determining these normal loads requires knowing the tension carried around each pulley. The remaining difficulty is that this tension is not generally constant around the pulley.

As established above, a pulley transfers torque through tangential traction distributed around its wrap. As the belt passes through the contact, this traction changes how strongly the belt is being pulled. The tension leaving the wrap therefore generally differs from the tension entering it. In the simplest steady, inertia-free picture, the transmitted torque is equal to the pulley radius multiplied by this tension difference. Retaining the inertia of the wrapped belt modifies that balance, but not the basic interpretation: torque transfer tells us how the tension changes through the contact.

The quantity needed to determine the normal loading is therefore itself changing as the belt moves through the wrap. Knowing the entry and exit tensions tells us only how that process begins and ends, not what tension is carried at each point between them. The normal contact must therefore be determined locally as well.

### 3.4.2 Local Equations of Motion on a Pulley Wrap

The previous subsection leaves one local question: how is the tension carried at a particular point in the wrap related to the contact force acting there? The next step is therefore to build the free-body diagram of an infinitesimal piece of belt within the wrap of either pulley  $j \in \{p, s\}$ .

Let  $\theta$  increase through the wrap in the positive belt-transport direction defined in [Section 2.3.1](#), and isolate the piece bounded by cuts at  $\theta$  and  $\theta + d\theta$ . Let  $T_j(\theta)$  denote the tension carried by the belt at angular position  $\theta$ . The tensions at the two cuts are therefore  $T_j(\theta)$  and  $T_j(\theta + d\theta)$ . To keep the labels in the free-body diagram compact, define

$$T_j \equiv T_j(\theta), \quad dT_j \equiv T_j(\theta + d\theta) - T_j(\theta). \quad (3.4.2)$$

The tension at the second cut can then be written exactly as  $T_j + dT_j$ . Thus,  $dT_j$  is the signed change in tension across the element: it is positive if the tension increases with  $\theta$  and negative if the tension decreases. No direction of change has yet been assumed.

Once the piece is cut free, the surrounding belt pulls on its two exposed ends. Under **Assumption 8 (Prescribed Planar Belt Path)**, the belt is perfectly flexible and follows the prescribed planar path. Each cut therefore carries only a longitudinal tension force directed along the local belt tangent; there is no transverse shear force or bending moment. At both cuts, the surrounding belt pulls away from the isolated piece.

Because the belt curves around the pulley, those two tension forces do not point in the same direction. The belt tangent turns through  $d\theta$  from one cut to the other. It is therefore convenient to draw the local directions at the midpoint of the element. Using the radial and tangential convention introduced in **Section 2.3.3**,  $\hat{\mathbf{e}}_r$  points radially outward from the pulley axis, while  $\hat{\mathbf{e}}_\theta$  points with increasing  $\theta$ . Each end tangent is inclined by  $d\theta/2$  from the midpoint tangent. The left-hand view in **Figure 3.11** shows this in-plane geometry.

The pulley also acts on the element through both sheave faces. To complete the free-body diagram, these contact forces must be added to the cut-end tensions. Their geometry cannot be represented clearly in the pulley plane, so the right-hand view in **Figure 3.11** shows the cross-section of the same element. Under **Assumption 4 (Fixed, Uniform Belt Cross-Section)**, its shape and mass centroid remain fixed. Under **Assumption 12 (Symmetric Sheave-Face Sharing)**, the two faces carry equal shares of the normal contact. Let  $dN_j$  denote the sum of the magnitudes of the two face-normal forces; each face therefore exerts a force of magnitude  $dN_j/2$ . Their axial components oppose one another and cancel, whereas their outward radial components add to  $dN_j \sin \beta$ .

The remaining contact action is tangential. Under **Assumption 10 (Gross Tangential Coulomb Traction)**, let  $dF_{t,j}$  denote the resultant tangential force exerted by both sheave faces on the element. It is defined as positive when it acts in the direction of increasing  $\theta$ ; its actual sign will be determined by the contact state. Because this force acts in the pulley plane, it appears in the left-hand view of **Figure 3.11**. The two views therefore complete a single free-body diagram: the left shows the forces acting around the wrap, while the right shows how the two face-normal forces produce the radial contact force appearing on the left.

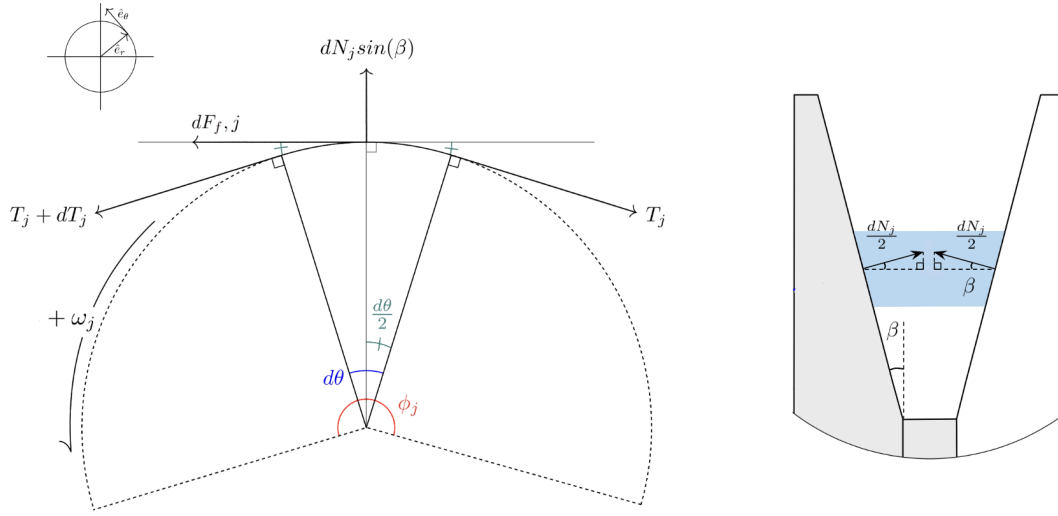


Figure 3.11: Two views of the same infinitesimal belt element, bounded by  $\theta$  and  $\theta + d\theta$  within the wrap of pulley  $j$ . The in-plane free-body diagram on the left shows the cut-end tensions  $T_j = T_j(\theta)$  and  $T_j + dT_j = T_j(\theta + d\theta)$ , the midpoint directions, the half-angle  $d\theta/2$ , the signed tangential contact force  $dF_{t,j}$ , and the outward radial contact force  $dN_j \sin \beta$ . The cross-section on the right shows the two face-normal forces of magnitude  $dN_j/2$ , whose axial components cancel and whose radial components add.

**Equations of motion for the infinitesimal element** With the free-body diagram complete, Newton's second law can now be assembled in the two local directions. The radial direction comes first because it contains the normal contact that motivated the local analysis.

In the radial direction, the two sheave-face forces supply the combined outward component  $dN_j \sin \beta$ . The surrounding belt pulls inward at both cuts. Since each end tangent is inclined by  $d\theta/2$  from the midpoint tangent, the corresponding inward components are  $T_j \sin(d\theta/2)$  and  $(T_j + dT_j) \sin(d\theta/2)$ . Taking outward as positive, Newton's second law therefore gives

$$dN_j \sin \beta - T_j \sin\left(\frac{d\theta}{2}\right) - (T_j + dT_j) \sin\left(\frac{d\theta}{2}\right) = dm_j a_{r,j}, \quad (3.4.3)$$

where  $dm_j$  is the mass of the element and  $a_{r,j}$  is its signed acceleration in the outward radial direction.

The same element must also satisfy Newton's second law in the tangential direction. The face-normal forces have no tangential component. At the cut at  $\theta$ , the tension pulls opposite increasing  $\theta$ , giving the component  $-T_j \cos(d\theta/2)$ . At the cut at  $\theta + d\theta$ , the tension pulls with increasing  $\theta$ , giving  $(T_j + dT_j) \cos(d\theta/2)$ . Including the signed tangential contact force  $dF_{t,j}$ , the tangential equation is therefore

$$(T_j + dT_j) \cos\left(\frac{d\theta}{2}\right) - T_j \cos\left(\frac{d\theta}{2}\right) + dF_{t,j} = dm_j a_{\theta,j}, \quad (3.4.4)$$

where  $a_{\theta,j}$  is the signed acceleration in the direction of increasing  $\theta$ .

The force side of each equation is now fully determined by the free-body diagram. Their right-hand sides still express the belt element's response only in the general form of mass times acceleration. Reducing them requires knowing how much belt mass lies within the angle  $d\theta$ , and how that material accelerates while it travels around a pulley whose wrap radius can simultaneously move as the CVT shifts. These quantities follow from the belt cross-section and the radius kinematics already established in [Section 3.1](#).

**Mass and motion of the belt element** The mass and acceleration of the element must be evaluated at the belt's mass centroid. As established in [Equations \(3.1.8\)](#) and [\(3.1.14\)](#),  $r_j$  locates the outer surface used to describe where the belt is seated, whereas  $r_{j,\text{cm}}$  locates the mass centroid. Under [Assumption 4 \(Fixed, Uniform Belt Cross-Section\)](#), the centroid remains a fixed radial distance from the outer surface. Its radial velocity and acceleration are therefore the radius rate and acceleration already derived for the seated belt:

$$\dot{r}_{j,\text{cm}} = \dot{r}_j, \quad \ddot{r}_{j,\text{cm}} = \ddot{r}_j.$$

The element spans the angle  $d\theta$ , so its arc length at the mass centroid is  $r_{j,\text{cm}}d\theta$ . Multiplying this length by the belt cross-sectional area  $A_b$  from [Equation \(3.1.12\)](#) and the belt density  $\rho_b$  gives the element mass:

$$dm_j = \rho_b A_b r_{j,\text{cm}} d\theta. \quad (3.4.5)$$

The element can move in two directions at once. It moves radially with the shifting wrap at speed  $\dot{r}_j$ , while the belt's transport carries it tangentially at speed  $v_b$ . Its velocity is therefore

$$\mathbf{v}_{b,j} = \dot{r}_j \hat{\mathbf{e}}_r + v_b \hat{\mathbf{e}}_\theta. \quad (3.4.6)$$

Acceleration is the time derivative of this velocity. Both scalar speeds can change, but the local directions also change as the element travels around the pulley. Differentiating while retaining both effects gives

$$\begin{aligned} \mathbf{a}_{b,j} &= \frac{d}{dt} (\dot{r}_j \hat{\mathbf{e}}_r + v_b \hat{\mathbf{e}}_\theta) \\ &= \ddot{r}_j \hat{\mathbf{e}}_r + \dot{r}_j \dot{\hat{\mathbf{e}}}_r + \dot{v}_b \hat{\mathbf{e}}_\theta + v_b \dot{\hat{\mathbf{e}}}_\theta. \end{aligned} \quad (3.4.7)$$

The two basis-vector derivatives follow from how quickly the element moves around the wrap. At the centroidal radius, its tangential speed and angular rate are related by

$$v_b = r_{j,\text{cm}} \dot{\theta}, \quad \dot{\theta} = \frac{v_b}{r_{j,\text{cm}}}. \quad (3.4.8)$$

As  $\theta$  increases, the outward radial direction turns toward  $\hat{\mathbf{e}}_\theta$ , while the tangential direction turns toward  $-\hat{\mathbf{e}}_r$ . Multiplying these changes per unit angle by the angular rate in [Equation \(3.4.8\)](#) gives

$$\dot{\mathbf{e}}_r = \frac{v_b}{r_{j,\text{cm}}} \hat{\mathbf{e}}_\theta, \quad \dot{\mathbf{e}}_\theta = -\frac{v_b}{r_{j,\text{cm}}} \hat{\mathbf{e}}_r. \quad (3.4.9)$$

Substituting these basis rates into Equation (3.4.7) and collecting the radial and tangential terms gives

$$\mathbf{a}_{b,j} = \left( \ddot{r}_j - \frac{v_b^2}{r_{j,\text{cm}}} \right) \hat{\mathbf{e}}_r + \left( \dot{v}_b + \frac{\dot{r}_j v_b}{r_{j,\text{cm}}} \right) \hat{\mathbf{e}}_\theta. \quad (3.4.10)$$

The acceleration components required by the two equations of motion are therefore

$$a_{r,j} = \ddot{r}_j - \frac{v_b^2}{r_{j,\text{cm}}}, \quad (3.4.11)$$

$$a_{\theta,j} = \dot{v}_b + \frac{\dot{r}_j v_b}{r_{j,\text{cm}}}. \quad (3.4.12)$$

The radial component contains the outward acceleration of the shifting wrap and the familiar inward centripetal acceleration produced as the belt turns around the pulley. The tangential component contains the change in belt speed and an additional contribution from the radial velocity changing direction as the element moves through the wrap. Together with Equation (3.4.5), these two components now provide the complete inertial sides of the radial and tangential balances.

**Radial equation of motion** We first return to the radial balance in Equation (3.4.3). Substituting the element mass from Equation (3.4.5) and the radial acceleration from Equation (3.4.11), then collecting the two inward tension components, gives

$$dN_j \sin \beta - (2T_j + dT_j) \sin \left( \frac{d\theta}{2} \right) = \rho_b A_b (r_{j,\text{cm}} \ddot{r}_j - v_b^2) d\theta. \quad (3.4.13)$$

The desired local quantity is the normal contact carried per unit wrap angle,  $dN_j/d\theta$ . Dividing Equation (3.4.13) by  $d\theta$  gives

$$\sin \beta \frac{dN_j}{d\theta} - (2T_j + dT_j) \frac{\sin(d\theta/2)}{d\theta} = \rho_b A_b (r_{j,\text{cm}} \ddot{r}_j - v_b^2).$$

Only the tension term still depends on the finite angular width of the element. To put its sine factor into the standard small-angle form, write

$$\frac{\sin(d\theta/2)}{d\theta} = \frac{1}{2} \frac{\sin(d\theta/2)}{d\theta/2}.$$

Redistributing this factor of 1/2 through  $2T_j + dT_j$ , the radial balance can be written directly as

$$\sin \beta \frac{dN_j}{d\theta} - \left( T_j + \frac{dT_j}{2} \right) \frac{\sin(d\theta/2)}{d\theta/2} = \rho_b A_b (r_{j,\text{cm}} \ddot{r}_j - v_b^2).$$

The infinitesimal element is obtained by taking  $d\theta \rightarrow 0$ . Because  $dT_j = T_j(\theta + d\theta) - T_j(\theta)$ , continuity of the tension field gives  $dT_j \rightarrow 0$ . At the same time, the small-angle ratio tends to one:

$$T_j + \frac{dT_j}{2} \longrightarrow T_j, \quad \frac{\sin(d\theta/2)}{d\theta/2} \longrightarrow 1.$$

The complete local radial balance is consequently

$$\boxed{\sin \beta \frac{dN_j}{d\theta} = T_j - \rho_b A_b (v_b^2 - r_{j,\text{cm}} \ddot{r}_j)} . \quad (3.4.14)$$

To see the physical meaning of this result most clearly, it is useful first to consider a stationary belt on a fixed wrap. Setting  $v_b = 0$  and  $\ddot{r}_j = 0$  gives

$$\sin \beta \frac{dN_j}{d\theta} = T_j. \quad (3.4.15)$$

We now recover the result anticipated from the physical picture: increasing the belt tension increases the normal loading carried per unit wrap angle. This remains true even when the tension is uniform through the contact, so there is no tension difference and no torque is transmitted. The normal loading therefore depends on the absolute tension level, not only on the tension difference across the wrap.

Returning to the full balance, motion modifies this stationary relationship. As the belt turns around the pulley, part of the inward tension is required to provide its centripetal acceleration, reducing the outward sheave reaction at a given tension. An outward wrap acceleration has the opposite effect and increases that reaction.

**Tangential equation of motion** Finally, we complete the tangential equation of motion introduced in Equation (3.4.4). Substituting the element mass from Equation (3.4.5) and the tangential acceleration from Equation (3.4.12) gives the complete balance

$$(T_j + dT_j) \cos\left(\frac{d\theta}{2}\right) - T_j \cos\left(\frac{d\theta}{2}\right) + dF_{t,j} = \rho_b A_b r_{j,\text{cm}} d\theta \left( \dot{v}_b + \frac{\dot{r}_j v_b}{r_{j,\text{cm}}} \right). \quad (3.4.16)$$

The desired result is again a local relation per unit wrap angle. Cancelling the common  $T_j$  contribution from the two end tensions, simplifying the acceleration term, and dividing through by  $d\theta$  gives

$$\frac{dT_j}{d\theta} \cos\left(\frac{d\theta}{2}\right) + \frac{dF_{t,j}}{d\theta} = \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b).$$

The remaining cosine records that the end tensions are not quite parallel on a finite element. In the infinitesimal limit, the two end tangents coincide and

$$\lim_{d\theta \rightarrow 0} \cos\left(\frac{d\theta}{2}\right) = \cos(0) = 1.$$

The cosine therefore becomes a factor of one; the tension-gradient term remains. The local tangential equation of motion is

$$\frac{dT_j}{d\theta} + \frac{dF_{t,j}}{d\theta} = \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b). \quad (3.4.17)$$

This balance identifies the two tangential forces acting per unit wrap angle: the change in belt tension and the traction supplied by the pulley. Their sum produces the tangential acceleration of the belt. What remains is to relate that tangential contact force to the normal contact already present in the radial balance.

Under **Assumption 10 (Gross Tangential Coulomb Traction)**, the sheave contact obeys Coulomb friction. While the belt sticks, the tangential force is a reaction that can act in either direction and take any magnitude up to the static limit. Once the belt slides, its magnitude is fixed by kinetic friction and its direction opposes the relative sliding. Both cases relate the tangential force on the element to the same normal-force increment  $dN_j$ , but with different rules for the allowable proportionality.

Rather than carrying a separate magnitude condition and direction sign through the remaining equations, define the signed traction utilization  $\lambda_j$  by

|                                                                                                                                                                                                                          |                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Signed Traction Utilization</b>                                                                                                                                                                                       | <b>(3.4.18)</b> |
| $dF_{t,j} = \lambda_j dN_j, \quad \lambda_j = \begin{cases} \lambda_{j,\text{st}}, &  \lambda_{j,\text{st}}  \leq \mu_s, & \text{sticking,} \\ \sigma_j \mu_k, & \sigma_j \in \{-1, +1\}, & \text{sliding.} \end{cases}$ |                 |

The magnitude of  $\lambda_j$  gives the tangential force used per unit normal load, while its sign gives the direction of that force. Thus,  $\lambda_j > 0$  acts with increasing  $\theta$ ,  $\lambda_j < 0$  acts against it, and  $\lambda_j = 0$  represents normal contact with no tangential force. In sliding,  $\sigma_j$  selects the direction that opposes the relative motion, while  $\mu_k$  fixes the magnitude. Under **Assumption 11 (Wrap-Level Contact Resolution)**, one value of  $\lambda_j$  represents the complete wrap at a given instant.

Dividing  $dF_{t,j} = \lambda_j dN_j$  by  $d\theta$  and substituting it into **Equation (3.4.17)** gives the complete local tangential balance:

|                                                                                                                   |                 |
|-------------------------------------------------------------------------------------------------------------------|-----------------|
| $\frac{dT_j}{d\theta} + \lambda_j \frac{dN_j}{d\theta} = \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b).$ | <b>(3.4.19)</b> |
|-------------------------------------------------------------------------------------------------------------------|-----------------|

To expose the physical meaning of this result, it is useful again to remove the effects of motion. For a fixed wrap and steady belt speed,  $\dot{r}_j = 0$  and  $\dot{v}_b = 0$ , so the tangential inertia vanishes and the balance becomes

$$\frac{dT_j}{d\theta} = -\lambda_j \frac{dN_j}{d\theta}. \quad (3.4.20)$$

We now recover the expected local relationship: the pulley traction must be balanced by an equal and opposite tension gradient. Traction therefore forces the tension to evolve around the wrap, and integrating this local evolution produces the tension difference associated with transmitted torque.

Since the normal contact is compressive,  $dN_j/d\theta \geq 0$ : positive traction makes the tension fall as  $\theta$  increases, while traction in the opposite direction makes it rise. Retaining the full balance, some of the traction may be used to accelerate the belt, but the same underlying link between traction and tension evolution remains.

### 3.4.3 Tension Evolution Through a Pulley Wrap

The balances derived above describe the forces on one infinitesimal belt element at an arbitrary position within a pulley wrap. The next step is to carry that local description across the complete contact, so that the effect of every element from the wrap entrance to the wrap exit can be accumulated.

For pulley  $j$ , let  $\theta$  increase from 0 at the entrance to  $\phi_j$  at the exit. The belt enters with tension  $T_{j,\text{in}}$ , carries the tension  $T_j(\theta)$  at an intermediate position, and leaves with tension  $T_{j,\text{out}}$ :

$$0 \leq \theta \leq \phi_j, \quad T_j(0) = T_{j,\text{in}}, \quad T_j(\phi_j) = T_{j,\text{out}}.$$

Both  $T_j(\theta)$  and the normal loading  $dN_j/d\theta$  are needed throughout this interval. Tension provides the natural starting point:  $T_{j,\text{in}}$  is the entrance boundary value, and following its evolution through the wrap produces  $T_{j,\text{out}}$ . Once that tension field is known, the radial balance recovers the normal loading at every position.

The tangential balance, Equation (3.4.19), contains the desired tension gradient but remains coupled to  $dN_j/d\theta$ . Rearranging the radial balance, Equation (3.4.14), gives

$$\frac{dN_j}{d\theta} = \frac{1}{\sin \beta} [T_j - \rho_b A_b (v_b^2 - r_{j,\text{cm}} \ddot{r}_j)]. \quad (3.4.21)$$

Substituting this expression into the tangential balance, then expanding and rearranging, gives

$$\begin{aligned} \frac{dT_j}{d\theta} + \frac{\lambda_j}{\sin \beta} [T_j - \rho_b A_b (v_b^2 - r_{j,\text{cm}} \ddot{r}_j)] \\ = \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) \\ \frac{dT_j}{d\theta} + \frac{\lambda_j}{\sin \beta} T_j = \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) + \rho_b A_b \frac{\lambda_j}{\sin \beta} (v_b^2 - r_{j,\text{cm}} \ddot{r}_j). \end{aligned}$$

Factoring the common  $\rho_b A_b$  on the right gives the local equation that governs tension evolution through either pulley wrap:

|                                                                                                                                                                                                               |                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Pulley-Wrap Tension Evolution</b>                                                                                                                                                                          | <b>(3.4.22)</b> |
| $\frac{dT_j}{d\theta} + \frac{\lambda_j}{\sin \beta} T_j = \rho_b A_b [r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b + \frac{\lambda_j}{\sin \beta} (v_b^2 - r_{j,\text{cm}} \ddot{r}_j)], \quad j \in \{p, s\}.$ |                 |

The two local balances have now been reduced to one first-order equation for  $T_j(\theta)$ .

**Solving the tension equation** Before Equation (3.4.22) can be integrated through the wrap, its coefficients must be considered along the spatial coordinate  $\theta$ . At a fixed instant, the circular wrap prescribed by Assumption 8 (Prescribed Planar Belt Path) gives one  $r_{j,\text{cm}}$ ,  $\dot{r}_j$ , and  $\ddot{r}_j$  for pulley  $j$ . Likewise, Assumption 7 (Single Belt Transport Speed) makes  $v_b$  and  $\dot{v}_b$  common throughout the belt, while Assumption 11 (Wrap-Level Contact Resolution) assigns one  $\lambda_j$  to the complete wrap. None of these quantities therefore varies with  $\theta$  during the integration.

To expose the structure of the equation, define the two constant coefficients

$$\begin{aligned} a_j &= \frac{\lambda_j}{\sin \beta}, \\ c_j &= \rho_b A_b \left[ r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b + a_j (v_b^2 - r_{j,\text{cm}} \ddot{r}_j) \right]. \end{aligned} \quad (3.4.23)$$

Then Equation (3.4.22) becomes

$$\frac{dT_j}{d\theta} + a_j T_j = c_j. \quad (3.4.24)$$

To integrate Equation (3.4.24), it is useful to combine the two terms on the left into a single derivative. The exponential  $e^{a_j \theta}$  is chosen because its derivative is  $a_j e^{a_j \theta}$ , matching the coefficient multiplying  $T_j$ . The product rule then gives

$$\frac{d}{d\theta} \left( e^{a_j \theta} T_j \right) = e^{a_j \theta} \left( \frac{dT_j}{d\theta} + a_j T_j \right). \quad (3.4.25)$$

Multiplying Equation (3.4.24) by  $e^{a_j \theta}$  therefore gives

$$\frac{d}{d\theta} \left( e^{a_j \theta} T_j \right) = c_j e^{a_j \theta}. \quad (3.4.26)$$

The entrance condition  $T_j(0) = T_{j,\text{in}}$  now supplies the lower boundary. Integrating from the wrap entrance to an arbitrary position  $\theta$  gives

$$\int_0^\theta \frac{d}{d\vartheta} \left( e^{a_j \vartheta} T_j(\vartheta) \right) d\vartheta = c_j \int_0^\theta e^{a_j \vartheta} d\vartheta. \quad (3.4.27)$$

For  $a_j \neq 0$ , evaluating the integrals yields

$$e^{a_j \theta} T_j(\theta) - T_{j,\text{in}} = \frac{c_j}{a_j} \left( e^{a_j \theta} - 1 \right), \quad (3.4.28)$$

and isolating  $T_j(\theta)$  gives

$$T_j(\theta) = T_{j,\text{in}} e^{-a_j \theta} + \frac{c_j}{a_j} \left( 1 - e^{-a_j \theta} \right). \quad (3.4.29)$$

Substituting the constant coefficients back into Equation (3.4.29) and rearranging gives the tension carried at every position through the wrap:

$$T_j(\theta) = T_{j,\text{in}} e^{-\frac{\lambda_j}{\sin \beta} \theta} + \rho_b A_b \left( v_b^2 - r_{j,\text{cm}} \ddot{r}_j + \frac{\sin \beta}{\lambda_j} (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) \right) \left( 1 - e^{-\frac{\lambda_j}{\sin \beta} \theta} \right), \quad \lambda_j \neq 0. \quad (3.4.30)$$

At the wrap exit,  $\theta = \phi_j$  and  $T_j(\phi_j) = T_{j,\text{out}}$ . Evaluating Equation (3.4.30) at that endpoint gives

$$T_{j,\text{out}} = T_{j,\text{in}} e^{-\frac{\lambda_j}{\sin \beta} \phi_j} + \rho_b A_b \left( v_b^2 - r_{j,\text{cm}} \ddot{r}_j + \frac{\sin \beta}{\lambda_j} (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) \right) \left( 1 - e^{-\frac{\lambda_j}{\sin \beta} \phi_j} \right), \quad \lambda_j \neq 0. \quad (3.4.31)$$

The closed form is written for  $\lambda_j \neq 0$  because its derivation divided by  $\lambda_j$ , but zero utilization presents no difficulty. For completeness, setting  $\lambda_j = 0$  directly in the tangential balance of Equation (3.4.19) gives

$$\frac{dT_j}{d\theta} = \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b). \quad (3.4.32)$$

Integrating from the wrap entrance to  $\theta$ , and then evaluating the result at  $\theta = \phi_j$ , gives

$$\begin{aligned} T_j(\theta) &= T_{j,\text{in}} + \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) \theta, \\ T_{j,\text{out}} &= T_{j,\text{in}} + \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) \phi_j. \end{aligned} \quad (3.4.33)$$

**Interpreting the wrap solution** The local balances have now been carried across the complete wrap. For a specified entrance tension and instantaneous wrap state, Equation (3.4.30) gives the tension at every position, while Equation (3.4.31) gives the resulting exit tension. Within the constant-wrap description established by Assumption 11 (Wrap-Level Contact Resolution), the solution has therefore retained every transient term admitted by the model: belt acceleration, radial migration, radial acceleration, and centripetal inertia. These instantaneous quantities are uniform with respect to  $\theta$ , even though the resulting tension and normal loading vary through the wrap. Improving the contact resolution further would therefore mean allowing local belt speed, radial motion, and traction state to vary within the wrap itself, for which the constant-coefficient solution would be replaced by a spatially varying model (Kong and Parker, 2006; Messick, 2018; Julió and Plante, 2011).

The full expression describes the wrap even while the belt speed and pulley radius are changing. When those changes stop, the transient terms disappear, leaving a direct connection to the familiar steady belt relation. Holding the pulley radius fixed and the belt transport speed steady gives  $\dot{r}_j = \ddot{r}_j = \dot{v}_b = 0$ . Substituting these conditions directly into the entry-to-exit result, Equation (3.4.31), and rearranging gives

$$\frac{T_{j,\text{out}} - \rho_b A_b v_b^2}{T_{j,\text{in}} - \rho_b A_b v_b^2} = e^{-\frac{\lambda_j \phi_j}{\sin \beta}}. \quad (3.4.34)$$

This is the inertia-corrected structure used by Gerbert (Gerbert, 1996), written in the same in/out convention as the full wrap solution. The correction has a direct physical meaning:  $\rho_b A_b v_b^2$  is the

portion of the tension needed simply to turn the moving belt, whereas the remainder produces normal contact and is therefore called the tractive tension. The ratio shows that it is this tractive portion, rather than the total tension, that friction causes to evolve across the steady wrap. At the traction limit,  $|\lambda_j| = \mu$ , while the sign of  $\lambda_j$  determines whether the tension rises or falls from the wrap entrance to the wrap exit.

The same comparison leads directly to the familiar classical limit. If belt inertia is neglected and the V-groove is removed by taking  $\beta = 90^\circ$ , then the centrifugal correction vanishes and  $\sin \beta = 1$ , so Equation (3.4.34) becomes

$$\frac{T_{j,\text{out}}}{T_{j,\text{in}}} = e^{-\lambda_j \phi_j}. \quad (3.4.35)$$

This is the Euler–Eytelwein equation in the same signed in/out convention (Euler, 1769; Eytelwein, 1808). No new law has been introduced: the classical result is simply what remains of the entry-to-exit balance after transient acceleration, centrifugal inertia, and V-groove amplification are removed. Read in reverse, the full solution restores each of those effects while preserving the same underlying idea that distributed traction changes the tension through the wrap. With that tension now known at every position, Equation (3.4.21) can recover the corresponding normal loading, allowing the pulley load and transmitted torque to be assembled from the same contact distribution.

#### 3.4.4 Pulley Contact Loads and Torque

With  $T_j(\theta)$  now known, the same distributed contact can supply the two belt actions left in the pulley equations: the normal resultant  $N_j$  and the contact torque  $\tau_j$ . Begin with the radial balance. Equation (3.4.21) gives the normal loading per unit wrap angle, so accumulating that distribution from the entrance at  $\theta = 0$  to the exit at  $\theta = \phi_j$  gives

$$N_j = \frac{1}{\sin \beta} \int_0^{\phi_j} [T_j(\theta) - \rho_b A_b (v_b^2 - r_{j,\text{cm}} \ddot{r}_j)] d\theta. \quad (3.4.36)$$

Here,  $N_j$  is the total normal load integrated over both sheave faces. Since  $T_j(\theta)$  is supplied by Equation (3.4.30), this integral now determines the normal resultant required in Equations (3.2.14) and (3.2.33).

It is important to note that, under Assumption 8 (Prescribed Planar Belt Path), this mathematical solution is physically admissible only while

$$T_j(\theta) \geq 0, \quad \frac{dN_j}{d\theta} \geq 0, \quad 0 \leq \theta \leq \phi_j. \quad (3.4.37)$$

A negative tension indicates slack, while a negative normal-loading gradient indicates local lift-off; neither represents a physical negative load.

Although Equation (3.4.36) determines  $N_j$ , evaluating it requires the complete tension field. The tangential balance gives the same resultant directly from the wrap boundary values. Under Assumption 11 (Wrap-Level Contact Resolution),  $\lambda_j$  is constant throughout the wrap, so integrating Equation (3.4.19) from  $\theta = 0$  to  $\theta = \phi_j$  gives

$$T_{j,\text{out}} - T_{j,\text{in}} + \lambda_j N_j = \rho_b A_b \phi_j (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b). \quad (3.4.38)$$

For  $\lambda_j \neq 0$ , rearranging gives the normal resultant in its compact entry-to-exit form:

|                                                                                                                                                       |                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Pulley-Wrap Normal Resultant</b>                                                                                                                   | <b>(3.4.39)</b> |
| $N_j = \frac{T_{j,\text{in}} - T_{j,\text{out}} + \rho_b A_b \phi_j (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b)}{\lambda_j}, \quad \lambda_j \neq 0.$ |                 |

The division by  $\lambda_j$  only excludes zero traction from this compact form. When  $\lambda_j = 0$ , the tangential balance no longer determines  $N_j$ , but the radial integral remains valid. Substituting the zero-traction tension field from [Equation \(3.4.33\)](#) into [Equation \(3.4.36\)](#) gives

$$\begin{aligned} N_j = & \frac{\phi_j}{\sin \beta} [T_{j,\text{in}} - \rho_b A_b (v_b^2 - r_{j,\text{cm}} \ddot{r}_j)] \\ & + \frac{\rho_b A_b \phi_j^2}{2 \sin \beta} (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b). \end{aligned} \quad (3.4.40)$$

Thus  $\lambda_j = 0$  removes the tangential contact force, not the normal contact: the pulley can remain clamped even though the modeled contact transmits no torque.

The axial derivation already established how the normal contact enters each movable-sheave free body. Under [Assumption 12 \(Symmetric Sheave-Face Sharing\)](#), the movable face carries  $N_j/2$ ; projecting that load onto the local shaft axis gives the previously introduced belt-opening term

$$F_{\text{belt},x_j} = \frac{N_j}{2} \cos \beta, \quad j \in \{p, s\}.$$

With  $N_j$  now known, this determines the belt-opening term left unknown in [Equations \(3.2.14\)](#) and [\(3.2.33\)](#).

The remaining unresolved belt term is the contact torque  $\tau_j$  in [Equations \(3.3.5\)](#) and [\(3.3.13\)](#). Recall from [Section 3.4.1](#) that  $dF_{t,j} = \lambda_j dN_j$  acts on the belt, while the equal-and-opposite force acts on the pulley. Using the cord-line lever arm  $r_{j,\text{eff}}$  from [Equation \(3.1.9\)](#), and then using the integrated tangential balance in [Equation \(3.4.38\)](#), gives

$$\begin{aligned} \tau_j &= -r_{j,\text{eff}} \int_0^{\phi_j} \lambda_j \frac{dN_j}{d\theta} d\theta \\ &= -r_{j,\text{eff}} \lambda_j N_j \\ &= r_{j,\text{eff}} [T_{j,\text{out}} - T_{j,\text{in}} - \rho_b A_b \phi_j (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b)], \quad j \in \{p, s\}. \end{aligned} \quad (3.4.41)$$

The final line makes the transient correction explicit: the pulley torque is set by the entry-to-exit tension change only when the tangential acceleration of the wrapped belt vanishes.

For a specified wrap state, the local mechanics are now complete, but the absolute tension level is not. [Equation \(3.4.30\)](#) maps the entrance tension  $T_{j,\text{in}}$  to the exit tension  $T_{j,\text{out}}$ , but an isolated wrap cannot determine either boundary value on its own. Those tensions continue into the adjoining

straight spans and, ultimately, around the same closed belt. The next step is therefore to reconnect the two pulley wraps through the straight spans and require the tension to be compatible after one complete circuit of the belt loop.

### 3.4.5 Reassembling the Closed Belt

Dividing the complete belt at the four in-out-of-wrap points in [Figure 3.12](#) produces four regions: the primary wrap, the secondary wrap, and the upper and lower straight spans. Newton's second law has already been applied throughout both wrapped regions. The two straight spans remain. They carry no distributed pulley contact, but they still contain belt mass, so their end tensions need not be equal while the belt accelerates.

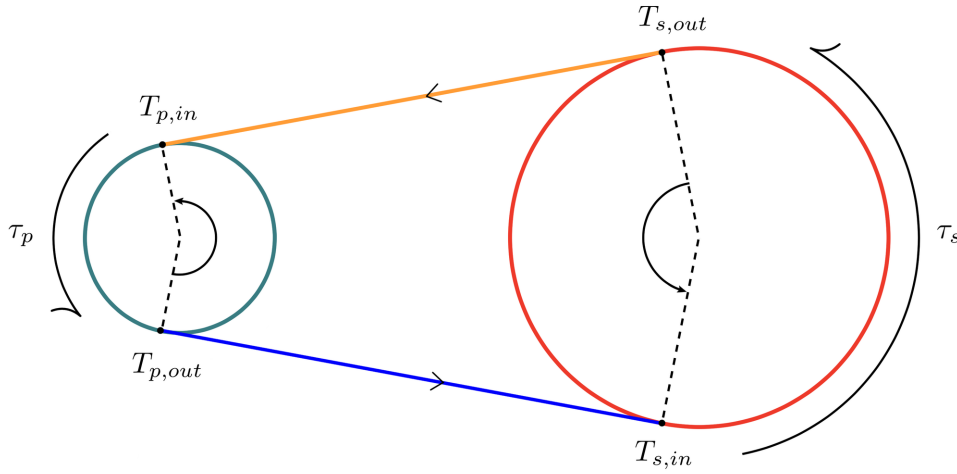


Figure 3.12: Complete belt divided into two wrapped regions and two straight spans. The arrows show the positive belt-transport direction. The upper span runs from the secondary wrap exit,  $T_{s,out}$ , to the primary wrap entrance,  $T_{p,in}$ ; the lower span runs from  $T_{p,out}$  to  $T_{s,in}$ . The torque arrows identify the contact actions on the wrapped regions already treated above.

As the in-out-of-wrap points move during a shift, each straight span also rotates. [Assumption 8 \(Prescribed Planar Belt Path\)](#) prescribes this changing path but explicitly omits the transverse inertia associated with its reorientation. Accordingly, Newton's second law is applied only along the instantaneous straight span.

Take this direction as positive with belt travel, and let  $p_u$  and  $p_l$  be the tangential momenta of the belt particles occupying the two spans at the instant considered; Newton's second law follows those same particles. The downstream tension acts positively and the upstream tension negatively, so

$$\begin{aligned} T_{p,in} - T_{s,out} &= \frac{dp_u}{dt}, \\ T_{s,in} - T_{p,out} &= \frac{dp_l}{dt}. \end{aligned} \tag{3.4.42}$$

At that instant, a straight span  $i \in \{u, l\}$  has mass  $\rho_b A_b l_i$ . Under [Assumption 7 \(Single Belt Transport Speed\)](#) and [Assumption 4 \(Fixed, Uniform Belt Cross-Section\)](#), its stored tangential momentum is  $p_i = \rho_b A_b l_i v_b$ , so the obvious first step is to differentiate:

$$\frac{d}{dt}(\rho_b A_b l_i v_b) = \rho_b A_b (l_i \dot{v}_b + \dot{l}_i v_b).$$

The term  $\rho_b A_b \dot{l}_i v_b$  exposes the complication. Each station marks where the belt enters or leaves the current pulley geometry; it is not fixed to a particular belt particle. As the stations move, belt particles pass between the wraps and straight span, so the region contains a changing mass. The derivative above therefore describes the momentum stored within a moving region, not yet the material momentum rate required by Newton's second law. The momentum crossing its boundaries must also be included.

Let  $v_i^-$  and  $v_i^+$  denote the velocities of the upstream and downstream stations, respectively, projected along the straight span. The belt crosses those boundaries at the relative speeds  $v_b - v_i^-$  and  $v_b - v_i^+$ . Adding the outgoing momentum flux and subtracting the incoming flux gives

$$\begin{aligned} \frac{dp_i}{dt} &= \frac{d}{dt}(\rho_b A_b l_i v_b) \\ &\quad + \rho_b A_b v_b [(v_b - v_i^+) - (v_b - v_i^-)]. \end{aligned} \tag{3.4.43}$$

These same station velocities determine the changing distance between them:

$$\dot{l}_i = v_i^+ - v_i^-. \tag{3.4.44}$$

Expanding the stored momentum and substituting [Equation \(3.4.44\)](#) gives

$$\begin{aligned} \frac{dp_i}{dt} &= \rho_b A_b (l_i \dot{v}_b + \dot{l}_i v_b) + \rho_b A_b v_b (v_i^- - v_i^+) \\ &= \rho_b A_b (l_i \dot{v}_b + \dot{l}_i v_b - \dot{l}_i v_b) \\ &= \rho_b A_b l_i \dot{v}_b. \end{aligned} \tag{3.4.45}$$

The cancellation is exact. The station motion changes the momentum stored within the geometric span by  $\rho_b A_b \dot{l}_i v_b$ , but the same motion produces an equal and opposite momentum flux through its boundaries. Once that flux is included, the moving stations leave no additional term in the tangential balance; only the acceleration of the belt transport remains.

The aligned-pulley geometry gives both straight spans the same length  $l$ , as shown in [Equation \(3.1.19\)](#). Substituting [Equation \(3.4.45\)](#) into the two Newton balances therefore yields

$$\begin{aligned} T_{p,\text{in}} - T_{s,\text{out}} &= \rho_b A_b l \dot{v}_b, \\ T_{s,\text{in}} - T_{p,\text{out}} &= \rho_b A_b l \dot{v}_b. \end{aligned} \tag{3.4.46}$$

Because the right-hand sides of the two straight-span balances are identical, their tension differences must be equal:

$$T_{p,\text{in}} - T_{s,\text{out}} = T_{s,\text{in}} - T_{p,\text{out}} \quad (3.4.47)$$

Rearranging this result places the two tensions belonging to each pulley on the same side:

$$T_{p,\text{in}} + T_{p,\text{out}} = T_{s,\text{in}} + T_{s,\text{out}}. \quad (3.4.48)$$

The closed belt therefore requires the sum of the primary boundary tensions to match the sum of the secondary boundary tensions. The remaining task is to evaluate that sum from the mechanics already obtained for a generic pulley wrap.

The integrated tangential balance in Equation (3.4.38) gives the boundary-tension difference as

$$T_{j,\text{out}} - T_{j,\text{in}} = \rho_b A_b \phi_j (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) - \lambda_j N_j. \quad (3.4.49)$$

This and the entry-to-exit solution in Equation (3.4.31) are two relations for the same two boundary tensions. Solving them for the entrance tension gives

$$\begin{aligned} T_{j,\text{in}} = \rho_b A_b \left[ v_b^2 - r_{j,\text{cm}} \ddot{r}_j + \frac{\sin \beta}{\lambda_j} (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) \right] \\ + \frac{\lambda_j N_j - \rho_b A_b \phi_j (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b)}{1 - e^{-\frac{\lambda_j}{\sin \beta} \phi_j}}, \quad \lambda_j \neq 0. \end{aligned} \quad (3.4.50)$$

The required sum can now be written as  $2T_{j,\text{in}} + (T_{j,\text{out}} - T_{j,\text{in}})$ . Substituting Equations (3.4.49) and (3.4.50) and simplifying gives

$$\begin{aligned} T_{j,\text{in}} + T_{j,\text{out}} = 2\rho_b A_b \left[ v_b^2 - r_{j,\text{cm}} \ddot{r}_j + \frac{\sin \beta}{\lambda_j} (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) \right] \\ + \frac{1 + e^{-\frac{\lambda_j}{\sin \beta} \phi_j}}{1 - e^{-\frac{\lambda_j}{\sin \beta} \phi_j}} [\lambda_j N_j - \rho_b A_b \phi_j (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b)], \quad \lambda_j \neq 0. \end{aligned} \quad (3.4.51)$$

Applying this expression to the primary and secondary sides of Equation (3.4.48) eliminates all four temporary boundary tensions:

$$\begin{aligned}
& 2\rho_b A_b \left[ v_b^2 - r_{p,\text{cm}} \ddot{r}_p + \frac{\sin \beta}{\lambda_p} (r_{p,\text{cm}} \dot{v}_b + \dot{r}_p v_b) \right] \\
& + \frac{1 + e^{-\frac{\lambda_p}{\sin \beta} \phi_p}}{1 - e^{-\frac{\lambda_p}{\sin \beta} \phi_p}} [\lambda_p N_p - \rho_b A_b \phi_p (r_{p,\text{cm}} \dot{v}_b + \dot{r}_p v_b)] \\
& = 2\rho_b A_b \left[ v_b^2 - r_{s,\text{cm}} \ddot{r}_s + \frac{\sin \beta}{\lambda_s} (r_{s,\text{cm}} \dot{v}_b + \dot{r}_s v_b) \right] \\
& + \frac{1 + e^{-\frac{\lambda_s}{\sin \beta} \phi_s}}{1 - e^{-\frac{\lambda_s}{\sin \beta} \phi_s}} [\lambda_s N_s - \rho_b A_b \phi_s (r_{s,\text{cm}} \dot{v}_b + \dot{r}_s v_b)], \quad \lambda_p \lambda_s \neq 0.
\end{aligned}$$

The displayed form assumes nonzero utilization only because the analytical wrap solution contains factors of  $1/\lambda_j$ . The zero-utilization limit remains regular. Using the zero-traction relations from [Equations \(3.4.33\) and \(3.4.40\)](#) gives

$$T_{j,\text{in}} + T_{j,\text{out}} = 2\rho_b A_b (v_b^2 - r_{j,\text{cm}} \ddot{r}_j) + \frac{2N_j \sin \beta}{\phi_j}, \quad \lambda_j = 0. \quad (3.4.53)$$

This expression replaces the corresponding side of [Equation \(3.4.52\)](#) whenever  $\lambda_p = 0$  or  $\lambda_s = 0$ . The four boundary tensions have therefore disappeared from the closed-loop equation: the straight spans require the two boundary-tension sums to match, and the local wrap mechanics express each sum in terms of the pulley normal load, traction utilization, and retained belt kinematics. Once the coupled mechanical equations determine those quantities, [Equations \(3.4.31\) and \(3.4.50\)](#) recover the boundary tensions and [Equation \(3.4.30\)](#) recovers the tension through each wrap.

### 3.4.6 Belt-Mechanics Closure and Scope

The closed-loop compatibility in [Equation \(3.4.52\)](#) is the final relation required from the belt itself. Rather than fixing the absolute tension level through a prescribed tight- or slack-side boundary condition, it makes that level emerge from the two pulley-wrap solutions together with the momentum balances of the two straight spans. This way of closing the loop also defines the level of belt resolution retained by the present model.

This treatment combines ideas that more commonly appear at different levels of resolution. Classical belt relations describe the tension change through a single wrap once an endpoint tension and contact state are supplied ([Gerbert, 1996](#)). Steady whole-belt boundary-value models couple the pulley contacts and free spans, but typically require operating-point boundary data such as a prescribed slack-side tension ([Kong and Parker, 2006](#); [Messick, 2018](#)). Transient discretized rubber-belt models instead enforce whole-loop compatibility through the states of many belt elements ([Julió and Plante, 2011](#); [Ballew, 2015](#)). The present formulation occupies a reduced middle ground: it retains one global belt-transport coordinate and an analytical closed-loop compatibility without introducing spatial or nodal belt states.

The chain-CVT model of [Duan et al. \(2016\)](#) provides the closest comparison for the retained transient physics. Both formulations account for centripetal inertia, tangential belt or chain acceleration, and the additional tangential acceleration produced as the contact radius changes. Duan also couples the

pulley axial and rotational dynamics through the contact solution, as is done here. One difference is that Duan neglects radial acceleration when reducing the final spatial wrap equations, whereas the present radial balance retains it.

Duan resolves considerably more physics within each pulley wrap. The chain trajectory may depart locally from the nominal pitch circle; radial and circumferential relative motion determine the sliding direction; friction varies continuously with sliding speed; and pulley deformation changes the contact geometry with angular position. Tension and normal loading are therefore obtained from coupled spatial equations. In the present model, [Assumption 8 \(Prescribed Planar Belt Path\)](#) fixes a circular contact path on rigid pulley geometry, [Assumption 7 \(Single Belt Transport Speed\)](#) assigns one transport speed to the belt, and [Assumption 11 \(Wrap-Level Contact Resolution\)](#) assigns one traction utilization  $\lambda_j$  to each wrap. The present formulation consequently retains the mechanism-level transient dynamics while sacrificing within-wrap variation of migration, microslip, traction reversal, and pulley deformation.

The complete belt loop is also closed differently. Duan concentrates the chain’s longitudinal deformation in the tight span, obtains its absolute tension from a lumped spring–damper law, and assumes that tension is equal at the driving-pulley entrance and driven-pulley exit. The tension difference across the slack span then accelerates the mass of the complete chain. Under [Assumption 6 \(Fixed Belt Length\)](#), the present formulation has no longitudinal stretch law from which to obtain an absolute tension. Instead, both straight spans are treated as massive belt regions, including the momentum carried through their moving in–out-of-wrap boundaries. Combining those balances with the two wrap solutions closes the loop without prescribing a tight- or slack-side tension.

The comparison is therefore best understood as a difference in where each model places its resolution. Duan resolves the local contact, friction, chain elasticity, and pulley deformation in greater detail. The present formulation uses a coarser contact description, but carries radial acceleration, both straight-span inertias, moving-boundary momentum transport, movable-sheave dynamics, and pulley rotation through the complete belt loop. The distributed tangential action within each wrap is reduced to the single signed utilization  $\lambda_j$ , allowing these mechanism-level transients to be retained without introducing a spatially discretized belt state.

### 3.5 Tangential Contact-State Closure

The fully engaged CVT has now been followed through every retained motion and free body. [Section 3.1](#) relates the sheave positions and pulley radii, [Section 3.2](#) balances the two movable assemblies, [Section 3.3](#) balances the two rotating assemblies, and [Section 3.4](#) balances the belt around both wraps and straight spans. No additional free body or degree of freedom remains to be introduced.

Looking across those relations, only two quantities remain without relations that fix their values at the current instant:  $\lambda_p$  and  $\lambda_s$ . The reason lies in how  $\lambda_j$  was introduced in [Equation \(3.4.18\)](#). There, the separate Coulomb conditions on traction magnitude, direction, and whether the contact is sticking or sliding were collected into the single signed utilization  $\lambda_j$ . That compression let the remainder of the belt derivation carry one contact quantity through the wrap tension, normal loading, pulley torque, and closed-loop belt relations without reopening the piecewise contact law at every step.

Importantly, this is different from adding one more balance. The equations derived so far keep the same form as the CVT moves. Changing a belt–pulley contact between sticking and sliding,

however, is a change of mechanical regime: the actual relation used to determine  $\lambda_j$  changes with the contact state. This is the type of regime change introduced in [Section 2.5](#), where a change in the active contact or constraint state changes the equations governing the mechanism. What remains is therefore not another motion or free-body balance, but the choice of which tangential contact relation governs each wrap.

### 3.5.1 Relative Motion at the Pulley Contacts

With the remaining ambiguity reduced to the two Coulomb branches, the first useful distinction is the contact motion itself. Sticking and sliding differ in whether the belt has relative tangential motion with respect to the pulley contact represented by a wrap. Before that relative motion can be written, the surface speed represented by each reduced contact must therefore be identified.

For the primary, this comparison is direct. Both sheave faces rotate with the primary shaft, so their angular speed is  $\omega_p$ . The secondary is different. Under [Assumption 11 \(Wrap-Level Contact Resolution\)](#), the complete wrap carries one traction state rather than separate states for the fixed and movable faces, while the helix kinematics already established that the two physical faces do not generally have the same speed during a shift. The fixed face rotates at  $\omega_s$ , whereas [Equation \(3.2.24\)](#) gives the movable-face speed  $\omega_{s,M} = \omega_s + \dot{\theta}_s$ .

One representative secondary surface speed is therefore required. Its value is fixed by the same face-torque reduction already used in [Assumption 12 \(Symmetric Sheave-Face Sharing\)](#). In particular, [Equation \(3.2.21\)](#) gives  $\tau_{s,M} = \tau_s/2$ , and therefore  $\tau_{s,F} = \tau_s/2$ . The instantaneous power delivered by the belt torque to the two faces is

$$\begin{aligned} P_{s,\text{faces}} &= \tau_{s,F}\omega_s + \tau_{s,M}\omega_{s,M} \\ &= \frac{\tau_s}{2}\omega_s + \frac{\tau_s}{2}(\omega_s + \dot{\theta}_s) \\ &= \tau_s\left(\omega_s + \frac{1}{2}\dot{\theta}_s\right). \end{aligned} \tag{3.5.1}$$

Let  $\bar{\omega}_s$  be the angular speed assigned to the single aggregate secondary contact. The same total torque represented by that contact carries power  $\tau_s\bar{\omega}_s$ . Equating this power with [Equation \(3.5.1\)](#) fixes the representative speed. The primary and secondary contact speeds can therefore be written together as

$$\boxed{\bar{\omega}_p = \omega_p, \quad \bar{\omega}_s = \omega_s + \frac{1}{2}\dot{\theta}_s = \frac{\omega_s + \omega_{s,M}}{2}.} \tag{3.5.2}$$

Thus the average in [Equation \(3.5.2\)](#) is not an additional empirical approximation. Once one secondary traction state and the equal face-torque split have been adopted, power consistency selects this representative surface speed. The arithmetic mean appears specifically because the two torque shares are equal. It remains an aggregate quantity: the model still does not determine the separate sliding states of the fixed and movable faces.

Returning now to relative motion, look at pulley  $j$ . The belt moves through the wrap with signed transport speed  $v_b$ , while the represented pulley surface moves tangentially at  $r_{j,\text{eff}}\bar{\omega}_j$ . Their difference defines the relative tangential speed,

$$v_{\text{rel},j} = v_b - r_{j,\text{eff}}\bar{\omega}_j, \quad j \in \{p, s\}. \quad (3.5.3)$$

The sign follows the common belt-transport and rotational directions defined in [Section 2.3.1](#). Positive  $v_{\text{rel},j}$  means the belt moves ahead of the represented pulley contact in the positive belt direction, while a negative value means it moves the other way.

If  $v_{\text{rel},j} \neq 0$ , the contact is already sliding. Its relative motion fixes the direction of friction, and kinetic Coulomb friction fixes the magnitude. The sliding branch of [Equation \(3.4.18\)](#) therefore gives

$$\begin{aligned} \sigma_j &= -\text{sgn}(v_{\text{rel},j}), \\ \lambda_j &= \sigma_j \mu_k, \quad v_{\text{rel},j} \neq 0. \end{aligned} \quad (3.5.4)$$

Sliding therefore closes  $\lambda_j$  directly from the current contact motion. Sticking begins at the opposite kinematic condition,

$$v_{\text{rel},j} = 0 \quad \Longleftrightarrow \quad v_b = r_{j,\text{eff}}\bar{\omega}_j. \quad (3.5.5)$$

The important point is that zero relative speed does not imply zero tangential force. It only says that the belt and represented pulley contact move together at that instant. Static friction may still supply a nonzero reaction, with  $\lambda_j$  taking whatever signed value is required to maintain sticking so long as the static limit in [Equation \(3.4.18\)](#) is not exceeded.

That speed match must also persist. If the belt speed and represented pulley-surface speed begin changing at different rates, relative motion appears immediately. Differentiating [Equation \(3.5.2\)](#) gives

$$\begin{aligned} \dot{\bar{\omega}}_p &= \dot{\omega}_p, \\ \dot{\bar{\omega}}_s &= \dot{\omega}_s + \frac{1}{2}\ddot{\theta}_s \\ &= \dot{\omega}_s + \frac{1}{2} \left[ \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2 \right], \end{aligned} \quad (3.5.6)$$

where the final line follows directly from [Equation \(3.2.25\)](#). Differentiating [Equation \(3.5.3\)](#) then gives the relative tangential acceleration,

$$\begin{aligned} a_{\text{rel},j} &\equiv \dot{v}_{\text{rel},j} \\ &= \dot{v}_b - r_{j,\text{eff}}\dot{\bar{\omega}}_j - \dot{r}_{j,\text{eff}}\bar{\omega}_j. \end{aligned} \quad (3.5.7)$$

The first two terms are the direct comparison between belt acceleration and the acceleration of a represented pulley surface at fixed radius. The final term appears because the contact radius is not fixed during a shift. As established in [Equation \(3.1.58\)](#), the effective radius moves with the pulley radius, so  $r_{j,\text{eff}}\omega_j$  can change even when  $\omega_j$  is momentarily constant. A sticking belt must follow that change as well.

Established sticking therefore requires

$$a_{\text{rel},j} = 0. \quad (3.5.8)$$

The two Coulomb branches now leave different things to determine. Sliding already supplies  $\lambda_j$ . Sticking instead supplies the condition that the mechanics must satisfy,  $a_{\text{rel},j} = 0$ , while  $\lambda_j$  remains the static reaction required to make that condition hold within the available friction limit.

### 3.5.2 Determining the Sticking Traction

For a sticking contact, the previous subsection leaves a simple-looking pair:  $\lambda_j$  is unknown, while the contact must satisfy  $a_{\text{rel},j} = 0$ . At first, this seems no different from any other unknown and equation already added to the model. If  $\lambda_j$  entered the mechanics linearly, it could simply be solved algebraically alongside the other instantaneous unknowns.

Whether that is possible can be read directly from the equations already derived. The pulley torque in [Equation \(3.4.41\)](#), for example, contains

$$\tau_j = -r_{j,\text{eff}} \lambda_j N_j, \quad (3.5.9)$$

so an unknown  $\lambda_j$  multiplies the unknown normal resultant  $N_j$ . The equation is therefore no longer linear in the unknowns. If this product were the only complication, however, another rearrangement might still isolate  $\lambda_j$ . The stronger obstacle comes from the wrap relation carried into the closed-loop belt compatibility. Through [Equation \(3.4.51\)](#), the same  $\lambda_j$  also appears reciprocally and inside terms such as

$$\frac{\sin \beta}{\lambda_j}, \quad \frac{1 + e^{-\frac{\lambda_j \phi_j}{\sin \beta}}}{1 - e^{-\frac{\lambda_j \phi_j}{\sin \beta}}}. \quad (3.5.10)$$

With the same unknown appearing both in products with other unknowns and inside reciprocal and exponential terms, there is no useful algebraic rearrangement that isolates the required static utilization.

There is one important simplification. This nonlinear dependence comes from the unknown static  $\lambda_j$  itself; the other remaining accelerations and contact loads appear linearly. If the static utilization were known, those quantities could be obtained directly from the balances already derived.

That suggests turning the problem around. Rather than trying to isolate  $\lambda_j$  explicitly, choose a candidate static utilization and determine the response it would produce. The resulting accelerations give  $a_{\text{rel},j}$  through [Equation \(3.5.7\)](#). Since sticking requires the relative acceleration to vanish,  $a_{\text{rel},j}$  can be used directly as the residual for the static-traction solve,

$$a_{\text{rel},j} = 0. \quad (3.5.11)$$

This is the same basic idea used for the geometric compatibility roots in [Section 3.1](#). A nonzero residual means that the candidate traction would make the belt and pulley surface begin accelerating relative to one another. The required static utilization is the value for which the residual vanishes. Finding that value is therefore a root solve in the unknown static traction.

If only one wrap is sticking, the other utilization is already fixed by its sliding branch. One static  $\lambda_j$  and one sticking residual remain, so the root is one-dimensional. If both wraps are sticking, the two static utilizations influence the same mechanical response and must be varied together until both residuals vanish,

$$a_{\text{rel},p}(\lambda_p, \lambda_s) = 0, \quad a_{\text{rel},s}(\lambda_p, \lambda_s) = 0. \quad (3.5.12)$$

This leaves only a one- or two-dimensional nonlinear search in the static traction utilizations. For each candidate value or pair, everything else is obtained from the linear balances already derived.

### 3.5.3 The Common Mechanical Solve

The two contact branches now reduce to the same mechanical calculation. Sliding supplies  $\lambda_j$  directly, while sticking tests candidate values as its root is found. In either case, evaluating the instantaneous response begins with one traction utilization for each wrap. The remaining step is to solve the mechanics for that pair.

With  $\lambda_p$  and  $\lambda_s$  specified, the equations already derived can now be read simply for the quantities they still have to determine. The axial balances expose the first of these unknowns. The primary balance in Equation (3.2.38) relates the shared shift acceleration  $\ddot{s}$  to the primary normal resultant  $N_p$ . The secondary balance in Equation (3.2.39) contains the same  $\ddot{s}$ , the secondary normal resultant  $N_s$ , and also the secondary shaft acceleration  $\dot{\omega}_s$  and belt torque  $\tau_s$ .

The rotational balances add the shaft dynamics. The primary and secondary relations in Equations (3.3.5) and (3.3.13) contain  $\dot{\omega}_p$  and  $\dot{\omega}_s$ , together with the belt torques  $\tau_p$  and  $\tau_s$ ; the secondary relation also contains the same  $\ddot{s}$  through the helix kinematics. The torques may look like two more unknowns, but the specified traction pair removes them immediately. From Equation (3.4.41),

$$\tau_j = -r_{j,\text{eff}} \lambda_j N_j, \quad j \in \{p, s\}, \quad (3.5.13)$$

so each pulley torque is set by the same normal resultant already appearing in the axial balances.

The complete-belt transport balance in Equation (3.4.1) brings in the belt acceleration  $\dot{v}_b$ . Its two tangential pulley actions are again the belt torques, so after the relation above is substituted the transport equation involves  $\dot{v}_b$ ,  $N_p$ , and  $N_s$  for the specified traction pair.

Finally, the closed-loop tension compatibility in Equation (3.4.52) relates the same belt acceleration and normal resultants to the radial accelerations of the two wraps. Those radial accelerations introduce no new degree of freedom: Equations (3.1.66) and (3.1.83) already express them in terms of the current geometry,  $\dot{s}$ , and the same shift acceleration  $\ddot{s}$ . The four temporary boundary tensions have likewise already disappeared from the closed-loop relation.

No new unknown appears as these equations are brought together. The quantities left to determine are exactly

$$\mathbf{z} = [\dot{\omega}_p \quad \dot{\omega}_s \quad \dot{v}_b \quad \ddot{s} \quad N_p \quad N_s]^\top. \quad (3.5.14)$$

At this point, every other term in the six equations is fixed by the current state, the specified traction pair, and the component relations already derived. Each equation is also linear in the six entries of  $\mathbf{z}$ . Collecting them therefore gives

$$\mathbf{A}(\lambda_p, \lambda_s) \mathbf{z} = \mathbf{b}(\lambda_p, \lambda_s). \quad (3.5.15)$$

To write  $\mathbf{A}$  and  $\mathbf{b}$  out cleanly, a small amount of temporary notation is useful. For this matrix only, primes shorten derivatives:  $x'_s$ ,  $r'_j$ ,  $q'_f$ , and  $J'_{f,p}$  denote derivatives with respect to  $s$ , while  $\theta'_s$  denotes differentiation with respect to  $x_s$ ; double primes follow the same convention. The only additional symbols are the two analytical wrap factors repeated in the closed-loop row,

$$\begin{aligned} \mathcal{C}_{v,j} &\equiv \begin{cases} \frac{2 \sin \beta}{\lambda_j} - \phi_j \frac{1 + e^{-\frac{\lambda_j \phi_j}{\sin \beta}}}{1 - e^{-\frac{\lambda_j \phi_j}{\sin \beta}}}, & \lambda_j \neq 0, \\ 0, & \lambda_j = 0, \end{cases} \\ \mathcal{C}_{N,j} &\equiv \begin{cases} \lambda_j \frac{1 + e^{-\frac{\lambda_j \phi_j}{\sin \beta}}}{1 - e^{-\frac{\lambda_j \phi_j}{\sin \beta}}}, & \lambda_j \neq 0, \\ \frac{2 \sin \beta}{\phi_j}, & \lambda_j = 0. \end{cases} \end{aligned} \quad (3.5.16)$$

These two symbols are temporary assembly notation only. Their  $\lambda_j = 0$  branches are the finite limits already established in [Equation \(3.4.53\)](#).

The coefficient matrix is then

$$\mathbf{A} = \begin{bmatrix} I_{B,p} + I_{p,\text{const}} + J_{f,p} & 0 & 0 & 0 & r_{p,\text{eff}} \lambda_p & 0 \\ 0 & I_{B,s} + I_{s,F} + I_{s,M} & 0 & I_{s,M} \theta'_s x'_s & 0 & r_{s,\text{eff}} \lambda_s \\ 0 & 0 & 0 & m_p + I_f (q'_f)^2 & \frac{1}{2} \cos \beta & 0 \\ 0 & I_{s,M} \theta'_s & 0 & [m_s + I_{s,M} (\theta'_s)^2] x'_s & 0 & \frac{1}{2} (\cos \beta + r_{s,\text{eff}} \lambda_s \theta'_s) \\ 0 & 0 & m_b & 0 & -\lambda_p & -\lambda_s \\ 0 & 0 & \rho_b A_b (r_{p,\text{cm}} C_{v,p} - r_{s,\text{cm}} C_{v,s}) & 2 \rho_b A_b (r_{s,\text{cm}} r'_s - r_{p,\text{cm}} r'_p) & C_{N,p} & -C_{N,s} \end{bmatrix}. \quad (3.5.17)$$

The corresponding known-side vector is

$$\mathbf{b} = \begin{bmatrix} \tau_{B,p} - J'_{f,p} \dot{s} \omega_p \\ \tau_{B,s} - I_{s,M} [\theta'_s x''_s + \theta''_s (x'_s)^2] \dot{s}^2 \\ \frac{1}{2} \omega_p^2 J'_{f,p} - I_f q'_f q''_f \dot{s}^2 - k_p (x_{p,\text{pre}} + s) \\ \theta'_s k_{s,\theta} (\theta_{s,\text{pre}} - \theta_s) + k_{s,x} (x_{s,\text{pre}} - x_s) - \left\{ [m_s + I_{s,M} (\theta'_s)^2] x''_s + I_{s,M} \theta'_s \theta''_s (x'_s)^2 \right\} \dot{s}^2 \\ 0 \\ 2 \rho_b A_b (r_{p,\text{cm}} r''_p - r_{s,\text{cm}} r''_s) \dot{s}^2 + \rho_b A_b v_b (\dot{r}_s C_{v,s} - \dot{r}_p C_{v,p}) \end{bmatrix}. \quad (3.5.18)$$

The row order follows the derivation: the first two rows are the pulley rotational balances, the next two are the axial balances, the fifth is the complete-belt transport balance, and the sixth is the closed-loop tension compatibility.

For a specified traction pair, every entry of  $\mathbf{A}$  and  $\mathbf{b}$  is fixed by the current state and the component relations already derived. Solving [Equation \(3.5.15\)](#) then gives the two shaft accelerations, belt

acceleration, shift acceleration, and the two normal resultants. A sliding  $\lambda_j$  reaches this solve directly from kinetic friction; a sticking  $\lambda_j$  reaches the same solve as a candidate value whose resulting  $a_{\text{rel},j}$  is returned to the root described in [Section 3.5.2](#).

The four contact combinations can now be read off directly:

|                         | Secondary sliding                                                                                                                                                                                        | Secondary sticking                                                                                                                                                                                       |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary sliding</b>  | Both $\lambda_p$ and $\lambda_s$ are fixed by kinetic friction. One $6 \times 6$ solve gives the instantaneous response.                                                                                 | $\lambda_p$ is fixed by kinetic friction, while $\lambda_s$ is found from the one-dimensional root $a_{\text{rel},s} = 0$ . Each candidate $\lambda_s$ is evaluated through the same $6 \times 6$ solve. |
| <b>Primary sticking</b> | $\lambda_s$ is fixed by kinetic friction, while $\lambda_p$ is found from the one-dimensional root $a_{\text{rel},p} = 0$ . Each candidate $\lambda_p$ is evaluated through the same $6 \times 6$ solve. | $\lambda_p$ and $\lambda_s$ are found together from $a_{\text{rel},p} = 0$ and $a_{\text{rel},s} = 0$ . Each candidate pair is evaluated through the same $6 \times 6$ solve.                            |

The six mechanical balances are therefore common to all four cases. What changes with the contact regime is only how the traction pair supplied to those balances is obtained: directly from kinetic friction, from one static-traction root, or from the two sticking conditions together.

### 3.5.4 Contact Admissibility and Regime Transitions

The four contact combinations above are enough to obtain an instantaneous response for any assumed tangential regime. They do not yet tell us whether that regime can actually be maintained, or what should happen when it cannot.

For a sticking contact,  $\lambda_j$  is the static reaction required to keep the belt and pulley moving together, as introduced in [Equation \(3.4.18\)](#). The root in [Section 3.5.2](#) finds the value of that reaction that satisfies  $a_{\text{rel},j} = 0$ . Sticking is physically available only while the required reaction lies within the static Coulomb limit,

$$|\lambda_j| \leq \mu_s. \quad (3.5.19)$$

If this condition is satisfied, the contact can supply the traction needed to remain stuck. If the required  $|\lambda_j|$  would exceed  $\mu_s$ , no admissible static reaction can preserve zero relative motion and the sticking regime must end.

Sliding is simpler away from zero relative speed. While  $v_{\text{rel},j} \neq 0$ , its sign identifies the sliding direction and [Equation \(3.5.4\)](#) fixes the corresponding kinetic traction. The next ambiguity appears only when  $v_{\text{rel},j}$  reaches zero. At that instant the belt and pulley are momentarily moving together, so the first possibility to test is sticking. The required static  $\lambda_j$  is found from  $a_{\text{rel},j} = 0$  and checked against [Equation \(3.5.19\)](#). If it is admissible, the contact captures into sticking.

If sticking is not admissible, the contact must leave  $v_{\text{rel},j} = 0$  on a sliding branch. For a contact that arrived at zero while already sliding, two continuations are possible: the relative velocity may grow again in the same direction as before, or the sliding direction may reverse and grow in the opposite direction. At the event itself,  $v_{\text{rel},j} = 0$ , so the relative velocity cannot distinguish between those two possibilities. The relative acceleration does: its sign tells which way  $v_{\text{rel},j}$  begins to move away from zero.

For a candidate kinetic sign  $\sigma_j$ , consistency therefore requires

$$\sigma_j a_{\text{rel},j} < 0, \quad v_{\text{rel},j} = 0, \quad (3.5.20)$$

because  $\sigma_j$  is defined opposite to the direction of sliding. The valid kinetic branch is the one whose solved acceleration produces the relative-motion direction assumed by that branch. The same check selects the initial sliding direction when a sticking contact first releases.

Together, these checks also make the order of a regime update clear. Consider a sliding–sliding solution in which the primary relative velocity reaches zero. The secondary contact can initially be left on its current kinetic branch while the primary is tested as sticking. If the required static  $\lambda_p$  is admissible, the candidate regime becomes sticking–sliding. If it is not, the primary remains sliding and the kinetic branch consistent with Equation (3.5.20) is selected instead. The resulting traction pair changes the common mechanical response, so that response is solved again and the contact conditions at both wraps are checked once more. If both contacts reach a regime boundary at the same instant, the same logic is applied to the combined candidate regime rather than updating the two wraps independently.

These tangential contact changes are nonimpulsive regime transitions in the sense introduced in Section 2.5. The belt and pulley reach the event with continuous position and velocity; changing the active contact law changes the acceleration that follows rather than resetting the state instantaneously. The remaining structural regime changes associated with engagement and shift boundaries require a separate treatment.

### 3.6 Shift-Regime Dynamics and Boundary Transitions

The freely shifting engaged CVT is now mechanically closed. For any admissible tangential contact regime, Section 3.5 determines how the pulley speeds, belt transport, and shift coordinate evolve from the current state. But that solution does more than describe the mechanism at one instant: as it is integrated forward in time, it moves the hardware.

Consider what that motion can do. Continued backshift closes the secondary until it can move no farther; continued primary opening can then separate the primary sheave from the belt. In the opposite direction, continued upshift eventually carries the primary to the end of its closing travel. The equations developed so far can bring the CVT to each of these configurations, but once one is reached, the physical problem has changed. A sheave that was free to move may now be supported by a stop, or a belt–sheave interaction that previously transmitted force may no longer exist.

Reaching one of these boundaries therefore changes the shift regime of the CVT, in the same sense that a change between sticking and sliding changes a tangential contact regime in Section 3.5. That structural change has two separate consequences. First, the active contacts, supports, and permitted motions are different, so the continuous governing equations must change with the regime. Second, the state arriving at the boundary may contain a velocity that the new regime cannot support, so the incoming motion must be transferred into a compatible outgoing state.

These are different mechanical problems. The first determines how the CVT accelerates once the new regime is active. The second determines how its velocity changes at the instant the regime is entered. Both are required if the model is to pass through a shift-regime boundary without replacing the mechanics by an imposed reset.

### 3.6.1 Shift Domain and Structural Regimes

Before the governing equations can change with the shift regime, the boundaries between those regimes must first be located along the shift motion. The shift coordinate already traces the primary through its full travel, so the geometry derived in [Section 3.1](#) identifies where contact is gained or lost, where hardware begins to support the motion, and which arrangement exists on either side of each boundary. Following the primary from fully open to fully closed makes those changes explicit.

Consider first the primary closing during engaged upshift. Its groove narrows, moving the belt outward. The fixed belt length requires the secondary radius to decrease, so the secondary opens. This motion can continue until the primary reaches its closing stop at  $x_p = x_{p,\max}$ . Further upshift would require the primary to close farther, into the stop. The belt remains between both pairs of sheave faces, so the mechanism can be held at this ratio with both pulley contacts present.

Opening the primary reverses these motions. The belt moves inward on the primary and outward on the secondary, bringing the secondary toward its fully closed stop. The secondary reaches that stop when the primary is at  $x_p = x_{p,dz}$ , as given by [Equation \(3.1.32\)](#). Now try to continue the same engaged backshift: the belt would have to move farther outward on the secondary, which would require its sheave to close beyond the stop. With shift motion at rest and primary belt contact maintained, the secondary stop can therefore hold the mechanism at its minimum ratio. This is the *engaged low-ratio seat*.

Yet the primary has not reached its own opening stop. Its groove can open wider than the belt, leaving the clearance introduced in [Equation \(3.1.16\)](#). The primary can therefore open farther if it separates from the belt. That opening no longer moves the belt inward or requires the secondary to close farther. Under the deadzone mapping, the secondary stays fully closed and both prescribed belt radii remain fixed while the primary crosses its clearance. Only when the primary reaches  $x_p = 0$  does its own opening stop prevent further travel.

The secondary stop thus limits the engaged backshift before the primary exhausts its opening travel. Closing the primary through the same clearance first brings it back to the belt; further closure then moves the belt outward and opens the secondary away from its stop.

Since  $s \equiv x_p$  by [Equation \(3.1.31\)](#), the same travel and boundary positions in the generalized shift coordinate are

$$0 \leq s \leq s_{\max}, \quad s_{dz} \equiv x_{p,dz}, \quad s_{\max} \equiv x_{p,\max}. \quad (3.6.1)$$

These motions divide the full primary travel into the five structural arrangements summarized in [Table 3.1](#).

| Structural regime          | Primary belt contact | Axial motion or support                                                        |
|----------------------------|----------------------|--------------------------------------------------------------------------------|
| Fully open primary stop    | Absent               | Primary held at $s = 0$ ; secondary remains fully closed.                      |
| Free deadzone              | Absent               | Primary free for $0 < s < s_{dz}$ ; secondary remains fully closed.            |
| Engaged low-ratio seat     | Present              | Fully closed secondary stop holds the engaged mechanism at $s = s_{dz}$ .      |
| Free engaged shift         | Present              | Free shift for $s_{dz} < s < s_{max}$ ; both sheaves follow the belt geometry. |
| Engaged upper primary stop | Present              | Primary closing stop holds the engaged mechanism at $s = s_{max}$ .            |

Table 3.1: Structural arrangements over the primary travel. The secondary retains belt contact in all five arrangements.

A boundary position does not, by itself, determine which shift regime is active. At  $s = s_{dz}$ , the secondary may be held against its fully closed stop while the primary remains in contact with the belt. The primary may instead begin closing, opening the secondary away from that stop, or it may begin opening and separate from the belt into the clearance. All three continuations pass through the same value of  $s$ . What changes is which contacts and supports are active and which motions they permit. The shift regime must therefore be carried as part of the mechanical description; it cannot be inferred from the shift coordinate alone. This is the same regime-dependent logic introduced in [Section 2.5](#), now applied to the changing mechanical arrangement of the CVT.

The same boundary also shows that shift regime and tangential contact state describe different parts of the mechanics. While the secondary stop holds the engaged mechanism at  $s = s_{dz}$ , the two shafts and the belt can continue moving. The stop determines whether further axial motion is available, but it does not determine whether the belt sticks to either pulley or slides relative to it. Each engaged shift regime must therefore still be combined with the corresponding tangential contact regimes from [Section 3.5](#).

The three boundary positions and the five arrangements in [Table 3.1](#) therefore define where structural regime changes may occur and which arrangements are physically admissible there. They do not assign a unique regime to each value of  $s$ ; at a shared boundary, the active contacts, supports, and direction of continuation still matter. The active shift regime and tangential contact regimes together specify the mechanical regime that the model must track as the CVT evolves.

### 3.6.2 Governing Equations in Each Shift Regime

Knowing which shift regime is active tells us which contacts and supports exist, but it does not yet determine how the CVT moves within that regime. The continuous equations must reflect those physical conditions. The useful starting point is the mechanical system already assembled for free engaged motion in [Equation \(3.5.15\)](#). Its six rows are the primary and secondary rotational balances, the primary and secondary axial balances, the complete-belt transport balance, and the closed-loop tension compatibility. A different shift regime does not require these mechanics to be derived again. It changes which of those balances remain active, which contact terms disappear, and which motions are prescribed by the hardware.

## Disengaged Primary and Deadzone Motion

At the fully open end of travel, the primary sheaves do not touch the belt, and the same remains true while the primary crosses the clearance toward  $s = s_{\text{dz}}$ . The immediate consequence is

$$N_p = 0, \quad \tau_p = 0. \quad (3.6.2)$$

Putting this contact loss back into the assembled mechanics shows exactly what changes on the primary side. The rotational balance in Equation (3.3.5) and the axial balance in Equation (3.2.38) still describe the same primary hardware, but their belt terms vanish. The primary rotational row therefore becomes

$$\tau_{B,p} = [I_{B,p} + I_{p,\text{const}} + J_{f,p}(s)] \dot{\omega}_p + \frac{dJ_{f,p}}{ds} \dot{s} \omega_p, \quad (3.6.3)$$

while the primary axial row becomes

$$\begin{aligned} \left[ m_p + I_f \left( \frac{dq_f}{ds} \right)^2 \right] \ddot{s} &= \frac{1}{2} \omega_p^2 \frac{dJ_{f,p}}{ds} \\ &\quad - I_f \frac{dq_f}{ds} \frac{d^2 q_f}{ds^2} \dot{s}^2 - k_p (x_{p,\text{pre}} + s). \end{aligned} \quad (3.6.4)$$

The belt side changes more substantially. During engaged shift, the secondary axial balance in Equation (3.2.39) participates in the shared shift motion, while Equation (3.4.52) closes the taut belt path through both pulley contacts. Neither role remains in the deadzone. The secondary stays fully closed while  $s$  moves only the primary, so its axial balance no longer supplies an equation for  $\ddot{s}$ . Primary separation also breaks the two-contact tension loop, so the engaged closed-loop compatibility cannot be retained.

This is where the reduced deadzone belt model is needed. The fully closed secondary cannot close farther to take up the remaining belt slack and establish the taut path used by the engaged model. The belt model also has no longitudinal strain or pretension state from which a residual tension could be recovered while that slack remains. Although the belt may still rest on the minimum-radius primary support, that contact is not represented as a clamped belt contact. Resolving the loads produced by a slack or pretensioned belt as it is taken up would therefore require additional belt elasticity and slack-span geometry beyond the present reduced description.

Under Assumption 9 (Reduced Belt Motion While the Primary Is Disengaged), the disengaged belt is instead kept seated on the fully closed secondary and carried with its represented contact surface. Because the secondary has no axial or helix-relative motion,  $\dot{\theta}_s = 0$ , and Equation (3.5.2) reduces to  $\bar{\omega}_s = \omega_s$ . Hence,

$$v_b = r_{s,\text{eff,dz}} \bar{\omega}_s = r_{s,\text{eff,dz}} \omega_s, \quad r_{s,\text{eff,dz}} \equiv r_{s,\text{eff}}(s_{\text{dz}}). \quad (3.6.5)$$

The radius is fixed, so

$$\dot{v}_b = r_{s,\text{eff,dz}} \dot{\omega}_s. \quad (3.6.6)$$

The secondary rotational balance remains active, but at fixed secondary geometry it takes the form given by [Equation \(3.3.14\)](#). The complete-belt transport balance in [Equation \(3.4.1\)](#) also remains active. The deadzone lock ties these two motions together, so their internal secondary contact torque can be eliminated by combining the balances. This gives

$$\tau_{B,s} = [I_{B,s} + I_{s,F} + I_{s,M} + m_b r_{s,\text{eff,dz}}^2] \dot{\omega}_s. \quad (3.6.7)$$

The final term is the complete belt transport inertia reflected through the fixed secondary radius. Primary disengagement removes the primary belt load without removing the belt mass that the secondary must still accelerate.

The active free-deadzone system is now visible. Its unknown accelerations are

$$\mathbf{z}_{\text{dz}} = [\dot{\omega}_p \quad \dot{\omega}_s \quad \dot{v}_b \quad \ddot{s}]^T, \quad (3.6.8)$$

and the four active relations are [Equations \(3.6.3\)](#), [\(3.6.4\)](#), [\(3.6.6\)](#) and [\(3.6.7\)](#). They may therefore be assembled, just as in the engaged case, as

$$\mathbf{A}_{\text{dz}} \mathbf{z}_{\text{dz}} = \mathbf{b}_{\text{dz}}. \quad (3.6.9)$$

For  $0 < s < s_{\text{dz}}$ , this reduced system determines  $\dot{\omega}_p$ ,  $\dot{\omega}_s$ ,  $\dot{v}_b$ , and  $\ddot{s}$  while the primary moves freely through the clearance.

**Fully open primary stop.** The boundary at  $s = 0$  introduces a different kind of regime change. Primary belt contact is already absent there, so no additional interaction disappears. Instead, the hardware prevents the primary from moving beyond the end of its travel. A sustained stop condition requires

$$s = 0, \quad \dot{s} = 0, \quad \ddot{s} = 0. \quad (3.6.10)$$

Inside the free deadzone, the primary axial balance in [Equation \(3.6.4\)](#) determines  $\ddot{s}$ . At the stop, that acceleration is no longer unknown because the hardware prescribes it to be zero. The force balance must still hold, so a new quantity appears instead: the reaction exerted by the stop on the primary.

Let  $F_{\text{stop},0} \geq 0$  denote the magnitude of this reaction in the  $+s$  direction, the only direction in which the stop can push the primary. The primary axial balance becomes

$$0 = F_{\text{fw},x_p} - F_{\text{spring},x_p} + F_{\text{stop},0}, \quad (3.6.11)$$

and therefore

$$F_{\text{stop},0} = F_{\text{spring},x_p} - F_{\text{fw},x_p}. \quad (3.6.12)$$

The axial balance has not been removed; its role has changed. During free deadzone motion it determines the primary acceleration. At the stop the acceleration is known, so the same balance

determines the support force. Equivalently, the free-deadzone system replaces its primary axial row by  $\ddot{s} = 0$ , and the original axial balance is evaluated afterward to recover  $F_{\text{stop},0}$ .

The fully open stop can sustain the shift regime only while

$$F_{\text{stop},0} \geq 0. \quad (3.6.13)$$

A negative value would require the stop to pull the primary in the  $-s$  direction, which a unilateral contact cannot do. Once the required reaction reaches zero and the unconstrained primary tendency points into the deadzone, the stop no longer supports the motion.

### Engagement and the Low-Ratio Seat

Closing through the deadzone leaves the belt radii unchanged until the primary reaches  $s = s_{\text{dz}}$ . At that point the movable primary sheave reaches the belt, so the primary normal load and contact torque can reappear and the engaged belt mechanics become active again. The secondary is still at its fully closed position, so the first engaged shift regime is the low-ratio seat rather than free shift.

For a sustained low-ratio seat,

$$s = s_{\text{dz}}, \quad \dot{s} = 0, \quad \ddot{s} = 0. \quad (3.6.14)$$

The fixed shift also gives  $\dot{\theta}_s = 0$ , so [Equation \(3.5.2\)](#) again reduces to  $\bar{\omega}_s = \omega_s$ . The shafts and belt may nevertheless continue to rotate and accelerate, and the two pulley contacts retain the tangential contact regimes developed in [Section 3.5](#).

The stop mechanics are the same type just introduced at  $s = 0$ , but the supported member is now the secondary. The secondary axial balance in [Equation \(3.2.39\)](#) would determine the shared shift acceleration during free engagement. At the low-ratio seat,  $\ddot{s} = 0$  is prescribed instead, and the fully closed secondary stop supplies the reaction needed to satisfy the original secondary free body in [Figure 3.8](#).

Let  $F_{\text{stop,LR}} \geq 0$  denote the magnitude of that reaction in the secondary opening direction. Using [Equations \(3.2.30\)](#) and [\(3.2.32\)](#), the held secondary balance is

$$0 = F_{\text{helix},x_s} + F_{\text{spring},x_s} - F_{\text{belt},x_s} - F_{\text{stop,LR}}, \quad (3.6.15)$$

so

$$F_{\text{stop,LR}} = F_{\text{helix},x_s} + F_{\text{spring},x_s} - F_{\text{belt},x_s}. \quad (3.6.16)$$

As before, the stopped axial balance changes role rather than disappearing. The only additional complication is that the reaction cannot be evaluated before the coupled engaged response is known. The belt term contains the unknown  $N_s$ , while the helix term contains the transmitted secondary torque and  $\dot{\omega}_s$ .

For a trial traction pair  $(\lambda_p, \lambda_s)$ , the mechanical core in [Equation \(3.5.15\)](#) normally determines  $\ddot{s}$  together with  $\dot{\omega}_p$ ,  $\dot{\omega}_s$ ,  $\dot{v}_b$ ,  $N_p$ , and  $N_s$ . At the seat, the secondary axial row is replaced by  $\ddot{s} = 0$ . The primary axial row in [Equation \(3.2.38\)](#) remains active because the primary itself is not supported

by a stop. The rotational balances in [Equations \(3.3.5\) and \(3.3.13\)](#), the complete-belt transport balance in [Equation \(3.4.1\)](#), and the closed-loop tension compatibility in [Equation \(3.4.52\)](#) retain their engaged forms. Together these active rows determine the fixed-geometry response for the chosen tangential contact regimes.

The original secondary axial balance is then evaluated with the solved quantities to recover  $F_{\text{stop,LR}}$  from [Equation \(3.6.16\)](#). The low-ratio seat is supportable only while

$$F_{\text{stop,LR}} \geq 0. \quad (3.6.17)$$

The stop reaction is a hardware load path, not an additional belt normal force. Any spring or helix loading carried directly by the stop does not add to  $N_s$  and therefore does not increase the secondary tangential traction capacity.

Once the secondary moves away from its closing stop while primary belt contact is retained, its axial row becomes active again. All six rows of [Equation \(3.5.15\)](#) are then the free engaged system already developed in [Section 3.5](#). No additional continuous equation is required in the interior shift regime.

### Upper Primary Stop

Free upshift continues until the primary reaches  $s = s_{\text{max}}$ . Belt contact remains present, but the primary can no longer move farther in the  $+s$  direction. This is the same unilateral support mechanism as the previous two stops, now applied to the primary axial row of the engaged system. A sustained upper-stop state satisfies

$$s = s_{\text{max}}, \quad \dot{s} = 0, \quad \ddot{s} = 0. \quad (3.6.18)$$

Let  $F_{\text{stop,max}} \geq 0$  denote the magnitude of the primary stop reaction in the  $-s$  direction. Using the primary axial balance in [Equation \(3.2.38\)](#),

$$0 = F_{\text{fw},x_p} - F_{\text{spring},x_p} - F_{\text{belt},x_p} - F_{\text{stop,max}}, \quad (3.6.19)$$

so

$$F_{\text{stop,max}} = F_{\text{fw},x_p} - F_{\text{spring},x_p} - F_{\text{belt},x_p}. \quad (3.6.20)$$

The constrained solve follows the same pattern as the low-ratio seat. The primary axial row is replaced by  $\ddot{s} = 0$ , while the secondary axial, rotational, belt-transport, and closed-loop compatibility rows remain active. The solved engaged response is then substituted into [Equation \(3.6.20\)](#) to recover  $F_{\text{stop,max}}$ . The upper stop can sustain the shift regime only while

$$F_{\text{stop,max}} \geq 0. \quad (3.6.21)$$

As at low ratio, the recovered reaction is carried by the hardware rather than added to the belt normal load.

The deadzone and stop-supported regimes therefore modify the assembled mechanics in two distinct ways. Primary separation removes the primary belt interaction and replaces the engaged belt closure with the reduced secondary-carried belt dynamics of Equation (3.6.9). A travel stop instead preserves the remaining contacts while prescribing the stopped axial motion.

The three stop-supported shift regimes then follow the same rule: impose  $\ddot{s} = 0$  in place of the axial row belonging to the supported member, solve the remaining active mechanics, and evaluate the original axial balance to recover the unilateral stop reaction. The resulting cases are summarized in Table 3.2.

| Supported shift regime     | Stopped member | Reaction direction | Balance used to recover the reaction |
|----------------------------|----------------|--------------------|--------------------------------------|
| Fully open primary stop    | Primary        | $+s$               | Primary axial balance                |
| Engaged low-ratio seat     | Secondary      | Opening secondary  | Secondary axial balance              |
| Engaged upper primary stop | Primary        | $-s$               | Primary axial balance                |

Table 3.2: Hardware reactions for the three stop-supported shift regimes. In each case, the stopped member’s original axial balance is evaluated after the corresponding constrained continuous response has been solved.

Each stop is a unilateral contact, so all three supported shift regimes require

$$F_{\text{stop},0} \geq 0, \quad F_{\text{stop},\text{LR}} \geq 0, \quad F_{\text{stop},\text{max}} \geq 0. \quad (3.6.22)$$

The continuous mechanics are now defined for the fully open stop, free deadzone, low-ratio seat, free engaged shift, and upper primary stop. Within each shift regime, the active balances determine the instantaneous accelerations and any support reaction required by the hardware.

### 3.6.3 Shift-Regime Transitions and Momentum Transfer

The continuous equations now describe the motion within every shift regime, but they do not by themselves complete the trajectory through the boundaries between them. At such a boundary, the mechanics on either side may already be known, yet it remains to determine when the current constraints should be released and whether the existing velocity can be carried into the new regime unchanged.

A stop-supported regime exposes this missing boundary logic directly. While the stop is active, its continuous equations impose  $\ddot{s} = 0$ . As the surrounding loads evolve, that prescribed acceleration cannot indicate when the mechanism should begin moving away from the stop. The release condition must instead come from the quantity that measures whether the constraint is still mechanically required: the reaction needed to enforce  $\ddot{s} = 0$ .

The stop reactions derived in Section 3.6.2 provide that condition directly. For any of the three stops, the supported regime remains admissible while

$$F_{\text{stop}} \geq 0. \quad (3.6.23)$$

Release occurs when the required reaction reaches zero and crosses toward the inadmissible side. The reaction is precisely the force needed to hold the supported member at  $\ddot{s} = 0$ ; once that support vanishes, restoring the free axial balance gives the acceleration away from the stop. Thus the same quantity that closes the constrained mechanics also supplies its breakaway condition.

For the fully open primary stop, this occurs when  $F_{\text{stop},0}$  crosses zero, releasing the primary into the free deadzone. At the upper primary stop,  $F_{\text{stop},\text{max}}$  crossing zero similarly releases the mechanism into free engaged shift.

The low-ratio seat is different because two unilateral interactions are present at the same configuration: the secondary movable sheave is supported at its fully closed stop, while the primary remains in contact with the belt. There are therefore two physically distinct ways to leave the seat.

The first is the same stop-release mechanism just described. When

$$F_{\text{stop,LR}} = 0 \quad (3.6.24)$$

is crossed toward inadmissibility, the fully closed secondary stop no longer needs to carry load. The secondary is then free to move away from the stop, and the CVT returns to free engaged shifting.

The second possibility is for the primary to separate from the belt and move into the deadzone while the secondary remains fully closed. Here a vanishing primary normal resultant is only part of the release condition. Reaching  $N_p = 0$  establishes that the belt is no longer supplying a compressive load to the primary, but it does not tell us whether the primary would actually open away from the belt or immediately close back into contact.

While the low-ratio seat is active,  $\ddot{s} = 0$  is prescribed, so that direction cannot be read from the seated solution itself. Instead, evaluate the contact-free primary axial equation in [Equation \(3.6.4\)](#) at the same instantaneous state. Let  $\ddot{s}_{\text{cf}}$  denote the acceleration returned by this contact-free evaluation. It is not used to evolve the seated regime; it is only the acceleration the primary would have if belt contact were removed at that instant. Since positive  $s$  closes the primary, separation into the deadzone requires

$$N_p = 0, \quad \ddot{s}_{\text{cf}} < 0. \quad (3.6.25)$$

The contrast with a metal stop is important. A stop reaction crossing zero already identifies the direction of the free motion because that reaction is the force suppressing the corresponding free axial acceleration. Loss of primary normal load does not provide the same information, so the contact-free acceleration must be checked separately before primary separation is admitted.

At the instant a stop releases, the mechanism is already stationary in the constrained shift direction. The stop has held  $\dot{s} = 0$  up to the moment its reaction vanishes; removing that support does not require the hardware to acquire a new velocity instantaneously. It only allows the restored free axial balance to determine what acceleration follows. The same is true when the primary separates from an established low-ratio seat. The seated state already has  $\dot{s} = 0$ , and the deadzone can begin from that same instantaneous velocity once belt contact is lost.

Position and velocity therefore remain continuous through these releases, as in [Equation \(2.5.2\)](#). Their effect is only to change which continuous equations govern the next acceleration. In this

sense the release is nonimpulsive: the outgoing regime already admits the velocity with which the constrained regime ends.

The reverse crossing need not have this property. A free regime can reach a new boundary carrying a velocity that the newly active constraint or kinematic mapping does not admit.

### Finite-Speed Arrival at a Boundary

The release conditions determine when a stop can cease supporting the mechanism. Shift is then already at rest, so the free axial balance determines the acceleration away from the stop. For the reverse crossing, arrival is detected simply by reaching the stop position. The difficulty is the shift velocity with which the sheaves arrive. Bringing that motion to rest requires an impulse, and the motion immediately after the event must be determined before the continuous solution can proceed. This is the finite-speed case distinguished in [Section 2.5](#): reaching the boundary does not cause a jump in the sheave positions, but the incoming motion may not be compatible with the new constraint.

At the upper primary stop, for example, free upshift can reach  $s = s_{\max}$  while the primary is still closing. The movable sheave strikes the fixed sheave, metal on metal. Under the perfectly inelastic impact model of [Assumption 13 \(Instantaneous, Perfectly Inelastic Constraint Activation\)](#), this collision arrests its axial motion without rebound. With superscripts  $-$  and  $+$  denoting values immediately before and after the event, an arrival with  $\dot{s}^- > 0$  therefore gives

$$\dot{x}_p^+ = \dot{s}^+ = 0. \quad (3.6.26)$$

The belt remains seated between both pairs of sheaves. The secondary cannot keep opening while the primary holds its radius fixed: the two radii must still fit the same belt length. The belt therefore brings the secondary's axial travel to rest with the primary's.

During this travel, however, the movable secondary has also been turning relative to its shaft as its pin follows the helix. With axial travel blocked, the pin can no longer move along the inclined track. Continued relative rotation would drive it into the side of the track, which pushes back on the pin. The force exerts a torque on the movable sheave; the pin exerts an equal and opposite force on the track attached to the shaft. Unlike the axially fixed stop, the shaft and fixed sheave remain free to change their angular speed. The reaction therefore changes their rotation too, until the movable sheave and shaft turn together.

To see how this exchange changes the shaft speed, follow the angular momentum through the collision. During this upshift, the movable sheave arrives rotating more slowly than the shaft, as established in [Section 3.3.3](#). If the belt supplies no angular impulse during the collision, the faster shaft must supply the angular momentum that brings the movable sheave up to their common speed. The movable assembly gains exactly what the shaft and its attached rotating hardware lose. The shaft slows while the movable assembly speeds up, conserving their combined angular momentum. Their common speed lies between the two incoming speeds.

The incoming state supplies the shaft speed  $\omega_s^-$  and shift rate  $\dot{s}^-$ . The helix relation then gives the known angular speed of the movable assembly before impact:

$$\omega_{s,M}^- = \omega_s^- + \frac{d\theta_s}{ds} \dot{s}^-. \quad (3.6.27)$$

After impact the relative rotation has ended, and the two assemblies share the unknown speed  $\omega_s^+$ :

$$\omega_{s,M}^+ = \omega_s^+. \quad (3.6.28)$$

The stop condition has fixed the axial velocities, but this common rotation speed still has to be found from the incoming angular momentum and the inertias that now turn together.

The belt can take part in the same exchange. If an existing secondary sticking contact remains stuck, changing the pulley speed while keeping the belt speed unchanged would create slip. A tangential contact impulse must change their momenta together so that  $v_b^+ = r_{s,\text{eff}}\omega_s^+$ . If the primary contact remains stuck too, the same belt speed must also satisfy  $v_b^+ = r_{p,\text{eff}}\omega_p^+$ , extending the exchange to the primary. These are the retained sticking conditions of [Assumption 14 \(Tangential Contact Through Boundary Events\)](#); sliding interfaces and newly formed contacts receive no automatic impulsive sticking condition. The shaft and belt speeds must therefore be found together, with the stop and every retained contact condition satisfied by the same outgoing state.

Arrival at the engaged low-ratio seat or the fully open primary stop requires the same accounting. The stop position locates the event, and the stop condition specifies the motion that must end. The remaining velocities depend on which bodies exchange momentum as that motion ends. A complete stop transition must supply those velocities before the continuous balances can be used again.

**Changing the coupling** At these stops, the connections between the bodies remain in place. Primary separation introduces a related difficulty: the primary can keep moving while its connection to the secondary is lost. At the low-ratio boundary  $s = s_{\text{dz}}$ , the secondary is fully closed, but the primary still has room to open. If the primary separates from the belt, it can continue through the deadzone while the secondary remains at its seat. The shift rate still gives  $\dot{x}_p = \dot{s}$ , but no longer produces secondary translation, helix-relative rotation, or changing belt radii.

For a backshift that reaches this boundary at finite speed, simply carrying all the state velocities across unchanged would keep the primary moving while secondary translation and radial belt motion ceased. The movable secondary would also lose the helix contribution to its angular speed. The same state-velocity values would thus describe different physical motions on the two sides. Those changes still involve momentum: the secondary seat arrests its translation, while the helix can transfer angular momentum into the continuing shaft rotation. The event balance must use the actual body velocities produced by the connections on each side of the boundary.

The same separation can impose a velocity restriction as well. In the deadzone, the belt and secondary must satisfy the lock  $v_b^+ = r_{s,\text{eff,dz}}\omega_s^+$  from [Equation \(3.6.5\)](#). If the incoming belt and secondary surface speeds differ, bringing them into agreement requires an impulse even when the shift rate is zero. Entry into the deadzone can change the coupling between bodies and impose a tangential restriction at the same instant.

**The general boundary problem** A boundary can thus change how the state velocities move the bodies, restrict which velocities are allowed, or do both. These two effects account for all five structural regimes. Free engaged shift, the low-ratio seat, and the upper primary stop share the engaged relations between state velocities and body motion. Free deadzone motion and the fully open primary stop share the deadzone relations. There are only two sets of these relations; the stops impose  $\dot{s} = 0$  within them, and tangential locks impose conditions on the shaft and belt speeds.

These examples leave two separate pieces of mechanics that a general boundary law must carry together. The state velocities do not by themselves specify the physical momentum; the current connections determine which masses and inertias those velocities move. The outgoing regime can also restrict the velocity state beyond those connections. A finite-speed transition must therefore

carry the physical momentum from the incoming connection to the outgoing one and, when the outgoing state is more restricted, account for the impulse that establishes those restrictions.

The two effects identified above enter the boundary problem differently. Momentum is carried by the physical hardware in motion, so its contribution can be made explicit before any additional stop or contact restriction is imposed. That accounting begins with the velocities already retained by the model.

**Momentum carried through the boundary** The first part of the boundary problem is to identify the momentum carried by the moving hardware. For each retained translation or rotation, that requires its physical velocity together with the mass or inertia moving at that speed. The model describes these motions through two shaft speeds, a belt transport speed, and the shift rate, collected as

$$\boldsymbol{\nu} = [\omega_p \quad \omega_s \quad v_b \quad \dot{s}]^T. \quad (3.6.29)$$

The upper-stop example already showed that the movable secondary's angular speed contains both shaft rotation and helix-relative rotation, while its axial travel carries a separate linear momentum. The separation example established the complementary point: the physical motion produced by a given  $\boldsymbol{\nu}$  depends on which kinematic connection is active. Applying the established kinematics to the remaining translations and rotations gives the physical velocities in [Table 3.3](#), with the corresponding mass or inertia beside each one.

| Physical motion                 | Mass or inertia                             | Engaged velocity                          | Deadzone velocity         |
|---------------------------------|---------------------------------------------|-------------------------------------------|---------------------------|
| Primary shaft-carried rotation  | $I_{B,p} + I_{p,\text{const}} + J_{f,p}(s)$ | $\omega_p$                                | $\omega_p$                |
| Primary movable axial motion    | $m_p$                                       | $\dot{s}$                                 | $\dot{s}$                 |
| Primary flyweight pivot motion  | $I_f$                                       | $\frac{dq_f}{ds} \dot{s}$                 | $\frac{dq_f}{ds} \dot{s}$ |
| Secondary shaft-side rotation   | $I_{B,s} + I_{s,F}$                         | $\omega_s$                                | $\omega_s$                |
| Secondary movable-side rotation | $I_{s,M}$                                   | $\omega_s + \frac{d\theta_s}{ds} \dot{s}$ | $\omega_s$                |
| Secondary movable axial motion  | $m_s$                                       | $\frac{dx_s}{ds} \dot{s}$                 | 0                         |
| Complete-belt transport         | $m_b$                                       | $v_b$                                     | $v_b$                     |
| Primary-wrap radial motion      | $\rho_b A_b r_{p,\text{cm}} \phi_p$         | $\frac{dr_p}{ds} \dot{s}$                 | 0                         |
| Secondary-wrap radial motion    | $\rho_b A_b r_{s,\text{cm}} \phi_s$         | $\frac{dr_s}{ds} \dot{s}$                 | 0                         |

Table 3.3: Retained physical motions and their momentum-carrying masses and inertias. Each row identifies one physical motion, the mass or inertia carried with it, and the velocity produced by the engaged and deadzone kinematics.

Each velocity in the table is a linear combination of the same four state entries. For the engaged movable secondary, for example,

$$\omega_{s,M} = \omega_s + \frac{d\theta_s}{ds}\dot{s} = \begin{bmatrix} 0 & 1 & 0 & \frac{d\theta_s}{ds} \end{bmatrix} \boldsymbol{\nu}. \quad (3.6.30)$$

In the deadzone, the same row becomes  $\begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix}$ : the shaft still turns the movable assembly, but primary travel no longer adds helix-relative rotation. Stacking the nine rows in table order gives the engaged and deadzone velocity maps

$$J_{\text{eng}}, J_{\text{dz}} \in \mathbb{R}^{9 \times 4}. \quad (3.6.31)$$

For either topology,  $J\boldsymbol{\nu}$  therefore contains the nine physical velocities produced by the four state velocities. Read by columns, the same matrix shows where each retained velocity appears through the mechanism; in particular, the fourth column gives the coefficients multiplying  $\dot{s}$  in each retained physical velocity.

The linear or angular momentum carried by each motion is obtained by multiplying its velocity by the corresponding mass or inertia. Placing the masses and inertias from [Table 3.3](#) on the diagonal of  $W \in \mathbb{R}^{9 \times 9}$ , in the same row order, performs these multiplications together:

$$\mathbf{p} = WJ\boldsymbol{\nu} \in \mathbb{R}^9, \quad J \in \{J_{\text{eng}}, J_{\text{dz}}\}. \quad (3.6.32)$$

The entries are the linear and angular momenta of the individual translations and rotations in the table; they are not the components of one spatial momentum vector.

The same construction gives the momentum immediately before and after the boundary event. Positions remain continuous through the instantaneous transition, so the masses and inertias are evaluated at the same configuration and  $W$  is common to both sides. The incoming and outgoing connections determine  $J_-$  and  $J_+$ , respectively, so both maps are known for the transition being considered:

$$J_- \in \{J_{\text{eng}}, J_{\text{dz}}\}, \quad J_+ \in \{J_{\text{eng}}, J_{\text{dz}}\}. \quad (3.6.33)$$

A crossing between engaged and deadzone motion uses different maps, whereas arrival at the upper primary stop has  $J_- = J_+ = J_{\text{eng}}$ . At the fully open primary stop, both sides instead use  $J_{\text{dz}}$ .

The incoming state supplies the known velocity  $\boldsymbol{\nu}^-$ , while  $\boldsymbol{\nu}^+$  remains to be found. The corresponding physical momenta are therefore

$$\mathbf{p}^- = WJ_- \boldsymbol{\nu}^-, \quad \mathbf{p}^+ = WJ_+ \boldsymbol{\nu}^+. \quad (3.6.34)$$

Because each row follows the same physical translation or rotation on both sides, subtracting the two gives its momentum change through the event:

$$\Delta \mathbf{p} = WJ_+ \boldsymbol{\nu}^+ - WJ_- \boldsymbol{\nu}^- \in \mathbb{R}^9. \quad (3.6.35)$$

If  $J_- \neq J_+$ , even identical state-velocity vectors can correspond to different physical motions and therefore different momenta.

**Velocities allowed after the event** With the momentum on both sides now expressed through  $J_-$  and  $J_+$ , the effect of changing the kinematic connection is accounted for in the momentum inventory. The boundary examples established a second possibility as well: even after the appropriate outgoing map  $J_+$  is known, the new regime can impose additional relations on the outgoing velocities  $\boldsymbol{\nu}^+$  themselves. At the upper primary stop, for example, the engaged connection remains valid, so  $J_- = J_+ = J_{\text{eng}}$ , while the already-established stop condition is

$$\dot{s}^+ = 0. \quad (3.6.36)$$

The shift rate  $\dot{s}^+$  is the fourth entry of  $\boldsymbol{\nu}^+$ . Writing the stop condition's coefficients in that state order gives a row that selects this entry, so the required relation becomes

$$c_{\text{stop}} = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}, \quad c_{\text{stop}} \boldsymbol{\nu}^+ = 0. \quad (3.6.37)$$

The engaged low-ratio seat and fully open primary stop arrest the same shift coordinate and therefore use the same row. The engaged low-ratio seat uses  $J_{\text{eng}}$ , while the fully open primary stop uses  $J_{\text{dz}}$ .

A belt–pulley contact can restrict the outgoing velocities for a different physical reason. If a contact that was already sticking remains stuck through the collision, the two represented surface speeds must still agree afterward. At the primary, both sheave faces rotate with the shaft, so retained sticking requires

$$v_b^+ - r_{p,\text{eff}} \omega_p^+ = 0. \quad (3.6.38)$$

In the state ordering of [Equation \(3.6.29\)](#), this condition is

$$c_{p,\text{stick}} = \begin{bmatrix} -r_{p,\text{eff}} & 0 & 1 & 0 \end{bmatrix}, \quad c_{p,\text{stick}} \boldsymbol{\nu}^+ = 0. \quad (3.6.39)$$

The same no-slip requirement applies at the secondary, but the represented surface speed there also contains the movable sheave's helix-relative rotation. Using the representative contact speed from [Equation \(3.5.2\)](#), retained secondary sticking requires

$$v_b^+ - r_{s,\text{eff}} \left( \omega_s^+ + \frac{1}{2} \frac{d\theta_s}{ds} \dot{s}^+ \right) = 0, \quad (3.6.40)$$

which gives

$$c_{s,\text{stick}} = \begin{bmatrix} 0 & -r_{s,\text{eff}} & 1 & -\frac{r_{s,\text{eff}}}{2} \frac{d\theta_s}{ds} \end{bmatrix}, \quad c_{s,\text{stick}} \boldsymbol{\nu}^+ = 0. \quad (3.6.41)$$

These rows apply only to sticking contacts retained through the collision. An incoming sliding contact contributes no sticking relation, and a contact created by the event is not assigned one automatically.

Retained sticking through the event is treated here as an ideal impulsive constraint under [Assumption 14 \(Tangential Contact Through Boundary Events\)](#). The resulting tangential impulse is therefore not re-tested against a separate impulsive Coulomb bound; doing so would require the normal and tangential contact impulses to be resolved together during the event.

For outgoing deadzone motion, the reduced model's prescribed belt–secondary lock replaces the engaged secondary sticking condition. The outgoing map is  $J_+ = J_{\text{dz}}$ , and the belt remains locked to the fully closed secondary after primary separation. The closure already derived in [Equation \(3.6.5\)](#) requires

$$v_b^+ - r_{s,\text{eff,dz}} \omega_s^+ = 0, \quad (3.6.42)$$

so an outgoing deadzone state must satisfy

$$c_{s,dz} = \begin{bmatrix} 0 & -r_{s,\text{eff},dz} & 1 & 0 \end{bmatrix}, \quad c_{s,dz}\boldsymbol{\nu}^+ = 0. \quad (3.6.43)$$

The lock applies even when entry into the deadzone must first bring unequal incoming belt and secondary surface speeds into agreement.

At the upper primary stop, if both belt contacts remain sticking, the stop condition and the two sticking conditions must all hold for the same  $\boldsymbol{\nu}^+$ . Their rows can therefore be imposed together by stacking them into a single matrix. More generally, for the  $n_c$  restrictions present in a given event,

$$C_+ = \begin{bmatrix} c_1 \\ \vdots \\ c_{n_c} \end{bmatrix} \in \mathbb{R}^{n_c \times 4}, \quad C_+\boldsymbol{\nu}^+ = 0. \quad (3.6.44)$$

**Constraint impulses** These relations determine which combinations of outgoing velocities are allowed, but they do not themselves supply the momentum change required to establish them. At the upper primary stop, that distinction has a direct physical source: the movable sheave strikes the axially fixed sheave, and the metal contact develops whatever axial reaction is required to prevent further closing. Over the instantaneous event, that reaction appears as a finite impulse associated with the stop constraint.

Let  $\Gamma_{\text{stop}}$  denote this impulse, signed positive toward increasing  $s$ . Since  $c_{\text{stop}}$  identifies the retained direction in which the stop acts, its impulse is

$$c_{\text{stop}}^\top \Gamma_{\text{stop}} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \Gamma_{\text{stop}} \end{bmatrix}. \quad (3.6.45)$$

A retained sticking condition has the same structure. The condition  $c_{p,\text{stick}}\boldsymbol{\nu}^+ = 0$  requires the belt and primary surface to leave with zero relative velocity; the tangential contact can supply the impulse required to maintain that relation through the collision. Let  $\Gamma_p$  be positive when the impulse on the belt acts in the positive  $v_b$  direction. The belt then receives  $+\Gamma_p$ , while the equal and opposite action on the primary gives angular impulse  $-r_{p,\text{eff}}\Gamma_p$ :

$$\begin{bmatrix} -r_{p,\text{eff}}\Gamma_p \\ 0 \\ \Gamma_p \\ 0 \end{bmatrix} = c_{p,\text{stick}}^\top \Gamma_p. \quad (3.6.46)$$

When both contacts remain sticking, both tangential impulses contribute to the belt's momentum change. Contributions from the active restrictions therefore add wherever they act on the same retained motion. Associate each row  $c_i$  of  $C_+$  with an unknown signed impulse amplitude  $\Gamma_i$ , and collect these amplitudes in

$$\boldsymbol{\Gamma} = [\Gamma_1, \dots, \Gamma_{n_c}]^\top \in \mathbb{R}^{n_c}, \quad (3.6.47)$$

Their combined generalized impulse is then

$$C_+^T \mathbf{\Gamma} \in \mathbb{R}^4. \quad (3.6.48)$$

Each row therefore carries both the velocity restriction it imposes and the unknown impulse required to enforce it:  $c_i \mathbf{v}^+ = 0$  specifies the outgoing velocity relation, while  $c_i^T \mathbf{\Gamma}_i$  gives the corresponding generalized impulse.

**Momentum balance in the retained velocity directions** The impulses have known places to act, but their magnitudes and the resulting speeds remain unknown. Determining them requires balancing those impulses against the momentum changes of the moving hardware. The difficulty is that  $\Delta \mathbf{p} \in \mathbb{R}^9$  still lists the nine physical translations and rotations separately, whereas  $C_+^T \mathbf{\Gamma} \in \mathbb{R}^4$  already acts in the four retained velocity directions. The body momentum changes must therefore be gathered into those same four directions before the two can be balanced.

That gathering is already encoded in the outgoing velocity map. Each column of  $J_+$  states how one retained velocity appears among the nine outgoing body velocities; the same coefficients therefore state how the corresponding body momentum changes contribute back to that retained direction. For secondary rotation, for example, the  $\omega_s$  column has coefficient one in both the shaft-side and movable-side rotation rows. Denote their angular-momentum changes by  $\Delta H_{s,F}$  and  $\Delta H_{s,M}$ , with the shaft-side term including the boundary inertia  $I_{B,s}$ . These changes therefore enter together:

$$\left[ J_+^T \Delta \mathbf{p} \right]_{\omega_s} = \Delta H_{s,F} + \Delta H_{s,M}. \quad (3.6.49)$$

The upper-stop exchange described at the outset reappears here: the pin-track reaction transfers angular momentum between the two secondary assemblies, and its equal and opposite angular impulses cancel from their combined rotational balance. That transfer remains present in the individual momentum changes. The shift balance, however, also contains the reaction's axial contribution, so the angular and axial impulses must be expressed in the same shift direction before they can be combined.

The compatible-work relation already derived for the helix in [Equation \(3.2.28\)](#) supplies this conversion. A torque contributes to the shift balance through the relative rotation produced per unit shift,  $d\theta_s/ds$ . Because the geometry is fixed through the instantaneous event, integrating the balance preserves this factor: the movable secondary's angular-momentum change contributes

$$\frac{d\theta_s}{ds} \Delta H_{s,M}$$

to the shift-direction balance. The other rotational and translational momentum changes enter with their corresponding motion ratios, which are precisely the entries of the shift column.

For the ideal pin-track reaction, the same compatible-work relation makes its weighted angular and axial impulse contributions equal and opposite in this balance. Their cancellation here, together with the angular cancellation in the secondary rotational balance, explains why the helix reaction requires no additional component of  $\mathbf{\Gamma}$ , even though its momentum transfer remains present in  $\Delta \mathbf{p}$ .

Applying this operation to all four columns collects the nine physical momentum changes into

$$J_+^T \Delta \mathbf{p} \in \mathbb{R}^4, \quad (3.6.50)$$

with one balance associated with each of  $\omega_p$ ,  $\omega_s$ ,  $v_b$ , and  $\dot{s}$ .

Taken together, the gathered balance has a useful interpretation. Motion that exists only on the incoming side is still retained in  $\Delta \mathbf{p}$  through  $W J_- \boldsymbol{\nu}^-$ , while reactions internal to the ideal kinematic connections cancel when the body balances are combined, as the helix exchange does above. The stop and contact impulses enforce additional restrictions rather than those kinematic connections, so they remain explicitly as  $C_+^\top \boldsymbol{\Gamma}$ .

**Coupled event solve** The physical momentum changes and the explicit constraint impulses are now expressed in the same four retained directions. They can therefore be balanced directly, while the outgoing velocities must simultaneously satisfy the restrictions already collected in  $C_+$ :

$$\begin{aligned} J_+^\top \Delta \mathbf{p} &= C_+^\top \boldsymbol{\Gamma}, \\ C_+ \boldsymbol{\nu}^+ &= 0. \end{aligned} \quad (3.6.51)$$

The first relation determines the velocity response to an impulse exchange, but the amplitudes remain unknown until the second relation is satisfied. The second fixes the required speed relationships, but in the upper-stop example their common value still depends on the incoming momentum in the first.

These two relations contain the complete event mechanics, but they are not yet written directly in the unknowns. Substituting the momentum difference from Equation (3.6.35) and placing the known incoming term on the right collects both sets of unknowns:

$$\begin{bmatrix} J_+^\top W J_+ & -C_+^\top \\ C_+ & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{\nu}^+ \\ \boldsymbol{\Gamma} \end{bmatrix} = \begin{bmatrix} J_+^\top W J_- \boldsymbol{\nu}^- \\ 0 \end{bmatrix}. \quad (3.6.52)$$

At this point the event can be solved in one step. For a known incoming state and a selected outgoing regime, every block in Equation (3.6.52) is known; solving the system returns the outgoing velocities  $\boldsymbol{\nu}^+$  and the constraint impulses  $\boldsymbol{\Gamma}$  together.

The block matrix has the standard saddle-point, or KKT, structure of a constrained mechanical system (Benzi et al., 2005). Here the constraint multipliers are impulses rather than continuous reaction forces, because the balance is taken across an instantaneous transition (Stewart, 2000). The block  $J_+^\top W J_+$  is the generalized inertia seen in the outgoing velocity directions,  $C_+^\top$  places the unknown constraint impulses into those directions, and  $C_+$  imposes the required outgoing velocity relations. The right-hand side carries the known incoming momentum into the same directions.

The resulting event state must still respect the unilateral character of any active stop. With  $\Gamma_{\text{stop}}$  signed positive toward increasing  $s$ , an impulse at the fully open stop or engaged low-ratio seat must satisfy  $\Gamma_{\text{stop}} \geq 0$ , whereas the upper primary stop can act only toward decreasing  $s$  and therefore requires  $\Gamma_{\text{stop}} \leq 0$ . A solution requiring the opposite sign cannot be supplied by that stop, so the proposed outgoing constraint is not mechanically admissible.

**Completing the upper primary stop** The final system can now resolve the secondary-speed question posed at the outset. Take  $J_- = J_+ = J_{\text{eng}}$  and first consider the case in which the belt supplies no angular impulse to the secondary. The shaft-side and movable-side secondary rotations contribute the following  $(\omega_s, \dot{s})$  block to the generalized inertia matrix:

$$\left[ J_{\text{eng}}^\top W J_{\text{eng}} \right]_{\omega_s, \dot{s}}^{\text{secondary rotation}} = \begin{bmatrix} I_{B,s} + I_{s,F} + I_{s,M} & I_{s,M} \frac{d\theta_s}{ds} \\ I_{s,M} \frac{d\theta_s}{ds} & I_{s,M} \left( \frac{d\theta_s}{ds} \right)^2 \end{bmatrix}. \quad (3.6.53)$$

The rows identify the retained direction in which momentum is counted, while the columns identify the retained velocity contributing that momentum. The upper-left entry therefore describes the contribution from  $\omega_s$  to momentum in the  $\omega_s$  direction: both secondary assemblies rotate with the shaft. The upper-right entry describes the contribution from  $\dot{s}$  to that same momentum, because shift rotates the movable assembly through the helix. The lower-left entry is the reciprocal coupling, in which shaft rotation contributes to the shift-direction momentum. The lower-right entry describes the contribution from  $\dot{s}$  to the shift-direction momentum, giving the movable secondary's rotational inertia reflected through the helix into shift.

At the upper stop, the constraint row in [Equation \(3.6.52\)](#) imposes  $\dot{s}^+ = 0$ . The outgoing terms multiplied by  $\dot{s}^+$  therefore vanish, while the incoming shift-induced rotation remains in the known right-hand side. With no secondary belt impulse, the  $\omega_s$  row of the same system becomes

$$(I_{B,s} + I_{s,F} + I_{s,M}) \omega_s^+ = (I_{B,s} + I_{s,F} + I_{s,M}) \omega_s^- + I_{s,M} \frac{d\theta_s}{ds} \dot{s}^-. \quad (3.6.54)$$

The final term on the right is the angular momentum carried into the event by the movable secondary's helix-relative rotation. Using

$$\omega_{s,M}^- = \omega_s^- + \frac{d\theta_s}{ds} \dot{s}^-, \quad (3.6.55)$$

the same row gives

$$\omega_s^+ = \frac{(I_{B,s} + I_{s,F}) \omega_s^- + I_{s,M} \omega_{s,M}^-}{I_{B,s} + I_{s,F} + I_{s,M}}. \quad (3.6.56)$$

The common secondary speed introduced at the beginning of the boundary example has therefore emerged directly from [Equation \(3.6.52\)](#). The incoming helix-relative motion is carried into the solve through the off-diagonal coupling, while the stop condition removes that relative motion from the outgoing state. The pin-track impulse that transfers angular momentum between the shaft-side and movable-side assemblies is internal to the engaged connection; its equal-and-opposite actions cancel from the combined  $\omega_s$  balance rather than appearing as another component of  $\mathbf{\Gamma}$ .

### Remark (Energy consistency check)

The completed event solution also provides a useful consistency check. Because the physical velocities on both sides are already known from  $J_- \boldsymbol{\nu}^-$  and  $J_+ \boldsymbol{\nu}^+$ , the kinetic energy before and after the transition follows directly:

$$T^- = \frac{1}{2} (J_- \boldsymbol{\nu}^-)^\top W (J_- \boldsymbol{\nu}^-), \quad T^+ = \frac{1}{2} (J_+ \boldsymbol{\nu}^+)^\top W (J_+ \boldsymbol{\nu}^+). \quad (3.6.57)$$

For the perfectly inelastic event model, the solved transition should not increase the total kinetic energy. This follows directly from the event balance. Premultiplying the first relation in Equation (3.6.51) by  $(\boldsymbol{\nu}^+)^\top$  gives

$$(J_+ \boldsymbol{\nu}^+)^\top \Delta \mathbf{p} = (C_+ \boldsymbol{\nu}^+)^\top \boldsymbol{\Gamma} = 0,$$

because the outgoing solution satisfies  $C_+ \boldsymbol{\nu}^+ = 0$ . Substituting the momentum change from Equation (3.6.35) then gives

$$T^- - T^+ = \frac{1}{2} (J_+ \boldsymbol{\nu}^+ - J_- \boldsymbol{\nu}^-)^\top W (J_+ \boldsymbol{\nu}^+ - J_- \boldsymbol{\nu}^-) \geq 0. \quad (3.6.58)$$

The right-hand side is a sum of one half of each retained mass or inertia times the square of its physical velocity change. The result therefore provides a direct check that the event solution is dissipative even when the kinematic connection changes. Individual bodies can still gain kinetic energy through the momentum exchange, as the movable secondary does at the upper stop; only the total must not increase.

## 3.7 Evaluation of the Complete System

The derivation is now complete. Beginning with a single pulley's geometry, we have brought the shafts, sheaves, and belt into one coupled description, including the changes in contact and support encountered as they move. We can now step back from the individual balances and see how they work together to follow the CVT through time.

**Establishing the initial condition.** We first choose  $\mathbf{x}_0$  to describe the starting configuration and motion for the case being studied. That choice must respect the travel limits in Equation (3.6.1) and the velocity constraints of the contacts and supports present. In the deadzone, for example, the belt must move with the secondary:  $v_b = r_{s,\text{eff,dz}} \omega_s$ , as required by Equation (3.6.5). The shift position restricts the arrangements in Table 3.1, while nonzero relative speed identifies an existing contact as sliding. Zero relative speed alone does not establish sticking; the required traction must still be available. Likewise, a sheave at rest against a stop can remain there only if the stop can supply the necessary reaction. The contact and support checks in Sections 3.5.4 and 3.6.3 settle these remaining choices under the initial shaft loading, establishing the initial regime  $q_0$  that accompanies the prescribed state.

**Evaluating the current regime.** With  $q$  established, we hold it fixed while calculating how the CVT accelerates. The geometry and actuator maps are evaluated at the current state, while the shaft interfaces supply the applied torques and attached inertias. For engaged motion, the  $6 \times 6$  system in Equation (3.5.15) gathers the mechanics of Sections 3.1 to 3.4 into one coupled response. In the deadzone, Equation (3.6.9) is used instead. At a supported stop,  $\ddot{s} = 0$  is prescribed, so

the supported member's axial balance determines the force needed to hold it there rather than its acceleration, as summarized in [Table 3.2](#).

Within the engaged system, two sliding contacts supply both traction utilizations through [Equation \(3.5.4\)](#), leaving one linear solve. One sticking contact requires the scalar root in [Equation \(3.5.11\)](#), and two require the coupled root in [Equation \(3.5.12\)](#). Each trial is evaluated through the same active linear system, and the root is reached when the returned accelerations preserve the speed match at every sticking contact. The solved accelerations and the stored shift velocity  $\dot{s}$  then complete the state derivative  $\dot{\mathbf{x}}$  in [Equation \(2.4.2\)](#).

**Checking whether the regime can continue.** As the integrator advances the state using this derivative, the contact and support conditions determine whether the same regime can be retained. The tangential checks in [Section 3.5.4](#) watch for a sliding relative speed reaching zero or a sticking traction reaching its static limit. The shift checks in [Section 3.6.3](#) locate arrivals at travel boundaries and monitor the stop reactions. A stop releases when its required reaction crosses zero, since holding the mechanism there would then require the stop to pull. Primary separation requires both zero primary normal load and an opening tendency in the contact-free primary balance, as in [Equation \(3.6.25\)](#). The engaged belt must also satisfy [Equation \(3.4.37\)](#): negative tension or local normal loading would indicate slack or lift-off, neither of which this belt representation describes.

If a regime event occurs, the integrator advances only to that instant. Before it can continue, the transition must establish both the motion immediately afterward and the regime that can sustain it. If the proposed regime admits the incoming physical motion, the state remains continuous; otherwise, [Equation \(3.6.52\)](#) supplies the outgoing velocities and impulses using the connections on both sides. Any stop impulse must act in the direction in which the stop can push. The continuous mechanics are then evaluated at that outgoing state to test whether the proposed contacts and supports can be maintained. At zero relative speed, sticking is tested first; if it is unavailable, the sliding branch must produce the direction of relative motion it assumes ([Equation \(3.5.20\)](#)). Simultaneous changes are checked as one combined regime, since they affect the same mechanical response.

**Advancing and repeating.** With no intervening event, the integrator advances the state using evaluations of  $\dot{\mathbf{x}}$  and retains  $q$ . After a transition, it resumes at the event time from the accepted outgoing state and regime. In either case, the next evaluation begins with an updated state and an established regime, and the cycle repeats. Together, the continuous balances and transition laws complete the process introduced in [Sections 2.5 to 2.6](#), determining the CVT's motion from the prescribed initial condition and shaft loading as its contacts and supports change.

## Chapter 4

# Results and Discussion

The formulation in [Chapter 3](#) determines the CVT response from its internal mechanics and the loading at its shafts. The numerical investigation begins with verification of this coupled model and comparison with an independent transient formulation. It then examines when the retained actuator and belt dynamics change the predicted behaviour. Full transient simulations finally bring these mechanisms together, following how the CVT engages, shifts, and responds to changing loads.

### 4.1 Computational Implementation and Reproducibility

CINDER is the simulator developed in this work to implement the formulation and carry out these numerical studies. Its numerical methods are documented in [Appendix B](#).

The model was developed around a Baja SAE CVT, whose hardware forms the basis of the reference configuration in [Appendix C](#). Much of the investigation therefore concerns the operating range of a small off-road vehicle. The quantitative importance of the effects examined here requires separate assessment for substantially different machines.

Complete study definitions, including the physical configurations and numerical settings, are collected in [Appendix D](#). The external systems that drive and load the CVT, including simplified engine and vehicle models, are defined separately in [Appendix C.2](#).

The results reported here use a frozen CINDER release and a separate frozen study record containing the case definitions, analysis scripts, and outputs. Both records are identified in [Table 4.1](#). The correspondence between the archived outputs and the manuscript figures and tables is given in ??.

Table 4.1: Frozen software and study records used for the results in this chapter.

| Record                | Release / source revision                                                           | Source / archive                                                             |
|-----------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| CINDER implementation | <code>cinder-v1.1.2</code><br><code>7637a38b4fb9ec21dfb953c1c80a27ec5f389654</code> | <a href="#">Tagged source</a>                                                |
| Study record          | <code>develop (working)</code><br><code>TBD-at-results-freeze</code>                | <a href="#">Working Results source</a><br><code>TBD-at-results-freeze</code> |

## 4.2 Internal Verification

The same belt–pulley contact forces enter the belt transport, shaft rotation, and axial sheave balances. Their numerical evaluation must preserve these coupled effects as contacts stick or slide and travel constraints engage or release. Verification therefore covers both the individual mechanical conditions and their combined response: the operating-domain audit and work–energy balance check local and global consistency, while refinement of the time solution and examination of the instantaneous contact closure assess the numerical reliability of that response.

### 4.2.1 Mechanical Consistency Across Operating Conditions

The formulation is intended to follow motion both with and without primary belt contact, with either pulley sticking or sliding, and with the sheaves shifting freely or held at a travel boundary. The cases in [Table 4.2](#) examine this range of contact and shift conditions. Reversing rotation and sliding direction also tests the force signs.

Most cases apply constant torques to shafts with attached inertias, leaving the dynamics to determine the shaft speeds, belt speed and shift motion. The vehicle case instead uses the engine and vehicle boundaries of [Appendix C.2](#). It starts with the vehicle and belt at rest and the engine running, then follows full-throttle acceleration on level ground for 10 s.

| Operating condition    | Selected cases                                                                                       | Examination |
|------------------------|------------------------------------------------------------------------------------------------------|-------------|
| Rest                   | All velocities and applied shaft torques zero; primary at its lower stop                             | 50 ms       |
| Primary disengaged     | One interior deadzone state, with the belt moving with the secondary                                 | One state   |
| Both contacts sticking | Forward rotation; reverse rotation                                                                   | 30 ms each  |
| One contact sliding    | Primary sliding in + or – with secondary sticking; secondary sliding in + or – with primary sticking | 30 ms each  |
| Both contacts sliding  | Primary/secondary sliding signs (+, +), (+, –), (–, +) and (–, –)                                    | 30 ms each  |
| Free shifting          | Initially closing; initially opening                                                                 | 20 ms each  |
| Boundary arrivals      | Approaching the lower stop; primary engagement; approaching the low-ratio seat                       | 30 ms each  |
| Vehicle acceleration   | Full-throttle, level-ground start; reaches the upper travel stop                                     | 10 s        |

Table 4.2: Cases examined with the reference CVT. Each alternative is a separate case, followed for the stated time. Sliding signs refer to belt velocity relative to the pulley surface along the positive belt direction. Exact inputs and numerical settings are given in [Appendix D.1](#).

For each case, the checks ask whether the calculated loads account for the motion and can be sustained by the belt, contacts and supports. At engaged states, the calculated forces and accelerations are substituted back into the coupled equations. Contact consistency is tested separately. For a sticking contact, matching surface speeds must be preserved by the accelerations, and the required traction must lie within the static-friction limit. For a sliding contact, friction must oppose relative motion and remove energy from it. A supported sheave must remain stationary with a reaction in the direction its stop can exert. Finally, the belt must carry tension rather than compression, and

the normal load must remain nonnegative everywhere along each engaged wrap. A positive total clamping force alone does not establish that last condition.

These checks are applied along each continuation, including both ends of every continuous segment. When contact or support changes, the new forces can differ immediately, so the exact outgoing state is also checked using its new contact and support conditions. All listed continuations completed, and the applicable checks satisfied the numerical tolerances in [Appendix D.1](#). [Table 4.3](#) reports the observed residuals and limiting values.

| Mechanical requirement               | Observed result                                                                                                                                                         |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Engaged force/torque balance         | Force residuals below $4 \times 10^{-11}$ N; torque residuals below $4 \times 10^{-11}$ N m.                                                                            |
| Sticking motion and traction         | Largest speed mismatch 0.13 mm/s; largest relative-acceleration residual $1 \times 10^{-8}$ m/s <sup>2</sup> . Minimum static-traction margin $-8.88 \times 10^{-16}$ . |
| Sliding removes energy               | Largest signed net friction power $+1.31 \times 10^{-8}$ W; negative power denotes dissipation.                                                                         |
| Belt tension and wrap contact        | Minimum tension 0.17 N; minimum local normal loading 0.0482 N/rad on the primary and 0.877 N/rad on the secondary.                                                      |
| Motion held by a stop                | Zero shift acceleration; speed at most $3.71 \times 10^{-21}$ m/s. Minimum signed stop reaction $-2.27 \times 10^{-13}$ N.                                              |
| Geometric and disengaged constraints | Largest belt-length mismatch $8.85 \times 10^{-13}$ m; largest deadzone belt-secondary speed mismatch $2.13 \times 10^{-14}$ m/s.                                       |

Table 4.3: Mechanical results over the cases in [Table 4.2](#), including the states immediately after transitions. Each quantity is checked only where its contact or constraint applies. The static-traction margin is the remaining allowance in traction utilization; a negative value exceeds a friction limit. Net friction power sums the work rates on the belt and pulley.

The balance residuals are at floating-point scale. The small negative traction margin and stop reaction, and small positive sliding power, also lie within their respective tolerances of  $1 \times 10^{-9}$ ,  $1 \times 10^{-6}$  N and  $1 \times 10^{-3}$  W. To this accuracy, sticking stays within available traction, the stops do not have to pull on the sheaves, and sliding adds no energy. Positive tension and local normal loading also show that the belt remains taut and in contact throughout each wrap at the engaged states examined. [Appendix D.1](#) describes the local-loading calculation in more detail, including how the loading changes when sliding reverses.

The speed check asks whether the belt and pulley surface speeds remain equal during sticking. Matching their accelerations preserves equal speeds in the exact dynamics. Numerical integration of the belt speed, pulley speeds and shift position can nevertheless introduce a small mismatch. The 1 mm/s audit tolerance limits this numerical drift; it does not enter the friction law. The largest mismatch, 0.13 mm/s, occurs during sticking in the case that begins with  $(-, +)$  sliding. The belt is then moving at 18.3 m/s, so even the audit limit is only 0.005 45% of its speed. Repeating this case with both relative and absolute integration tolerances reduced tenfold lowers the maximum sampled mismatch to 0.0164 mm/s, with the same sequence of contact and support changes. The mismatch is small compared with the belt speed, and its reduction with tighter integration supports the interpretation as numerical drift.

At rest, no spurious motion developed. The initial-state checks rejected positions outside the travel

range and a deadzone state whose belt and secondary speeds did not match. At the engagement boundary, the model also distinguished a primary closing onto the belt from one moving away.

Across these tests, the loads that account for the CVT's accelerations also satisfy the limits imposed by the belt, contacts and supports. The agreement extends across sticking and sliding, forward and reverse rotation, and free or supported sheave motion, including the checked states immediately after contact or support changes. Engagement, slip, shifting and travel-stop arrival can therefore be followed within one consistent mechanical description. Within the tested conditions and numerical accuracy, the internal loads explain both the motion and how it changes when different contacts or supports carry the load.

#### 4.2.2 Energy Consistency

The preceding checks show that the calculated forces account for the motion. The corresponding energy balance asks where the energy supplied over that motion goes. The level-ground acceleration introduced in [Section 4.2.1](#) provides a test of this balance: it combines increasing speeds and spring deformation with sliding during engagement and the arrest of shift motion at the upper travel stop. The same 10 s case is used here, first with the nominal integration settings and then with tighter settings, as specified in [Appendix D.2](#).

During this acceleration, the difference between engine work and work done against the vehicle load is not all a loss. It also accelerates the moving parts and changes the spring deformations. Stored energy therefore includes the retained CVT motions, the attached engine and referred vehicle inertias, and the axial- and torsional-spring potentials. In particular, the energy used to accelerate the vehicle remains in the account; it is not treated as energy lost through the secondary shaft. The external shaft torques supply the signed net work

$$W_B(t) = \int_0^t (\tau_{B,p}\omega_p + \tau_{B,s}\omega_s) dt, \quad (4.2.1)$$

with the resisting vehicle load contributing negative work.

At a sliding contact, traction acts with equal magnitude and opposite directions on the belt and pulley, but their surface speeds differ. Since power is force times velocity, these two work rates do not cancel. Friction opposes the relative motion, so their sum is negative: mechanical energy is removed at a rate equal to the traction magnitude times the relative sliding speed. At a sticking contact the speeds match, and traction transfers power without this sliding loss.

Engagement and travel-stop arrival can also remove kinetic energy when a new constraint arrests motion. In the inelastic event model of [Equation \(3.6.58\)](#), positions remain continuous while the velocities change. The corresponding kinetic-energy drop must therefore be included separately from the work accumulated between events. The balance is

$$W_B(t) = \Delta E_{\text{stored}}(t) + E_{\text{slip}}(t) + E_{\text{capture}}(t) + R_E(t), \quad (4.2.2)$$

where  $R_E$  is the work left unaccounted for by the calculated storage and losses. Work and sliding loss are integrated between events; capture losses are added at the events themselves. Retaining the states immediately before and after each event counts an instantaneous energy drop once, without integrating power across a velocity jump.

[Figure 4.1\(a\)](#) separates the two shaft-work contributions from the increase in kinetic energy. By 10 s, the engine has supplied 66.78 kJ, while 14.61 kJ has been spent against the resisting vehicle

load. Most of the remaining 52.17 kJ appears as kinetic energy. The smaller contributions are given in [Table 4.4](#): the springs store a further 48.78 J, sliding dissipates 1.588 kJ, and inelastic captures remove 0.254 J. Thus the difference between supplied work and kinetic-energy increase has distinct physical causes, each evaluated from the calculated motion and loads.

| Quantity                | Nominal [J] | Tighter integration [J] |
|-------------------------|-------------|-------------------------|
| Primary boundary work   | +66 778.132 | +66 778.104             |
| Secondary boundary work | −14 609.239 | −14 609.246             |
| Net shaft work          | 52 168.893  | 52 168.858              |
| Kinetic-energy increase | 50 531.492  | 50 531.505              |
| Spring-energy increase  | 48.782      | 48.782                  |
| Sliding loss            | 1 588.340   | 1 588.339               |
| Inelastic capture loss  | 0.253859    | 0.253884                |
| Signed remainder $R_E$  | +0.025445   | −0.022841               |

Table 4.4: Energy balance at 10 s, using the finest power-integration spacing. The boundary works sum to the net work; the rows below the dividing line account for that sum. Remainders use unrounded values. Only the numerical settings in [Table D.8](#) change between runs.

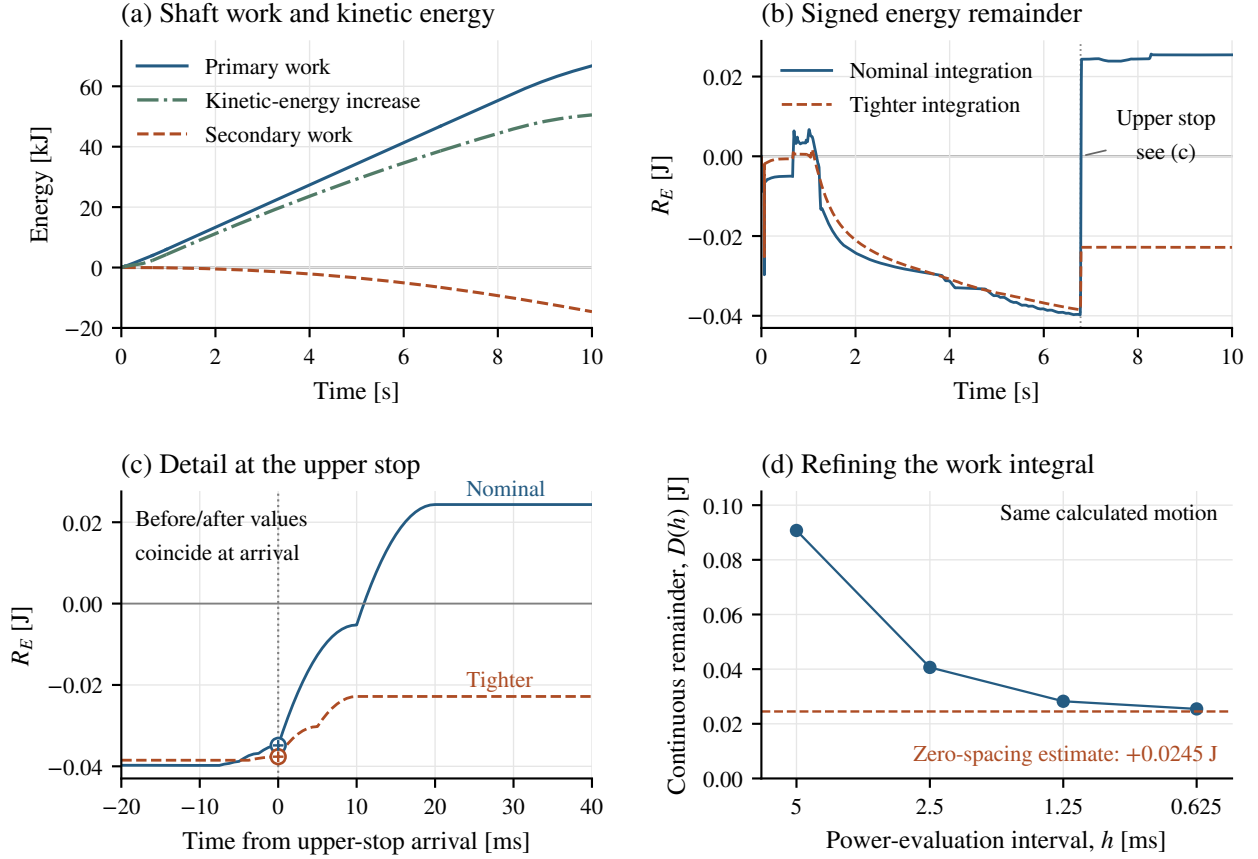


Figure 4.1: Energy accounting during the reference acceleration. (a) Nominal shaft work and kinetic-energy increase. Negative secondary work removes energy through the vehicle load; smaller terms are in Table 4.4. (b) Signed remainders for the nominal and tighter integrations. (c) Detail using each calculation’s upper-stop arrival as time zero. Circles and crosses mark values immediately before and after arrest; each pair coincides. Lines do not bridge events. (d) Fixed nominal motion with successively halved maximum intervals between power evaluations. The dashed line estimates the zero-spacing remainder from the two finest intervals. Panels (b,c) use the finest interval, 0.625 ms.

Agreement at the kilojoule scale cannot by itself establish the accuracy of this balance. Panel (b) shows the signed remainder: it stays below 0.040 J in magnitude at the sampled states in both integrations. The nominal final value is +0.0254 J, or  $4.88 \times 10^{-7}$  of the net work.

The apparently abrupt change near 6.78 s needs closer examination because it occurs near the upper-stop arrival. Arresting the shift motion reduces stored kinetic energy. The capture loss must increase by the same amount, leaving  $R_E$  unchanged at the instant of arrest; omitting that loss would instead produce a jump in the remainder. Panel (c) enlarges this part of panel (b), with the nominal and tighter calculations identified separately. In each, the recorded values immediately before and after arrest coincide. The fast change visible in panel (b) takes place over the adjoining continuous motion, not across the instantaneous capture. The stop’s energy removal is therefore included without introducing a discontinuity in the balance.

Two refinements examine the small remainder. First, the calculated motion is held fixed, but power is evaluated more frequently to improve the work and sliding-loss integrals. Panel (d) shows

the result as the maximum interval between these evaluations is halved successively from 5 ms to 0.625 ms. The summed continuous remainder falls from 0.0907 J to 0.0254 J, with progressively smaller changes. The two finest intervals give a zero-spacing estimate of +0.0245 J (Equation (D.2.1)). Finer integration of power therefore reduces the discrepancy but does not appear to remove it. This estimate concerns the work integrated along the calculated motion; it is not an error bound on that motion.

Second, the motion itself is recomputed with tighter time-integration settings. This changes the final remainder to  $-0.0228$  J, while leaving the final motion nearly unchanged: the shaft speeds change by less than  $6.1 \times 10^{-5}$  rad/s and the belt speed by less than  $3.9 \times 10^{-6}$  m/s. The small remainder is therefore sensitive to how the motion is calculated, as well as to how power is integrated along it. Its final magnitude decreases with each tested refinement, although these comparisons do not identify a unique source of the remaining discrepancy. Signed work associated with numerical drift during sticking is reported separately in Appendix D.2; it is not subtracted to improve this balance.

The energy account therefore holds to a small absolute remainder through the acceleration, shifting and checked changes of constraint. Changes in the retained kinetic and spring energy, together with the calculated sliding and capture losses, account for the net shaft work over the complete transient, including the instantaneous changes of velocity. The final energy balance closes with remainder magnitudes of  $4.88 \times 10^{-7}$  and  $4.38 \times 10^{-7}$  of the net shaft work in the nominal and tighter integrations, respectively.

### 4.2.3 Solver Convergence

The reliability of a transient calculation also depends on how accurately it follows the motion in time. Satisfying the mechanical and energy balances does not, by itself, establish that accuracy. The numerical check is therefore whether the motion and its changes of contact and support approach the same response as the integration is refined.

The full-throttle, level-ground acceleration provides a concrete example of how the integration settings affect the calculated motion. It includes primary engagement, low-ratio seating, free shifting and upper-stop arrival within 10 s.

Figure 4.2 compares loose, intermediate and tight integrations of this same case. The physical inputs and contact rules remain fixed; only the integration controls change. Relative and absolute tolerances control the solver’s estimates of local error, while the maximum step caps its adaptive integration step. The tight calculation supplies the numerical reference.

The complete shift curves nearly coincide in panel (a). Panel (b) resolves the difference: the loose integration gives shallower returns below the engagement position and earlier seating, while the intermediate and tight calculations agree closely on this motion. Below the engagement position, the primary supplies no belt force or torque, so these differences change when it carries belt load. Resolving the short contact changes therefore requires more than agreement on the overall upshift.

The complete response is therefore compared through all five state variables—the two shaft speeds, belt speed, shift position and shift speed—together with the order and timing of contact and support changes. When the same changes occur, motion is compared at equal fractional progress between corresponding events, keeping incoming and outgoing states separate. Otherwise, a small timing offset could compare a velocity before capture with one already changed by capture. Event times are checked separately on the actual clock, so this alignment does not hide a timing difference.

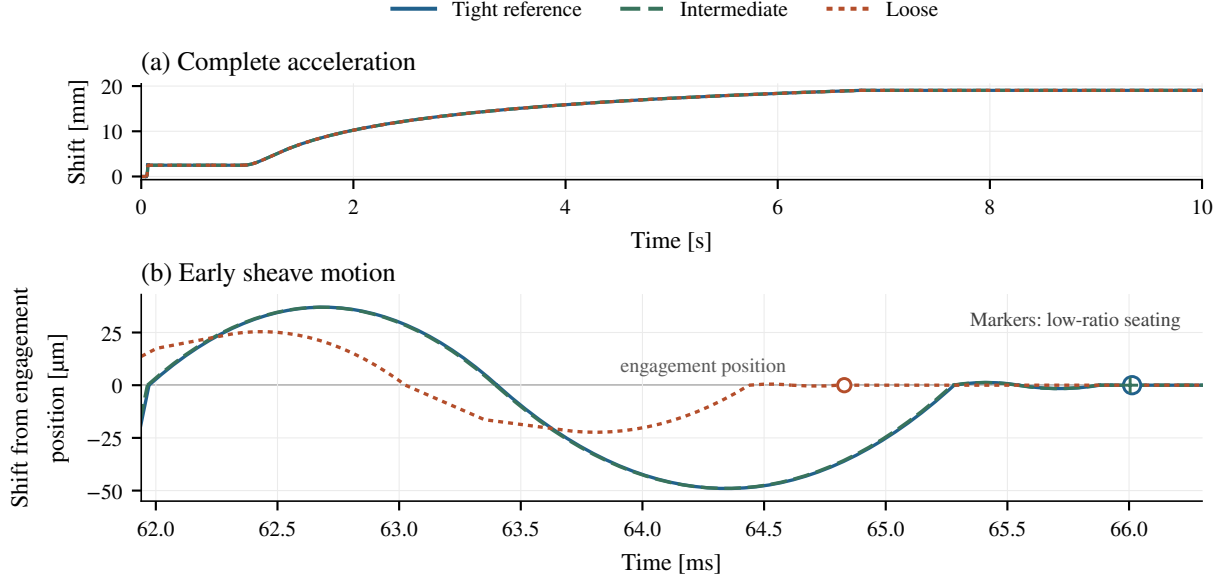


Figure 4.2: Three integrations of the same acceleration. The control triples ( $\text{rtol}, \text{atol}, \Delta t_{\max}$ ) are approximately  $(1.27 \times 10^{-2}, 1.27 \times 10^{-5}, 25.0 \text{ ms})$  for the loose run,  $(10^{-4}, 10^{-7}, 10 \text{ ms})$  for the intermediate run and  $(10^{-7}, 10^{-10}, 2.5 \text{ ms})$  for the tight reference. In (b), negative displacement indicates primary separation. Markers identify low-ratio seating; the reference circle and intermediate cross nearly coincide. Curves retain separate continuous segments. Exact settings and the fixed numerical seating rule are documented in [Appendix D.3](#).

To compare quantities with different units, each state difference is divided by a fixed scale based on the reference motion. Their duration-weighted root-mean-square (RMS) difference measures agreement over the complete transient. Brief discrepancies can contribute little to this average, so the largest sampled difference is also checked. These are differences from a numerical reference, not an exact solution.

The wider refinement sweep combines eight relative tolerances with five maximum-step caps, reducing absolute tolerance in proportion to relative tolerance. The reference is tighter than every combination in this grid. Shift speed supplies the largest scaled peak in every completed run; panel (b) of [Figure 4.3](#) shows it in physical units. Panel (c) shows the largest event-time difference. Both bands span the completed step caps at each tolerance.

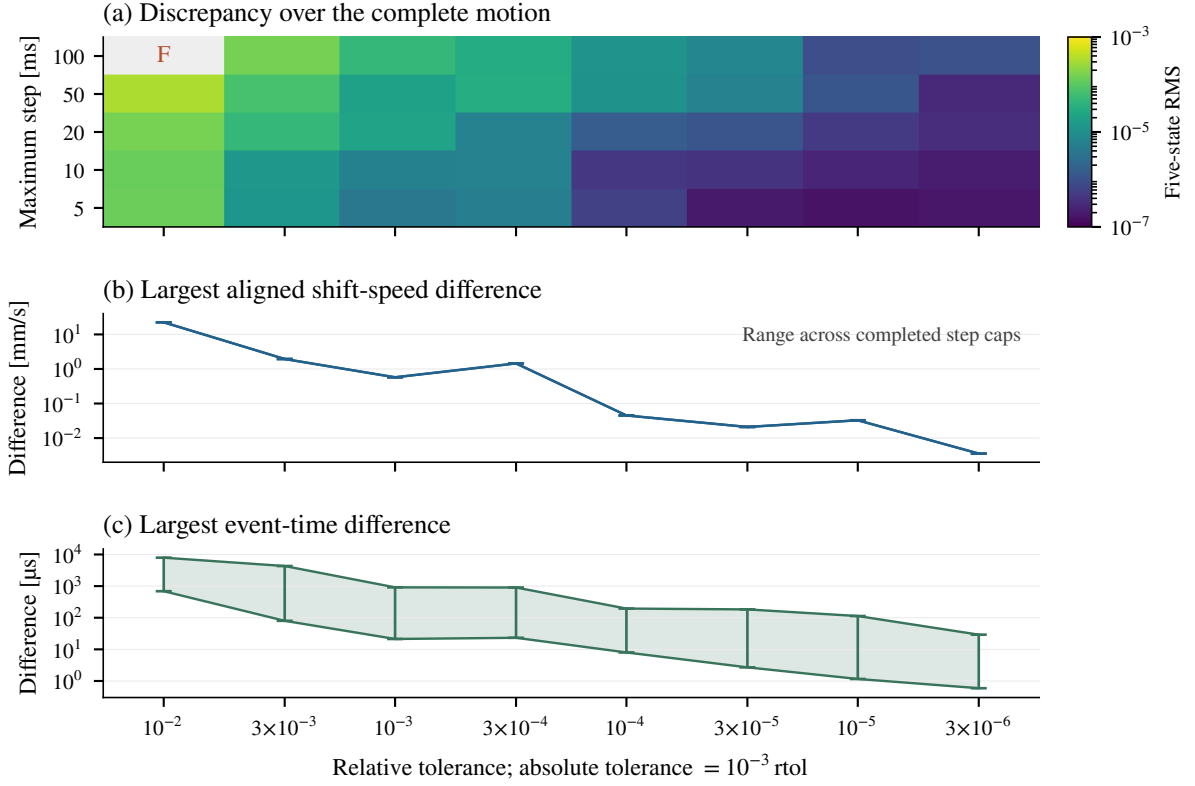


Figure 4.3: State and timing differences from the tight numerical reference across the main refinement sweep. In (a), each cell is one tested tolerance/step-cap combination; F denotes the sole failed integration. Panels (b) and (c) span the minimum to maximum across the completed step caps at each tolerance: four at  $10^{-2}$  and five elsewhere. The edges nearly coincide in (b). These bands describe variation between numerical settings, not statistical uncertainty. Lines connect tested tolerance levels; tighter tolerances lie to the right.

The heatmap shows a substantial reduction in discrepancy as the controls are tightened. Across the completed step caps, the largest RMS falls from  $3.23 \times 10^{-4}$  at  $\text{rtol} = 10^{-2}$  to  $1.03 \times 10^{-6}$  at  $3 \times 10^{-6}$ . The local and timing measures show that this improvement extends beyond the average motion. The peak shift-speed difference falls from approximately 22.4 mm/s to 0.003 56 mm/s, while the largest event-time difference across caps falls from 7.93 ms to 29.2  $\mu$ s. Every completed main-grid calculation reproduces the reference order of contact and support changes.

Only the coarsest combination,  $\text{rtol} = 10^{-2}$  with a 100 ms cap, fails to complete and is marked F. At these loose settings, a force evaluation would require the ramp to pull on the primary flyweight, violating its one-sided contact condition. The finer integrations complete without this failure, supporting its interpretation as a numerical failure rather than a prediction of physical flyweight lift-off.

The bands also distinguish the roles of the two controls. At a given tolerance, changing the step cap has little effect on the peak shift-speed difference, but produces a wider spread in event-time differences. Both controls therefore matter when assessing the complete transient, even when one local measure has become insensitive to the step cap.

At the tightest main-grid tolerance, all five step caps give aligned shift-position differences below  $0.068\text{ }\mu\text{m}$ , compared with the approximately 20 mm travel scale, and event-time differences below  $30\text{ }\mu\text{s}$ . Together with the reduction in the local speed difference, this supports numerical convergence of the calculated acceleration through engagement, shifting and stop arrival. Agreement extends to the motion between changes of constraint and to when those changes occur, under the fixed contact and seating rules.

#### 4.2.4 Closure Conditioning and Root Structure

The time integration relies on a force and acceleration calculation at each instant. Its reliability therefore also depends on whether the mechanical balances remain solvable as the operating conditions change, and whether the tractions required for sticking can be found consistently.

For a prescribed  $(\lambda_p, \lambda_s)$ , [Section 3.5.3](#) reduces these balances to a  $6 \times 6$  linear system. Sliding supplies its traction directly from kinetic friction; sticking uses this same system with trial tractions. The implementation retains the two pulley torques among the unknowns, as detailed in [Appendix B.2](#). This gives an  $8 \times 8$  system, but the mechanical solve remains the same: eliminating those torques recovers the six-variable formulation.

First consider this  $8 \times 8$  system with the traction pair prescribed. Four states are examined: low-ratio seating, mid-shift and upper-stop motion from the level-ground acceleration, and a constructed opening state at 70% engaged travel. At each state, the position, velocities and applied shaft loads are held fixed while the traction pair is varied over the static-friction range  $[-0.65, 0.65]^2$  and the wider  $[-2.6, 2.6]^2$  diagnostic range.

The mechanical system has full rank at every sampled pair in both ranges, so each trial gives a unique set of contact loads and accelerations. [Figure 4.4\(a\)](#) shows the condition number of this  $8 \times 8$  system at the upper stop, after equilibration. Its value at the operating pair is  $8.56 \times 10^3$ , much lower than the static-map maximum of  $1.05 \times 10^6$ , indicating much less numerical sensitivity there. The maximum lies among trials with negative belt loading. Those linear systems still have a unique solution, but the belt cannot sustain the calculated forces. Full rank and traction within the static-friction range therefore do not replace the physical contact checks.

For sticking, a further question remains: which traction pair makes that mechanical response preserve matching tangential accelerations of the belt and pulley surfaces? Panel (b) shows these acceleration mismatches at the mid-shift state. Each point still uses one fixed-traction mechanical solve; the solid and dashed zero contours mark the pairs that preserve sticking at the primary and secondary, respectively. Their intersection satisfies both conditions and determines the sticking traction pair.

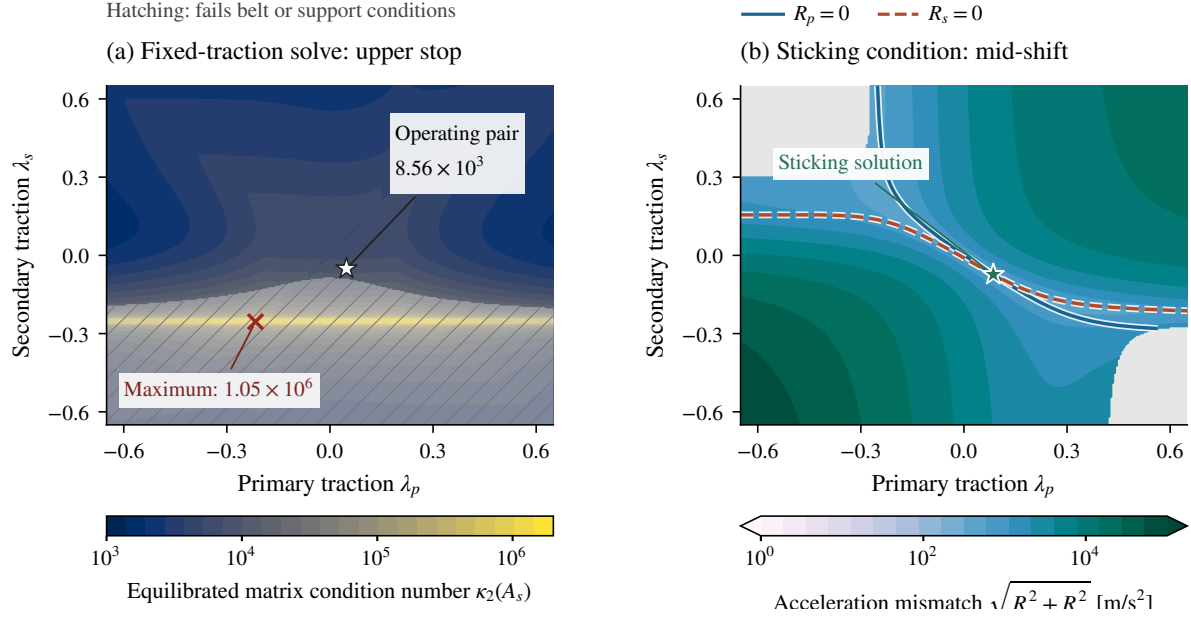


Figure 4.4: The common mechanical solve and the additional sticking condition. Both panels use  $241 \times 241$  static-capacity grids. (a) Equilibrated mechanical condition number at the upper-stop state; hatching marks trials that fail belt-loading or support conditions. The cross is the sampled maximum and the star is the operating pair. (b) Dimensional acceleration mismatch at mid-shift, with  $R_j = a_{\text{rel},j}$ . The solid and dashed lines are the primary and secondary zero contours. Grey regions fail contact checks and are excluded from these contours.

Recovery of that intersection was checked from different starting guesses at 15 operating states. Four come from the acceleration history; eleven constructed cases cover forward and reverse rotation, low, middle and high travel, increased shaft loading, and free opening and closing at three travel positions. Their individual results are listed in [Table D.17](#).

At each state, seven starting values for each traction are combined into 49 starting pairs spanning the static-friction range. Repeating the search from these different parts of the traction plane checks whether it returns to the same solution. Of the 735 searches, 666 reach sticking, and every accepted result at a given state recovers the same operating solution.

The remaining 69 attempts stop with nonzero acceleration mismatches. Other starting guesses recover an admissible sticking solution at those same states, so these failures concern finding an available root, rather than insufficient friction to sustain sticking. A wider starting grid at the four mapped states also recovers only their existing solutions. The same sticking response is therefore recovered repeatedly at each state, although the numerical search remains sensitive to its starting guess.

The applied shaft torques also affect the acceleration mismatches. To examine how the sticking solution changes with loading, a constructed opening state is held at 95% engaged travel,  $-40$  mm/s shift speed and 3000 rpm primary speed. The primary boundary torque is fixed at  $-81.15$  N m while the secondary torque varies. Both shafts rotate positively, so at the selected positive secondary torque the secondary supplies power and the primary absorbs it. These are fixed-state bench loads, rather than another vehicle trajectory.

At +189.96 N m secondary torque, [Figure 4.5\(a\)](#) has two intersections. Both pairs satisfy sticking and pass the friction, belt-loading and actuator-contact checks. They give different responses: the secondary normal resultant is 2.401 kN at root 1 and 1.669 kN at root 2, with different accelerations as well. Root 2 is closer to loss of wrap contact: its minimum secondary local normal loading is 0.243 N/rad, compared with 20.84 N/rad at root 1.

Panel (b) shows how both pairs belong to one connected sticking branch. As secondary traction increases, the required secondary torque first rises and then falls, turning near 197.39 N m. The selected load consequently intersects both sides of the turn. Increasing traction changes the normal loads and accelerations through the coupled balances, so it need not increase the compatible boundary torque monotonically. The mechanical matrix remains full rank throughout this branch: each prescribed pair still gives one force response, while two different pairs can satisfy the sticking conditions under the same applied load.

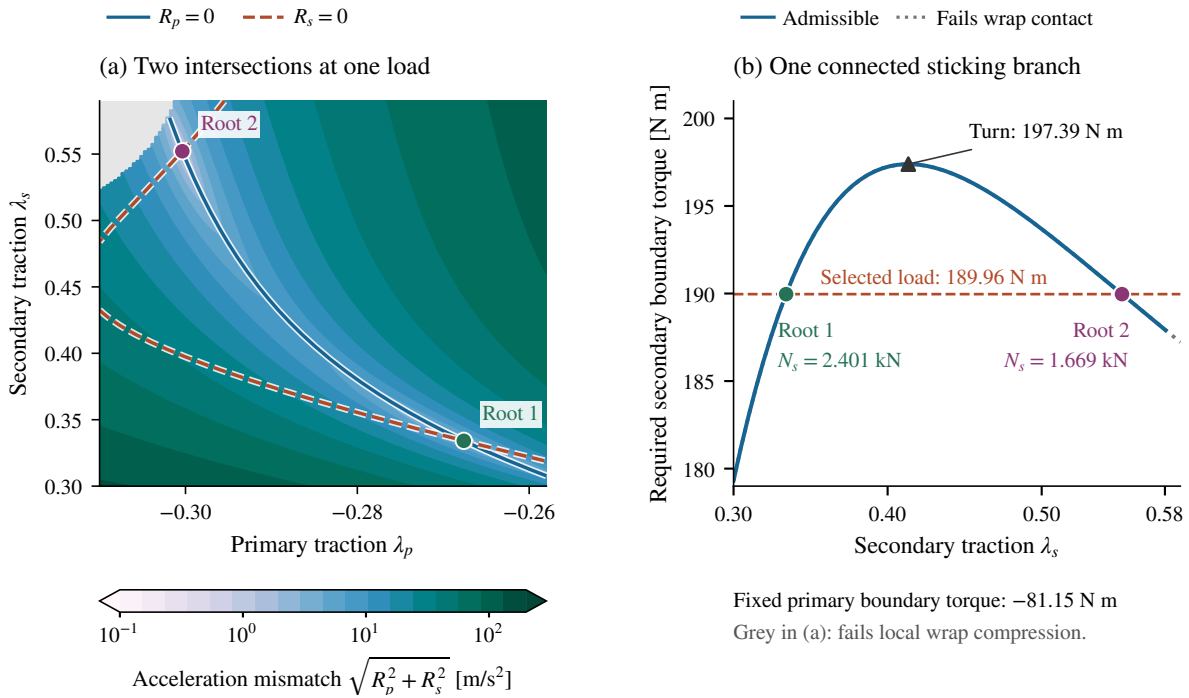


Figure 4.5: Two sticking responses at the same opening state and applied load. (a) Dimensional acceleration mismatch and the two zero contours at the selected shaft torques. The marked roots use the same identifiers in both panels; grey marks loss of local wrap compression. (b) Secondary boundary torque required to maintain sticking along the connected branch, with primary boundary torque fixed. The selected load intersects the branch twice. The dotted extension fails local wrap compression and is not an admissible continuation. Secondary traction is a branch coordinate, not time or shift position. Exact state, loads and signed responses are given in [Appendix D.4](#).

This distinction also matters when only one contact sticks. The targeted example in [Appendix D.4](#) admits two secondary sticking tractions with primary traction fixed at its kinetic value, at the onset of primary sliding. Reducing the search to one unknown therefore does not by itself ensure a unique sticking solution.

Within the sampled static and expanded ranges, the mechanical solve remains robust, while the operating-state searches repeatedly recover a single sticking root. The selected alternatives are locally unique away from the fold. Using the preceding accepted traction pair helps CINDER follow the nearby branch and avoid root jumps, while rechecking compatibility and contact limits. This guides the numerical choice without establishing a physical selection law. No root jump or effect of an alternative root was detected in the other reported calculations.

### 4.3 Independent Transient-Model Benchmark

The preceding checks establish the consistency of CINDER’s calculated motion. A comparison with an independent formulation can now examine how that motion changes when the belt is described in more detail. Ballew’s discretized rubber-belt model provides a five-second ATV acceleration case with published shaft-speed histories, primary clamping force and controller gains (Ballew, 2015). Its speeds are produced under feedback, however, so the comparison must consider both the motion and the force used to obtain it. Reconstructing the feedback allows the controller to adjust CINDER’s clamp demand; imposing Ballew’s published force instead tests the response to a common force history.

#### 4.3.1 Benchmark Reconstruction

Both protocols use the same reconstructed belt geometry, shaft inertias and ATV road-load boundary, with a constant primary torque of 18 N m and secondary closing force of 2000 N. The initial shaft speeds are Ballew’s tabulated 2500 and 1136 rpm; a compatible belt speed and shift position follow from the reconstructed geometry. These inputs belong to the published benchmark rather than the Baja reference. Appendix D.5 gives the source quantities, their translation into CINDER and the numerical settings. The accompanying [reconstruction record](#) identifies the source locations, digitization procedure and choices needed to re-create the case. No physical parameter or controller gain is fitted to the reference histories.

The translations preserve the published belt mass and combined shaft inertias. Friction coefficients are converted between the two normal-force conventions so that the same clamp gives the same gross tangential capacity. Both literal moving-sheave masses are set to zero because Ballew obtains radial position from an algebraic axial-force balance. CINDER’s belt inertia and reduced shift dynamics remain active. Ballew also initializes internal belt-deformation states, including zero longitudinal tension, that have no direct counterpart in CINDER. The reconstruction therefore matches the specified external conditions without making the internal initial states identical.

#### 4.3.2 Closed- and Open-Loop Comparisons

In the closed-loop protocol, the primary clamp regulates speed toward 2500 rpm. Ballew reports the PI and feed-forward gains but does not fully specify the controller equation. The reconstruction uses those gains in the force law of Equation (C.3.7), with zero initial integral error and no added saturation. The published force is an output reference for this comparison.

The open-loop protocol instead prescribes that force. Ballew’s digitized Figure 45 history is applied by piecewise-linear interpolation, with its first visible value held back to  $t = 0$ . All other reconstructed inputs are unchanged. The two columns of Figure 4.6 therefore compare the same shaft quantities while distinguishing a force calculated by feedback from one supplied as an input.

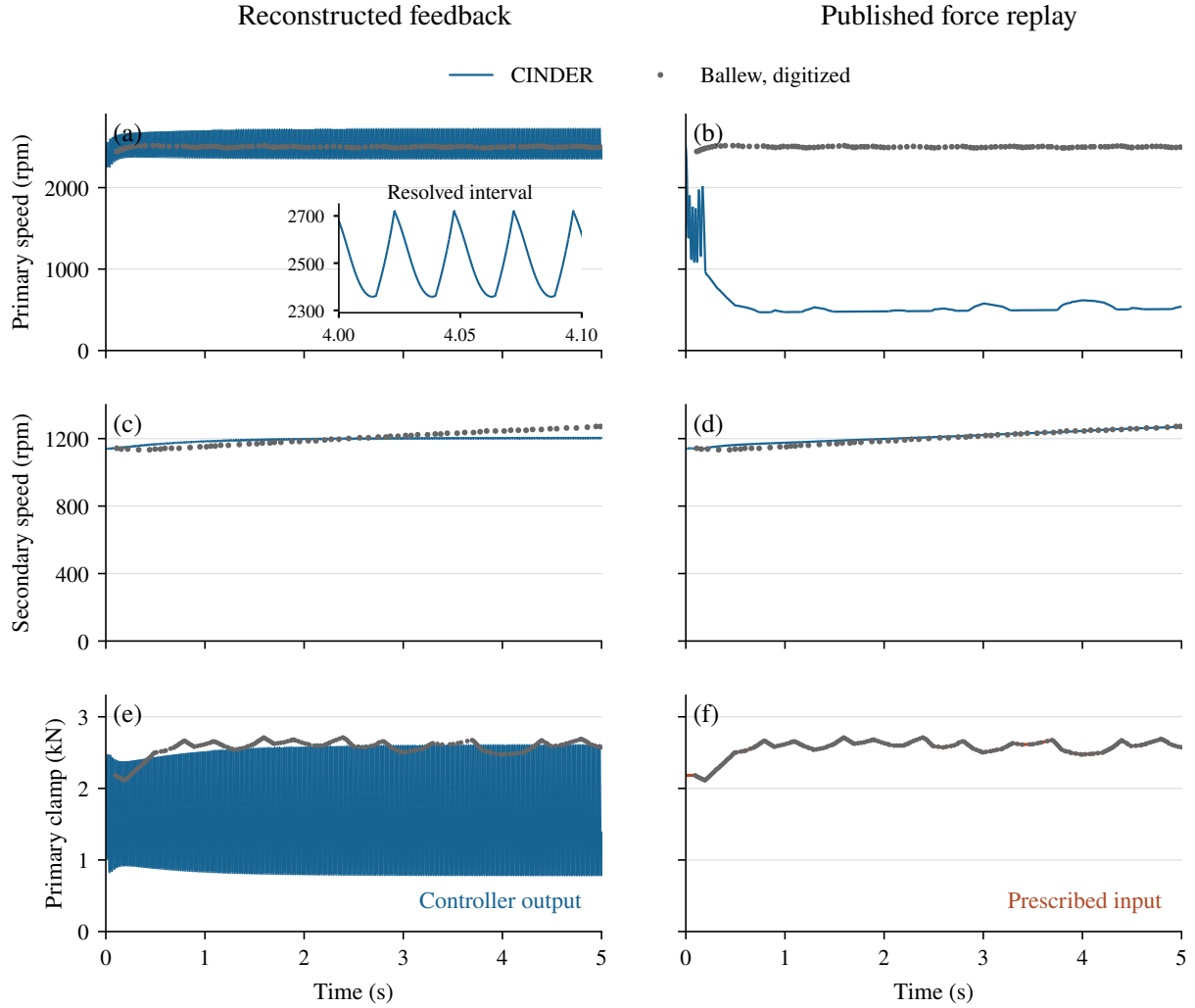


Figure 4.6: Reconstructed feedback and prescribed-force replay against Ballew’s simulated histories. Corresponding panels use the same scales. Points are digitized from Figures 41 and 45 (Ballew, 2015); the CINDER traces are unsmoothed, and the inset resolves the primary-speed oscillation over 0.1 s. The primary force is a controller output in (e) and a prescribed input in (f). Both protocols share the secondary clamp and shaft boundaries.

With feedback, the primary speed remains near the target but oscillates much more strongly than the published history. The secondary speed also differs in shape: CINDER levels off near 1200 rpm, while Ballew’s result continues toward 1270 rpm. Their speed RMSE values are 110 and 33 rpm, respectively (Table 4.5). The primary-clamp RMSE is much larger relative to its reference, at 1.20 kN, or 46.66%. Thus the modest shaft-speed errors coexist with a substantially different force demand. Because clamping governs both the shift balance and available traction, that difference matters to the transmission response.

| Quantity          | Closed loop      | Force replay      |
|-------------------|------------------|-------------------|
| Primary speed     | 110 rpm (4.42%)  | 1946 rpm (77.89%) |
| Secondary speed   | 32.6 rpm (2.71%) | 13.1 rpm (1.09%)  |
| Shaft-speed ratio | 0.100 (4.81%)    | 1.633 (78.31%)    |
| Primary clamp     | 1196 N (46.66%)  | Prescribed input  |

Table 4.5: RMSE against the digitized reference histories. Parentheses give RMSE as a percentage of the mean absolute reference value at the comparison samples. The shaft-speed ratio is  $\omega_p/\omega_s$ . Speeds and force are compared at their native digitization times; the ratio uses the union of the two speed grids over their common interval. The initial force hold used for replay is excluded from the closed-loop force comparison.

Removing the controller’s adjustment makes the primary-speed discrepancy much larger. Under force replay, CINDER’s primary speed falls to roughly 500 rpm after the initial transient, rather than remaining near 2500 rpm. Its primary-speed and shaft-ratio errors approach 78%, even though the secondary speed is closer to Ballew’s result than in the controlled case. Agreement at the output shaft alone would therefore conceal a very different relation between the two shaft motions.

### 4.3.3 Internal Motion and Slip

The controlled case also illustrates why similar shaft speeds need not imply similar belt motion. This benchmark has no relative sheave-rotation mechanism, so  $\omega_p/\omega_s$  would equal  $r_{s,\text{eff}}/r_{p,\text{eff}}$  if both contacts rolled without slip. [Figure 4.7\(a,b\)](#) compares these ratios within CINDER. The shaft-speed ratio stays near two, while the radius ratio repeatedly moves between approximately 0.46 and 2.49, reaching the low-ratio seat. The difference between them shows that the shaft speeds do not track the large excursions of the belt along the pulleys. Slip accommodates that relative motion and dissipates energy. Panel (c) shows 19.7 kJ of cumulative modeled slip loss over the five seconds, contributed by both contacts. These internal quantities describe CINDER’s response; corresponding histories are not available from the published benchmark for comparison.

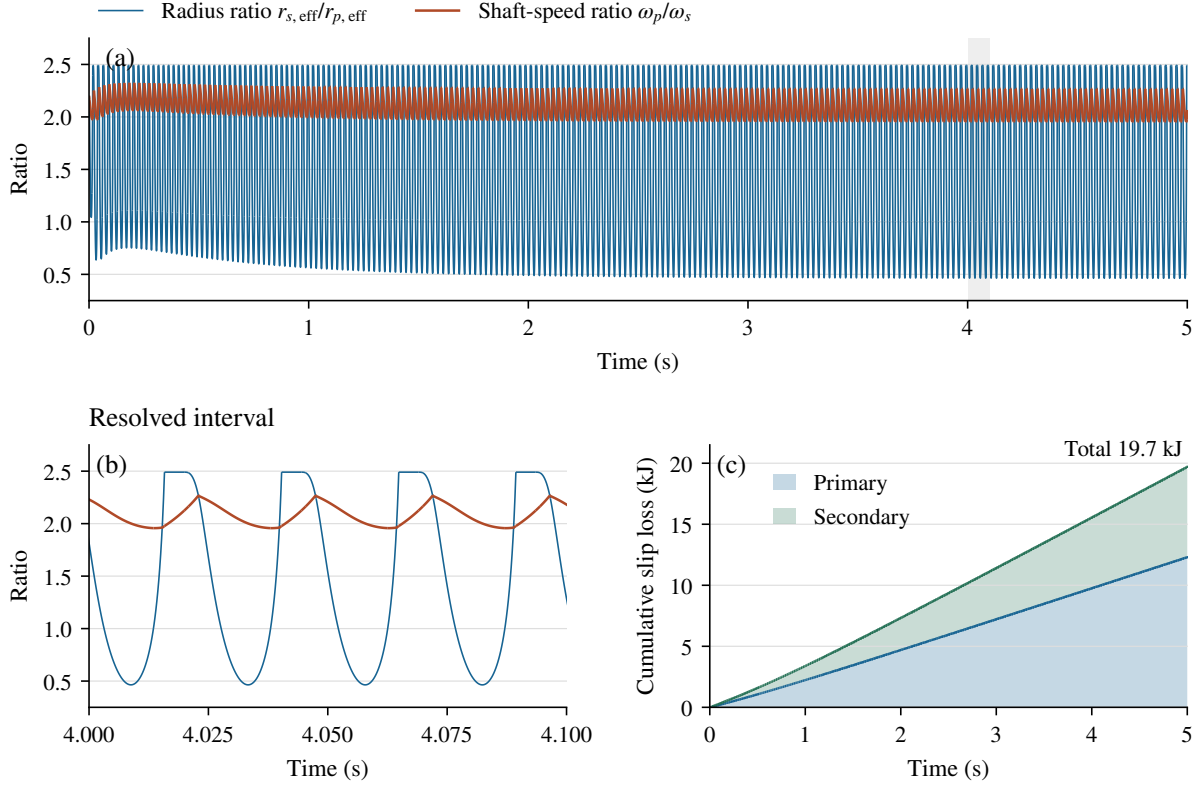


Figure 4.7: Internal response under reconstructed feedback. (a) Shaft-speed and effective-radius ratios over the complete run. The shaded interval is expanded in (b), showing the actual motion rather than an averaged ratio. (c) Stacked cumulative slip losses at the primary and secondary contacts; the upper boundary is their sum. Continuous segments retain their separate event endpoints.

Because the oscillations occur over a short timescale, their numerical convergence was checked. Reducing the maximum step from 1 ms to 0.5 ms and 0.25 ms, and tightening both integration tolerances at the 0.5 ms limit, leaves these oscillations essentially unchanged. Across the three refinements, the largest same-time RMS differences from the nominal run are only 0.032 rpm in primary speed and 0.160 N in clamp force. The sticking/sliding and support changes also retain their order, although the number of recorded slip-direction updates differs. [Appendix D.5.4](#) gives the refinement results and event comparison. The benchmark discrepancy therefore persists under these changes in numerical resolution.

#### 4.3.4 Interpretation of the Model-Form Difference

Force replay shows that the discrepancy does not arise solely from the reconstructed controller: the models also respond differently when its adjustment is removed. The principal difference in the belt descriptions is the motion each can represent. Ballew resolves internal deformation and local nodal motion, whereas CINDER uses a fixed belt length, one transport speed and a prescribed path controlled by the shift coordinate. Local deformation and force redistribution can therefore alter the response to a given clamp. The unmatched initial tension and other necessary reconstruction choices still limit how directly this difference can be assigned to any one belt mechanism.

Compliance and damping have different roles in that explanation. Compliance allows energy storage and redistribution of motion; damping removes energy. Figures 41 and 45 are Ballew’s case without the added linear inter-node damper, although the source model retains angular damping. The mismatch cannot be attributed to the added linear damper used in his separate comparison (Ballew, 2015, Figures 40–45). Isolating the remaining causes would require controlled changes to the belt descriptions.

Taken together, the two protocols show that matching a controlled speed is an incomplete test of a CVT model. The required clamp and the response when that clamp is imposed provide additional, distinct information. Here they reveal a substantial difference in the reconstructed relation between clamp force, shift and slip, despite the closer controlled shaft histories. The benchmark therefore identifies where CINDER’s reduced belt description needs further physical assessment; it is an independent model comparison, not experimental validation of either formulation.

## 4.4 Dynamic Mechanics Retained by CINDER

The preceding consistency checks, numerical refinements and comparison with Ballew provide a basis for examining the additional dynamics retained by CINDER. The primary flyweights, secondary helix and belt are considered in turn to determine how large these contributions become and how much they change the predicted motion. Their force and torque contributions are evaluated along the full-model motion; separately integrating a reduced model establishes the effect of omitting them.

### 4.4.1 Primary Actuation Dynamics

For the primary, the question is how much the flyweights’ pivot motion changes the closing force and the resulting shift. The quasi-static flyweight force, denoted  $F_{\text{fw},x_p}^{\text{QS}}$ , is the centrifugal closing force at a given shaft speed and sheave position. As the sheave moves, the flyweights also swing about their pivots, and accelerating this motion changes the force delivered to the sheave. With  $x_p = s$ , Equation (3.2.11) separates into

$$\begin{aligned} F_{\text{fw},x_p} &= F_{\text{fw},x_p}^{\text{QS}} + \Delta F_{\text{fw},x_p}, & F_{\text{fw},x_p}^{\text{QS}} &= \frac{1}{2} \omega_p^2 \frac{dJ_{f,p}}{ds}, \\ \Delta F_{\text{fw},x_p} &= -I_f \left( \frac{dq_f}{ds} \right)^2 \ddot{s} - I_f \frac{dq_f}{ds} \frac{d^2 q_f}{ds^2} \dot{s}^2. \end{aligned} \quad (4.4.1)$$

The first correction opposes shift acceleration. Its coefficient  $I_f(dq_f/ds)^2$  is the flyweight pivot inertia expressed as a mass acting on the shift coordinate; it ranges from 0.519 kg to 0.572 kg over the reference mechanism’s engaged travel. The second correction arises because the flyweight rotation per unit sheave travel changes along the ramp.

The comparison retains the same flyweights, ramp and spring. The quasi-static primary omits these two force corrections and holds its shaft-axis inertia at the initial value, while the secondary remains dynamic. It therefore also omits the changing angular-momentum contribution derived in Section 3.3.2. The exact reduction and study settings are given in Appendices C.3.1 and D.6.

A first measure of these effects comes from the reference level-ground acceleration. It starts with the vehicle and belt at rest and the primary at 1800 rpm, under the full-throttle engine and vehicle boundaries of Appendix C.2. In Figure 4.8(a), retaining the primary dynamics lowers the closing speed at first belt contact from about 673 mm/s to 565 mm/s, and delays that contact by 1.93 ms.

Immediately afterward, the flyweights decelerate and their inertial contribution adds 74.8 N to the closing action. Panel (b) shows why this is appreciable: the spring opposes most of the 1404 N centrifugal force. The flyweight and spring contributions then sum to 244.5 N, compared with 169.7 N when evaluated without the pivot terms at the same state. These are the actuator forces before the belt reaction, not the impulse at capture.

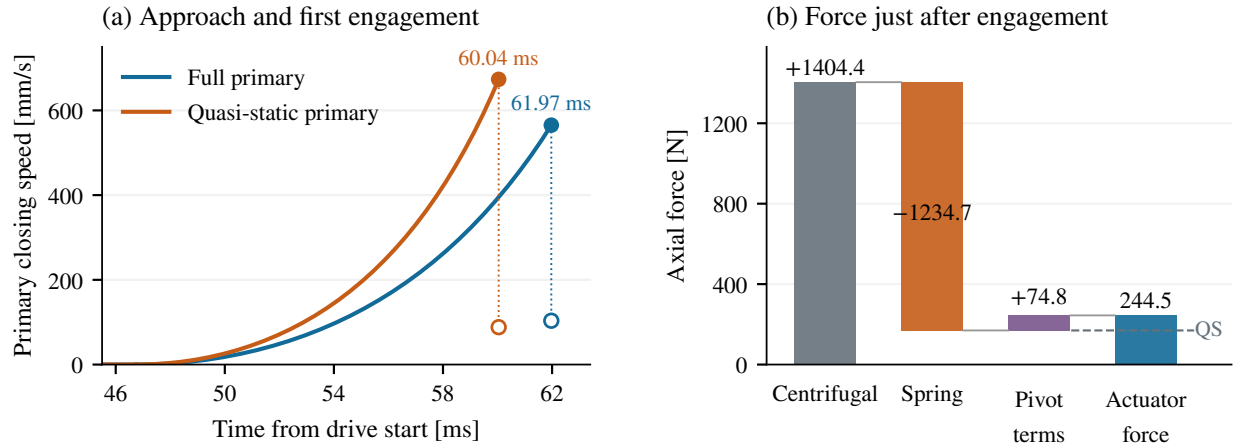


Figure 4.8: Primary dynamics during the approach to first engagement. (a) Independently integrated full and quasi-static primary models. Filled and open markers give the speeds immediately before and after capture; dotted lines mark the discontinuity. (b) Signed force contributions in the full model immediately after its first engagement. The first three bars accumulate the signed contributions; the final bar gives their sum. The dashed level is the quasi-static actuator force at that same state.

The later shifting is much gentler. Over the sampled engaged motion from 0.1 s to 10 s, the largest pivot-force correction is only 0.045 N, and the largest magnitude relative to the simultaneous centrifugal force is 0.001 47%. The distinction is the motion of the flyweights: the same mechanism that changes the engagement response operates very close to its quasi-static axial-force law during the subsequent shift.

Changing the applied torque tests whether this remains true away from that smooth drive. The primary torque is increased or decreased by 5 N m, 10 N m or 20 N m, with rise times of 5 ms, 20 ms, 100 ms and 250 ms, starting near 20%, 50% and 80% of engaged travel. All these cases remain in free shifting with both contacts sticking. Figure 4.9(a) shows that faster changes increase the force correction, but it remains small throughout this tested range. In the selected  $-20$  N m, 5 ms case at mid travel, the resolved peak is  $-1.83$  N, or 0.0510% of the centrifugal force. The acceleration term accounts for almost all of it; the motion-ratio term is below 0.001 N at that instant. Even this rapid torque change produces far less flyweight acceleration than engagement does.

Both models are already shifting before the added torque is applied. Each is therefore also continued from its own starting state with the original engine boundary and no torque increment. Subtracting that motion from the ramped case leaves the change caused by the torque ramp.

The torque reduction initially lowers the belt's opening reaction more than the actuator's closing force, so the primary closes further than it would without the added torque. Panel (b) shows

a shift response of about 0.82 mm, with a maximum full–quasi-static difference of 2.53  $\mu\text{m}$ . The corresponding primary-speed difference is 0.670 rpm. These response differences remain stable when both integration tolerances are tightened tenfold ([Appendix D.6](#)). The small force correction is therefore accompanied by a small, numerically resolved change in the motion.

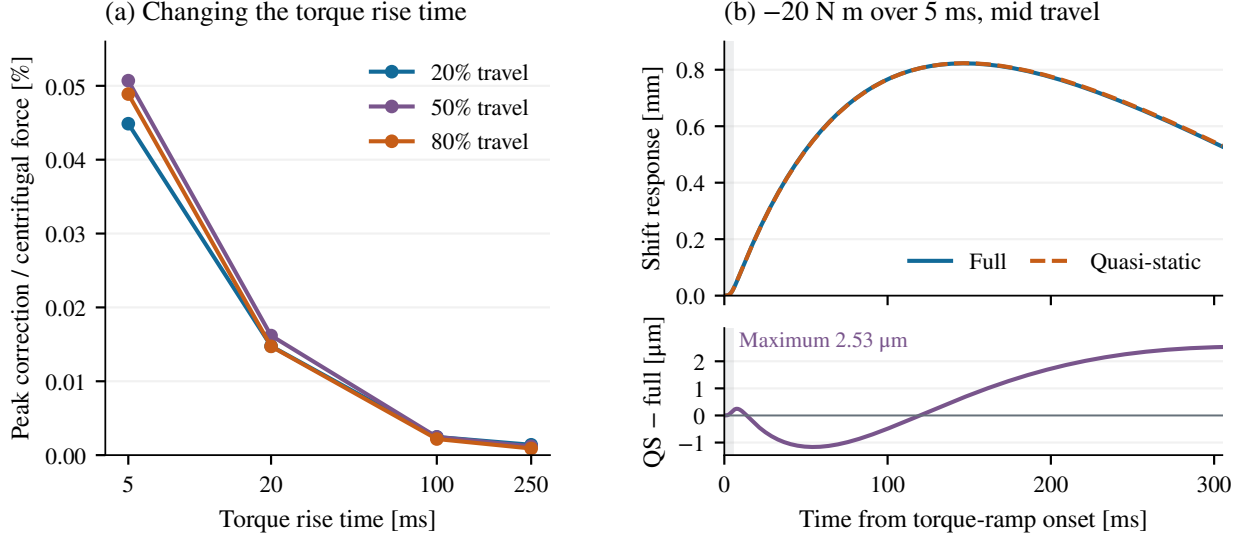


Figure 4.9: Primary response to smooth torque changes. (a) Each point is the largest sampled magnitude of  $\Delta F_{\text{fw},x_p}/F_{\text{fw},x_p}^{\text{QS}}$  over the six signed torque increments at the stated starting travel and rise time. (b) Each model’s extra shift in the selected mid-travel case, relative to its continuation without the added torque. Shading marks the 5 ms ramp; the lower trace resolves the difference between the two responses. The secondary remains dynamic in every comparison.

For this primary mechanism, the centrifugal approximation is consequently well supported for the smooth shifting and torque changes examined here. Engagement requires a different level of detail: the rapid flyweight motion changes both the approach to contact and the force subsequently delivered to the sheave. Retaining the primary dynamics allows these parts of the same drive to be represented without extending the quasi-static approximation beyond the motion for which it is accurate.

#### 4.4.2 Secondary Helix Dynamics and Transient Sensitivity

A torque-reactive secondary need not deliver the same closing force during acceleration as it does at steady speed. Its movable assembly rotates with the shaft and also turns relative to it as the sheave shifts. The torque required to accelerate that assembly changes the reaction carried through the helix, and therefore the force available to close the pulley.

With the symmetric face-torque share used here, the quasi-static reaction is

$$\tau_{s,\text{helix}}^{\text{QS}} = \frac{\tau_s}{2} + k_{s,\theta}(\theta_{s,\text{pre}} - \theta_s). \quad (4.4.2)$$

Writing the helix force from [Equation \(3.2.30\)](#) as

$$F_{\text{helix},x_s} = \frac{d\theta_s}{dx_s} \left[ \tau_{s,\text{helix}}^{\text{QS}} - I_{s,M}(\dot{\omega}_s + \ddot{\theta}_s) \right] \quad (4.4.3)$$

makes the two sources of acceleration explicit: a change in shaft speed and a change in the movable assembly's rotation relative to the shaft. The latter accompanies axial motion. The same rotational inertia consequently also contributes

$$m_{h,\text{ref}} = I_{s,M} \left( \frac{d\theta_s}{ds} \right)^2 \quad (4.4.4)$$

to the inertia resisting motion in the shared shift coordinate. This inertia therefore enters both the helix force and the resistance to shifting. It stores kinetic energy; it does not dissipate energy like a damper.

To examine the effect on the motion, the full secondary is compared with a quasi-static reduction. The reduction retains the geometry, springs and total inertia for rigid shaft rotation, but assigns the movable rotational inertia to the fixed secondary shaft and uses  $\tau_{s,\text{helix}}^{\text{QS}}$  alone in the helix force. Both models retain the full primary dynamics and the same bilateral helix slot. The reduction is defined in [Appendix C.3.1](#); the exact inputs, component estimates and numerical checks are given in [Appendix D.7](#).

### Rapid loading and approach to a traction limit

The reference level-ground acceleration provides a first measure of the secondary correction. Between 0.1 s and 10 s, its largest sampled magnitude is only 0.75% of the simultaneous quasi-static helix force. Thus this part of the reference drive stays close to the quasi-static force law, even though the shafts and sheaves are moving.

A short load change tests a different acceleration demand. Starting near 30% of active shift travel, the reference secondary with its 300° torsional preload is subjected to an additional opposing secondary torque of  $-120 \text{ N m}$ , applied over 2 ms. The full and quasi-static models start from the same state and receive the same torque history. [Figure 4.10\(a\)](#) shows how the two acceleration contributions combine during the initial response. Slowing the shaft adds to the helix reaction, while accelerating the relative sheave motion initially subtracts from it. For example, 1.8 ms after load onset the contributions are approximately  $0.444 \text{ N m}$  and  $-0.891 \text{ N m}$ . Their sum,  $-0.448 \text{ N m}$ , is small beside the  $40.43 \text{ N m}$  quasi-static reaction at that instant.

The resulting motion also changes little. The models backshift by about 5.2 mm, while their largest shift separation is only 0.051 mm, resolved in panel (b); negative values mean more backshift in the full model. Local wrap loading remains positive at the checked engaged states, including the later seating and re-engagement transitions. Rapid forcing has therefore activated the inertial terms without producing a large difference in this response. The two terms must be combined with their signs before their size is judged.

Near a traction limit, agreement in shift position is not the only relevant comparison. Changing the helix reaction changes clamping, while changing the reflected inertia changes how the sheaves accelerate. Both can affect when the required belt traction reaches the available capacity. To examine that approach, the reference secondary is given 270° torsional preload and the primary applied torque is reduced to  $-28 \text{ N m}$  over 50 ms. This pair again starts from one common state near 30% travel, with both contacts sticking. It changes the loading direction as well as the preload, rather than isolating preload alone.

Panel (c) follows the primary's traction demand late in this reversal. The fraction  $|\lambda_p|/\mu_s$  reaches unity when sticking uses all available static traction. Panel (d) separates the direct shift-inertia coefficient into its helix and other contributions at the full model's arrival.

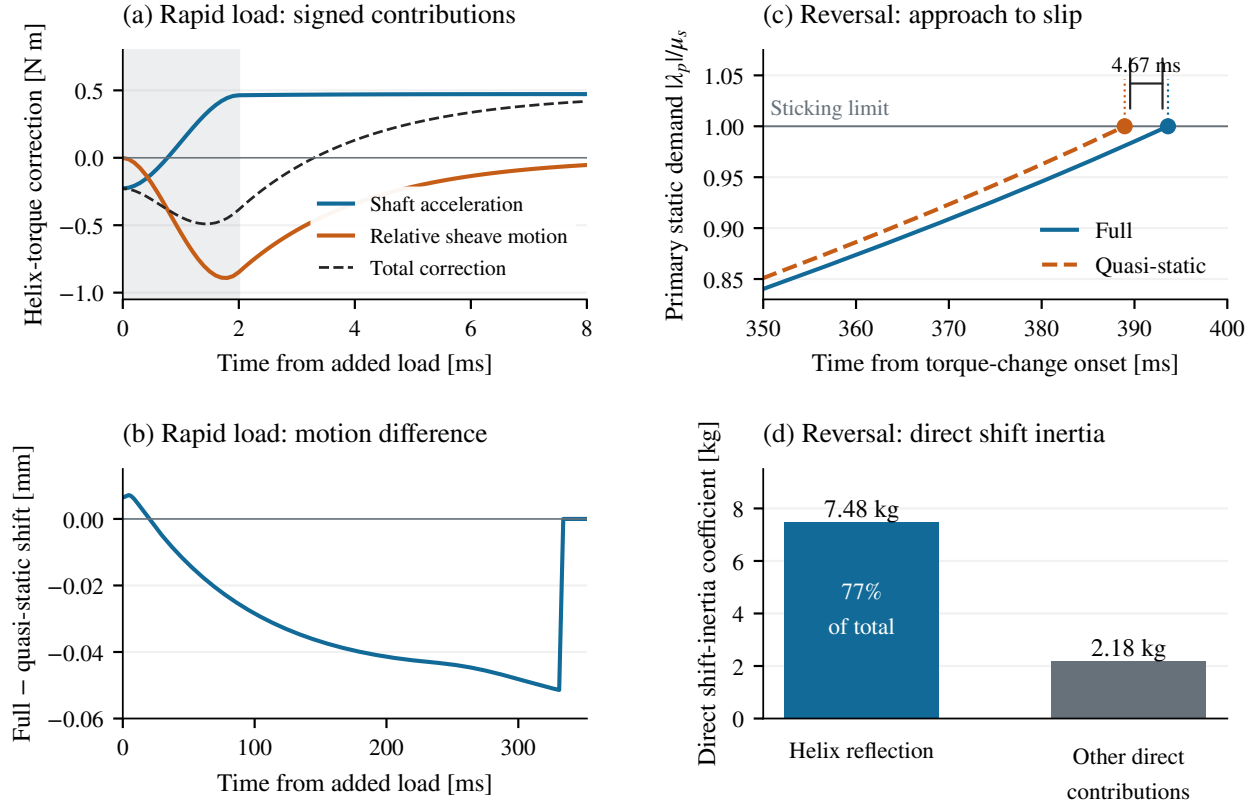


Figure 4.10: Reference-secondary transients. (a,b) Added secondary torque of  $-120$  N m over 2 ms, with  $300^\circ$  preload: signed torque corrections on the full trajectory and independently integrated shift difference. Shading marks the ramp. (c,d) Primary torque reduced to  $-28$  N m over 50 ms, with  $270^\circ$  preload: traction demand and the direct shift-inertia budget. Curves in (c) end at their incoming traction-limit states; (d) uses the full model's incoming state. The outgoing states fail local belt contact. Time is measured from each torque-change onset.

The quasi-static model reaches the traction limit 4.67 ms earlier than the full model, although their shift positions differ by at most 0.0417 mm before the earlier event. The change is therefore visible in the contact timing even while the positions remain close. At the full model's incoming event state, the helix contributes 7.48 kg of the 9.67 kg direct shift-inertia coefficient: approximately 77%. It supplies the majority of the direct inertia in the shift direction, with sheave translation and flyweight reflection supplying the remaining 2.18 kg. The denominator is the direct coefficient multiplying shift acceleration, before eliminating the coupled shaft and contact equations; it is not physical sheave mass.

Removing the relative helix dynamics therefore removes a substantial contribution to shift inertia as well as the acceleration-dependent reaction. These effects belong to the same coupled motion: shift acceleration changes the helix reaction, and the resulting clamp and traction loads in turn affect the accelerations.

The reversal also reaches a limit of the retained belt description. Immediately after each traction transition, the calculated primary loading becomes negative at one end of the wrap, despite a positive integrated normal force. The contact would have to pull on the belt there, which the

prescribed wrap cannot support. This violation persists with tighter integration. The comparison consequently establishes the approach and arrival timing, not an admissible post-slip response. [Appendix D.7.3](#) records the event-side forces and the excluded continuation. The small rapid-load shift difference and the reversal’s timing difference are both stable under the numerical refinements reported in [Table D.27](#).

### How the hardware changes the dynamic contribution

These tests concern the reference Baja secondary. In a different assembly, a larger rotational inertia requires more torque for the same angular acceleration, while the helix geometry controls how that torque becomes axial force. The small correction seen in the reference drive therefore need not carry over to other hardware.

[Table 4.6](#) compares the reference with an engineering estimate for a secondary using a  $31^\circ$  helix. The nominal movable rotational inertia is 5.12 times as large. Its smaller motion ratio partly offsets that increase: the coefficient converting shaft acceleration into axial force grows by 2.76, while the local reflected axial inertia grows by 1.49. Rotational inertia alone therefore does not determine the axial effect. The mass properties are estimated from component dimensions, not measured manufacturer data; the source dimensions and estimation method are identified in [Appendix D.7](#).

| Quantity                                         | Reference | OTS estimate | Ratio |
|--------------------------------------------------|-----------|--------------|-------|
| $I_{s,M}$ [kg m <sup>2</sup> ]                   | 0.002514  | 0.012875     | 5.12  |
| $H = d\theta_s/dx_s$ [rad/m]                     | 61.8      | 33.3         | 0.539 |
| Shaft-acceleration coefficient $I_{s,M}H$ [kg m] | 0.155     | 0.429        | 2.76  |
| Local reflected inertia $I_{s,M}H^2$ [kg]        | 9.60      | 14.3         | 1.49  |

Table 4.6: Reference and nominal off-the-shelf (OTS) secondary estimate. Ratios are estimate/reference. The reference helix is  $20^\circ$ ; the estimate uses  $31^\circ$ . These axial coefficients refer to motion in  $x_s$ ; reflection into the shared shift coordinate also includes  $(dx_s/ds)^2$ .

To see what these coefficients produce under load, the estimated helix geometry and movable inertia are inserted into the otherwise reference system. An additional opposing secondary torque of  $-120$  N m is applied over 25 ms, starting near the middle of the shift range. This is a component comparison in the reference vehicle model, not a reconstruction of a complete commercial vehicle.

Both models are already shifting when the torque changes. Each is also continued from its own starting state without the added torque. Subtracting that motion from its loaded motion leaves the extra shift and speed change caused by the load. This keeps their separately conditioned starting motions from being counted as part of the response.

As the added load slows the secondary, the shaft-acceleration term changes sign and increases the closing force. [Figure 4.11\(a\)](#) shows the signed contributions. Following the ramp, the shaft term supplies almost all of the dynamic correction. The total correction reaches 89.3 N, and the largest fraction of the simultaneous quasi-static helix force is 7.87%. The quasi-static reaction stays well away from zero. Panels (b,c) show the corresponding extra shift and primary-speed change; negative shift response means that the load causes more backshift than in that model’s unforced motion.

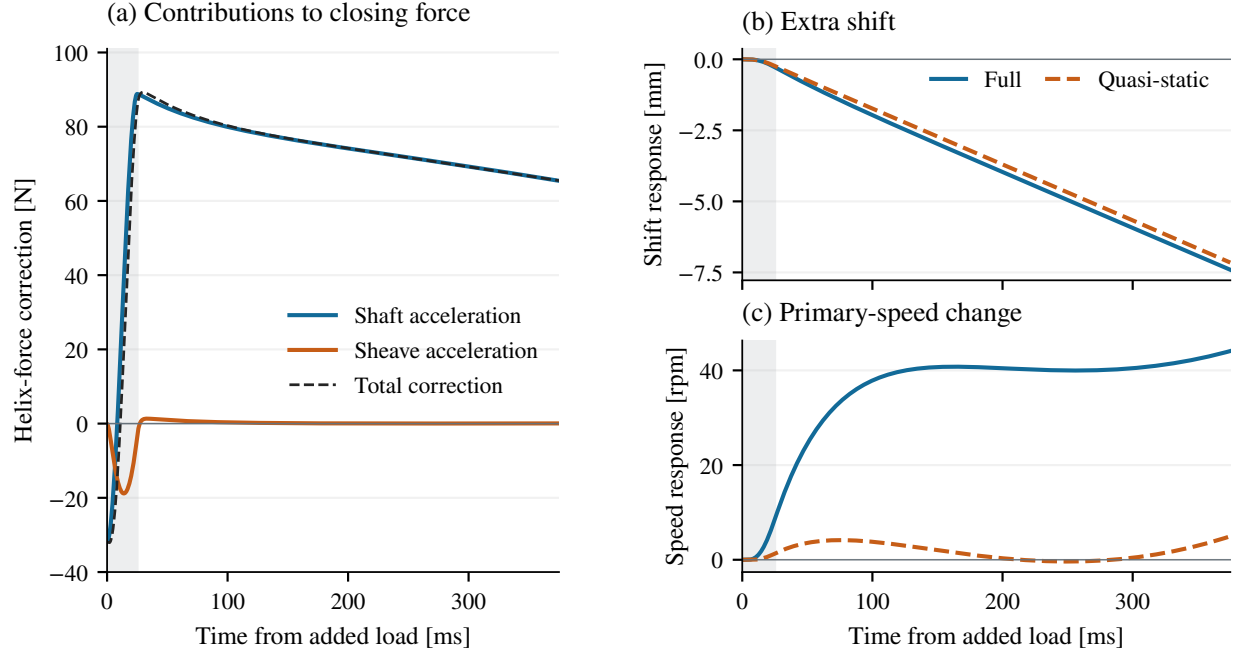


Figure 4.11: Response with the nominal estimated secondary to an added torque of  $-120 \text{ N m}$  over 25 ms, with the ramp shaded. (a) Signed helix-force corrections evaluated on the full model’s motion. (b,c) Each model’s loaded motion minus its own unforced continuation. Both contacts stick and the sheaves remain free to shift throughout all four continuations. The force correction and the independently integrated response are different comparisons.

The additional closing action is accompanied by more backshift in the full model. At the largest shift-response separation, the extra backshift is 4.35 mm in the full model and 4.08 mm in the quasi-static model: a difference of 0.274 mm. The primary-speed responses differ by up to 40.4 rpm. The difference remains modest beside the total response, but develops during continuous sticking with positive local wrap loading, rather than requiring a contact transition.

The low and high component estimates retain this hardware dependence. They span 2.21–8.14 times the reference rotational inertia and 0.45–2.92 times its local reflected axial inertia. Under the same added load, their recorded peak force corrections are 3.15% and 13.6%, respectively. The helix radius and inertia vary together, so this range reflects the component estimate as a whole rather than inertia alone. [Table D.26](#) gives the associated motion differences.

The quasi-static relation is recovered when the acceleration terms vanish, but its usefulness during a drive depends on more than whether the motion looks smooth or the applied load changes quickly. Shaft and sheave accelerations can partly cancel in the reaction, while a change in hardware alters the coefficients multiplying both. Close shift histories can also conceal a change in when a contact reaches its limit. A static torque-to-clamp characteristic alone cannot settle these transient questions: the inertia, helix geometry and available traction must be considered with the motion they produce. Retaining the secondary dynamics makes that joint assessment possible within the belt-contact conditions the formulation can sustain.

### 4.4.3 Reduced-Belt Transient Mechanics

#### Judge each term in the equation it must close

The belt retains two different balances. The global transport equation Equation (3.4.1) relates the net pulley traction to  $m_b \dot{v}_b$ . The closed tension compatibility Equation (3.4.52) instead requires the two wraps and straight spans to form one consistent tension field. A small global inertia force need not imply that acceleration is unimportant to that second balance.

For the term comparison, the final tension compatibility is collected as

$$F_{\dot{s}} + F_{\dot{s}^2} + F_{\dot{v}_b} + F_{\dot{s}v_b} + F_N = 0. \quad (4.4.5)$$

Here the terms are the surviving contributions in Table 4.7; the common  $2\rho_b A_b v_b^2$  contribution has already cancelled between the two wrap-boundary sums. That cancellation is specific to this compatibility equation and does not remove centrifugal loading from the local contact conditions.

To state the coefficients compactly, let  $r'_j = dr_{j,\text{cm}}/ds$  and  $r''_j = d^2 r_{j,\text{cm}}/ds^2$ . For nonzero  $\lambda_j$ , define

$$\begin{aligned} G_j &= \lambda_j \frac{1 + e^{-\lambda_j \phi_j / \sin \beta}}{1 - e^{-\lambda_j \phi_j / \sin \beta}}, \\ H_j &= \frac{2 \sin \beta}{\lambda_j} - \phi_j \frac{1 + e^{-\lambda_j \phi_j / \sin \beta}}{1 - e^{-\lambda_j \phi_j / \sin \beta}}. \end{aligned} \quad (4.4.6)$$

The continuous limits are  $G_j = 2 \sin \beta / \phi_j$  and  $H_j = 0$  at  $\lambda_j = 0$ . These factors collect the existing wrap solution; they do not add a contact law.

Table 4.7: Surviving contributions to the reduced tension compatibility. Radius derivatives are with respect to the shared shift coordinate. The contact term is the difference of the two mapped normal-load contributions, not either gross pulley load.

| Contribution       | Final-equation term                                                                      |
|--------------------|------------------------------------------------------------------------------------------|
| Shift acceleration | $F_{\dot{s}} = -2\rho_b A_b (r_{p,\text{cm}} r'_p - r_{s,\text{cm}} r'_s) \dot{s}$       |
| Path curvature     | $F_{\dot{s}^2} = -2\rho_b A_b (r_{p,\text{cm}} r''_p - r_{s,\text{cm}} r''_s) \dot{s}^2$ |
| Belt acceleration  | $F_{\dot{v}_b} = \rho_b A_b (H_p r_{p,\text{cm}} - H_s r_{s,\text{cm}}) \dot{v}_b$       |
| Moving radius      | $F_{\dot{s}v_b} = \rho_b A_b (H_p r'_p - H_s r'_s) \dot{s} v_b$                          |
| Net contact        | $F_N = G_p N_p - G_s N_s$                                                                |

The distinction between gross and net loading is central to the result.  $G_p N_p$  and  $G_s N_s$  can each be hundreds or thousands of newtons while nearly cancelling. A sub-newton transient term can then be important to the remaining compatibility. Its scale should not be dismissed by comparing it only with gross clamp.

The reported activity share uses the absolute contributions to this final balance. For a time window  $\mathcal{W}$ ,

$$a_i(t) = \frac{|F_i(t)|}{\sum_k |F_k(t)|}, \quad A_i(\mathcal{W}) = \frac{\int_{\mathcal{W}} |F_i(t)| dt}{\int_{\mathcal{W}} \sum_k |F_k(t)| dt}, \quad (4.4.7)$$

where the sum contains the five terms in Equation (4.4.5). Windows with vanishing total activity carry no such percentage interpretation. Signed terms and dimensional magnitudes remain necessary:

### Existing term data: two distinct belt balances

Compare the global transport row with the closed tension balance on the same reference interval. Show gross contact cancellation, signed surviving terms, and equation-activity shares using separate denominators.

`figures/results/dynamics/belt_balance_hierarchy.pdf`

Figure 4.12: Term hierarchy in the global transport and closed tension compatibility equations. The global inertia term is compared with the two opposing pulley reactions; the local acceleration terms are compared within the surviving tension compatibility. These are distinct force balances, so their relative contributions need not have the same hierarchy.

activity describes participation in the equation, not an energy fraction or the trajectory error caused by deleting a term.

### Rapid changes in shift motion activate radial belt inertia

During smooth free shifting, belt acceleration and net contact provide the main opposing contributions to the tension compatibility. The shift-acceleration contribution grows when sheave motion is strongly accelerated, arrested, or reversed. This is the distinction tested by applying the same  $+18^\circ$  road-load change over different rise times. The finite-rise cases produce peak  $|F_s|$  values of approximately 0.0360, 0.00941, and 0.00269 N for 50, 200, and 800 ms, respectively.

Figure 4.13 separates this rate effect from load magnitude. Across the three smooth rises, the fitted peak varies approximately as  $T_{\text{rise}}^{-0.94}$ , close to inverse rise time over that range. This is a response of the coupled system to those forcing programmes, not a universal power law or a prescribed scaling of sheave motion. The instantaneous-load case is shown separately; only the boundary load is stepped, not a CVT state.

The wider twenty-eight-case envelope connects the controlled experiment to other operating states. A rapid uphill case near 64% active shift, with about  $+20.7^\circ$  loading applied over 56 ms, reaches approximately 24% shift-acceleration activity over the load-rise window. Increased road load does not always produce backshift: the tested responses include continued upshift, arrest, and reversal. The better predictor of this term's importance is the realized shift acceleration, not the name assigned to the maneuver.

The same mechanism remains active when power flow is reversed. A separate overrun case combines downhill loading with a resisting primary torque and verifies secondary-to-primary power flow. Its shift-acceleration contribution accounts for approximately 17% of integrated tension-equation activity in the reported transition window. The result is therefore not limited to uphill backshift: rapid changes in the shift motion also matter during controlled overrun. The ordinary downhill cases, by contrast, retain full-throttle drive and represent unloading rather than this reverse-power condition.

### Existing results: loading rate and realized shift acceleration

Pair the controlled rise-time curve with the envelope's activity versus actual shift acceleration. Add a short trace of the rapid-shift exemplar to connect the percentage to arrest/reversal of sheave motion.

`figures/results/dynamics/belt_rate_dependence.pdf`

Figure 4.13: Activation of the shift-acceleration term by rapid loading. The controlled  $+18^\circ$  family isolates boundary rise time, while the wider envelope links the contribution to the solved sheave acceleration. Activity is evaluated over the stated load-rise window using Equation (4.4.7); it is not a percentage change in vehicle acceleration or shift position.

### A sensitive coefficient need not produce an important term

The ordinary load envelope reaches only about 19% of the static traction limit, so it does not alone test whether contact asymmetry could amplify the moving-radius contribution. Independent evaluation of the coefficients toward 95% utilization shows that it can. This motivates the high-contact comparisons rather than treating small terms in the nominal launch as sufficient evidence for omission.

The subsequent clamp-stress cases reach sticking states near 40, 60, 80, and 95% utilization without altering the friction coefficients. More severe cases enter primary-slip/secondary-stick, and one also enters both-slip. The largest realized moving-radius coefficient is approximately  $0.928 \text{ N}/(\text{m}^2 \text{ s}^{-2})$ , yet the largest realized moving-radius force is only 0.0182 N. Its greatest instantaneous activity in these stress cases is approximately 0.13%.

Figure 4.14 should be read as a product, not as three independent rankings. The large coefficient tends to appear near engagement, when belt speed is still low, or in heavily utilized seated states, when shift speed is near zero. Either condition suppresses  $\dot{s}v_b$ . Multiplying the largest coefficient from one instant by the largest driver from another would describe a state the trajectory did not visit. The mixed-slip example retains the same distinction: high contact asymmetry does not by itself supply the missing kinematic driver.

The path-curvature term remains very small for a related but distinct reason: the realized  $\dot{s}^2$  is too small to activate its finite coefficient strongly. Belt acceleration does not share that suppression and remains a substantial partner to net contact, including at high traction demand and in overrun. These observations support a hierarchy of complete terms, not a claim that the small coefficients vanish or that every contact branch has been sampled. In particular, the opposite mixed branch, primary-stick/secondary-slip, did not arise naturally in this belt-study set.

### What can be reduced without breaking the coupled description

The global transport inertia presents a different reduction question. Across the investigated envelope,  $m_b \dot{v}_b$  is small relative to the two opposing pulley reactions. Scaling only that coefficient to 10% or

### New composition from existing high-contact and overrun data

Show the moving-radius coefficient, its simultaneous driver, and their product through high utilization and mixed slip. Contrast with the significant shift-acceleration contribution in true overrun.

`figures/results/dynamics/belt_coefficient_driver.pdf`

Figure 4.14: Why coefficient sensitivity alone overstates the realized moving-radius effect. The complete term uses coefficient and driver at the same state. High utilization can enlarge the coefficient while small belt speed or nearly stationary shift suppresses its contribution. Overrun supplies an additional test with verified reverse power flow.

3%, however, prevents admissible engagement in the flat, fast-backshift, and fast-unloading tests. This isolated modification leaves the local wrap inertia unchanged and therefore does not represent a consistent change in belt mass.

Scaling belt density instead changes the global mass and local wrap inertia together. Both the 10% and 3% density continuations complete all three cases and preserve their respective hybrid transition counts. [Figure 4.15](#) compares the trajectory differences from this coherent modification with the isolated-row outcomes. This is evidence for investigating a coupled low-inertia reduction, rather than deleting a small coefficient from one equation on its own. It does not solve the exact zero-density limit.

#### Evidence to complete: R1 — reconcile the continuation error scale

The source recap's graph and narrative report approximately 0.81–0.86% worst five-state normalized RMSE at 3% density; its page-10 table prints approximately 81–86%. Match the archived `inertia_continuation_comparison.csv` to the normalization and plot before inserting the quantitative error bound. The completion and unchanged transition-count statements above do not resolve this discrepancy.

#### Evidence to complete: R2 — test the proposed small-term omission

Re-integrate a consistently assembled model omitting  $F_{s2}$  and  $F_{svb}$ , first jointly and separately only if attribution is needed. Use the flat reference, rapid backshift, rapid unloading, high-contact mixed-slip, and true-overrun cases. Compare state errors, event histories, and admissibility at resolved numerical settings. This is the evidence needed to turn small observed equation activity into a quantitative bound on the reduced model's response; no passing outcome is presumed.

The resulting approximation guidance is summarized in [Table 4.8](#). Belt acceleration belongs in the present finite-inertia tension compatibility. Shift-acceleration inertia must also be retained when rapid changes in sheave motion are part of the intended use. Curvature and moving-radius terms are the strongest candidates for omission over the investigated Baja envelope, because their complete realized contributions remain small even in the targeted high-contact and overrun cases. Their

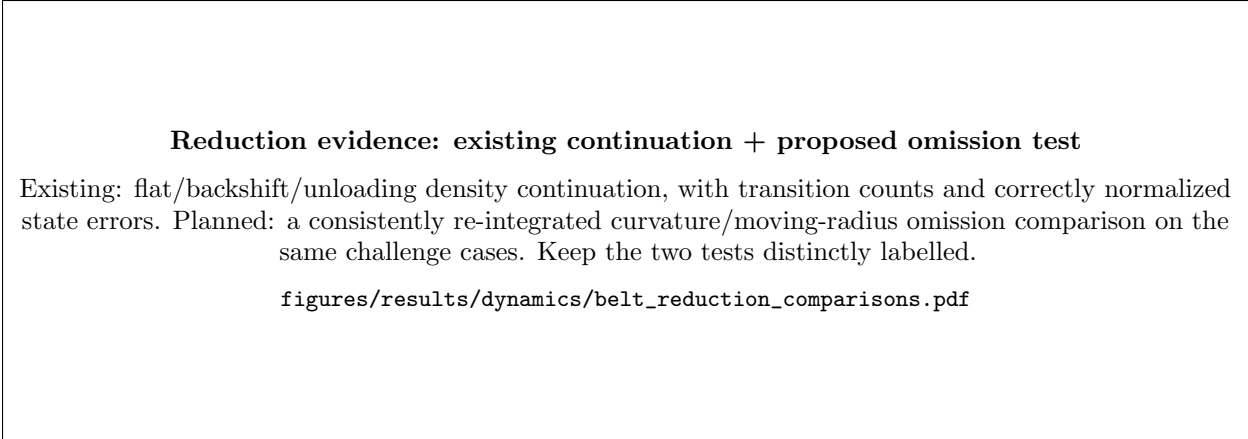


Figure 4.15: Tests of further belt-model reduction. The completed continuation changes density in all retained belt-inertia contributions; isolated transport-row changes are shown separately. A second, explicitly planned comparison tests omission of the two small kinematic terms. Its results remain to be generated, and the density-continuation error scale requires the source check identified in the draft.

actual omission error must still be established by the reduced comparison. Global low-belt-inertia modeling is a separate, coupled limit.

Table 4.8: Reduction guidance for the investigated belt model and operating envelope. Equation-level smallness, a completed continuation, and a quantitatively verified reduced trajectory are different levels of evidence.

| Contribution                         | Disposition                 | Condition / evidence                                                                                                             |
|--------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Belt acceleration in tension closure | Retain                      | Substantial contribution throughout the tested finite-inertia balance.                                                           |
| Shift acceleration                   | Retain for rapid transients | Appreciable during arrest/reversal and overrun; smaller in smooth shifting.                                                      |
| Path curvature                       | Candidate omission          | Small realized squared-speed contribution; re-integrated omission error pending.                                                 |
| Moving radius                        | Candidate omission          | Small simultaneous coefficient–driver product, including tested high utilization and slip; re-integrated omission error pending. |
| Global transport inertia             | Reduce coherently           | Density continuation completes; local and global belt inertia must change together. Exact massless limit not evaluated.          |

The distinction across these results is not simply between a dynamic model and a static one. Some retained mechanisms participate throughout ordinary motion, some become important only when the corresponding acceleration is excited, and others remain small because their coefficient and driver do not become large together. That separation explains both why a relatively simple approximation can succeed in smooth operation and why it can miss the response to a rapid disturbance. It also identifies which simplifications should be tested first, with the remaining coupled balances preserved.

## 4.5 Coupled Operation and Tuning on a Common Off-Road Course

The preceding studies examined the formulation in controlled pieces: whether its coupled balances remain mechanically and numerically consistent, how its transient response compares with an independent model, and when the retained actuator and belt dynamics alter the predicted motion. Those questions are necessary, but they stop short of the problem that motivated the complete CVT formulation. If the engine and vehicle supply the loading at the two shafts, can the same internal mechanics carry the transmission through the changing conditions of a drive, and can the resulting behaviour be followed back to the hardware that produced it?

To answer that question, each configuration is placed in the same vehicle, started from the same state, and driven at full throttle over the same continuous road. The drive is not divided into independent launch, hill, and backshift tests. The shaft speeds, ratio, contact state, and hardware motion left by one part of the road are carried into the next. Several different CVTs can therefore meet the same sequence of external loads while the model is left to determine how each one engages, shifts, transmits torque, and uses its available supports and traction.

Every configuration therefore operates between the same engine and the same vehicle. At the primary shaft, every run uses the same full-throttle Kohler CH440 boundary described in [Appendix C.2.2](#). At the secondary, the same locked-final-drive vehicle boundary of [Appendix C.2.3](#) supplies the vehicle inertia and road load; road grade enters that load through [Equation \(C.2.8\)](#). Every run also begins from the state in [Equation \(C.1.1\)](#). The engine map, vehicle, final drive, shaft-boundary inertias, and starting condition are therefore unchanged from one hardware case to another.

The road itself is also common to every run. It is prescribed as a function of travelled distance rather than clock time, so a faster vehicle reaches a road feature sooner but does not encounter a different feature. The complete course is shown in [Figure 4.16](#).

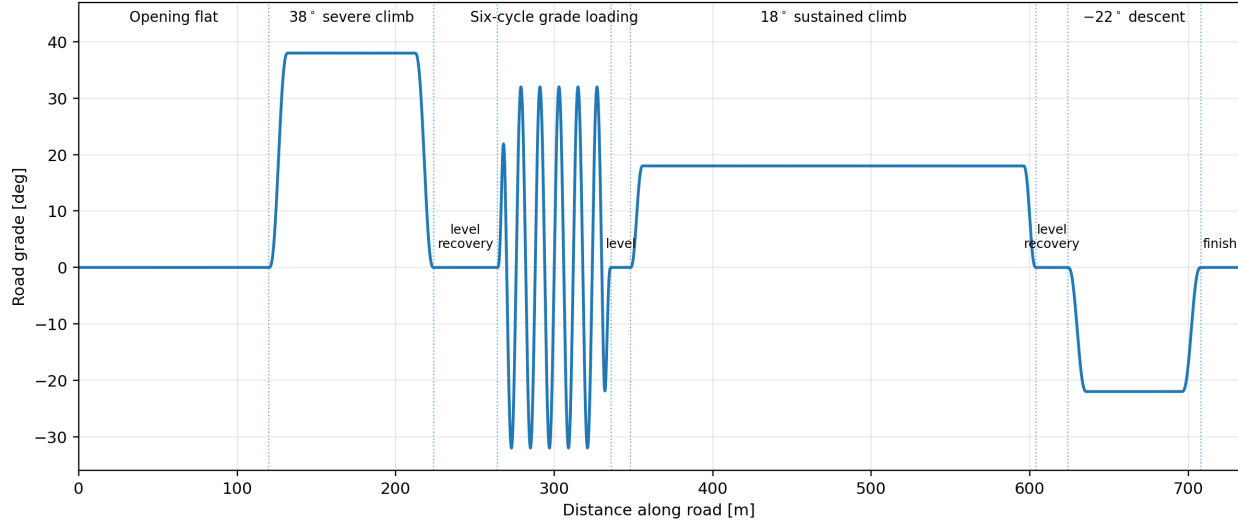


Figure 4.16: Distance-indexed road grade used by every configuration on the common 732 m course. The first 120 m are level. A smooth 12 m entry leads to an 80 m hold at  $38^\circ$ , followed by a 12 m exit and 40 m of level recovery. From 264 m to 336 m, the road completes six 12 m grade cycles with a peak amplitude of  $\pm 32^\circ$ , followed by 12 m of level road. The second climb uses an 8 m entry, a 240 m hold at  $18^\circ$ , and an 8 m exit, followed by 20 m of level recovery. The final load change is a  $-22^\circ$  descent with 12 m entry and exit transitions around a 60 m constant-grade section, followed by a 24 m level finish. The cyclic region changes only the longitudinal road grade; suspension motion, wheel unloading, tire normal-load variation, and airborne vehicle dynamics are not represented.

Each part of the road is included to ask a different mechanical question. The opening flat gives the CVT room to engage and move through its ratio range before a large road load is introduced. The  $38^\circ$  climb then places a much larger demand on the drivetrain, allowing limitations in available ratio or belt traction to appear if the hardware drives the transmission toward them. After a level recovery, the six-cycle region applies repeated changes in longitudinal load and tests how much shift motion remains available as those changes continue. The long  $18^\circ$  climb serves a different purpose: it holds the load long enough to separate the initial backshift from whatever interior ratio, engine speed, and climbing speed the coupled system approaches. The descent reverses the road-grade contribution and allows any corresponding reverse transfer of power through the drivetrain to emerge from the same signed mechanics.

Every vehicle has now been given the same engine boundary, vehicle boundary, starting state, and road. What changes from run to run is selected hardware inside the CVT. Those configurations are defined in [Table 4.9](#). The nominal case, R00, is the unmodified Baja reference CVT of [Appendix C.1](#), using the bilateral secondary helix topology of [Appendix C.1.1](#). Every other row lists only the hardware changed from that baseline; unlisted quantities retain the reference values. The mass percentages refer to the replaceable flyweight tip mass rather than the total flyweight mass, and the corresponding mass moments are changed consistently.

Table 4.9: Selected CVT hardware configurations used on the common course. Only changes from the nominal R00 reference are listed for the remaining cases. Ramp distances are physical coordinates along the ramp, not movable-sheave travel.

| Case     | Change from R00                                                                                                                                                                                             |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R00      | Unmodified reference configuration; no hardware change.                                                                                                                                                     |
| W85      | Replaceable primary flyweight tip mass reduced to 85%.                                                                                                                                                      |
| P300     | Primary spring stiffness increased to $3k_p$ ; installed compression reduced so that the spring force matches R00 at the low-ratio engagement geometry.                                                     |
| RC10     | Reference ramp retained for the first 10 mm, followed by a 5 mm smooth blend into an arc ending at $10^\circ$ .                                                                                             |
| R26B7    | Reference ramp retained for the first 10 mm, followed by a 7 mm smooth blend into an arc ending at $26^\circ$ .                                                                                             |
| RC40L    | Reference ramp retained for the first 18 mm, followed by a 5 mm smooth blend into an arc ending at $40^\circ$ .                                                                                             |
| H28      | Secondary helix angle increased from $20^\circ$ to $28^\circ$ from the circumferential direction.                                                                                                           |
| U55      | Replaceable primary flyweight tip mass reduced to 55%.                                                                                                                                                      |
| D01      | Tip mass increased to 130%; primary and secondary installed spring compressions reduced to 80%; secondary torsional preload reduced from $300^\circ$ to $240^\circ$ ; helix angle increased to $28^\circ$ . |
| D02      | Tip mass reduced to 65% and primary installed spring compression increased to 115%.                                                                                                                         |
| D02_M    | D02 with the flyweight tip mass restored to the reference value.                                                                                                                                            |
| D02_P    | D02 with the primary installed spring compression restored to the reference value.                                                                                                                          |
| D02_M170 | D02 with the flyweight tip mass increased to 170% of the reference value; elevated primary spring compression retained.                                                                                     |

The cases span several ways of changing the forces and motion inside a mechanically actuated CVT. W85 changes the flyweight mass properties that produce the primary centrifugal action. P300 changes the primary spring rate while matching its force to R00 at the low-ratio engagement geometry, allowing the spring characteristic to be changed without simply beginning from a larger or smaller opposing force. RC10, R26B7, and RC40L alter different portions of the physical ramp followed by the roller, so the effect of changing that contact geometry can be examined without replacing the flyweight mechanism itself. H28 makes the corresponding comparison on the secondary by changing the helix geometry that couples reacted torque to axial motion.

U55, D01, and D02 extend the comparison farther from the nominal hardware. U55 continues the flyweight-mass change to a more severe value, while D01 and D02 combine larger changes in the primary and secondary actuation. They are included to expose whether the more demanding parts of the road bring out limitations that are not apparent near the reference configuration. D02\_M and D02\_P then separate two of the primary-side changes contained in D02 by restoring, one at a time, the nominal flyweight mass and the nominal primary spring preload. D02\_M170 extends the mass correction to test how stronger primary closing affects both clamping and backshift. Together, the set provides several distinct mechanical routes away from the reference before any of their course responses are interpreted.

#### 4.5.1 Launch and the Shape of the Shift Curve

The road begins with 120 m of level ground. Before the later changes in grade disturb the drivetrain, this opening lets each CVT engage and begin its upshift from the common initial state. Plotting

primary speed against secondary speed gives the familiar shift-curve view used in practical CVT tuning (Aaen, 2007; Skinner, 2020), but here no speed curve is prescribed. Each trajectory follows from the actuator geometry and spring loading acting through the belt, secondary, and accelerating vehicle. Figure 4.17 shows this response for six primary-side configurations, from launch at zero secondary speed to just before the first hill, including their low-ratio motion, free shift, and any high-ratio motion reached along the way.

The low-ratio limit is the lower end of engaged shift travel, where the secondary is fully closed against its stop. The high-ratio limit is reached when the primary closes against its upper stop (Section 3.6.2). The dashed lines show the no-slip shaft-speed relations at those two fixed geometries. While a stop holds the geometry and both contacts stick, the shafts accelerate along its line; the disengaged and slipping parts of launch need not follow either relation.

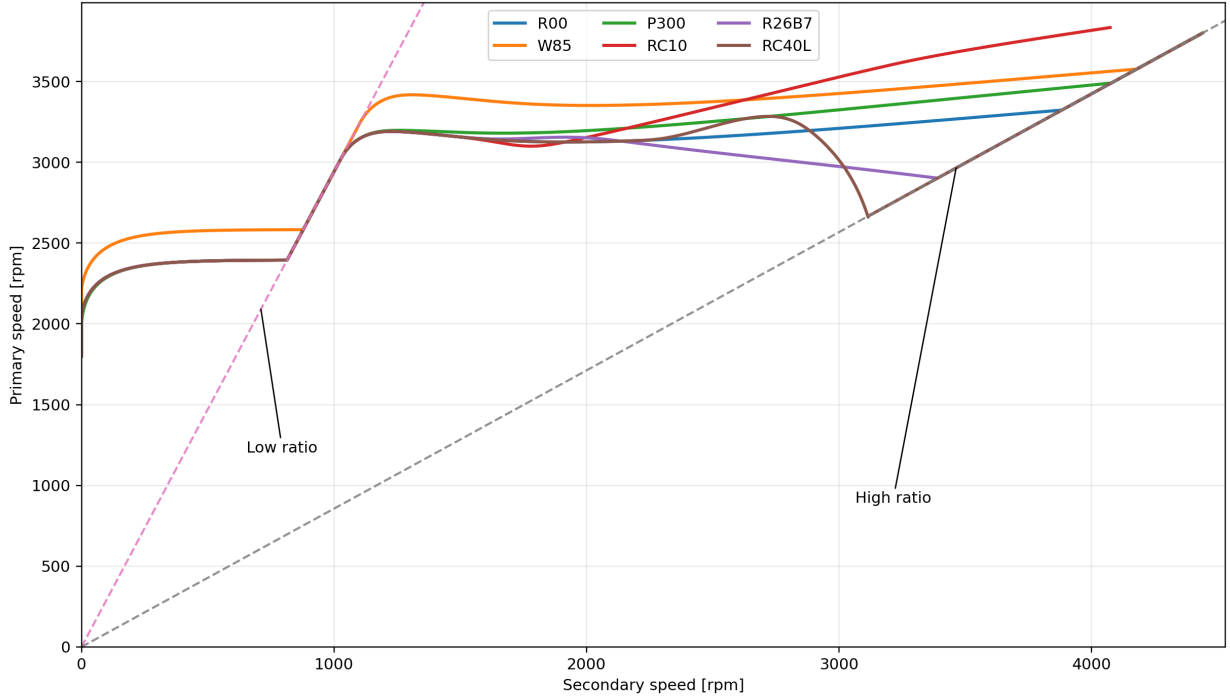


Figure 4.17: Complete opening-flat shift curves for the six selected primary hardware configurations, from the initial state through the last recorded point before 120 m. The solid curves retain the initial disengaged motion, launch slip, supported motion, and free shift. The dashed lines are the no-slip speed relations at the geometric low- and high-ratio limits.

W85 gives the most familiar change. Reducing the replaceable flyweight tip mass weakens the speed-dependent centrifugal closing action at a given primary speed. The transmission therefore has to reach a higher primary speed before a similar sequence of axial balances can be recovered. The result is not an exact vertical translation, because every force and inertia term continues to evolve with the trajectory, but much of the R00 progression is reproduced at a higher working speed. Between 20% and 80% of the available engaged shift travel, W85 changes from approximately 3393 to 3447 rpm, compared with approximately 3169 to 3205 rpm for R00. This is the classical lighter-flyweight effect. The leading speed-dependent contribution appears explicitly in Equation (3.2.11) through  $\frac{1}{2}\omega_p^2 dJ_{f,p}/dx_p$ .

P300 begins much closer to the reference but separates for a different reason. Its installed compression is reduced so that Equation (3.2.12) gives the same primary spring opening force as R00 at the low-ratio engagement geometry, as shown in Figure 4.18(a). Early in the shift, the two cases therefore begin from nearly the same competition between flyweight closing action and spring opposition. As the primary continues to close, however, the three-times-greater spring rate makes that opposition grow more rapidly. The later sheave motion consequently slows relative to R00. Secondary speed can continue to rise while less ratio change is produced, giving the primary and its attached engine more opportunity to accelerate relative to the secondary. The curve therefore peels upward as the shift proceeds, reaching approximately 3336 rpm by 80% travel even though it began near the R00 operating point. Matching the spring force at one position has not matched the spring characteristic through the rest of the shift.

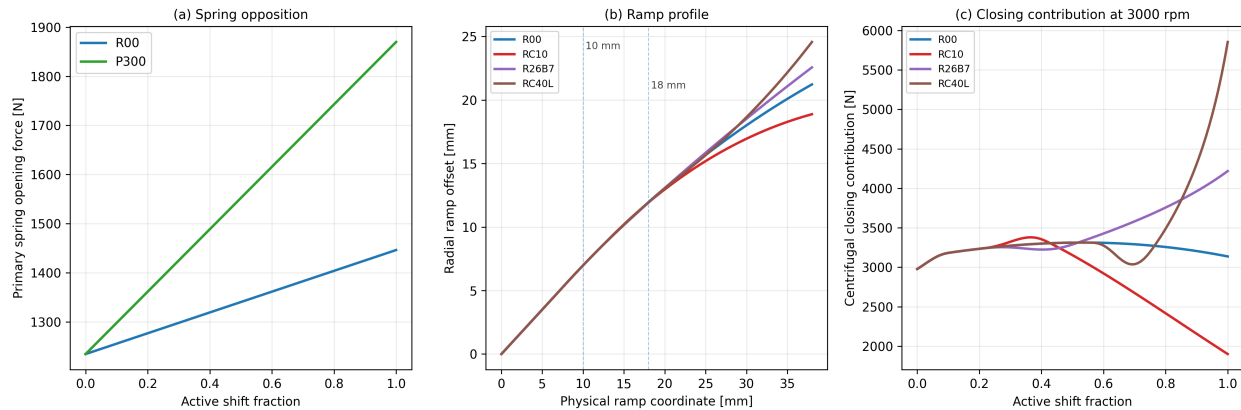


Figure 4.18: Mechanical origin of the selected shift-curve changes. (a) Primary spring opening force for R00 and P300, matched at the low-ratio engagement geometry. (b) Physical ramp profiles: RC10 and R26B7 preserve the first 10 mm of the reference ramp, while RC40L preserves the first 18 mm; these distances are ramp coordinates rather than movable-sheave travel. (c) Leading centrifugal closing contribution at the same 3000 rpm primary speed. The complete actuator also retains the corresponding motion-ratio and flyweight-inertia terms, so panel (c) isolates one contribution rather than replacing the full dynamic force law. In (a) and (c), active shift fraction runs from zero at the low-ratio seat to one at the upper primary stop.

The spring case changes a force that varies linearly with displacement. The ramp cases instead change the geometry through which the roller moves, so the relationship between flyweight motion and axial closing force can vary nonlinearly as different parts of the profile are reached. The changed profiles in Figure 4.18(b) therefore produce different closing-force variations even at the same primary speed, as panel (c) illustrates at 3000 rpm.

That difference in how the hardware varies through travel is visible in the three ramp trajectories. RC10 and R26B7 preserve the same early ramp as R00, so their initial free-shift response remains close to the reference. Once the roller reaches the modified region, the closing contribution no longer has to change monotonically or even in the same direction as the reference. RC10 first drops slightly and then turns sharply upward, reaching approximately 3655 rpm at 80% travel. R26B7 develops the opposite tendency and falls to approximately 3013 rpm. RC40L leaves a longer portion of the reference profile untouched; its trajectory remains essentially coincident with R00 through the earlier shift before developing a distinct late shoulder and downturn. Unlike the linear spring-rate

change, a ramp change can therefore reshape the working-speed characteristic differently in different parts of the same shift.

The shift curves also show that the upper primary stop is reached at different points in the opening acceleration. RC10 is still freely shifting when the first hill arrives, whereas the other five curves have already joined the high-ratio line. With the stop geometry unchanged, these different arrivals follow from the actuator characteristics acting through the same belt, secondary, engine boundary, and vehicle.

One ambiguity remains when a case has not reached the upper stop before the first hill arrives. The common course deliberately changes grade after 120 m, so an incomplete opening shift by itself cannot distinguish a slow approach to high ratio from an interior balance that would persist on level ground. The more extreme flyweight-mass reduction in U55 also leaves its opening shift incomplete. To see whether it simply needs a longer level road, R00 and U55 were also run over a separate 800 m flat.

R00 again reaches the 19.05 mm upper stop near 64.9 m. U55 instead approaches an interior shift position near 14.90 mm, changing by less than 0.001 mm over the final 10 s. Under the observed flat-road load, the reduced primary closing action therefore approaches a balance with the opposing spring, belt, and secondary reactions before the high-ratio stop is reached. The finite run does not prove that U55 could never move farther, but it shows that its incomplete shift on the common course is not simply caused by the 38° hill arriving a few metres too soon.

The opening-flat cases therefore make several familiar tuning ideas more precise. A lighter flyweight can make much of the same shift progression occur at a higher primary speed. A stiffer primary spring can leave the beginning of that progression nearly unchanged while increasingly slowing the later ratio motion, allowing engine speed to rise relative to the secondary. Changing the ramp can alter the same balance nonlinearly and only over the portions of travel where the new geometry is encountered. Those different states are then carried directly into the first severe climb, where the question changes from how the hardware shapes the upshift to what limits the transmission when the road demands a large backshift.

#### 4.5.2 A Severe Climb Separates Poor Shifting from Loss of Traction

The first climb increases the gravitational resistance to forward motion, so the wheels need more torque to sustain the vehicle's speed. If the ratio stays fixed while the belt sticks, slowing the vehicle also slows the engine, potentially taking it away from the speeds at which it supplies useful power. Gearing down allows the same transmitted power to provide more wheel torque at a lower wheel speed, while keeping the engine turning faster relative to the vehicle. Backshifting serves that purpose by moving the belt inward on the primary and outward on the secondary. The secondary helix provides a physical connection between torque loading and this motion: the torque it reacts produces axial closing thrust alongside the spring force (Equation (3.2.29)). At the same time, falling primary speed weakens the centrifugal action holding the primary closed. Through the belt, these changes can open the primary and close the secondary, producing the required backshift.

How much this helps the vehicle depends on both the ratio reached and the traction available to transmit torque through the belt. Figure 4.19 therefore brings together shift position, primary speed, and vehicle speed for R00 and the two adverse cases, D01 and D02. The road-position axis places them under the same local grade, while retaining the different states inherited from the opening flat. R00 provides the reference response: it leaves the high-ratio stop, backshifts to the low-ratio

seat, and continues climbing. After the brief seating transition, both contacts stick and the vehicle approaches approximately 3.77 m/s. Reaching the end of backshift travel has limited the ratio, but has not prevented torque transmission.

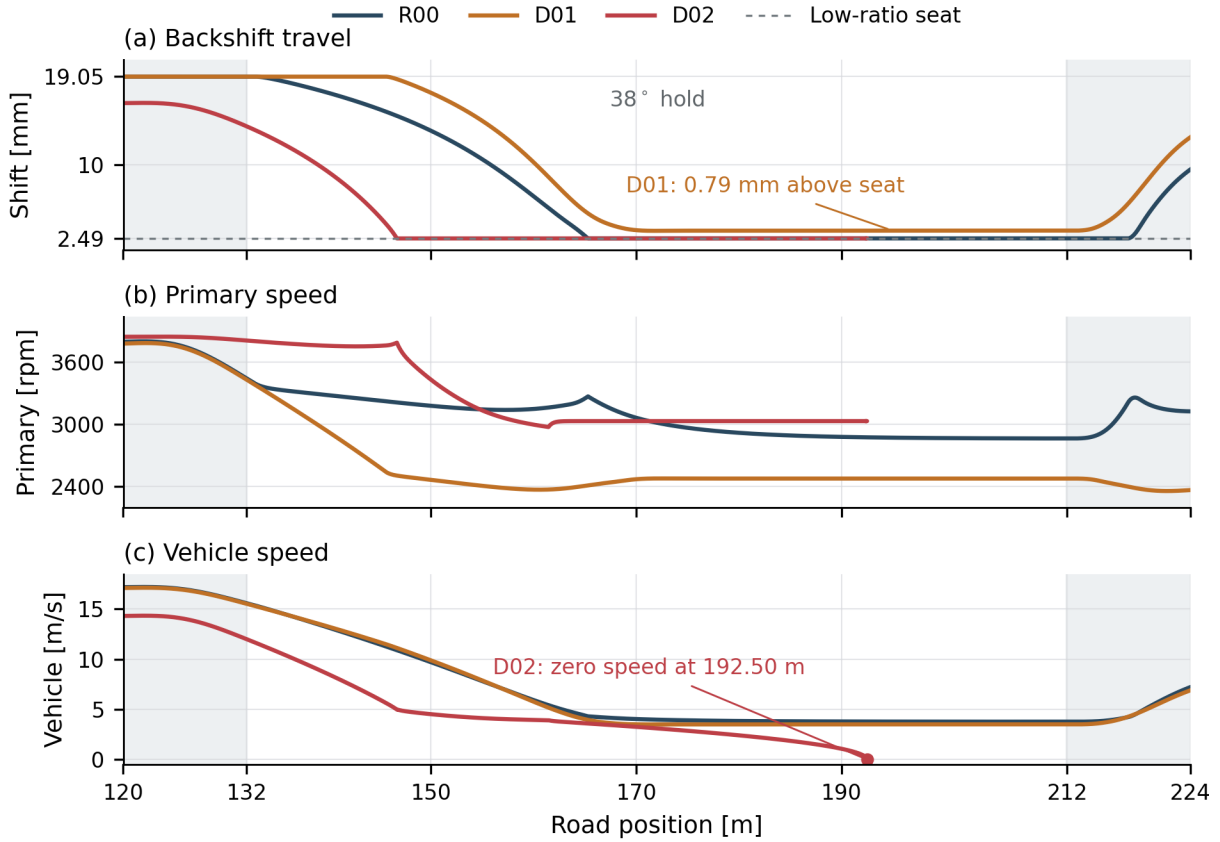


Figure 4.19: Response to the first severe climb. The shaded bands mark the entry and exit tapers; the intervening road holds 38°. The dashed line in (a) marks the low-ratio seat. All three D02 traces end at its first zero vehicle speed, near 192.50 m. Equal road position means equal imposed grade, not equal elapsed time or inherited state.

D01 remains at the high-ratio stop farther into the climb. While the geometry is held and the belt sticks, engine speed must fall with vehicle speed. Its heavier primary tips and reduced primary spring opposition favour primary closure, while the reduced secondary spring loading and changed helix weaken the secondary closing action that favours backshift. Near 200 m, D01 still has approximately 0.79 mm of backshift travel available and runs near 2477 rpm, where the supplied engine boundary delivers less power than in R00. Both belt contacts continue sticking throughout the hill, and the vehicle still climbs, but at a lower speed. Its difficulty is the ratio and engine operating condition reached under load; primary traction has not been exhausted.

D02 reaches the low-ratio seat near 146.68 m, so its later loss of progress begins after the available backshift has already been used. The secondary has completed its closing travel, but the primary must still squeeze the belt strongly enough to transmit torque. D02's light tips provide less centrifugal closing action, while its elevated primary spring compression increases the opposing opening force. As established in [Section 3.6.2](#), the secondary is held against its closing stop while the primary

axial balance still determines clamping. At this fixed geometry, primary spring opposition stays constant, but the centrifugal closing contribution decreases as primary speed falls. The remaining closing force must balance the belt's opening reaction, so the primary normal load falls with it (Equations (3.2.11) and (3.2.13)). For D02, the normal load decreases from about 3.49 kN just after seating to 1.05 kN before sustained slip begins. Loading carried by the secondary stop does not add to the belt normal force or supply extra traction.

Sticking can continue only while the required tangential force remains within the static capacity supported by that normal load. While a contact sticks,  $|\lambda_j|/\mu_s$  measures the fraction of its available static traction required by the coupled motion; unity is the sticking limit. In Figure 4.20(a), the primary reaches this limit near 17.83 s after launch, while the secondary remains below its own limit and continues sticking. The primary then slides relative to the belt. Its speed recovers slightly toward 3032 rpm, but the vehicle keeps slowing, as already seen in Figure 4.19. Following the power in panel (b) explains how engine rotation can continue while forward progress is lost.

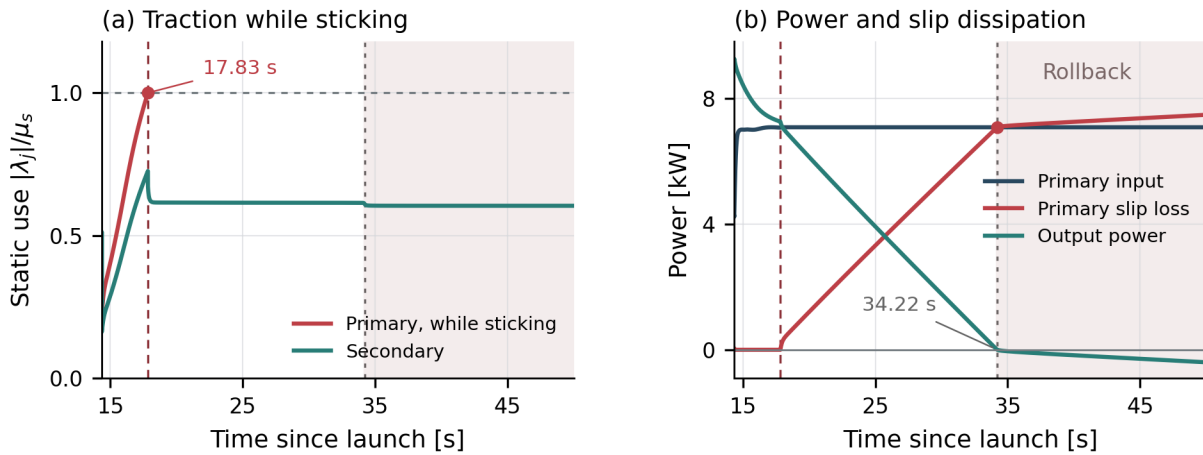


Figure 4.20: D02 from arrival at the low-ratio seat through the retained rollback. (a) Fraction of static traction used while each contact sticks; the primary trace ends when sustained sliding begins. (b) Primary boundary input, primary slip dissipation, and signed output power. Positive output power leaves the CVT; negative output power enters from the vehicle side. Vertical markers identify primary traction loss and the first zero vehicle speed, with rollback shaded. During transients, differences between input, output, and dissipation also include changes in stored mechanical energy.

Near 34.22 s, the vehicle reaches zero speed while the primary still turns near 3032 rpm. Approximately 7.07 kW enters through the primary boundary, and essentially the same power is dissipated in primary slip; useful output power has fallen to zero. The vehicle subsequently rolls back. Output power then becomes negative, because the descending vehicle supplies power into the CVT, and primary slip dissipation can exceed the engine input. For the remainder of the simulated interval, the vehicle continues rolling backward while the primary rotates forward and slides against the belt.

Preventing this loss of grip requires sufficient primary normal load at the low ratio D02 has already reached. At the seated geometry, the primary force balance relates that load to the difference between centrifugal closing and spring opposition. D02\_M restores the reference tip mass and its associated mass moments, strengthening centrifugal closing while retaining D02's elevated primary

spring compression. The same mass correction is also extended to 170% of the reference tip mass in D02\_M170, with corresponding mass moments and the same spring compression. D02\_P retains the light tips but restores the reference spring compression, reducing the opening force opposing the flyweights without changing the spring rate. A larger remaining closing force supports a larger primary belt normal load, and the unchanged friction coefficient then permits a larger tangential force before sticking is lost. These are two physical ways to correct the same clamping weakness.

The changed primary forces also alter the backshift shown in Figure 4.21(a). Both D02\_M and D02\_P reach the same low-ratio seat as D02 and remain there through the rest of the constant  $38^\circ$  climb, whereas D02\_M170 retains unused backshift travel.

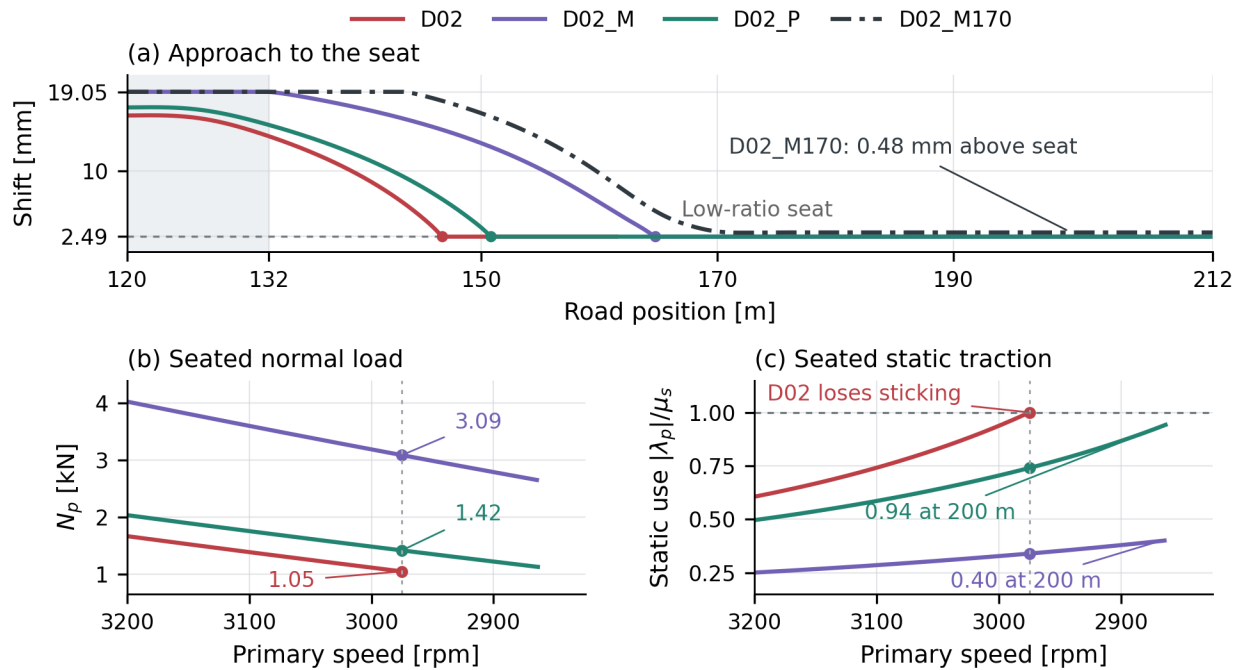


Figure 4.21: Clamping and backshift under the three primary-side corrections. (a) Approach to the common low-ratio seat, with the hill-entry taper shaded. Equal road position retains the different states inherited from the opening flat. (b) Total primary normal load and (c) static utilization during seated sticking on the  $38^\circ$  hold, after the seating transitions. Markers in (b,c) identify the comparison near 2975 rpm; the repair utilization annotations refer to states near 200 m. D02 traces end at sustained primary slip onset. D02\_M170 is absent from (b,c) because it does not reach the seat on the constant  $38^\circ$  climb.

The stronger mass correction in D02\_M170 maintains ample clamping, but its centrifugal closing also opposes the primary opening needed for backshift. Both contacts stick throughout the constant  $38^\circ$  climb, yet the transmission remains above the seat in panel (a). Near 200 m, the primary uses only about 0.20 of its static capacity, while the engine turns near 2660 rpm, compared with 2870 rpm in D02\_M. At this lower speed the supplied engine boundary delivers less power, and the vehicle climbs at approximately 3.66 m/s instead of 3.77 m/s. D02\_M170 still completes the course, but the stronger correction introduces the same kind of unfavourable ratio and engine operating condition seen in D01.

D02\_M and D02\_P restore grip while still allowing full backshift. Because they reach the same low-ratio seat as D02, their clamping can be compared at a common pulley geometry. Panels (b) and (c) make this comparison at matching primary speeds, showing normal load and the fraction of static capacity used as each case slows on the constant  $38^\circ$  slope. Matching states occur at different times and road positions in each run.

At the marked speed, the primary must transmit approximately 683 N of tangential force in each case. For D02, this demand has reached the static limit  $\mu_s N_p$ . Restoring the tip mass in D02\_M strengthens centrifugal closing, while reducing the spring compression in D02\_P leaves less opening force to oppose it. Both corrections therefore produce greater normal loads in panel (b), with the larger increase in D02\_M. This additional normal load raises the available static force, so the same tangential demand uses a smaller fraction of that capacity in panel (c). Both repairs retain grip below D02's slip-onset speed, even as their own normal loads fall and the fractions of capacity used rise.

The two repairs nevertheless leave different margins against slip. Near 200 m, D02\_M uses about 0.40 of its primary static capacity, leaving a substantial reserve, whereas D02\_P uses about 0.94 and is close to its traction limit. Yet both climb near 3.77 m/s at the same low-ratio seat. Both already transmit the required force without sliding. The larger normal load in D02\_M increases the force the primary could support at that state, but does not itself supply more power to the wheels.

The hill has separated an unfavourable ratio and engine speed with unused backshift travel from a loss of belt traction after full backshift. Following the latter failure through the primary force balance identifies two ways to restore enough clamping, with different remaining traction reserves. The stronger mass correction shows why that improvement cannot be judged from clamping alone: the same closing action also changes backshift, engine speed, and climbing speed.

### 4.5.3 Repeated Loading and Upper-Stop Support

After the first hill, the level recovery allows the vehicle to regain speed and the CVT to upshift. The next road feature repeatedly reverses the gravitational contribution to the load: six 12 m grade cycles alternate between climbing and descending. Increased resistance can again promote backshift through the torque-sensitive secondary, while the reduction in load gives primary closing action more opportunity to upshift. Repeating these changes tests whether the sheaves continue moving or become held at a travel limit as the vehicle accelerates. This remains a longitudinal-load test; suspension motion, wheel unloading and airborne dynamics are not represented.

The primary spring and ramp changes examined in [Section 4.5.1](#) also change the recovery from the hill and the state carried into these cycles. [Figure 4.22](#) compares R00 with P300 and the two ramp cases, R26B7 and RC10. R26B7 has already returned to the common 19.05 mm upper primary stop before the first cycle begins. The other three enter below it, with room for further closure. Shift position shows whether that room is used; the stop reaction shows when the hardware carries closing load instead of allowing further motion.

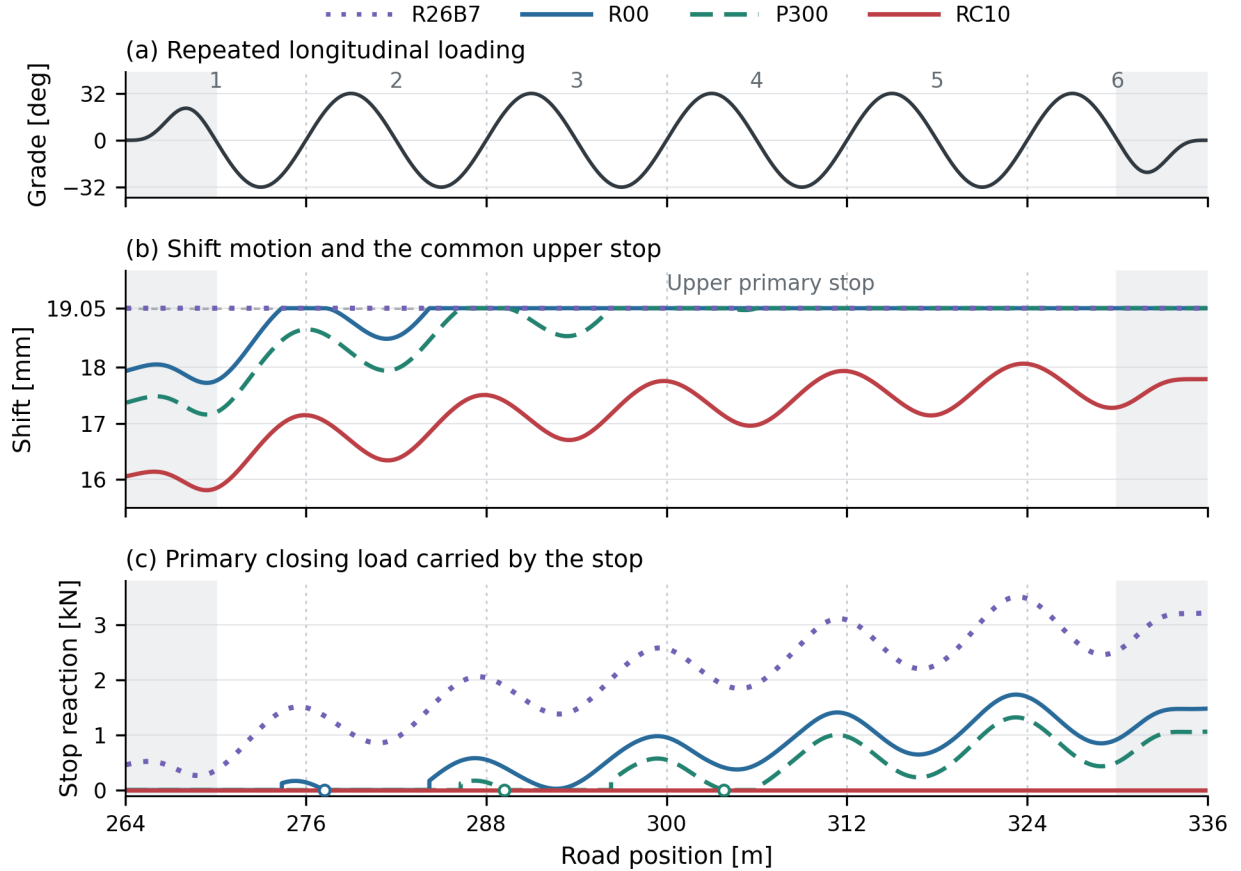


Figure 4.22: Repeated loading from 264 m to 336 m. (a) Road grade, with cycles numbered and the 6 m end tapers shaded; cycles 2–5 have the full  $\pm 32^\circ$  amplitude throughout. (b) Shift position; supported traces coincide at the common upper stop. (c) Primary stop reaction, zero while the primary is away from the stop. Open circles mark recorded releases from support. Equal road position aligns the imposed grade, not elapsed time or the states inherited from the preceding road.

R26B7 remains at the upper stop throughout all six cycles, but panel (c) shows that its loading is not constant. The stop carries the primary closing force left after spring opposition and the belt's opening reaction have been balanced. Its reaction varies with the drive yet remains positive, so the primary stays pressed against the stop. This does not mean that backshift has been mechanically blocked. The stop prevents further closure, but cannot pull the primary closed: if holding that position would require a tensile reaction, the primary must leave the stop and open.

R00 and P300 show that release directly. R00 reaches the stop during the first cycle, reopens during the second, and then returns to remain supported from the third cycle onward. With its greater spring opposition to closure, P300 continues releasing during the third and fourth cycles. Its largest backshift within a cycle falls from approximately 0.73 mm in cycle 2 to just 0.03 mm in cycle 4; by cycle 5 it too remains at the stop. The open circles in panel (c) connect those departures to the reaction falling to zero. Once the later load changes no longer unload the stop, shift position stays fixed even though the grade continues to vary. At that fixed geometry, the rising primary speed during continued acceleration strengthens centrifugal closing, increasing the load available to keep the primary pressed against the stop.

RC10's weaker late-travel centrifugal closing instead leaves the primary below the upper stop throughout this region. Its shift continues to rise overall, but each cycle still produces a distinct backshift: the largest fall from an earlier shift maximum is approximately 0.81 mm in cycle 2 and 0.79 mm in cycle 5. These opening excursions distinguish the repeated response from the underlying upshift: the primary continues to reopen appreciably even as its mean position moves closer to the high-ratio limit.

Both belt contacts stick throughout this region in all four cases, so the contrast is not caused by a loss of traction. Nor does the disappearing shift oscillation in R00 and P300, by itself, establish damping: the reaction and release history show how their motion gives way to sustained support. Following the load carried by the stop makes that distinction visible even when position alone becomes unchanging. The longer climb that follows asks a different question: where the coupled transmission and engine operate when the increased load is sustained rather than repeatedly reversed.

#### 4.5.4 Operating Conditions on a Sustained Grade

The next climb holds the increased gravitational resistance long enough to examine where the transmission operates after its initial backshift. The road rises to  $18^\circ$  between 348 m and 356 m, then maintains that grade for 240 m. As before, backshift allows the vehicle to slow without requiring the same reduction in engine speed. The useful comparison is therefore not just how far the CVT backshifts, but where that ratio places the engine and how much power it can supply for the climb.

**Figure 4.23** follows the resulting backshift and primary speed for R00 and five hardware variations. Each trajectory retains the state inherited from the preceding road. The adjoining engine characteristic relates the primary speed in panel (b) to the input power supplied for the climb.

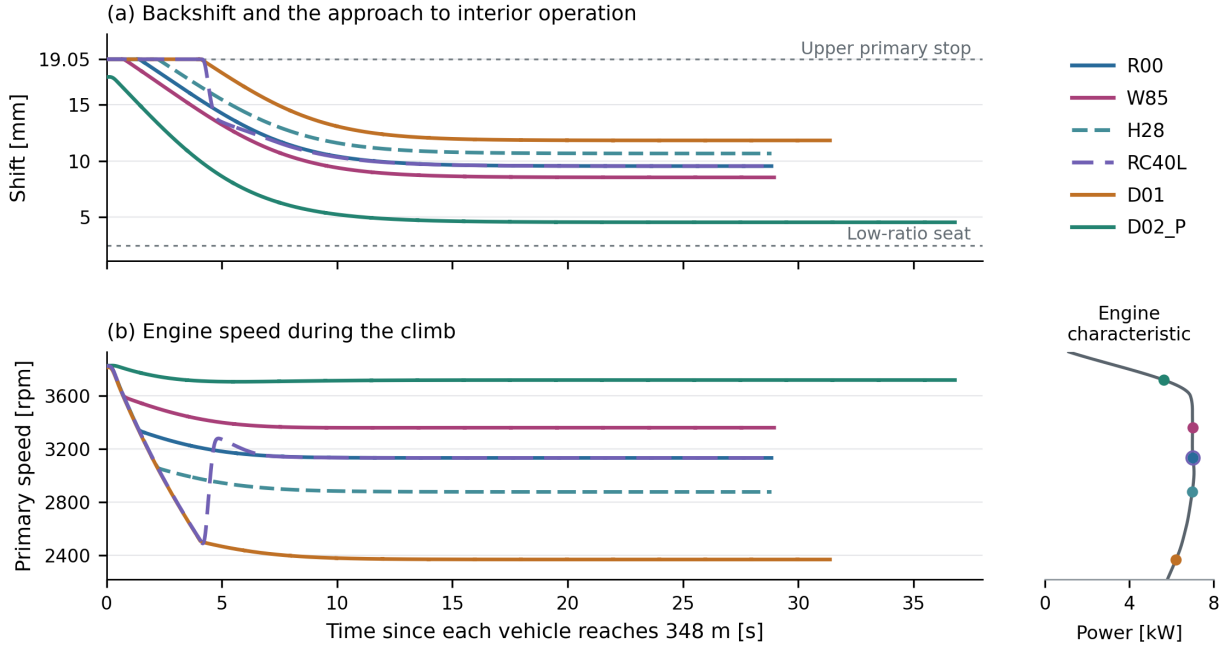


Figure 4.23: Approach to sustained climbing on the  $18^\circ$  hill. (a) Shift position, with the common upper primary stop and low-ratio seat marked. (b) Primary speed. The engine-power curve on the right shares its speed scale, with coloured markers at the late mean primary speeds. Time is measured from each vehicle's arrival at 348 m; each trace ends at 596 m, before the grade decreases. Equal elapsed time does not imply equal road position. The later R00 and RC40L traces nearly coincide in both panels; their power-curve markers also overlap.

R00 initially slows while still at the upper stop. Falling primary speed weakens centrifugal closing until the primary can reopen under the secondary's closing action. It then backshifts, limiting the further fall in engine speed. The motion becomes small well before reaching the low-ratio seat. Unlike the supported states in the preceding subsection, this nearly fixed geometry is maintained within the available travel: primary flyweight closing balances the spring and belt opening forces, while the secondary spring and helix closing balance its belt reaction. Both balances are satisfied without a stop reaction. The belt contacts remain sticking throughout the displayed climb in all six cases.

RC40L takes a different path to almost the same final condition. Its modified late ramp sustains primary closing farther into the hill, delaying departure from the upper stop to about 407 m, compared with 370 m for R00. Engine speed falls farther while the ratio stays fixed, then recovers as the rapid backshift begins. At their common later shift position of approximately 9.55 mm, the roller contact lies about 15 mm along the physical ramp, within the first 18 mm that RC40L leaves unchanged. The altered ramp delays backshift, but the later climbing geometry brings the roller back onto the reference profile and leaves the local force balance almost unchanged.

The remaining cases show what happens when the hardware change continues to affect that interior balance. Table 4.10 pairs the late shift positions and engine speeds with vehicle speed and engine input power. These quantities must be read together: once acceleration becomes small, the engine

must supply approximately the power consumed by gravity, rolling resistance and aerodynamic drag at the resulting climbing speed.

Table 4.10: Late operating conditions on the constant  $18^\circ$  grade. Values are time-weighted means over the final five-second window before 596 m; all six cases have free shift and sticking contacts throughout these windows.

| Case  | Shift [mm] | Primary [rpm] | Speed [m/s] | Input power [kW] |
|-------|------------|---------------|-------------|------------------|
| R00   | 9.55       | 3132          | 7.23        | 7.03             |
| RC40L | 9.55       | 3132          | 7.23        | 7.03             |
| W85   | 8.54       | 3360          | 7.20        | 7.00             |
| H28   | 10.67      | 2877          | 7.19        | 6.99             |
| D01   | 11.83      | 2368          | 6.42        | 6.22             |
| D02_P | 4.55       | 3719          | 5.85        | 5.64             |

W85 and H28 move to opposite sides of R00 in ratio and engine speed, yet climb at almost the same vehicle speed. W85’s lighter tips weaken primary centrifugal closing at a given speed and geometry, allowing more backshift. Its lower ratio keeps the engine turning faster for nearly the same output speed. H28 changes the opposing side of this balance: its larger helix angle reduces secondary axial closing for a given reacted torque, permitting greater primary closure and a higher ratio. Its engine consequently turns more slowly. Despite spanning approximately 2877–3360 rpm, these cases and R00 all operate in the nearly constant-power region of the engine curve, at about 7.0 kW. Their similar climbing speeds therefore coexist with distinctly different internal ratios and shaft speeds.

Moving farther in either direction does not preserve that power. D01’s combined changes leave it farther upshifted, with the engine near 2368 rpm, supplying only 6.22 kW. D02\_P retains the light flyweights and backshifts much farther, raising engine speed to about 3719 rpm. At that speed the supplied engine characteristic is already on its falling-power side, so the higher speed supplies only 5.64 kW. Their lower powers support lower climbing speeds of about 6.42 m/s and 5.85 m/s, respectively, while both contacts still stick. The difference is therefore attributable to where the coupled mechanism places the engine, rather than to exhausted belt traction or a travel stop. Additional backshift is useful when it preserves engine power; beyond that, a lower ratio alone does not make the vehicle climb faster.

#### 4.5.5 Reverse Power on the Descent

On the final  $-22^\circ$  descent, gravity can drive the vehicle and return power through the CVT to the engine. Both shafts continue to turn forward; the change is in which side drives the belt. [Figure 4.24](#) follows R00 from the level approach through the descent and onto the final level road. The two boundary powers use the same sign convention: positive power adds energy to the corresponding shaft-side system, while negative power removes it.

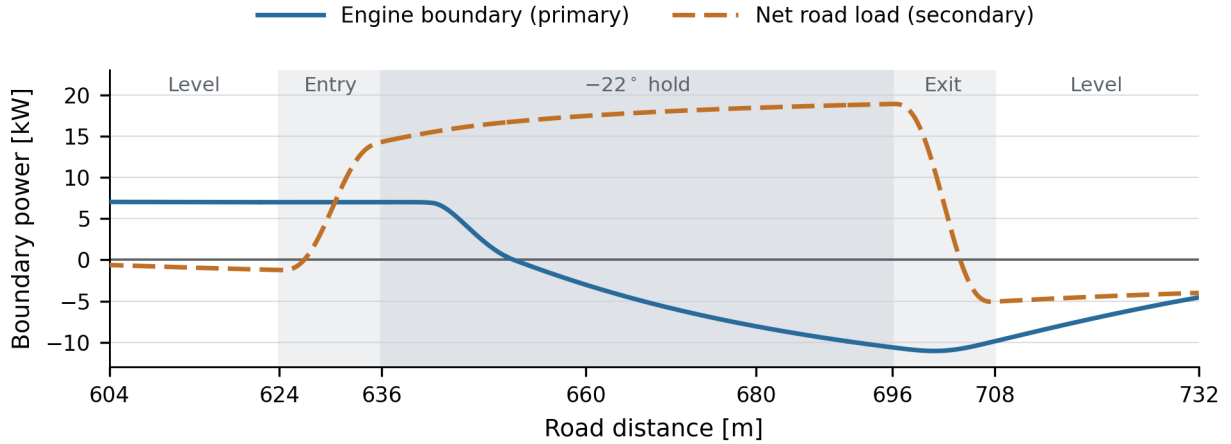


Figure 4.24: R00 boundary powers during the final descent. The primary trace is  $\tau_{B,p}\omega_p$ ; the secondary trace is  $\tau_{B,s}\omega_s$ , equal to the net road force times vehicle speed. Grey shading marks the descent, with darker shading for the constant  $-22^\circ$  grade. Negative primary power denotes absorption by the engine boundary. These powers need not balance while the vehicle and transmission gain or lose kinetic energy.

The road-side power becomes positive once gravity exceeds rolling resistance and aerodynamic drag. Initially the engine still supplies power as well. Near 652 m, the primary passes 4000 rpm, where the supplied engine torque changes sign. The engine now absorbs part of the energy supplied by the descent. The road supplies more power than the engine absorbs, so the vehicle continues to accelerate on the constant slope. The faster descent also releases more gravitational energy each second. Over the speeds reached here, the increase in gravitational power exceeds the increase in aerodynamic and rolling losses, so net road-side power continues to rise. On the final level road, both boundary powers become negative. The vehicle then slows, and its stored kinetic energy sustains the remaining return of power to the engine.

This return path still requires sufficient belt clamping. R00 reaches the upper primary stop near 637 m and remains there, but fixing the pulley geometry does not remove the traction limits. During reverse power, the belt resists the secondary and drives the primary. Both contacts remain sticking: over the descent entry, hold and exit, the primary uses at most about 0.19 of its static capacity and the secondary about 0.20. The existing clamping therefore carries the reversed torque with substantial capacity remaining.

Full throttle remains prescribed throughout this calculation. The negative engine torque comes from the extrapolated high-speed extension described in [Appendix C.2.2](#), not a measured closed-throttle CH440 braking characteristic. The absorption magnitude consequently depends on that supplied boundary. The CVT carries this reverse power through the same coupled shaft, belt and support mechanics used earlier for forward drive.

#### 4.5.6 What the Common Course Establishes

The common course has shown the CVT responding as a complete dynamic component between the engine and vehicle. Its motion and the internal conditions needed to sustain it have been determined together as the loading changes. Successful drive and loss of progress have therefore

been explained within the same formulation, through the forces carried by the hardware and the limits of its contacts and supports.

The sheaves that squeeze the belt also set its running radii, so the actuation responsible for grip participates in changing or holding the ratio. While both contacts stick, that ratio links engine and vehicle speeds. The resulting shaft speeds and transmitted torque influence primary centrifugal and secondary helix action, while sheave motion changes spring loading and actuator geometry. A hardware change therefore alters the forces and the operating conditions under which those forces are produced.

A tune must consequently be judged by the loaded drive it produces. Enough traction and a ratio that makes useful engine power available have to be achieved by the same hardware as it responds to changing loads. Within the adopted component and boundary laws, the formulation makes that connection traceable. A vehicle response can be followed back to the forces and constraints that produced it, and a proposed hardware change can be followed forward through the operating conditions it creates.

## Chapter 5

# Conclusion and Future Work

### 5.1 Conclusion

TODO

### 5.2 Future Work

The present formulation follows how the shafts, actuators, and belt respond together as the CVT changes operating condition. Its assumptions also identify where that description can be extended. Some omitted forces can be added to the existing free bodies; other changes would let the belt move and deform in ways that its present coordinates cannot describe. The extensions below examine how those effects change the transient response.

Much of the local physics needed for these extensions is already described in the literature reviewed in [Section 1.3](#). Adding those mechanisms must preserve the way the belt, pulleys, and actuators respond to one another over time. Some descriptions assume steady running, prescribe part of the motion, or determine a component's position through instantaneous force balance. Where these restrictions remove an interaction needed here, the local relations must be reconsidered and coupled to the equations of motion. Belt loading must affect how the hardware accelerates, while that motion changes the contact conditions and loading in return.

#### Actuator Friction and Directional Shifting

The actuator mechanisms can also resist motion in a direction-dependent way. The same mechanism position therefore need not produce the same clamping force while the CVT shifts in opposite directions. Acceleration can already contribute to this distinction in the present model because part of the applied loading accelerates the actuator itself. Friction in the ramps, helix, and guides would add resistance that changes direction with the relative motion. Static friction could also delay the start of motion or hold a sheave away from a travel stop. These effects are excluded by [Assumption 3 \(Idealized Actuator Mechanics\)](#); their forces would enter the primary free body and secondary helix balance in [Equations \(3.2.1\)](#) and [\(3.2.20\)](#).

In his tuning handbook, [Aaen \(2007\)](#) describes how reversal of friction at the secondary cam and sliding buttons delays backshift and allows engine speed to fall. [Kim et al. \(2002\)](#) include friction in their mechanical-actuator relations, while [Cammalleri \(2005\)](#) distinguishes its direction at the start

of upshift and downshift. Adding these forces to the dynamic balances would let us examine how much actuator friction contributes to the difference between upshift and backshift curves during complete maneuvers, including how engine speed recovers after a load increase.

That difference also depends on belt contact and inertia. Comparisons at different shift rates, with actuator friction characterised independently, could help distinguish these causes of delayed reversal. For the roller mechanisms considered here, the relevant resistance may come from roller bearings and rolling contact rather than the sliding buttons discussed by Aaen. In either case, friction changes the contact reaction that generates the clamping force, so it must be included in the actuator balance.

## Face Torque and Dynamic Helix Response

The secondary actuator also depends on how belt torque is divided between its two sheave faces. The movable face rotates relative to the shaft as it shifts, so the belt generally encounters a different surface velocity on each face. The current formulation includes this relative rotation in the actuator and shaft dynamics and uses the power-equivalent contact speed in [Equation \(3.5.2\)](#). What it does not determine is the division of torque between the two contacts: [Assumption 12 \(Symmetric Sheave-Face Sharing\)](#) supplies  $\tau_{s,M} = \tau_s/2$  in [Equation \(3.2.21\)](#).

[Sorge and Cammalleri \(2011\)](#) show why this division deserves closer attention. Their separate face-contact calculations produce substantially unequal torques, with circumferential tractions that can oppose one another over parts of the wrap. In some calculated conditions, even the total torque on the faster half-pulley is negative. Their analysis accounts for how relative face motion changes contact while neglecting inertia other than centrifugal loading. The present formulation includes the torque needed to accelerate the movable assembly but uses the prescribed equal share for the two face contacts. The two descriptions address different consequences of the same motion.

[Equation \(3.2.20\)](#) shows how the face torque enters the helix balance. The helix reacts the belt and spring torques remaining after the movable assembly has been accelerated. Replacing the prescribed half-share by the calculated movable-face torque changes that reaction and hence the axial force in [Equation \(3.2.30\)](#). The changed acceleration alters the relative face motion, which changes the contact conditions that supplied the torque. The shaft balance responds at the same time. Contact power would also be calculated from each face's torque and speed, replacing the equal-share reduction in [Equation \(3.5.1\)](#).

The changes in shift motion found in [Section 4.4.2](#) make this coupling worth investigating during smooth shifting, rapid backshift, and torque reversal. Comparing separate face contacts with and without the helix dynamics would show whether their interaction changes sheave motion or the approach to a contact or support transition. The separate torques and helix reaction could show changes that are not visible in total transmitted torque alone. This would help determine when the equal-share approximation is sufficient for predicting the shift response.

## Secondary Contact and Clearance

The helix reaction also raises a question about the contact that carries it. The reference mechanism uses a slot with no clearance, so either flank can carry the helix reaction ([Appendix C.1.1](#)). When the reaction changes direction, the load transfers immediately to the opposite flank. A one-sided helix could instead separate, while finite clearance in a slot would allow relative motion before the opposite flank is reached. Comparing these arrangements under the reaction reversals in [Section 4.4.2](#) would show how clearance or separation changes the response.

While flank contact is absent, the movable assembly’s rotation and axial motion would no longer be linked by the helix geometry. Allowing those motions to evolve independently would determine when contact returns and how clamp develops during the interval. The model would need to allow that extra motion; adding friction to the existing balance would leave the helix constraint in place. The opening of a helix contact remains distinct from belt lift-off, which depends on the normal loading at the belt–sheave interface.

## Belt Deformation, Transport, and Contact

Beyond the actuator mechanisms, the belt itself can be given more freedom to move and deform. In the present formulation, it couples the two pulleys through a fixed length, a single transport speed, and prescribed section geometry. With the fixed length of [Assumption 6 \(Fixed Belt Length\)](#), the compatibility relation in [Equation \(3.1.25\)](#) determines the secondary position from the shared shift coordinate. The belt could instead stretch to take up part of a sheave displacement. Tension would then depend on how much the belt stretches, and [Equation \(3.1.32\)](#) would no longer determine the secondary position from the primary position alone. Each movable sheave may then need its own position coordinate and equation of motion.

The dynamic belt model of [Ballew \(2015\)](#) provides a starting point for examining this coupling. It already calculates belt-node motion and both pulley speeds together, but its running-radius search matches belt axial reaction to applied clamp without calculating the movable sheave’s axial acceleration from its own force balance. Adding that balance would let sheave and actuator inertia affect the response. Such a comparison could help explain the different clamp-to-shift responses identified in [Section 4.3.4](#).

[Assumption 7 \(Single Belt Transport Speed\)](#) gives the whole loop one tangential transport speed. Allowing that speed to vary from place to place would add motion that need not come from belt stretch: [Carbone et al. \(2005\)](#) already describe local speed changes caused by changing geometry in a belt that does not stretch. Following this motion over time would require reconsidering the steady path and constant mass flow assumed by [Messick \(2018\)](#).

Those speed differences also change how contact can be determined. A single belt–pulley speed difference would no longer generally describe relative motion throughout the wrap. The present equations describe how tension and normal loading vary, but [Assumption 11 \(Wrap-Level Contact Resolution\)](#) assigns one contact state and one traction utilisation  $\lambda_j$  to each wrap through [Equation \(3.4.18\)](#). Different motions along the wrap could produce adhering and sliding regions, or regions with opposing traction, within the same wrap. A local Coulomb law could still represent friction, with contact states and traction determined along the wrap rather than assigned to it as a whole.

The relative velocity must be calculated between the contacting rubber and the sheave surface. Under [Assumption 4 \(Fixed, Uniform Belt Cross-Section\)](#), the section geometry and cord location are prescribed. Compression and bulging would change belt width and contact shape, altering the relation between sheave position and running radius. Shear and warping allow the rubber between the tension-carrying layer and the sheave to deform. As [Gerbert \(1996\)](#) shows, the cord and pulley can then have different speeds while the rubber surface adheres. Bending stiffness also affects seating and unseating as the belt enters and leaves a wrap; Gerbert shows that these processes can produce friction opposing the main torque-transmitting direction at the primary.

Sectional analysis offers a way to retain these effects without representing every cord or cog around

the loop. [Gerbert \(1996\)](#) uses it to construct an effective shear relation. Similar reductions could supply sectional forces, moments, and deformed contact geometry to the belt model. If a section adjusts much faster than the overall belt motion, the inertia of its internal motion may be negligible even though its deformation changes contact. Representing the distribution of cord and rubber mass more accurately could also improve the inertia calculation without resolving every local deformation.

Under [Assumption 10 \(Gross Tangential Coulomb Traction\)](#), friction acts only in the circumferential direction, leaving out resistance to radial migration. Traction along a sheave face can have both radial and circumferential components. Resistance to changing running radius and resistance to tangential slip therefore share the available contact traction. [Kim et al. \(2002, p. 645\)](#) use this connection to explain the different coefficients in their experimentally identified upshift and downshift laws. Their proposed pseudo-self-locking mechanism occurs at the primary belt–groove interface: outward radial friction during downshift can support a condition resembling steady running, whereas its reversal during upshift changes the radial and circumferential sliding. This supplies a belt-contact contribution to directional shifting alongside actuator friction.

The groove-contact model of [Kong and Parker \(2006\)](#) and shifting analysis of [Sorge \(2008\)](#) provide ways to calculate friction direction from local relative motion. Using these forces in the belt and sheave balances would show how radial resistance, torque transmission, and axial thrust develop together during shifting. A friction coefficient that varies with pressure or sliding speed would provide a further extension. Its effect on available traction could also be examined using the present belt coordinates.

These deformations and contact motions also change how mechanical work is stored and dissipated. Elasticity stores and returns energy; viscoelasticity adds dependence on deformation history and can dissipate energy during repeated stretching, compression, shear, and bending. Surface sliding dissipates work at the belt–sheave interface. The entry and exit processes considered by [Bertini et al. \(2014\)](#) recur as the belt circulates, so their contribution to loss continues after initial clutching. Messick measures temperature- and frequency-dependent rubber properties but uses an elastic approximation in the mechanical model; Ballew identifies independent damping measurements as unfinished work ([Messick, 2018](#); [Ballew, 2015](#)). A material law for this motion would need the deformation history of each belt section as it passes through the spans and wraps. Adding stored strain energy and material dissipation to the energy balance in [Equation \(4.2.2\)](#) would show how much of a drop in output power comes from energy being stored in the belt and how much is lost through sliding and material damping.

With compliance and inertia included, changes in tension and deformation could travel around the loop after a sudden load change. [Messick \(2018\)](#) identifies engine torque that varies with crankshaft angle as a possible source of vibration outside the scope of his steady model. A transient belt model could show how strongly these torque fluctuations affect contact loading and traction. Repeating the calculation with finer resolution along the belt and smaller time steps would help distinguish physical vibration from numerical oscillation, particularly given the stiffness difficulties reported by [Ballew \(2015\)](#).

## Belt Path, Support, and Re-engagement

The above belt motions can still be described while the loop remains on the prescribed circular wraps and straight spans. Relaxing the path itself is a separate extension. The circular wraps and straight spans of [Assumption 8 \(Prescribed Planar Belt Path\)](#) remove possible motions before the force balances are solved. The checks in [Equation \(3.4.37\)](#) can identify a negative tension or local

normal load. They cannot determine the shape of a slack belt or one that has partly lifted away from a pulley. Solving for the belt's shape and where it contacts the pulleys would allow the model to follow what happens next.

This would be especially useful while the primary is disengaged. Under [Assumption 9 \(Reduced Belt Motion While the Primary Is Disengaged\)](#), the belt is carried without slip by the fully closed secondary through [Equation \(3.6.5\)](#). A belt with its own deformation and contact motion could instead determine installed pretension, slack take-up, support at the primary hub, intermittent sheave contact, and possible secondary slip. Initial engagement would then depend on the belt's shape and motion at that time. These are changes to the physical description of engagement, beyond refining its event time.

Existing models already solve for belt paths and contact boundaries ([Kong and Parker, 2005, 2006; Julió and Plante, 2011](#)). The question here is how the belt and actuators respond to one another during disengagement and re-engagement when the belt is free to leave the prescribed path. The difficulty reported by [Julió and Plante \(2011\)](#) in obtaining stable results near the clutching-to-shifting transition makes low-load engagement a particularly useful test.

Planar motion may be sufficient for some studies of tension redistribution and slackness. Axial migration, skew, and section twist require out-of-plane motion. A floating secondary can also move along its shaft as a whole, separately from opening its groove. Pulley compliance and shaft misalignment are separate extensions of [Assumption 2 \(Rigid Pulley and Actuation Hardware\)](#) and [Assumption 5 \(Aligned Pulley Axes and Fixed Centre Distance\)](#). The metal-belt and chain-CVT studies of [Carbone et al. \(2005\)](#) and [Duan et al. \(2016\)](#) show that hardware deformation can alter contact. For the loads considered here, its effect would need to be compared with that of rubber deformation.

## Friction and Compliance During Finite-Speed Events

The extensions above concern how the CVT moves while its active constraints remain unchanged. Once engagement or a travel stop creates a new constraint at finite relative speed, the event itself introduces a different set of contact questions. The impulse-momentum balance in [Equation \(3.6.51\)](#) determines the resulting velocity change. This treatment allows the simulation to continue through the event. Two extensions would examine which contact details can change its outcome.

The first concerns friction during the instantaneous event. Under [Assumption 14 \(Tangential Contact Through Boundary Events\)](#), an existing sticking belt contact transmits the impulse required to remain stuck without a separate friction capacity check. A friction law relating tangential and normal impulses could determine whether sticking can be maintained or the event produces tangential slip. The constraints immediately after the event in [Equation \(3.6.44\)](#) would then depend on the contact impulses allowed by that law, following the frictional-impact framework discussed by [Stewart \(2000\)](#).

The second concerns contact duration. The instantaneous, perfectly inelastic treatment in [Assumption 13 \(Instantaneous, Perfectly Inelastic Constraint Activation\)](#) determines the motion and energy change across the event, but cannot resolve peak force, rebound, or vibration during capture. Following contact deformation over time would allow the model to calculate these forces and motions. Friction-limited impulses and finite-duration contact therefore answer different questions: whether the contact can remain stuck after the event, and how much contact deformation and duration

change the response. The instantaneous description may remain sufficient for the overall shift motion even where resolving peak loads requires compliance.

## Temperature During Transient Operation

The losses in the belt and its contacts can also change the properties that determine the motion that follows. Under [Assumption 1 \(Time-Invariant Material and Surface State\)](#), those properties remain fixed throughout a run. The measurements of [Messick \(2018\)](#) show that rubber stiffness and dissipation depend on temperature and loading frequency. The same maneuver may therefore produce a different response from a cold or warm belt. The question is whether fixed properties at the measured initial temperature are sufficient, or whether heating during the maneuver changes the mechanics enough that the model must follow temperature as it changes.

The temperature of the belt as a whole is only part of the question. Local contact can heat over much shorter distances and times than the bulk material. [Persson \(2006\)](#) describes how flash heating feeds back into rubber friction through the temperature-dependent material response. In a CVT, heat generated during clutching or brief slip may change the friction available as traction recovers. Its importance at the belt–sheave interface remains to be established. A nearly unchanged bulk temperature during a short maneuver would not, by itself, show that flash heating is negligible.

## Experimental Validation and Characterisation

These extensions also require experiments that can separate the effects they are intended to represent. Different operating conditions allow different parts of the model to be tested. While a travel stop holds the sheaves in place, loading can be varied without changing the pulley geometry. Free shifting then allows the predicted sheave motion to be tested. A travel stop also carries a reaction: the belt load need not equal the actuator force because the stop carries the remainder of the axial balance. [Grzegózek et al. \(2016\)](#) identify this limitation when inferring belt loads from regulator characteristics. The reactions in [Table 3.2](#) therefore require measurements beyond the actuator law if supported operation is to test the predicted belt loading.

Measurements during shifting should also include more than shaft speeds. In the model comparison of [Section 4.3.2](#), feedback brings the shaft speeds close to the reference traces while producing a substantially different primary clamp. Applying the reference force directly in [Section 4.3.2](#) removes the feedback adjustment and gives a different response. Sheave displacement distinguishes geometric shifting from the speed changes produced by slip; belt speed, torque, and selected actuator loads can further distinguish the causes. [Bertini et al. \(2014\)](#) measure both thrust and torque in their tests. For the secondary face-torque coupling, movable-face torque or helix reaction would be more informative than total shaft torque alone. If a measured force is supplied as an input, agreement in that force does not test the model’s ability to predict it.

The shaft boundaries defined in [Section 2.1](#) allow these comparisons to use a wide range of test arrangements. An engine or motor can drive the primary, while a brake, rotating inertia, or attached drivetrain loads the secondary. Their influence enters through the shaft interface in [Equation \(C.2.1\)](#), so changing the apparatus need not change the internal CVT formulation. With the applied loading and initial state specified, measurements can test predicted shaft and sheave motion through engagement, acceleration, upshift, backshift, and traction recovery. Different arrangements allow different loads and measurements, so the predictions that can be tested depend on what the rig applies and independently measures.

Material and component properties also need independent measurements. Geometry, inertia, and spring characteristics establish the mechanical inputs. Actuator friction requires measurements through holding, breakaway, and reversal, while belt tests must distinguish elastic storage from dissipation under relevant stretching, compression, shear, and bending. The tests described by [Balley \(2015\)](#) and [Messick \(2018\)](#) provide foundations, with further work needed on damping and dependence on loading history. Belt–sheave friction also requires independent characterisation over the relevant loads, speeds, and temperatures. These measurements would allow complete-CVT experiments to test how the belt, pulleys, and actuators respond together without adjusting the physical parameters to match each maneuver.

## Recovery and Assessment of the Reduced Model

The present formulation can serve as a common reference for these extensions. Wherever a reduction is well defined, applying the same assumptions and closure relations again should lead back to the equations used here. The reduced model should then reproduce the same response, within numerical accuracy, under corresponding loading and initial conditions.

With the added mechanisms active, the next question is how much they change the response. The comparisons in [Section 4.4.3](#) show why the equations need to be solved again: a term’s size in one run does not tell us how much the motion changes when that term is removed. Changes in clamp demand, shift timing, traction recovery, or contact transitions would show which predictions depend on the added mechanics. Experiments could then test whether the new predictions agree more closely with measurements. If the added mechanisms make little difference to the predictions needed for a component or design study, the simpler model may be enough. Each extension could then help show which questions the present formulation can answer and which need more detail.

## Author Contributions (CRediT)

**Kai Arseneau:** Conceptualization; Methodology; Software; Formal analysis; Visualization; Writing – original draft; Writing – review & editing.

**W. Spencer Smith:** Supervision; Writing – review & editing.

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## Appendix A

# Construction of Primary Flyweight Geometry Maps

The primary actuator dynamics derived in the main text do not require a particular flyweight shape directly. They require the mechanism maps

$$q_f = q_f(x_p), \quad \frac{dq_f}{dx_p}, \quad \frac{d^2q_f}{dx_p^2}, \quad J_{f,p} = J_{f,p}(x_p), \quad \frac{dJ_{f,p}}{dx_p}, \quad (\text{A.0.1})$$

together with the constant pivot-axis inertia  $I_f$ . This appendix constructs those quantities for the fixed-pivot roller flyweight used by the present primary mechanism.

The construction follows the physical mechanism directly. A ramp attached to the movable sheave defines the contact surface. The finite roller radius offsets that surface to a roller-centre locus. Intersecting that locus with the rigid flyweight-arm circle determines the assembled flyweight pose and therefore  $q_f(x_p)$ . Differentiating the same geometric constraint supplies the motion ratios required by the acceleration-level actuator model. Finally, the flyweight mass distribution is evaluated at the resulting pose to obtain  $I_f$ ,  $J_{f,p}(x_p)$ , and  $dJ_{f,p}/dx_p$ .

No independent initial flyweight angle is introduced. The assembled pose is a consequence of the pivot, ramp, roller, and arm geometry.

### A.1 Fixed-Pivot Flyweight Geometry

Consider the axial–radial plane of one primary flyweight. Let  $O$  denote the shaft centre, let  $+\mathbf{e}_x$  point in the positive primary closing direction, and let  $+\mathbf{e}_r$  point radially outward. The flyweight pivots about the fixed point

$$\mathbf{r}_P = \begin{bmatrix} x_P \\ r_P \end{bmatrix}, \quad (\text{A.1.1})$$

which is fixed to the pulley body. The movable sheave carries the physical ramp. The roller centre is denoted by  $C$ , the physical roller–ramp contact point by  $K$ , and the reference point at the start of the active ramp by  $A$ .

The rigid distance from the pivot to the roller centre is

$$\boxed{L_f = |PC|}, \quad (\text{A.1.2})$$

and the roller radius is  $R_{\text{roll}}$ . The flyweight angle  $q_f$  is measured from the positive axial reference to the line from  $P$  to  $C$ , with increasing  $q_f$  corresponding to outward flyweight rotation. Thus

$$\mathbf{r}_C(q_f) = \mathbf{r}_P + L_f \begin{bmatrix} \cos q_f \\ \sin q_f \end{bmatrix}. \quad (\text{A.1.3})$$

The arm is therefore constrained to a circle of radius  $L_f$  about  $P$ .

### A.1.1 Physical Ramp Representation

Introduce a scalar ramp coordinate  $\xi$ , with  $A$  corresponding to  $\xi = 0$ . Let  $h(\xi)$  be the radial profile measured from  $A$ , and let  $d_x \in \{-1, +1\}$  specify whether increasing  $\xi$  proceeds in the positive or negative physical axial direction. For movable-sheave closure  $x_p$ , a point on the physical ramp is

$$\mathbf{r}_K(\xi, x_p) = \begin{bmatrix} x_A + x_p + d_x \xi \\ r_A + h(\xi) \end{bmatrix}. \quad (\text{A.1.4})$$

The present Baja parameterization uses  $d_x = -1$ : increasing profile coordinate proceeds axially toward the pivot while the ramp rises radially outward.

Ramp angle is measured from the axial reference line, so  $0^\circ$  denotes an axial ramp. For a graph of the form in Equation (A.1.4), the local unsigned ramp angle satisfies

$$\alpha(\xi) = \tan^{-1} |h'(\xi)|. \quad (\text{A.1.5})$$

Accordingly, a  $20^\circ$  ramp is nearly axial rather than nearly radial.

The unnormalized tangent is

$$\mathbf{t}(\xi) = \begin{bmatrix} d_x \\ h'(\xi) \end{bmatrix}, \quad \|\mathbf{t}\| = \sqrt{1 + [h'(\xi)]^2}. \quad (\text{A.1.6})$$

With  $\sigma_n \in \{-1, +1\}$  selecting the physical side of the ramp on which the roller lies, a compatible unit normal is

$$\mathbf{n}(\xi) = \sigma_n \frac{\begin{bmatrix} -h'(\xi) \\ d_x \end{bmatrix}}{\sqrt{1 + [h'(\xi)]^2}}. \quad (\text{A.1.7})$$

The sign  $\sigma_n$  is fixed once from the assembled mechanism and is then used throughout the active ramp range.

The profile  $h$  may be assembled from straight, circular, spline, or other smooth segments. Those shapes define only the physical contact surface; they do not themselves prescribe the flyweight motion or force.

### A.1.2 Finite-Radius Roller-Centre Locus

The roller centre does not lie on the physical ramp. At contact it is offset by one roller radius along the selected ramp normal:

$$\mathbf{r}_C(\xi, x_p) = \mathbf{r}_K(\xi, x_p) + R_{\text{roll}} \mathbf{n}(\xi). \quad (\text{A.1.8})$$

This finite-radius offset is the roller-centre path used by the mechanism. The point  $K$  remains the physical contact point;  $C$  is the centre of the roller bearing.

The contact problem can now be viewed geometrically as the intersection of the roller-centre offset curve with the rigid circle of radius  $L_f$  about  $P$ . A point-contact or zero-radius follower approximation is not used for the present mechanism.

### A.1.3 Rigid-Arm Contact Constraint and Assembled Branch

A roller-centre location is compatible with the rigid flyweight when

$$\boxed{g(\xi, x_p) = \|\mathbf{r}_C(\xi, x_p) - \mathbf{r}_P\|^2 - L_f^2 = 0.} \quad (\text{A.1.9})$$

For any fixed  $x_p$ , every root of Equation (A.1.9) is a mathematical flyweight configuration that satisfies the rigid arm and roller contact geometry.

Define

$$a = x_C - x_P, \quad b = r_C - r_P. \quad (\text{A.1.10})$$

Then

$$a^2 + b^2 = L_f^2, \quad (\text{A.1.11})$$

and the corresponding flyweight angle is

$$\boxed{q_f(x_p) = \text{atan2}(b, a).} \quad (\text{A.1.12})$$

Thus the starting angle is recovered from the assembled geometry rather than specified as an independent parameter.

Multiple mathematical roots can exist at the same  $x_p$ . At the beginning of the declared operating interval,  $x_p = x_{p,\min}$ , enumerate the admissible contact configurations and initialize the assembled branch with

$$\boxed{q_f(x_{p,\min}) = \min_i q_{f,i}(x_{p,\min}).} \quad (\text{A.1.13})$$

This smallest- $q_f$  rule is an *initial assembly rule only*. For increasing or decreasing closure thereafter, the same solution branch is followed continuously. The flyweight is not permitted to jump to a disconnected arm orientation simply because another root has a smaller angle at a later position.

Alternate arm orientations that separately satisfy Equation (A.1.9) are not themselves a contact failure. They are mathematically distinct assembled configurations. A different condition occurs when the *same selected roller pose* contacts two distinct locations of the ramp simultaneously; that true two-point contact lies outside the present single-contact mechanism.

For the selected branch the geometry chain is therefore

$$\boxed{h(\xi) \longrightarrow \mathbf{r}_C(\xi, x_p) \longrightarrow g(\xi, x_p) = 0 \longrightarrow q_f(x_p).} \quad (\text{A.1.14})$$

## A.2 Contact-Branch Kinematics

The main actuator dynamics require  $q'_f(x_p)$  and  $q''_f(x_p)$ . These quantities follow from the same physical contact constraint rather than from an independent fitted motion law.

Along the selected branch,

$$g(\xi(x_p), x_p) = 0. \quad (\text{A.2.1})$$

Differentiating with respect to  $x_p$  gives

$$g_\xi \xi' + g_x = 0, \quad (\text{A.2.2})$$

where  $g_x \equiv \partial g / \partial x_p$ , and therefore

$$\boxed{\xi' = -\frac{g_x}{g_\xi}}. \quad (\text{A.2.3})$$

Differentiating once more gives

$$g_{xx} + 2g_{x\xi}\xi' + g_{\xi\xi}(\xi')^2 + g_\xi\xi'' = 0, \quad (\text{A.2.4})$$

and hence

$$\boxed{\xi'' = -\frac{g_{xx} + 2g_{x\xi}\xi' + g_{\xi\xi}(\xi')^2}{g_\xi}}. \quad (\text{A.2.5})$$

These relations require

$$\boxed{g_\xi \neq 0} \quad (\text{A.2.6})$$

along the selected branch. A zero value corresponds to a fold or dead-centre at which the local contact map cannot be continued smoothly in  $x_p$ .

For compactness write

$$\mathbf{c}(\xi, x_p) \equiv \mathbf{r}_C(\xi, x_p). \quad (\text{A.2.7})$$

Its total first and second derivatives along the selected branch are

$$\boxed{\mathbf{c}' = \mathbf{c}_x + \mathbf{c}_\xi \xi'}, \quad (\text{A.2.8})$$

and

$$\boxed{\mathbf{c}'' = \mathbf{c}_{xx} + 2\mathbf{c}_{x\xi}\xi' + \mathbf{c}_{\xi\xi}(\xi')^2 + \mathbf{c}_\xi\xi''}. \quad (\text{A.2.9})$$

The explicit  $x_p$ -dependence is particularly simple because the ramp carrier undergoes rigid axial translation, but the chain-rule form above keeps the construction general and makes the geometric origin of every derivative clear.

Using  $a$  and  $b$  from [Equation \(A.1.10\)](#), the first angle derivative is

$$\boxed{q'_f = \frac{ab' - ba'}{L_f^2}}. \quad (\text{A.2.10})$$

Because  $a^2 + b^2 = L_f^2$  is constant, the second derivative reduces to

$$\boxed{q''_f = \frac{ab'' - ba''}{L_f^2}}. \quad (\text{A.2.11})$$

The angle, motion ratio, and change of motion ratio are therefore all direct consequences of the ramp, roller, pivot, and arm geometry.

### A.2.1 Smoothness of the Dynamic Ramp Representation

The finite roller-centre curve contains the normal  $\mathbf{n}(h')$ . Its first profile derivative therefore contains  $h''$ , while its second profile derivative contains  $h'''$ . Since  $q_f''(x_p)$  depends on  $\mathbf{c}_{\xi\xi}$ , the acceleration-level mechanism map requires the active ramp representation to satisfy

$$\boxed{h \in C^3} \quad (\text{A.2.12})$$

through piecewise junctions used by the production dynamic map. A discontinuity in slope, curvature, or the third derivative can contaminate the acceleration terms even when the physical profile itself appears visually continuous.

When idealized or measured ramp pieces do not meet this requirement directly, a short artificial smoothing segment may be introduced explicitly as a model approximation. The provisional Baja ramp used during development consists of

$$20^\circ \text{ linear over } 5 \text{ mm} \longrightarrow 3 \text{ mm } C^3 \text{ blend} \longrightarrow 35^\circ \rightarrow 10^\circ \text{ circular over approximately } 30 \text{ mm}. \quad (\text{A.2.13})$$

The blend is represented by a sixth-order polynomial transition selected so that derivatives through third order match the neighbouring segments. The 3 mm blend length is a modelling choice, not a measured hardware dimension. A sharp-corner finite-roller construction may still be useful for geometry diagnostics, but it is not admitted directly into the production map while  $q_f''$  is retained.

## A.3 Flyweight Mass Properties

Once  $q_f(x_p)$  is known, the remaining mechanism quantities follow from the flyweight mass distribution. Introduce body-fixed coordinates relative to the pivot  $P$ :  $u$  along the nominal arm from the pivot toward the roller centre,  $v$  perpendicular to  $u$  in the axial–radial mechanism plane, and  $z$  circumferentially along the pivot axis.

For one flyweight, whose mass is  $m_f/n_f$ , define

$$\frac{m_f}{n_f} = \int dm, \quad (\text{A.3.1})$$

$$M_u = \int u dm, \quad M_v = \int v dm, \quad (\text{A.3.2})$$

$$S_u = \int u^2 dm, \quad S_v = \int v^2 dm, \quad (\text{A.3.3})$$

$$S_{uv} = \int uv dm, \quad S_z = \int z^2 dm. \quad (\text{A.3.4})$$

These first and second moments may be obtained from a simplified analytic mass model, direct measurement, or CAD.

For  $n_f$  identical flyweights, the inertia about their individual pivot axes is

$$\boxed{I_f = n_f(S_u + S_v)}. \quad (\text{A.3.5})$$

This is the inertia multiplying the relative pivot acceleration in the primary actuator balance.

At flyweight angle  $q_f$ , a mass element has radial shaft coordinate

$$r = r_P + u \sin q_f + v \cos q_f. \quad (\text{A.3.6})$$

Including its circumferential offset  $z$ , integration gives the total flyweight inertia about the primary shaft:

$$J_{f,p}(q_f) = n_f \left[ \frac{m_f}{n_f} r_P^2 + 2r_P(M_u \sin q_f + M_v \cos q_f) + S_u \sin^2 q_f + S_v \cos^2 q_f + 2S_{uv} \sin q_f \cos q_f + S_z \right]. \quad (\text{A.3.7})$$

Composition with the geometry map gives

$$J_{f,p}(x_p) = J_{f,p}(q_f(x_p)), \quad (\text{A.3.8})$$

and

$$\frac{dJ_{f,p}}{dx_p} = \frac{dJ_{f,p}}{dq_f} q'_f. \quad (\text{A.3.9})$$

The same mass description therefore supplies both the pivot inertia and the configuration-dependent shaft-axis inertia required by the dynamic actuator.

### A.3.1 Present Baja Mass Approximation

The present parameterization does not attempt to reproduce the exact outline and centroidal inertia of every flyweight component. For each flyweight, the arm/body mass  $m_a$  is represented as a uniform slender member extending from  $u = 0$  to  $u = L_f$ , while the hardware carried at the outer end is collected into a concentrated tip mass  $m_t$  at  $u = L_f$ ,  $v = z = 0$ . The tip package is

$$m_t = m_{\text{roller/bearing}} + m_{\text{bolt}} + m_{\text{nut/washer}} + m_{\text{fixed hardware}} + m_{\text{tuning}}. \quad (\text{A.3.10})$$

For the uniform arm,

$$M_{u,a} = \frac{m_a L_f}{2}, \quad S_{u,a} = \frac{m_a L_f^2}{3}, \quad (\text{A.3.11})$$

and for the concentrated tip package,

$$M_{u,t} = m_t L_f, \quad S_{u,t} = m_t L_f^2. \quad (\text{A.3.12})$$

Hence

$$M_u = \frac{m_a L_f}{2} + m_t L_f, \quad (\text{A.3.13})$$

and

$$S_u = \frac{m_a L_f^2}{3} + m_t L_f^2. \quad (\text{A.3.14})$$

with

$$M_v = S_v = S_{uv} = S_z = 0, \quad m_f = n_f(m_a + m_t). \quad (\text{A.3.15})$$

The resulting pivot inertia is

$$I_f = n_f \left( \frac{1}{3} m_a L_f^2 + m_t L_f^2 \right). \quad (\text{A.3.16})$$

The corresponding shaft-axis inertia map is

$$J_{f,p}(q_f) = n_f [(m_a + m_t)r_P^2 + 2r_P M_u \sin q_f + S_u \sin^2 q_f]. \quad (\text{A.3.17})$$

Its configuration derivative is

$$\frac{dJ_{f,p}}{dq_f} = 2n_f \cos q_f (r_P M_u + S_u \sin q_f), \quad (\text{A.3.18})$$

so the required axial derivative becomes

$$\frac{dJ_{f,p}}{dx_p} = 2n_f \cos q_f (r_P M_u + S_u \sin q_f) q'_f. \quad (\text{A.3.19})$$

This reduction is an explicit approximation to the hardware mass distribution. It neglects the centroidal inertia associated with the finite roller/bearing radius, bolt diameter and length, nut and washer dimensions, tuning-weight dimensions, arm thickness and nonuniform outline, holes and reliefs, and any off-centre placement of tip hardware not represented by the roller-centre station. These features are not assumed physically absent; they are omitted when reducing the real flyweight to the mass moments above.

For the present assembly, the tuning mass is of order  $> 100$  g, while the measured arm/body mass is approximately 13.646 g per flyweight. The parallel-axis contribution of the tip package is therefore expected to dominate many small centroidal corrections, although this observation does not make the omitted terms identically zero. If greater fidelity is needed, the same governing equations accept CAD- or measurement-derived  $M_u, M_v, S_u, S_v, S_{uv}, S_z$  without changing the actuator dynamics.

### A.3.2 Present Geometry Values

The provisional geometry used during development is summarized below. These values identify the mechanism currently being parameterized; the ramp blend and unverified tip masses should not be interpreted as final metrology.

| Quantity                             | Present value                                                                                        |
|--------------------------------------|------------------------------------------------------------------------------------------------------|
| Number of flyweights, $n_f$          | 3                                                                                                    |
| Pivot radius, $r_P$                  | 1.675 in = 42.545 mm                                                                                 |
| Pivot-to-roller-centre length, $L_f$ | 1.241 in = 31.5214 mm                                                                                |
| Roller radius, $R_{\text{roll}}$     | 6.5 mm                                                                                               |
| Ramp-reference offset from pivot     | $\Delta x_A = 1.5$ in, $\Delta r_A = 0.2776$ in                                                      |
| Measured arm/body mass, $m_a$        | 13.646 g per flyweight                                                                               |
| Profile direction                    | $d_x = -1$                                                                                           |
| Provisional active ramp              | 20° linear over 5 mm, 3 mm $C^3$ blend, then circular 35° $\rightarrow$ 10° over approximately 30 mm |

Table A.1: Present fixed-pivot flyweight geometry and provisional ramp data. The 3 mm smoothing length is a modelling choice rather than a measured hardware dimension.

The roller/bearing, bolt, nut/washer, fixed tip-hardware, and tuning masses are entered individually when measured and then collected into  $m_t$  through [Equation \(A.3.10\)](#).

## A.4 Resulting Mechanism Interface

The construction above supplies exactly the quantities required by the primary actuator dynamics:

$$\boxed{q_f(x_p), \quad q'_f(x_p), \quad q''_f(x_p), \quad I_f, \quad J_{f,p}(x_p), \quad J'_{f,p}(x_p).} \quad (\text{A.4.1})$$

For the fixed-pivot mechanism, the complete chain is

$$\boxed{\text{ramp geometry} \rightarrow \text{roller-centre locus} \rightarrow \text{rigid-arm branch} \rightarrow q_f, q'_f, q''_f \rightarrow \text{mass moments} \rightarrow I_f, J_{f,p}, J'_{f,p}.} \quad (\text{A.4.2})$$

Substitution into the main actuator result in [Equation \(3.2.11\)](#) gives

$$\boxed{F_{p,\text{fly}} = \frac{1}{2}\omega_p^2 J'_{f,p} - I_f(q'_f)^2 \ddot{x}_p - I_f q'_f q''_f \dot{x}_p^2.} \quad (\text{A.4.3})$$

The three terms are respectively the centrifugal generalized closing drive, the pivot inertia reflected into the axial coordinate, and the inertial term created by variation of the mechanism motion ratio.

The same configuration-dependent mass distribution contributes to the owning shaft through

$$\boxed{J_{f,p}\dot{\omega}_p + J'_{f,p}\dot{x}_p\omega_p,} \quad (\text{A.4.4})$$

which is the local  $x_p$ -coordinate form of the primary angular-momentum term used in [Equation \(3.3.4\)](#). Under the assumed circumferential symmetry, the corresponding flyweight kinetic energy may be written

$$\boxed{T_f = \frac{1}{2}J_{f,p}\omega_p^2 + \frac{1}{2}I_f(q'_f\dot{x}_p)^2.} \quad (\text{A.4.5})$$

Replacing the simplified mass moments with higher-fidelity CAD or measured moments changes only the supplied maps, not these governing relations.

## A.5 Mechanism Admissibility and Assumptions

The maps in [Equation \(A.4.1\)](#) are valid only while the geometric, mass, and unilateral-contact assumptions of the mechanism remain satisfied. The conditions below define the admissible mechanism used by the present model.

### A.5.1 Geometry and Branch Admissibility

All physical dimensions and masses must be finite, with

$$L_f > 0, \quad R_{\text{roll}} > 0, \quad r_P > 0, \quad n_f \geq 1, \quad (\text{A.5.1})$$

and nonnegative component masses. The declared ramp domain and primary travel interval must both be nonempty.

At  $x_p = x_{p,\text{min}}$ , at least one contact solution of [Equation \(A.1.9\)](#) must exist. The physical branch is selected there using [Equation \(A.1.13\)](#) and must then continue continuously through the entire declared travel. Loss of that branch or a jump to a disconnected arm orientation is inadmissible.

The selected branch must remain regular:

$$\boxed{g_\xi \neq 0, \quad 0 < q'_f(x_p) < \infty, \quad |q''_f(x_p)| < \infty.} \quad (\text{A.5.2})$$

The sign requirement corresponds to the present fixed-pivot mechanism, for which primary closure rotates the flyweight outward. A zero or negative motion ratio indicates a dead-centre, reversal, or incompatible branch for this mechanism definition.

The finite roller offset must also remain regular. Let  $\kappa_n$  denote the signed ramp curvature consistent with the selected normal. The normal-offset regularity factor must satisfy

$$\boxed{1 - R_{\text{roll}}\kappa_n \neq 0} \quad (\text{A.5.3})$$

through the selected contact range. Operation close to this singular condition should be treated as geometrically sensitive even before equality is reached.

The selected contact coordinate must remain within the physical ramp extent for the full primary travel. In practice, the active ramp should extend beyond the selected contact path with an appropriate geometric and manufacturing margin.

The rigid arm condition itself is exact up to numerical tolerance:

$$\boxed{\|\mathbf{r}_C - \mathbf{r}_P\| = L_f.} \quad (\text{A.5.4})$$

It is not an approximate fitted arm length.

### A.5.2 No Penetration or Simultaneous Double Contact

For the selected roller pose, no second portion of the physical ramp may penetrate the roller. Conceptually, for every distinct ramp coordinate  $\tilde{\xi}$  away from the selected contact,

$$\boxed{\|\mathbf{r}_C - \mathbf{r}_K(\tilde{\xi}, x_p)\| \geq R_{\text{roll}}.} \quad (\text{A.5.5})$$

A strict violation indicates interference. Equality at a second distinct ramp location indicates true simultaneous two-point contact and lies outside the present single-contact model. This condition is different from the existence of multiple arm orientations that separately solve the contact equation.

### A.5.3 Mass and Inertia Admissibility

The supplied mass description must satisfy

$$\boxed{I_f \geq 0, \quad J_{f,p}(x_p) > 0} \quad (\text{A.5.6})$$

throughout the selected branch. These conditions follow naturally for the nonnegative arm-plus-tip approximation. If arbitrary CAD moments are supplied, they must correspond to a physically realizable nonnegative mass distribution.

Flyweight mass and inertia represented here must not be included again in the movable-sheave translating mass, the constant rigid pulley rotational inertia, or another actuator inertia term. This separation prevents double counting of the same kinetic energy.

#### A.5.4 Unilateral Contact-Force Admissibility

Geometric compatibility alone does not guarantee that the ramp can maintain the constrained roller contact during every dynamic state. Under the sign convention used in the main text, compressive ramp contact requires

$$\boxed{F_{p,\text{fly}} \geq 0.} \quad (\text{A.5.7})$$

If  $F_{p,\text{fly}} < 0$ , the constrained solution would require the ramp to pull on the roller. That state is physically inadmissible. The result is not replaced by  $\max(0, F_{p,\text{fly}})$ , since clipping the force would hide a change in mechanism topology.

A complete treatment of such a state would introduce flyweight lift-off, free-flight, re-contact, and any associated impact dynamics. Those modes are not retained here. Until such an extension is introduced, negative contact force is treated as an explicit admissibility failure. The condition  $F_{p,\text{fly}} = 0$  marks the unilateral-contact boundary.

#### A.5.5 Mechanism and Contact Assumptions

The fixed-pivot construction uses the following assumptions:

- the flyweight pivot is fixed relative to the pulley body;
- the ramp translates rigidly with the movable sheave;
- the pivot-to-roller-centre length is rigid and constant;
- the roller has finite radius and the flyweight motion is planar in the axial–radial plane;
- repeated flyweights are identical and arranged with sufficient circumferential symmetry that omitted cross and gyroscopic terms cancel;
- structural deformation of the arm, pivot, roller, ramp, and carrier is neglected;
- backlash and manufacturing clearance in the kinematic constraint are neglected;
- the dynamic ramp representation is  $C^3$  over the active contact range;
- ramp contact is an ideal geometric, frictionless constraint in the flyweight force law;
- roller-bearing loss, roller spin inertia, local Hertzian compliance, and wear are neglected;
- only one active roller–ramp contact is retained; and
- the selected contact must remain compressive.

For the present Baja mass approximation, the arm/body is additionally treated as a uniform slender member and all outer hardware is concentrated at the roller-centre station. The finite dimensions and centroidal inertias listed in [Appendix A.3.1](#) are consequently omitted. A higher fidelity mass description may replace this reduction without changing the geometry or dynamics developed above.

### A.6 Relation to Existing Actuator Models

The geometry and mass reductions used here are not intended as novelty claims by themselves. Detailed CVT actuator models already use measured geometry, centre-of-mass locations, contact construction, and mechanical advantage to recover flyweight thrust. For example, [Messick \(2018\)](#) derives primary thrust from measured/CAD flyweight geometry under quasi-equilibrium, [Kim et al. \(2002\)](#) retain detailed roller/contact geometry and contact friction in an equilibrium actuator treatment, and [Skinner \(2020\)](#) resolves the flyweight, supporting link, and ramp reactions with explicit free bodies while representing the flyweight and link centrifugal loading through their mass and working radii. These approaches establish the use of reduced mass and geometry information for detailed actuator force prediction.

If the pivot inertia is neglected and the mass distribution is reduced to a point or centre-of-mass description at radius  $r_f(x_p)$ , then

$$J_{f,p} = m_f r_f^2 \quad (\text{A.6.1})$$

and the centrifugal term reduces to

$$\boxed{\frac{1}{2}\omega_p^2 \frac{dJ_{f,p}}{dx_p} = m_f r_f \omega_p^2 \frac{dr_f}{dx_p}.} \quad (\text{A.6.2})$$

This is the familiar centrifugal force-times-motion-ratio form used by quasi-static actuator descriptions. The extension developed here is that the same geometry and mass maps are consumed by the transient balance while retaining the flyweight's pivoting inertia, changing mechanism motion ratio, and configuration-dependent shaft angular momentum. The distinction is therefore dynamic coupling of the actuator to the shift and rotational equations, not the existence of curved ramps, finite rollers, or detailed flyweight geometry.

## Appendix B

# Numerical Implementation

*Implementation version: `cinder-v1.1.2` (commit `7637a38b4fb9ec21dfb953c1c80a27ec5f389654`).*

This appendix records the parts of the CINDER implementation that require a numerical procedure beyond the analytical equations themselves, together with cases where the implemented algebra differs from the form used in the formulation. The discussion therefore focuses on root solves, conditioning, contact closure, event handling, and the equivalent equation forms used in the code. Code references are included so that these numerical choices can be traced to the tagged implementation.

### B.1 Numerical Solution of Belt Geometry

The exact belt geometry derived in [Sections 3.1.2](#) and [3.1.3](#) is retained in the implementation. The quantities that remain implicit in the belt-length closure are found with SciPy's `scipy.optimize.brentq` routine ([Virtanen et al., 2020](#)).

The first such solve is performed once, during construction, to determine the fixed pulley centre distance  $d_c$  from the specified zero-shift radii  $r_{p,0}$  and  $r_{s,0}$ . A physical bracket can be obtained directly from the same belt geometry. The centre distance cannot be smaller than the sum of the two radii,

$$d_{c,lo} = r_{p,0} + r_{s,0}, \tag{B.1.1}$$

because the two pulley circles would otherwise overlap. The upper bound comes from the finite belt length. The total wrapped length is

$$\begin{aligned} r_p \phi_p + r_s \phi_s &= \pi(r_p + r_s) + 2(r_s - r_p) \sin^{-1} \left( \frac{r_s - r_p}{d_c} \right) \\ &\geq \pi(r_p + r_s). \end{aligned} \tag{B.1.2}$$

At the zero-shift geometry, let  $\ell_0$  denote the length of either straight span. The two straight spans can then use no more than

$$2\ell_0 \leq L_b - \pi(r_{p,0} + r_{s,0}). \tag{B.1.3}$$

From the exact span relation in [Equation \(3.1.19\)](#),

$$\ell_0^2 = d_c^2 - \Delta r_0^2, \quad \Delta r_0 = r_{s,0} - r_{p,0}. \quad (\text{B.1.4})$$

Combining these relations gives the upper bracket

$$d_{c,\text{hi}} = \sqrt{\Delta r_0^2 + \left[ \frac{L_b - \pi(r_{p,0} + r_{s,0})}{2} \right]^2}. \quad (\text{B.1.5})$$

The belt-length residual is checked at both endpoints before iteration. An endpoint is accepted when its absolute residual is no greater than  $10^{-12}$  m; otherwise a sign change is required and Brent iteration uses an absolute root-location tolerance of  $10^{-12}$  m. The bracket is therefore set by the geometry itself rather than by an arbitrary numerical search interval.

A second broad geometry solve is performed once at construction to make the repeated runtime solve cheaper. At  $s = s_{\text{max}}$ , the primary radius is known explicitly,

$$r_{p,\text{max}} = r_{p,0} + \frac{s_{\text{max}} - s_{\text{dz}}}{2 \tan \beta}, \quad (\text{B.1.6})$$

while the compatible secondary radius is still obtained from belt-length closure. This one setup solve uses the broad physical bracket

$$\begin{aligned} r_{s,\text{lo}}^{\text{setup}} &= h_b, \\ r_{s,\text{hi}}^{\text{setup}} &= \min \left( d_c - r_{p,\text{max}}, \frac{L_b}{\pi} - r_{p,\text{max}} \right). \end{aligned} \quad (\text{B.1.7})$$

The lower limit keeps the full belt section outside the pulley axis. The first upper limit leaves enough centre spacing for the primary and secondary radii to fit without intersecting, while the second follows from [Equation \(B.1.2\)](#): the two wrapped portions cannot by themselves require more belt than is available.

Once the secondary radius at maximum shift has been found, the two travel endpoints are stored as

$$r_s(s) \in [r_s(s_{\text{max}}), r_s(0)]. \quad (\text{B.1.8})$$

Every ordinary geometry evaluation during integration can then solve for  $r_s(s)$  on this much tighter interval instead of repeating the broad setup search. The extra work is paid once at construction so that the geometry root solved throughout the run uses the smallest precomputed physical bracket. The other radius definitions used by the model follow from fixed offsets from these solved radii. First and second radius derivatives are obtained directly from the differentiated belt-length closure, which avoids adding finite-difference error to quantities used by the dynamic equations. At  $s = s_{\text{dz}}$ , the radius itself is common to both sides of the boundary, but its governing branch changes; separate deadzone-side and engaged-side evaluations therefore provide the appropriate one-sided derivatives.

**Code reference** The geometry implementation is contained in [belt\\_length.py](#), [spec.py](#), and [belt\\_pulley.py](#).

## B.2 Fixed- $\lambda$ Engaged Mechanical Closure

For a proposed signed traction pair  $(\lambda_p, \lambda_s)$ , [Section 3.5.3](#) presents the freely shifting engaged mechanics after eliminating the two pulley torques  $\tau_p$  and  $\tau_s$ . The implementation instead retains those torques as simultaneous unknowns,

$$\mathbf{z}_{\text{impl}} = [\dot{\omega}_p \quad \dot{\omega}_s \quad \dot{v}_b \quad \ddot{s} \quad \tau_p \quad \tau_s \quad N_p \quad N_s]^\top, \quad (\text{B.2.1})$$

and keeps the two traction relations as explicit rows,

$$\frac{\tau_p}{r_{p,\text{eff}}} + \lambda_p N_p = 0, \quad \frac{\tau_s}{r_{s,\text{eff}}} + \lambda_s N_s = 0. \quad (\text{B.2.2})$$

Eliminating  $\tau_p$  and  $\tau_s$  from these rows recovers the reduced six-variable system used in the formulation, so the implementation does not add a physical degree of freedom. This form has two practical advantages: torque-reactive mechanisms remain coupled directly to the same simultaneous solve, and the normal resultants never have to be recovered by dividing a solved torque by  $\lambda_j$ . In particular, when  $\lambda_j = 0$ , the corresponding traction row simply gives  $\tau_j = 0$  while the remaining balances continue to determine  $N_j$ .

### B.2.1 Equilibrated linear solve

The closure combines accelerations, torques, forces, and normal resultants in one linear system. These quantities have different units and can produce very different coefficient magnitudes even when the underlying equations are well posed. Before the dense solve, CINDER therefore equilibrates the algebraically identical system. For  $\mathbf{A}\mathbf{z} = \mathbf{b}$ ,

$$\mathbf{A}_s = \mathbf{D}_r \mathbf{A} \mathbf{D}_c, \quad \mathbf{b}_s = \mathbf{D}_r \mathbf{b}, \quad \mathbf{z} = \mathbf{D}_c \mathbf{y}. \quad (\text{B.2.3})$$

Each row scale is the reciprocal of the largest absolute magnitude in that row or its right-hand side; column scales are then formed from the row-scaled matrix. The scaled system is solved for  $\mathbf{y}$  and mapped back to the physical unknowns. Residuals are still evaluated against the original, unscaled equations. Diagnostic solves additionally expose the rank and both raw and equilibrated condition numbers. The scaling therefore changes only the numerical conditioning of the solve, not the mechanical equations being enforced.

### B.2.2 Regular evaluation through zero traction

The wrap solution is analytically regular at  $\lambda_j = 0$ , as shown in [Equation \(3.4.33\)](#). A clamped contact can therefore pass continuously through zero signed traction while retaining  $N_j > 0$ ; zero traction is not a separate mechanical regime. The direct coefficient formulas are nevertheless numerically awkward near this limit because they subtract nearly equal quantities.

With  $z = \lambda_j \phi_j / \sin \beta$ ,

$$\Phi_-(z) = \frac{1 - e^{-z}}{z}, \quad \Psi_-(z) = \frac{z - 1 + e^{-z}}{z^2}. \quad (\text{B.2.4})$$

For  $|z| < 10^{-4}$ , CINDER evaluates the local series

$$\begin{aligned}\Phi_-(z) &= 1 - \frac{z}{2} + \frac{z^2}{6} - \frac{z^3}{24} + \frac{z^4}{120}, \\ \Psi_-(z) &= \frac{1}{2} - \frac{z}{6} + \frac{z^2}{24} - \frac{z^3}{120} + \frac{z^4}{720},\end{aligned}\tag{B.2.5}$$

and uses `expm1`-based expressions outside that neighbourhood. The change is only in how the same smooth analytical functions are evaluated; no artificial topology change is introduced at  $\lambda_j = 0$ .

**Code reference** The closure basis and linear solve are implemented in `closure.py` and `trial_system.py`; row assembly is in `equations.py`, and the regular wrap functions are evaluated in `equation_context.py`.

### B.3 Actuator Contributions

The actuator forces and torques derived in [Sections 3.2](#) and [3.3](#) enter the same coupled mechanical closure as the belt and pulley dynamics. The implementation preserves that coupling, but separates the actuator law from the closure assembly. A pulley actuator therefore supplies its contribution to the common force and torque balances through a shared interface rather than being built directly into those balances.

This separation allows the same CVT mechanics to be used with different clamping mechanisms without changing the closure itself. The derived flyweight, primary spring, secondary helix, and secondary spring models all use this interface, but so can a different mechanical actuator, a prescribed axial-force law, or a feedback-controlled actuator. This also allows study configurations to replace the native mechanical clamping without defining a separate CVT model. Only the actuator contribution changes; the belt, pulley, and contact equations continue to be solved as the same coupled system.

The interface places one numerical restriction on those contributions. Once the current state, geometry, and trial traction pair are fixed, each contribution must be affine in the instantaneous closure unknowns so that the common mechanical system remains linear. Thus a contribution is represented as

$$q(\mathbf{z}_{\text{impl}}) = q_0 + \mathbf{g}^\top \mathbf{z}_{\text{impl}}.\tag{B.3.1}$$

The affine requirement applies to the closure unknowns, not to the state on which the actuator law depends. Both  $q_0$  and  $\mathbf{g}$  may therefore depend nonlinearly on the current position and speed. Centrifugal flyweight action, for example, may retain its  $\omega_p^2$  dependence because  $\omega_p$  is known when the instantaneous closure is assembled.

Dependence on an unknown being solved at that instant is carried by  $\mathbf{g}$ . If a mounted mechanism uses a local pulley coordinate  $x_j(s)$ , for example,

$$\ddot{x}_j = x_j''(s)\dot{s}^2 + x_j'(s)\ddot{s}.\tag{B.3.2}$$

At the current state,  $x_j''(s)\dot{s}^2$  is known, while  $x_j'(s)\ddot{s}$  contributes a coefficient on the unknown global shift acceleration. The same interface can carry dependence on  $\dot{\omega}_j$ ,  $\tau_j$ , or  $N_j$  when those quantities appear in an actuator law, while preserving the linear simultaneous closure.

**Code reference** The affine closure objects are defined in [closure.py](#); the pulley element contract and local context are in [actuation/types.py](#), and mounted contributions are combined by [pulley\\_actuator.py](#).

## B.4 Numerical Solution of Tangential Contact

Sticking introduces the nonlinear traction-root conditions in [Equations \(3.5.11\) and \(3.5.12\)](#). CINDER solves these roots numerically around the fixed- $\lambda$  mechanical closure of [Appendix B.2](#).

### B.4.1 Search domain, Jacobian, and acceptance

The numerical root search is deliberately allowed to extend beyond the physical static-friction interval. This separates two questions that are physically distinct: first, what traction would be required to enforce stick; second, whether the available static friction can supply it. An above-capacity root can therefore be found numerically and then rejected by the static limit in [Equation \(3.5.19\)](#), rather than being hidden by the search bounds themselves.

When no continuation estimate is available, the initial guess is zero. CINDER uses SciPy’s bounded `scipy.optimize.least_squares` routine ([Virtanen et al., 2020](#)) with the trust-region reflective method, a three-point finite-difference residual Jacobian, and Jacobian-based scaling. The tagged defaults use `ftol=xtol=gtol=10-12` and at most 100 residual evaluations. A returned candidate is not accepted from the optimizer status alone: the mechanical system is evaluated once more at that traction, and the resulting sticking residual must satisfy the declared contact tolerance.

### B.4.2 Continuation between consecutive contact solves

Successive right-hand-side evaluations within one smooth contact topology are normally close in state space, so the required sticking traction is normally close as well. CINDER uses this local continuity to avoid restarting the full nonlinear solve from zero at every evaluation. The previous accepted traction root and its local residual Jacobian are stored, and one bounded Newton correction is attempted first. After a successful correction, the Jacobian is updated with a rank-one good-Broyden step.

If the continuation estimate is unavailable, reaches a search bound, cannot be evaluated mechanically, or fails the sticking-residual tolerance, the code falls back to the full bounded least-squares solve. Besides reducing repeated nonlinear work, following the nearby accepted solution helps the simulation remain on the locally connected traction branch if other roots exist elsewhere in the residual map. The stored numerical information is only a solver aid; it does not represent physical friction memory.

### B.4.3 Numerical hysteresis at reattachment

The physical static-capacity limit remains [Equation \(3.5.19\)](#). At finite numerical precision, however, using exactly the same zero-margin capacity for both loss and recovery of stick can cause repeated stick-slip switching at nearly the same state. CINDER therefore leaves stick loss at the physical zero-margin boundary but normally requires a positive restick reserve of  $10^{-3}$  in traction-utilization margin.

This reserve is a numerical switching margin, not a change to the friction law. If applying it would leave the event resolver with no admissible successor state, the sticking candidate is retried at the

physical static boundary. The extra margin therefore suppresses numerical chatter where it can do so without preventing an otherwise valid transition.

**Code reference** The nonlinear solve and continuation are implemented in [engaged\\_contact.py](#); contact residuals and tolerances are in [relative\\_motion.py](#) and [tolerances.py](#); hybrid switching is implemented in [cvt\\_contact\\_switching.py](#).

## B.5 Time Integration and Hybrid Boundary Events

### B.5.1 Continuous integration and host coupling

CINDER is implemented as a CVT component within a host simulation. The standard study configurations attach the surrounding engine, load, vehicle, or other required system through the shaft interfaces defined in [Section 2.1](#). At each interface the host supplies the applied torque and equivalent referred inertia seen by the CVT, while the shaft speed remains part of the coupled dynamic response.

The host layer is modular rather than fixed to those standard configurations. A different host can be attached without changing the internal CVT equations, and it may introduce additional states and boundary laws outside the transmission. Those states can be advanced together with the CVT in the same initial-value problem, allowing the transmission to be embedded in a different engine, vehicle, dynamometer, flywheel, controller, or larger drivetrain model while retaining the same shaft-port contract.

Within any fixed mechanical regime, the combined first-order initial-value problem is advanced through SciPy’s `scipy.integrate.solve_ivp` interface ([Virtanen et al., 2020](#)). The generic runner is method-configurable. The reported transient study drivers select LSODA, whose automatic selection between stiff and nonstiff multistep formulas follows the strategy of [Petzold \(1983\)](#). Relative and absolute tolerances and maximum-step limits are study settings and are therefore reported with the Results configuration rather than treated as constants of the mechanical model.

The integration mesh is adaptive. The right-hand side may be evaluated at several internal trial states before a single step is accepted, so a plotted or reported sampling interval is not an integration time step.

### B.5.2 Hybrid segmentation and event localization

The regime structure derived in [Section 3.6](#) is integrated one smooth regime at a time. Each active regime exposes only the boundary events that it can reach, together with the allowed crossing direction. When an event is localized, the current `solve_ivp` segment ends, the transition resolver determines the successor regime and any required state update, and integration restarts at the same physical time.

Events localized to the same endpoint within the configured association tolerance are resolved together. The adaptive integrator can also probe a trial state slightly beyond a valid boundary while attempting or rejecting a step. Roundoff-scale boundary snapping and protected evaluation of these trial states keep such numerical probes inside valid geometry; they do not move the accepted physical event surface.

### B.5.3 Integration mesh and reported output

The native adaptive trajectory is retained separately from the user-facing reporting grid. Uniform reports are generated with SciPy dense output within each smooth segment, and exact segment endpoints are kept explicitly. Dense output is therefore never used to interpolate across a discrete state reset.

**Code reference** The shaft ports are implemented in [ports.py](#); segmented integration is in [hybrid.py](#); and reporting materialization is in [reporting.py](#).

## B.6 Formulation-to-Implementation Correspondence

| Analytical presentation                         | cinder-v1.1.2 implementation                                                                              | Numerical purpose                                                                                                      |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Implicit exact belt geometry                    | SciPy <code>brentq</code> ; physical construction bounds and a cached runtime radius bracket              | Retain the exact belt-length closure while keeping the repeated radius solve tightly bounded.                          |
| Six-variable fixed- $\lambda$ closure           | Eight-variable closure retaining pulley torques and explicit traction rows                                | Keep torque-dependent mechanisms simultaneous and remain regular through zero traction.                                |
| $Az = b$                                        | Row/column equilibration followed by a dense direct solve                                                 | Reduce avoidable mixed-scale conditioning problems without changing the physical equations.                            |
| Continuous $\lambda_j \rightarrow 0$ wrap limit | Regular $\Phi_-/\Psi_-$ evaluation using local series and <code>expm1</code>                              | Avoid cancellation while preserving the same smooth contact solution.                                                  |
| Actuator laws entering the coupled balances     | Shared actuator interface with affine bias-plus-gain dependence on the closure unknowns                   | Allow actuator models to be exchanged without changing the CVT closure while preserving its linear simultaneous solve. |
| Static-traction root                            | Bounded trust-region <code>least_squares</code> , local continuation, and independent residual acceptance | Find the traction required for stick efficiently, then apply physical admissibility separately.                        |
| Continuous regime equations                     | Adaptive <code>solve_ivp</code> segments with explicit hybrid transitions                                 | Integrate smooth dynamics separately from changes in the active mechanical regime.                                     |

Table B.1: Correspondence between the analytical formulation and the numerical choices documented in this appendix.

# Appendix C

## Reference Inputs, Shaft Boundaries, and Study Actuators

This appendix records the shared Baja reference inputs used by the Results campaign and the shaft-boundary models attached to CINDER in the different simulation cases. Case-specific changes are listed with the corresponding study in [Appendix D](#).

The executable reference input is [defaults/baja/simulation\\_case.json](#), with the human-readable tuning quantities collected in [defaults/baja/tuning.json](#).

### C.1 Reference CVT Configuration

| Group                           | Quantity                                     | Symbol                          | Value                  |
|---------------------------------|----------------------------------------------|---------------------------------|------------------------|
| <b>Belt and pulley geometry</b> |                                              |                                 |                        |
| Belt                            | Outer-path length                            | $L_b$                           | 0.953 262 m            |
| Belt                            | Radial thickness                             | $h_b$                           | 0.015 570 2 m          |
| Belt                            | Outer-face width                             | $b_{\text{out}}$                | 0.021 336 m            |
| Belt                            | Inner-face width                             | $b_{\text{in}}$                 | 0.016 814 8 m          |
| Belt                            | Cord-layer depth from the outer surface      | $d_{\text{cord}}$               | 0.002 54 m             |
| Belt                            | Material density                             | $\rho_b$                        | 1100 kg/m <sup>3</sup> |
| Pulley geometry                 | Sheave half-angle                            | $\beta$                         | 0.200 712 864 rad      |
| Pulley geometry                 | Primary outer-surface radius at zero shift   | $r_{p,\text{outer},\text{min}}$ | 0.036 207 7 m          |
| Pulley geometry                 | Secondary outer-surface radius at zero shift | $r_{s,\text{outer},\text{max}}$ | 0.1016 m               |
| Pulley geometry                 | Primary deadzone travel                      | $s_{\text{dz}}$                 | 0.002 489 2 m          |
| Pulley geometry                 | Maximum shift coordinate                     | $s_{\text{max}}$                | 0.019 05 m             |
| <b>Belt–pulley contact</b>      |                                              |                                 |                        |
| Contact                         | Static traction coefficient                  | $\mu_s$                         | 0.65                   |
| Contact                         | Kinetic traction coefficient                 | $\mu_k$                         | 0.55                   |

Table C.1: Geometry, belt, and contact parameters of the shared Baja reference configuration.

| Group            | Quantity                                       | Symbol                  | Value                                         |
|------------------|------------------------------------------------|-------------------------|-----------------------------------------------|
| <b>Primary</b>   |                                                |                         |                                               |
| Primary          | Number of flyweights                           | $n_f$                   | 3                                             |
| Primary          | Total flyweight-set mass                       | $m_f$                   | 0.790 938 kg                                  |
| Primary          | Flyweight pivot radius                         | —                       | 0.042 545 m                                   |
| Primary          | Pivot-to-roller-centre distance                | —                       | 0.031 521 4 m                                 |
| Primary          | Roller radius                                  | —                       | 0.0065 m                                      |
| Primary          | Linear ramp angle                              | —                       | 35°                                           |
| Primary          | Linear ramp span                               | —                       | 0.005 m                                       |
| Primary          | $C^3$ transition span                          | —                       | 0.003 m                                       |
| Primary          | Circular ramp angle range                      | —                       | 35° → 20°                                     |
| Primary          | Circular ramp span                             | —                       | 0.03 m                                        |
| Primary          | Fixed rotating hardware inertia                | —                       | $3.220\,283\,8 \times 10^{-3} \text{ kg m}^2$ |
| Primary          | Moving-sheave translational mass               | $m_{p,m}$               | 1.0681 kg                                     |
| Primary          | Axial spring stiffness                         | $k_p$                   | 12 784 N/m                                    |
| Primary          | Axial spring initial compression               | $x_{p,0}$               | 0.094 090 49 m                                |
| <b>Secondary</b> |                                                |                         |                                               |
| Secondary        | Fixed rotating hardware inertia                | $I_{s,F}$               | $2.326\,26 \times 10^{-3} \text{ kg m}^2$     |
| Secondary        | Movable-member rotational inertia              | $I_{s,M}$               | 0.002 513 9 kg m <sup>2</sup>                 |
| Secondary        | Moving-sheave translational mass               | $m_{s,m}$               | 0.705 141 kg                                  |
| Secondary        | Axial spring stiffness                         | $k_{s,x}$               | 3532 N/m                                      |
| Secondary        | Axial spring initial compression               | $x_{s,0}$               | 0.11 m                                        |
| Secondary        | Torsional stiffness                            | $k_{s,\theta}$          | 3.476 N m/rad                                 |
| Secondary        | Initial torsional preload                      | $\theta_{s,\text{pre}}$ | 300°                                          |
| Secondary        | Helix reference radius                         | $r_h$                   | 0.044 45 m                                    |
| Secondary        | Helix angle from the circumferential direction | $\alpha_s$              | 20°                                           |
| Secondary        | Movable-face share of belt torque              | $\tau_{s,M}/\tau_s$     | 0.5                                           |

Table C.2: Actuation and inertia parameters of the shared Baja reference configuration.

### C.1.1 Reference secondary-helix contact topology

The Results reference CVT uses a zero-clearance bilateral slot at the secondary helix. The signed torque and axial reaction are those derived in [Section 3.2.2](#); a positive or negative required reaction is carried by the corresponding slot flank. The slot is idealized as having no backlash or free-play, so changing reaction sign does not introduce a separate kinematic state.

This is a physical reference-model assumption, not a change to the helix equations derived in the main text. The stock `cinder-v1.1.2` helix element additionally applies a one-sided contact-admissibility

check that requires the reaction on its implemented helix flank to remain compressive. For the reference Results model, the loader replaces that element by a bilateral subclass that preserves the same spring, torque-sharing, inertia, and helix kinematics while removing only that one-sided flank check. The implementation is in `defaults/reference_model/slotted_helix.py` and is applied by `defaults/reference_model/reference_case.py`.

The shared reference launch begins from

$$\omega_p(0) = 1800 \text{ rpm}, \quad \omega_s(0) = 0, \quad v_b(0) = 0, \quad s(0) = 0, \quad \dot{s}(0) = 0. \quad (\text{C.1.1})$$

### C.1.2 Basis of the reference parameters

The reference set is an engineering baseline representative of the custom CVT and vehicle used by McMaster Baja Racing. Its inputs combine direct hardware measurements, CAD- or geometry-derived quantities, tuning choices, manufacturer or published data, and engineering estimates. The principal basis of each parameter family is summarized in [Table C.3](#).

Table C.3: Basis of the shared Baja reference parameters.

| Parameter family                              | Basis used for the reference set                                                                                                                                                                                          |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CVT and pulley dimensions                     | Based on the team’s custom Baja CVT hardware, using direct measurements and derived geometry where available.                                                                                                             |
| Primary and secondary actuation               | Based on the team’s hardware and reference tune; preload and profile quantities are the values used by the frozen simulation input.                                                                                       |
| Secondary helix contact topology              | Zero-clearance bilateral slot used as the Results reference mechanism; this is a modelling assumption for the ideal signed helix relation, not a measured backlash/contact-compliance model.                              |
| Masses and inertias                           | Combination of known component masses, CAD/geometry-derived quantities, and engineering estimates where direct inertia measurements were unavailable.                                                                     |
| Belt density and traction coefficients        | Engineering estimates based on generic rubber/V-belt material and contact data; these quantities were not measured for the exact belt used by the team.                                                                   |
| Engine torque curve                           | Based on the Kohler CH440 torque/power data used in <a href="#">Figure 1.1 (Baja SAE, 2022)</a> ; interpolation and the out-of-range braking tails are numerical extensions described in <a href="#">Appendix C.2.2</a> . |
| Vehicle mass, wheel radius, and final drive   | Team vehicle/reference-drivetrain values.                                                                                                                                                                                 |
| Rolling resistance and aerodynamic parameters | Engineering estimates used to provide a representative longitudinal road load rather than an experimentally identified coast-down model.                                                                                  |

## C.2 Shaft-Boundary Interface

At each shaft, the host boundary supplies the signed external torque and the rotational inertia referred to that shaft. Using the shaft-interface notation introduced in [Section 2.1](#), a boundary evaluation returns

$$\mathcal{B}_j(t, \mathbf{x}, \mathbf{x}_h) \longmapsto (\tau_{B,j}, I_{B,j}), \quad j \in \{p, s\}, \quad (\text{C.2.1})$$

where  $\mathbf{x}_h$  denotes any additional host state required by the boundary. The shaft speed remains a CINDER state unless a separate kinematic constraint is introduced by the host.

Appendix C.2 distinguishes boundary components provided by CINDER v1.1.2 from adapters defined in the Results code and attached through the same interface.

| Boundary used in the Results campaign   | Origin        | Implementation                                                         | Boundary quantity                                                                                               |
|-----------------------------------------|---------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Full-throttle engine                    | CINDER v1.1.2 | <code>engine boundary</code> ; <code>torque curve</code>               | Speed-dependent engine torque and constant referred engine-side inertia.                                        |
| Locked final-drive vehicle              | CINDER v1.1.2 | <code>vehicle boundary</code> ; <code>road load</code>                 | Vehicle road-load torque and reflected vehicle/wheel inertia.                                                   |
| Constant torque–inertia boundary        | CINDER v1.1.2 | <code>constant boundary</code>                                         | Prescribed constant torque and referred inertia; shaft speed is not fixed.                                      |
| Time-programmed final-drive vehicle     | Results code  | <code>time-programmed boundary</code>                                  | Same vehicle and inertia as the decoded reference boundary, with grade supplied as $\gamma(t)$ .                |
| Additive torque wrapper                 | Results code  | <code>actuator-study wrapper</code> ; <code>helix-study wrapper</code> | Adds a prescribed smooth torque perturbation while retaining the wrapped boundary inertia.                      |
| Blend-to-target / torque-scale wrappers | Results code  | <code>torque-blend wrappers</code>                                     | Smoothly replaces or scales the torque returned by another boundary while retaining its inertia.                |
| Prescribed torque bench boundary        | Results code  | <code>prescribed-torque boundary</code>                                | Smooth transition between two prescribed torque values with a specified constant referred inertia.              |
| Controlled overrun boundary pair        | Results code  | <code>overrun boundary pair</code>                                     | Blends primary torque toward a prescribed resisting value while independently programming secondary road grade. |

Table C.4: Shaft-boundary implementations used by the Results studies.

### C.2.1 CINDER constant torque–inertia boundary

The CINDER class named `FixedShaftBoundary` fixes the *boundary inputs*, not the shaft speed. It returns

$$\tau_{B,j} = \tau_{B,j}^{(0)}, \quad I_{B,j} = I_{B,j}^{(0)}, \quad (\text{C.2.2})$$

while  $\omega_j$  remains part of the dynamically integrated CVT state. This boundary is used for constructed bench states in the verification studies and for source-constrained external cases such as the independent Ballew benchmark; the selected  $\tau_{B,j}^{(0)}$  and  $I_{B,j}^{(0)}$  values are listed with those cases in Appendix D.

### C.2.2 Primary full-throttle engine boundary

The reference primary boundary uses the CINDER v1.1.2 full-throttle engine component,

$$\tau_{B,p} = \tau_{\text{eng}}(\omega_p), \quad I_{B,p} = I_{\text{eng}}, \quad (\text{C.2.3})$$

with  $I_{\text{eng}} = 0.05 \text{ kg m}^2$ . The positive-drive points are based on the Kohler CH440 data used in [Figure 1.1](#) ([Baja SAE, 2022](#)) and are interpolated by CINDER using a shape-preserving piecewise cubic Hermite interpolant. The numerical boundary adds bounded low- and high-speed braking tails so that externally driven states remain finite outside the positive-drive range.

Table C.5: Reference positive-drive engine torque points.

| Engine speed [rpm] | Torque [N m] |
|--------------------|--------------|
| 1000               | 0.000        |
| 1800               | 24.405       |
| 2400               | 25.083       |
| 2600               | 24.540       |
| 2800               | 23.591       |
| 3000               | 22.507       |
| 3200               | 20.880       |
| 3400               | 19.659       |
| 3600               | 18.304       |
| 4000               | 0.000        |

The low-speed extension passes through  $(0, 0)$ ,  $(500 \text{ rpm}, -5 \text{ N m})$ , and  $(1000 \text{ rpm}, 0)$ . The high-speed extension begins at the final positive-drive point and transitions over 1500 rpm to a  $-28 \text{ N m}$  braking plateau; a second 1500 rpm interval is included in the PCHIP support before the plateau is held constant. These tails are extrapolated boundary behaviour rather than CH440 measurements.

### C.2.3 Secondary locked-final-drive vehicle boundary

The reference secondary boundary rigidly maps secondary-shaft speed to wheel speed through the final-drive reduction

$$G = \frac{\omega_s}{\omega_w}, \quad v = \frac{r_w}{G} \omega_s. \quad (\text{C.2.4})$$

For grade angle  $\gamma$ , the longitudinal forces are

$$F_{\text{grade}} = -m_v g \sin \gamma, \quad (\text{C.2.5})$$

$$F_{\text{roll}} = -C_{rr} m_v g \cos \gamma \frac{v}{\sqrt{v^2 + v_c^2}}, \quad (\text{C.2.6})$$

$$F_{\text{aero}} = -\frac{1}{2} \rho C_d A_f |v| v, \quad (\text{C.2.7})$$

and the external secondary torque is

$$\tau_{B,s} = \frac{r_w}{G} (F_{\text{grade}} + F_{\text{roll}} + F_{\text{aero}}). \quad (\text{C.2.8})$$

The vehicle translation and wheel rotation are reflected to the secondary as

$$I_{B,s} = I_{s,\text{dir}} + \frac{I_w}{G^2} + m_v \left( \frac{r_w}{G} \right)^2. \quad (\text{C.2.9})$$

Table C.6: Parameters of the reference vehicle boundary.

| Quantity                                                | Value                             |
|---------------------------------------------------------|-----------------------------------|
| Vehicle mass $m_v$                                      | 300 kg                            |
| Wheel rotational inertia $I_w$                          | 1.107 817 44 kg m <sup>2</sup>    |
| Wheel radius $r_w$                                      | 0.2794 m                          |
| Final-drive reduction $G$                               | 7.556                             |
| Direct secondary-shaft inertia $I_{s,\text{dir}}$       | 0.002 011 338 4 kg m <sup>2</sup> |
| Rolling-resistance coefficient $C_{rr}$                 | 0.015                             |
| Aerodynamic drag coefficient $C_d$                      | 0.6                               |
| Frontal area $A_f$                                      | 1.114 84 m <sup>2</sup>           |
| Air density $\rho$                                      | 1.225 kg/m <sup>3</sup>           |
| Gravity $g$                                             | 9.806 65 m/s <sup>2</sup>         |
| Rolling-speed regularization $v_\epsilon$               | 0.01 m/s                          |
| Total reflected secondary inertia from Equation (C.2.9) | 0.431 610 kg m <sup>2</sup>       |

### C.2.4 Results-side time-programmed and torque-modified boundaries

Several studies retain a CINDER boundary’s inertia and constitutive law while changing only the imposed external loading history. The Results adapters use a smooth transition coordinate

$$u(t) = \begin{cases} 0, & t \leq t_0, \\ 3\xi^2 - 2\xi^3, & t_0 < t < t_0 + T_r, \\ 1, & t \geq t_0 + T_r, \end{cases} \quad \xi = \frac{t - t_0}{T_r}. \quad (\text{C.2.10})$$

For a shaft  $j$ , the boundary forms used in the Results campaign include

$$\tau_{B,j}^{\text{add}}(t) = \tau_{B,j}^{\text{base}}(t) + u(t) \Delta\tau_j, \quad (\text{C.2.11})$$

$$\tau_{B,j}^{\text{target}}(t) = [1 - u(t)]\tau_{B,j}^{\text{base}}(t) + u(t)\tau_{j,\text{target}}, \quad (\text{C.2.12})$$

$$\tau_{B,j}^{\text{scale}}(t) = [1 + u(t)(k_\tau - 1)]\tau_{B,j}^{\text{base}}(t), \quad (\text{C.2.13})$$

$$\tau_{B,j}^{\text{prescribed}}(t) = [1 - u(t)]\tau_{j,i} + u(t)\tau_{j,f}. \quad (\text{C.2.14})$$

The additive, target, and scale forms retain the equivalent inertia returned by the wrapped boundary. The standalone prescribed-torque form uses a case-specified constant inertia. The time-programmed vehicle boundary retains Equations (C.2.8) and (C.2.9) and replaces the road profile by a prescribed  $\gamma(t)$ . Exact amplitudes, inertias, grades, and transition times are listed with the study cases that use them.

## C.3 Study-Specific Actuator Implementations

The implementation described in Appendix B.3 allows the pulley actuator attached to either side to be exchanged without changing the belt, pulley, or contact closure. Each actuator contribution is returned through the same affine closure interface. The Results studies use that interface both for the full mechanisms derived in the main text and for the study-specific alternatives summarized below.

| Actuator law                       | Used in                                  | Mechanical change relative to the main-text actuator                                                                                                                                                                                   | Origin                    |
|------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Bilateral dynamic helix            | Shared Baja-reference studies            | Same signed dynamic helix relation as <a href="#">Equation (3.2.27)</a> ; the physical contact is the zero-clearance bilateral slot of <a href="#">Appendix C.1.1</a> , rather than the stock v1.1.2 one-sided flank check.            | <a href="#">Results</a>   |
| Quasi-static fixed-pivot flyweight | <a href="#">Section 4.4.1</a>            | Retains the centrifugal term of <a href="#">Equation (3.2.11)</a> and removes the two pivot-acceleration terms. The associated rotating inertia is reassigned to the fixed primary inertia rather than deleted.                        | <a href="#">Results</a>   |
| Quasi-static helix                 | <a href="#">Sections 4.4.1 and 4.4.2</a> | Retains spring torque and reacted movable-face belt torque, but removes the movable-member inertial term from <a href="#">Equation (3.2.27)</a> . The removed movable rotational inertia is reassigned to the fixed secondary inertia. | <a href="#">Results</a>   |
| Constant axial force               | <a href="#">Section 4.3</a>              | Replaces the native mechanical actuator by a prescribed axial closing force independent of state and closure unknowns.                                                                                                                 | <a href="#">Benchmark</a> |
| Tabulated axial force              | <a href="#">Section 4.3.2</a>            | Replaces the primary mechanical actuator by a prescribed time history, evaluated by piecewise-linear interpolation of the prepared force data.                                                                                         | <a href="#">Benchmark</a> |
| PI-controlled axial force          | <a href="#">Section 4.3.2</a>            | Replaces the primary mechanical actuator by a feedback force law driven by primary speed and an integrated controller-error state carried by the benchmark host.                                                                       | <a href="#">Benchmark</a> |

Table C.7: Study-specific actuator laws used through the common CINDER pulley-actuator interface.

### C.3.1 Quasi-static actuator reductions

The quasi-static primary-flyweight law used for the actuator-dynamics comparison keeps only the first term of [Equation \(3.2.11\)](#),

$$F_{\text{fw},x_p}^{\text{QS}} = \frac{1}{2} \omega_p^2 \frac{dJ_{f,p}}{dx_p}. \quad (\text{C.3.1})$$

The terms proportional to  $\ddot{x}_p$  and  $\dot{x}_p^2$  are omitted from the actuator force. To avoid deleting physical rotating mass, the ablation moves the corresponding flyweight rotational inertia into the constant primary rotating inertia used by that comparison. The implementation is in [ablation\\_core.py](#).

For the secondary, define the quasi-static reacted torque

$$\tau_{s,\text{helix}}^{\text{QS}} = \tau_{s,M} + k_{s,\theta} (\theta_{s,\text{pre}} - \theta_s). \quad (\text{C.3.2})$$

The quasi-static helix closing force is then

$$F_{\text{helix},x_s}^{\text{QS}} = \frac{d\theta_s}{dx_s} \tau_{s,\text{helix}}^{\text{QS}}. \quad (\text{C.3.3})$$

Relative to [Equation \(3.2.27\)](#), the movable-member inertial contribution  $I_{s,M}\dot{\omega}_{s,M}$  is omitted. Its rotational inertia is moved to the fixed secondary inertia for the quasi-static comparison. The same reduction is used by the actuator-dynamics and helix-topology studies through [ablation\\_core.py](#) and [helix\\_mechanics.py](#).

### C.3.2 Ballew benchmark axial-force actuators

The Ballew benchmark uses prescribed clamping-force actuators rather than the mechanical flyweight, spring, and helix combination of the Baja reference CVT. The secondary actuator is constant in both benchmark protocols,

$$F_{s,\text{act}} = 2000 \text{ N}. \quad (\text{C.3.4})$$

The force-replay protocol sets the primary actuator equal to the prepared Figure 45 force history, with piecewise-linear interpolation between adjacent samples,

$$F_{p,\text{act}}(t) = \text{interp}_{\text{lin}}[F_{45}(t)]. \quad (\text{C.3.5})$$

The benchmark-local constant and tabulated force laws are implemented in [actuation.py](#).

For the reconstructed closed-loop protocol, let  $n_p = 60\omega_p/(2\pi)$  be primary speed in rpm and  $e = n_p - n^*$ , with  $n^* = 2500$  rpm. The host carries the controller integral  $\eta$ ,

$$\dot{\eta} = e, \quad (\text{C.3.6})$$

and the commanded primary closing force is

$$F_{p,\text{act}} = F_{\text{ff}} + K_p e + K_i \eta, \quad F_{\text{ff}} = K_{\text{ff}} F_{s,\text{act}}. \quad (\text{C.3.7})$$

The source-constrained reconstruction uses  $K_{\text{ff}} = 1.2$ ,  $K_p = 5$ ,  $K_i = 75$ , so  $F_{\text{ff}} = 2400$  N, with  $\eta(0) = 0$ . No saturation or anti-windup term is added. The controller force law and host state are implemented in [controller.py](#).

## Appendix D

# Results Configuration and Reproducibility

This appendix records exact case inputs, numerical settings, normalization definitions, and other reproduction details that are referenced by the Results chapter but are not needed in its main discussion. The working paper references the Results tree on `develop` (`working`); the single ref used by `\ResultsCode` and `\ResultsTree` will be replaced by the frozen Results revision before publication.

### D.1 Mechanical Invariants and Operating-Domain Audit

Exact configuration data supporting [Section 4.2.1](#) are defined by [study.json](#) and the shared accepted-case library [operating\\_cases.json](#).

| Audited case family              | Shaft-boundary environment                                                                                        | Case role                                                                         |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Reference Baja trajectory        | Full-throttle engine + locked final-drive vehicle from <a href="#">Appendices C.2.2</a> and <a href="#">C.2.3</a> | Ten-second reference launch/shift, including the observed upper-stop arrival.     |
| Static rest and deadzone states  | Constant torque-inertia boundaries from <a href="#">Appendix C.2.1</a>                                            | Zero-speed rest, deadzone lock, and classifier checks.                            |
| Canonical engaged contact states | Constant torque-inertia boundaries on both shafts; no vehicle host                                                | Stick-stick, both single-slip branches, and all four both-slip sign combinations. |
| Free-shift states                | Constant torque-inertia boundaries on both shafts; no vehicle host                                                | Positive and negative engaged shift speed.                                        |
| Structural arrivals              | Constant torque-inertia bench environment                                                                         | Engagement, low-ratio-seat, and lower-stop arrivals with exact successor audit.   |
| Geometry-domain audit            | No dynamic shaft boundary required                                                                                | Belt-length closure and radius derivatives over the physical shift domain.        |

| Audited case family | Shaft-boundary environment      | Case role                                                                                             |
|---------------------|---------------------------------|-------------------------------------------------------------------------------------------------------|
| Negative controls   | Initial-state/classifier checks | Deliberately invalid states required to be rejected rather than projected into the admissible domain. |

Table D.1: Case families used by the mechanical-invariants audit.

The standard constructed bench cases use [Appendix C.2.1](#) on both shafts with  $I_{B,p} = 0.03 \text{ kg m}^2$  and  $I_{B,s} = 0.05 \text{ kg m}^2$ . Their shaft speeds remain dynamic. The two targeted anchors use the explicit boundary values in [Table D.5](#).

| Required coverage class           | Case family              | Boundary environment                         |
|-----------------------------------|--------------------------|----------------------------------------------|
| Static zero-speed lower-stop rest | Static rest              | Constant torque–inertia, zero applied torque |
| Deadzone free                     | Deadzone state           | Constant torque–inertia bench                |
| Engaged stick–stick               | Canonical contact state  | Constant torque–inertia bench                |
| Primary slip / secondary stick    | Canonical contact state  | Constant torque–inertia bench                |
| Primary stick / secondary slip    | Canonical contact state  | Constant torque–inertia bench                |
| Both slip                         | Canonical contact state  | Constant torque–inertia bench                |
| Both-slip (+, +)                  | Canonical contact state  | Constant torque–inertia bench                |
| Both-slip (+, −)                  | Canonical contact state  | Constant torque–inertia bench                |
| Both-slip (−, +)                  | Canonical contact state  | Constant torque–inertia bench                |
| Both-slip (−, −)                  | Canonical contact state  | Constant torque–inertia bench                |
| Primary slip in both directions   | Canonical contact states | Constant torque–inertia bench                |
| Secondary slip in both directions | Canonical contact states | Constant torque–inertia bench                |
| Forward and reverse rotation      | Stick/contact states     | Constant torque–inertia bench                |
| Free shift closing                | Free-shift state         | Constant torque–inertia bench                |
| Free shift opening                | Free-shift state         | Constant torque–inertia bench                |
| Low-ratio seat                    | Structural state         | Constant torque–inertia bench                |
| Upper stop                        | Reference trajectory     | Full-throttle engine + locked final drive    |
| Deadzone lower stop               | Structural state         | Constant torque–inertia bench                |
| Engagement transition             | Structural arrival       | Constant torque–inertia bench                |
| Low-ratio arrival                 | Structural arrival       | Constant torque–inertia bench                |
| Upper-stop arrival                | Reference trajectory     | Full-throttle engine + locked final drive    |
| Lower-stop arrival                | Structural arrival       | Constant torque–inertia bench                |
| Negative controls                 | Invalid-state checks     | No accepted dynamic case                     |
| Engagement-side classifier        | Classifier control       | Constructed boundary state                   |
| Nominal Baja reference            | Reference trajectory     | Full-throttle engine + locked final drive    |

Table D.2: Required mechanical-invariant coverage classes and the case family used to exercise them.

The dedicated upper-stop bench search did not find a retained case within its specified candidate set. Upper-stop support and arrival are nevertheless covered by the reference launch, which reaches that stop at approximately 6.782 s. Coverage therefore concerns the arrangements actually observed across the audit, rather than success of every individual candidate search.

| Initial contact      | Initial-state source | Initial dwell  | First change in contact                               |
|----------------------|----------------------|----------------|-------------------------------------------------------|
| Stick–stick, forward | ordinary search      | $\geq 30$ ms   | none within 30 ms                                     |
| Stick–stick, reverse | ordinary search      | 13.114 ms      | primary static limit reached; both contacts slide     |
| Primary slip (+)     | ordinary search      | 26.225 ms      | primary sticks; secondary begins sliding              |
| Primary slip (–)     | ordinary search      | 5.624 ms       | primary reverses sliding; secondary begins sliding    |
| Secondary slip (+)   | supplied anchor      | 414.44 $\mu$ s | secondary sticks                                      |
| Secondary slip (–)   | ordinary search      | $\geq 30$ ms   | none within 30 ms                                     |
| Both slip (+, +)     | ordinary search      | 31.71 $\mu$ s  | secondary sticks                                      |
| Both slip (+, –)     | ordinary search      | $\geq 30$ ms   | none within 30 ms                                     |
| Both slip (–, +)     | supplied anchor      | 9.82 $\mu$ s   | secondary sticks                                      |
| Both slip (–, –)     | ordinary search      | 1.757 ms       | primary reverses sliding; secondary continues sliding |

Table D.3: Canonical engaged contact cases. Dwell is measured from the selected audit’s event records; no-exit cases give lower bounds. Slip signs describe the initial state. The final column describes the inspected continuation, not merely the name of the event that triggered its evaluation.

The two targeted reproduction anchors are frozen explicitly in the shared case library. Their initial CVT states are listed in [Table D.4](#); their constant shaft-boundary values are listed separately in [Table D.5](#).

Table D.4: Initial CVT states of the two targeted mechanical-invariant reproduction anchors.

| Quantity                | Secondary slip (+) | Both slip (–, +) |
|-------------------------|--------------------|------------------|
| Primary speed [rad/s]   | 272.786402         | 536.870017       |
| Secondary speed [rad/s] | 223.014459         | 173.988007       |
| Belt speed [m/s]        | 17.211387          | 17.245403        |
| Shift position [m]      | 0.01446320         | 0.00292236       |
| Shift speed [m/s]       | 0                  | 0                |

Table D.5: Constant torque–inertia boundary values of the two targeted mechanical-invariant reproduction anchors.

| Boundary quantity                               | Secondary slip (+) | Both slip (–, +) |
|-------------------------------------------------|--------------------|------------------|
| Primary external torque [N m]                   | -38.596922         | -105.580961      |
| Primary referred inertia [kg m <sup>2</sup> ]   | 0.01279093         | 0.06468014       |
| Secondary external torque [N m]                 | -17.984371         | -282.073589      |
| Secondary referred inertia [kg m <sup>2</sup> ] | 0.04127251         | 0.00250569       |

Table D.6: Numerical settings used by the mechanical-invariants case families.

| Case family                                   | rtol               | atol               | Max. step | Duration |
|-----------------------------------------------|--------------------|--------------------|-----------|----------|
| Reference Baja trajectory                     | $3 \times 10^{-5}$ | $3 \times 10^{-8}$ | 10 ms     | 10 s     |
| Selected contact-case audit                   | $3 \times 10^{-5}$ | $3 \times 10^{-8}$ | 1 ms      | 30 ms    |
| Targeted-anchor screening<br>before selection | $3 \times 10^{-5}$ | $3 \times 10^{-8}$ | 0.5 ms    | 30 ms    |
| Structural arrivals                           | $3 \times 10^{-5}$ | $3 \times 10^{-8}$ | 0.5 ms    | 30 ms    |
| Free-shift cases                              | $3 \times 10^{-5}$ | $3 \times 10^{-8}$ | 1 ms      | 20 ms    |
| Static rest                                   | $3 \times 10^{-5}$ | $3 \times 10^{-8}$ | 2 ms      | 50 ms    |

The geometry-domain audit evaluates 401 points over the engaged shift range and 100 points through the primary deadzone, for 501 geometry samples in total. A  $1 \times 10^{-6}$  m finite-difference displacement is used for the independent derivative checks away from one-sided boundaries. The two targeted contact anchors are first screened with the 0.5 ms maximum step shown in [Table D.6](#); once accepted, they are rerun by the main contact audit with the same 1 ms maximum step used for every selected contact case. The contact-case audit samples each continuous segment at 0.5 ms spacing and includes both of its endpoints, even when the segment lasts less than that spacing. Every exact post-transition state is also inspected separately. The  $9.82 \mu\text{s}$  branch is therefore not inferred from interpolation across a 0.5 ms output interval.

For each engaged state, the analytical wrap tension is monotone in wrap angle, so its minimum occurs at an endpoint. The radial inertial offset in [Equation \(3.4.21\)](#) is constant along that wrap, giving its least normal loading at the same endpoint. Figure ?? takes the minimum of these spatial minima over the audited engaged states in each 30 ms case. Deadzone intervals are excluded because the engaged full-wrap field is not defined there. Minima on different pulleys, or of local and integrated loads, need not describe the same state.

For the case initialized in both-slip  $(-, -)$ , the least primary local load,  $0.0481989 \text{ N/rad}$ , occurs on the outgoing side of the first transition at  $1.757127 \text{ ms}$ , where  $N_p = 410.851 \text{ N}$ . The smaller integrated minimum,  $374.309 \text{ N}$ , occurs on the incoming side of that same event. Both states are retained. The new traction changes the force solution even though the velocities are continuous at this sliding-direction reversal.

The selected figure is generated by the maintained [run.py](#) entry point, with `--plot-only` available for identified retained outputs. Its [publication\\_plots.py](#) postprocessor checks the saved input/output hashes, recomputes the state and successor guards, and records the case selection and engaged-state mask.

All 25 required coverage classes were reached across 6,257 sampled states and 81 transitions. The recorded mechanical checks passed at every sampled state and every exact post-transition state. The largest row-scaled closure residual was  $1.43 \times 10^{-14}$ , using the normalization below.

For the engaged closure, let  $\mathbf{r} = \mathbf{A}\mathbf{z} - \mathbf{b}$ . Each row is normalized by

$$s_i = \max \left( 1, |b_i|, \sum_k |A_{ik}| |z_k| \right), \quad \epsilon_{\text{cl}} = \max_i \frac{|r_i|}{s_i}. \quad (\text{D.1.1})$$

The “scaled closure-equation residual” in [Appendix D.1](#) is  $\epsilon_{\text{cl}}$ . The other residual thresholds retain their physical units as shown.

| Audit threshold                                       | Value                              |
|-------------------------------------------------------|------------------------------------|
| Belt-length residual                                  | $10^{-9}$ m                        |
| Scaled closure-equation residual                      | $10^{-10}$                         |
| Sticking velocity drift                               | $10^{-3}$ m/s                      |
| Sticking acceleration residual                        | $10^{-7}$ m/s <sup>2</sup>         |
| Allowed negative static-margin roundoff               | $10^{-9}$                          |
| Allowed positive kinetic pair power                   | $10^{-3}$ W                        |
| Allowed negative integrated normal load               | $10^{-6}$ N                        |
| Allowed negative belt tension                         | $10^{-6}$ N                        |
| Allowed negative distributed normal load              | $10^{-6}$ N/rad                    |
| Wrap-field normal-integral residual                   | $10^{-6}$ N                        |
| Allowed negative stop reaction                        | $10^{-6}$ N                        |
| Fixed-shift speed residual                            | $10^{-8}$ m/s                      |
| Fixed-shift acceleration residual                     | $10^{-7}$ m/s <sup>2</sup>         |
| Deadzone secondary-lock speed residual                | $10^{-7}$ m/s                      |
| Deadzone secondary-lock acceleration residual         | $10^{-7}$ m/s <sup>2</sup>         |
| Static-rest state drift                               | $10^{-10}$                         |
| Deadzone radius-derivative residual                   | $10^{-12}$                         |
| Geometry first-derivative finite-difference residual  | $10^{-7}$                          |
| Geometry second-derivative finite-difference residual | $5 \times 10^{-3}$ m <sup>-1</sup> |

Table D.7: Numerical review thresholds used by the mechanical-invariants audit.

## D.2 Mechanical-Energy Consistency

The study supporting [Section 4.2.2](#) follows the physical reference configuration of [Appendix C.1](#) with the engine and vehicle boundaries of [Appendices C.2.2](#) and [C.2.3](#). The engine starts at 1800 rpm, the vehicle and belt are at rest, and the primary is at its lower stop. Full throttle and zero road grade are maintained for 10s. Both calculations use LSODA; only the numerical settings in [Table D.8](#) change. The exact case and study settings are retained with [energy-consistency/study.json](#).

| Integration | Relative tolerance | Absolute tolerance | Maximum step |
|-------------|--------------------|--------------------|--------------|
| Nominal     | $3 \times 10^{-5}$ | $3 \times 10^{-8}$ | 10 ms        |
| Tighter     | $9 \times 10^{-6}$ | $9 \times 10^{-9}$ | 5 ms         |

Table D.8: Time-integration settings for the same physical energy-audit case.

Stored kinetic energy is evaluated with the released motion inventory of [Table 3.3](#) and [Equation \(3.6.57\)](#). This includes shaft and sheave rotation, the attached boundary inertias, axial sheave motion, the mounted primary actuator’s motion, belt transport, and radial motion of the wrapped belt. Conservative axial- and torsional-spring potentials complete the inventory. Changes are measured from the initial energy, which is nonzero because the engine is already rotating. In the nominal calculation, the primary boundary supplies 66 778.132 J, while the resisting secondary boundary contributes  $-14\,609.239$  J; their sum is the net work in [Table 4.4](#).

Sliding loss is the negative net work of contact traction on the belt and pulley, evaluated only while that contact is sliding. The secondary contact uses the representative surface speed defined in Equation (3.5.2). In the manuscript signs, the traction on the belt is  $\lambda_j N_j$  and its torque on the pulley is  $-r_{j,\text{eff}} \lambda_j N_j$ . Their combined power is therefore

$$\lambda_j N_j v_b - r_{j,\text{eff}} \lambda_j N_j \bar{\omega}_j = \lambda_j N_j v_{\text{rel},j}.$$

During kinetic sliding this is nonpositive, and its negative is integrated as sliding loss. The representative secondary speed accounts for power on both faces under the retained equal torque sharing. The present calculation attributes all 1588.340 J of sliding loss to the primary; no secondary sliding loss accumulates. This is a result for this acceleration, not a restriction imposed on which contact can dissipate energy.

Power is integrated by composite trapezoids, separately within each continuous interval of motion. For a requested maximum spacing  $h$ , an interval of duration  $\Delta t_i$  receives  $\max(\lceil \Delta t_i/h \rceil, 12)$  subintervals. Interior values use the retained numerical solution's dense output; the two endpoints use the exact states returned at the interval boundaries. Equal-time incoming and outgoing states are stored separately. Work and sliding loss do not change across the instantaneous event, while capture loss is added only on its outgoing side.

Table D.9 identifies the captures in the nominal calculation. The early sequence ends with the numerical low-ratio seating rule documented in ??; it is not a model of compliant rebound. Lower-stop release, the return to sticking and release from the low-ratio seat are also retained in the event ledger, but have no capture loss. The tighter calculation has the same counts in these categories.

| Capture                                       | Number | Total loss [J]         |
|-----------------------------------------------|--------|------------------------|
| Primary engagement                            | 6      | 0.214536               |
| Low-ratio stop arrival and primary separation | 5      | 0.039307               |
| Terminal numerical seating                    | 1      | $1.758 \times 10^{-9}$ |
| Upper travel-stop arrival                     | 1      | $1.604 \times 10^{-5}$ |

Table D.9: Nominal capture contributions. The first three rows occur between approximately 0.062 s and 0.066 s; the upper stop is reached at 6.781 921 s. Totals use the unrounded event records.

Across both calculations, the maximum difference between the stored-energy drop and the recorded capture loss is  $2.274 \times 10^{-13}$  J. Positions, and hence spring potentials, do not change during these projections. The recorded loss is itself calculated from the kinetic-energy drop, so this comparison checks the inventory and event bookkeeping rather than independently verifying the energy-loss law. The projection's momentum and active-constraint residuals are checked separately against the limits in Table D.10. At the upper stop the cumulative remainder has zero recorded jump in both calculations, despite the rapid change over the adjoining continuous intervals shown in Figure 4.1(c).

Let  $D(h)$  denote the sum of the continuous-interval remainders  $W_i - \Delta E_i - E_{\text{slip},i}$ . Four spacings are applied to the same nominal trajectory. At  $h = 5, 2.5, 1.25, 0.625$  ms, these remainders are respectively 0.090744, 0.040638, 0.028244, 0.025445 J. The reported second-order zero-spacing estimate is

$$D_0 = \frac{4D(h) - D(2h)}{3}, \quad h = 0.625 \text{ ms}, \quad (\text{D.2.1})$$

giving +0.024511 J. Only the two finest spacings define this extrapolate. It estimates the remainder of the fixed numerical trajectory as power-integration spacing tends to zero; it is neither a bound

on the solution error nor the integration error remaining at  $h$ . Short intervals already governed by the minimum of 12 subintervals do not refine in proportion to  $h$ . The four results therefore support reduced sensitivity to power sampling, rather than an exact determination of the limiting remainder.

The final balance fraction used by the study is

$$\epsilon_E = \frac{|R_E(t_f)|}{\max(|W_B(t_f)|, 1 \text{ J})}. \quad (\text{D.2.2})$$

The 1 J floor is inactive here. The observed fractions are  $4.88 \times 10^{-7}$  and  $4.38 \times 10^{-7}$  for the nominal and tighter calculations. Their maximum sampled absolute remainders are 0.039 779 J and 0.038 513 J, respectively. Signed work from velocity drift at contacts declared sticking is retained only as a diagnostic: the nominal value is +0.001016 J. It is not counted as frictional dissipation or subtracted from  $R_E$ .

The final-state comparison uses  $x_k \in \{\omega_p, \omega_s, v_b, s, \dot{s}\}$ , with

$$S_k = \max(|x_{k,\text{nom}}(t_f)|, |x_{k,\text{tight}}(t_f)|, S_{k,\text{min}}), \quad \epsilon_{x,f} = \max_k \frac{|x_{k,\text{tight}}(t_f) - x_{k,\text{nom}}(t_f)|}{S_k}, \quad (\text{D.2.3})$$

and floors (1 rad/s, 1 rad/s, 1 m/s,  $10^{-3}$  m,  $10^{-3}$  m/s). The observed value is  $1.30 \times 10^{-7}$ . The signed tighter-minus-nominal differences in primary speed, secondary speed and belt speed are  $+5.151 \times 10^{-5}$  rad/s,  $+6.017 \times 10^{-5}$  rad/s and  $+3.830 \times 10^{-6}$  m/s; final shift position and shift speed are unchanged. This final-state comparison does not test agreement of the complete trajectories or all event timings.

| Study review limit                                     | Value                |
|--------------------------------------------------------|----------------------|
| Final balance fraction $\epsilon_E$                    | $10^{-4}$            |
| Absolute event-energy defect                           | $1 \times 10^{-8}$ J |
| Impact momentum residual                               | $10^{-7}$            |
| Impact constraint residual                             | $10^{-8}$            |
| Normalized final-state difference $\epsilon_{x,f}$     | $10^{-2}$            |
| Allowed negative computed capture loss                 | $1 \times 10^{-9}$ J |
| Finest-to-previous absolute continuous-remainder ratio | 1.25                 |

Table D.10: Numerical review limits retained from the canonical energy study. All applicable checks pass. These limits are screening criteria, not derived bounds on numerical error; the main text reports the observed dimensional remainders and their response to refinement.

The maintained `energy-consistency/run.py` regenerates the numerical ledgers and publication figure using `cinder-cvt==1.1.2`. Its `--plot-only` option reuses the retained outputs after checking the executed input and output hashes, energy identities, both event sides and refinement quantities. Exact input snapshots, package versions and output hashes accompany the execution record. The study README gives the complete reproduction commands. No model term is fitted or subtracted to reduce the reported remainder.

### D.3 Solver Convergence

The study supporting [Section 4.2.3](#) uses the reference configuration and shaft boundaries of [Appendices C](#) and [C.2](#), with full throttle and zero road grade for ten seconds. The belt and vehicle

start at rest and the engine at 1800 rpm. The study overrides the shared case’s 5 s duration to include upper-stop arrival. Only the LSODA controls change; the contact, event and low-ratio seating rules remain fixed. The last of these is documented in ???. The maintained definition is [solver-convergence/study.json](#). Table D.11 separates the formal verification grid from the supplementary dense exploration.

| Calculation            | Relative tolerance                                                                                           | Absolute tolerance                                                                         | Maximum step          |
|------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------|
| Tight reference        | $10^{-7}$                                                                                                    | $10^{-10}$                                                                                 | 2.5 ms                |
| Intermediate run       | $10^{-4}$                                                                                                    | $10^{-7}$                                                                                  | 10 ms                 |
| Formal main grid       | $10^{-2}, 3 \times 10^{-3}, 10^{-3}, 3 \times 10^{-4}, 10^{-4}, 3 \times 10^{-5}, 10^{-5}, 3 \times 10^{-6}$ | $10^{-3}$ rtol                                                                             | 5, 10, 20, 50, 100 ms |
| Independent atol sweep | $10^{-4}$                                                                                                    | $10^{-5}, 3 \times 10^{-6}, 10^{-6}, 3 \times 10^{-7}, 10^{-7}, 3 \times 10^{-8}, 10^{-8}$ | 10 ms                 |
| Dense exploration      | $3 \times 10^{-6} - 3 \times 10^{-2}$ ; 115 levels                                                           | $10^{-3}$ rtol                                                                             | 5–150 ms; 43 levels   |

Table D.11: Solver-convergence designs. The dense grid contains 4,945 combinations and includes the formal main-grid axes. It does not replace the formal guards.

**Formal state and history comparison.** The archived transition signature is the ordered sequence of fired event names, outgoing mode strings and the Boolean `has_successor_state`. That flag records the provision of an outgoing state, not necessarily a nonzero velocity jump. The retained reference and intermediate caches were also checked directly on both sides of every event: each has 16 transitions and 13 velocity discontinuities, with the same locations in the event sequence. Stop release and primary resticking do not acquire fictitious jumps merely because an outgoing state is recorded. An absolute component difference above  $10^{-12}$  is used only to distinguish resolved jumps from endpoint roundoff in this supplementary audit; it is not a new integration tolerance.

For an exact signature match, corresponding smooth segments are parameterized by their own endpoint times,

$$u = \frac{t - t_m}{t_{m+1} - t_m} \in [0, 1]. \quad (\text{D.3.1})$$

Each retained dense interpolant is evaluated at the same 2,049 phase points. This separates differences within corresponding motion from differences in the times at which its constraints change. Incoming and outgoing event sides remain separate. A zero-duration segment retains its endpoint state, contributes to the maximum comparison, and has zero duration weight in the RMS. For each component, the scale is

$$S_k = \max \left( \max_i |x_{k,\text{ref}}(t_i)|, \text{range}_i x_{k,\text{ref}}(t_i), S_{k,\text{min}} \right), \quad (\text{D.3.2})$$

evaluated on the reference’s 1 ms segmentwise reporting trace with exact endpoints and the outgoing value retained at a shared timestamp. The scale floors are 100 rad/s for each shaft speed, 5 m/s for belt speed, 0.02 m for shift position and 0.05 m/s for shift speed. In that order and those units, the resulting scales are approximately

$$(S_{\omega_p}, S_{\omega_s}, S_{v_b}, S_s, S_{\dot{s}}) = (397.040, 463.848, 29.7486, 0.0200, 0.647974).$$

Thus

$$e_{k,m}(u) = \frac{x_{k,m}^{\text{cand}}(u) - x_{k,m}^{\text{ref}}(u)}{S_k}, \quad (\text{D.3.3})$$

and, using reference durations  $\Delta t_m$  and total duration  $T$ ,

$$\epsilon_{\text{RMS}} = \left[ \frac{1}{5T} \sum_k \sum_m \Delta t_m \int_0^1 e_{k,m}^2(u) du \right]^{1/2}. \quad (\text{D.3.4})$$

The integral is evaluated by the trapezoidal rule on the common phase grid;  $\epsilon_{\text{max}} = \max_{k,m,u} |e_{k,m}(u)|$  is its sampled maximum. Corresponding event-time errors are calculated separately. For regime mismatch, the union of all candidate and reference boundaries partitions absolute time into intervals of constant mode. Summing durations with different mode labels gives the mismatch fraction without a reporting-grid approximation. A different signature makes the aligned comparison undefined and fails acceptance even if other diagnostics are small.

| Metric                        | Guard          | Intermediate run         |
|-------------------------------|----------------|--------------------------|
| Aligned five-state RMS        | $\leq 10^{-4}$ | $4.27408 \times 10^{-7}$ |
| Aligned maximum               | $\leq 10^{-3}$ | $6.96223 \times 10^{-5}$ |
| Maximum event-time difference | $\leq 1$ ms    | 7.98615 $\mu\text{s}$    |
| Regime-mismatch fraction      | $\leq 10^{-3}$ | $8.45730 \times 10^{-6}$ |
| Transition signature          | Exact          | Exact, 16 transitions    |

Table D.12: Formal guards and the retained intermediate-run result. All guards must pass. The formal sweep has 39 completed integrations and 25 passes; the other main-grid run is an explicitly retained integration/domain failure.

The intermediate-run comparison is also given in physical units in [Table D.13](#). Its largest event-time difference is 7.986  $\mu\text{s}$ ; the combined duration of differing contact or support conditions is 84.573  $\mu\text{s}$  over the ten seconds.

| State variable        | Largest aligned difference |
|-----------------------|----------------------------|
| Primary shaft speed   | 0.002 36 rad/s             |
| Secondary shaft speed | 0.000 427 rad/s            |
| Belt speed            | 0.0335 mm/s                |
| Shift position        | 0.106 $\mu\text{m}$        |
| Shift speed           | 0.0451 mm/s                |

Table D.13: Intermediate-run differences from the tight reference, sampled within corresponding continuous intervals and on both event sides. These are event-aligned comparisons, not bounds against an exact solution or maxima at equal absolute times.

These limits screen the tested settings; they are not derived error bounds or physical contact thresholds. The peak guard corresponds to 20  $\mu\text{m}$  in shift position and 0.647 974 mm/s in shift speed. The observed intermediate differences in [Table D.13](#) are appreciably smaller. The main figure, [Figure 4.3](#), gives the RMS map and ranges of peak and timing differences across completed step caps. These ranges are the observed minima and maxima at each tested tolerance, with the failed row excluded. [Figure D.1](#) identifies the step caps underlying the timing range and retains the

independent absolute-tolerance sweep. All completed main-grid rows have the reference transition sequence. Twenty-five satisfy all limits, including all 20 combinations at  $\text{rtol} \leq 10^{-4}$ . The five  $\text{rtol} = 10^{-3}$  rows also pass, whereas the intervening  $3 \times 10^{-4}$  rows fail the peak-state check. No continuous or monotonic acceptance boundary is inferred between tested settings. The coarsest  $10^{-2}$ , 100 ms combination terminates during a deadzone force evaluation with a signed primary flyweight force of  $-8421.34 \text{ N}$ . This would require the ramp to pull on the flyweight. The frozen release rejects such an evaluation because it does not model flyweight separation and re-contact. Finer integrations of the same case complete; the failed row records a numerical failure, not a resolved physical lift-off event. The retained exception does not identify the offending internal step, so no particular overshoot is assigned as its cause.

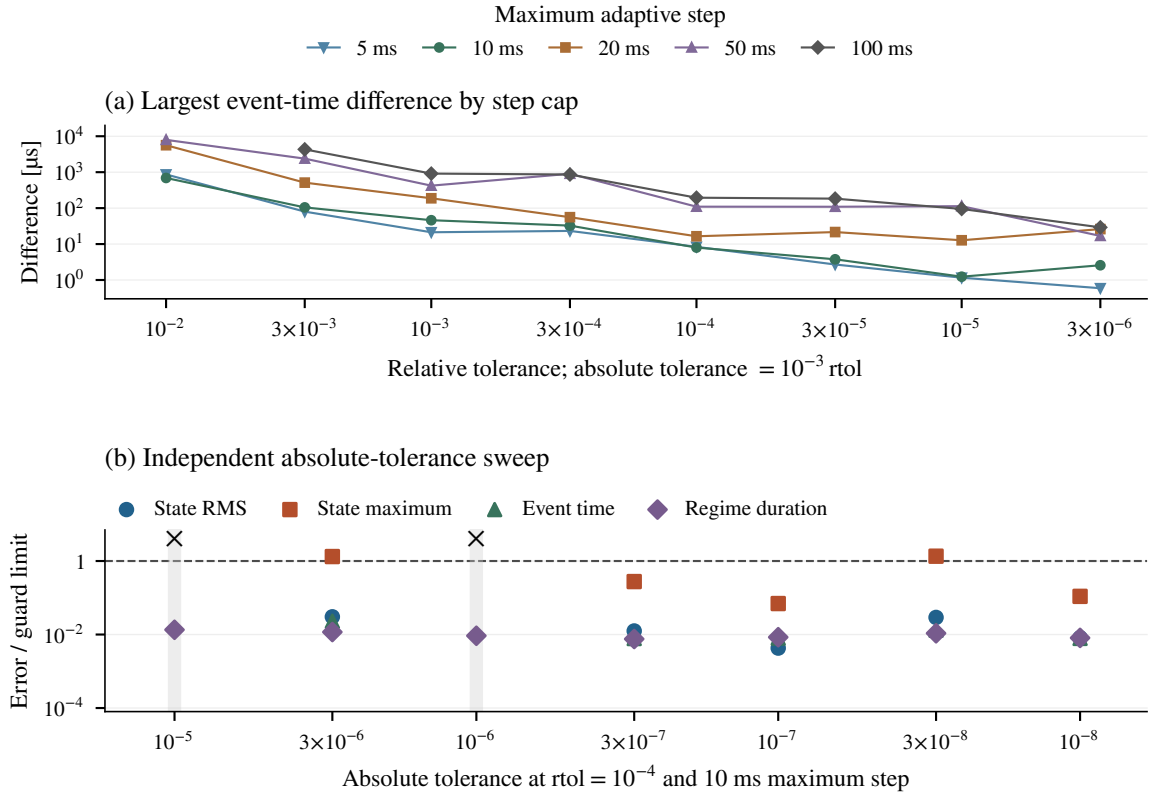


Figure D.1: Additional control comparisons. Panel (a) identifies the five step caps underlying the event-time range in Figure 4.3(c). The failed  $10^{-2}$ , 100 ms combination has no ordinate. Lines connect tested settings. Panel (b) varies absolute tolerance alone. Each defined measure is divided by its own limit, so values above one fail that check. Crosses and grey strips denote different transition sequences, for which aligned state and corresponding-event measures are undefined. The regime-duration comparison remains defined in absolute time.

Absolute tolerance sets the local-error scale near zero. It is therefore varied separately at fixed  $\text{rtol} = 10^{-4}$  and a 10 ms maximum step. In this sweep,  $10^{-5}$  and  $10^{-6}$  give different transition histories. The  $3 \times 10^{-6}$  and  $3 \times 10^{-8}$  rows retain the reference history but exceed the maximum-state guard, with shift-speed peaks of  $0.857448 \text{ mm/s}$  and  $0.876497 \text{ mm/s}$ . The other three rows satisfy all checks. The nonmonotonic peak at  $3 \times 10^{-8}$  is retained as a measured sensitivity; its native history was not archived, so no particular event or closure solve is assigned as its cause. The main-grid

intermediate row uses the proportional absolute tolerance stored as  $1.0000000000000001 \times 10^{-7}$ , whereas the retained intermediate cache uses literal  $10^{-7}$ . Their RMS values differ in late digits ( $4.2741159 \times 10^{-7}$  and  $4.2740764 \times 10^{-7}$ ). Figures and tables retain those execution identities rather than combining their unrounded metrics.

**Separate gross-motion diagnostic and selected history.** The exploratory population uses the archived absolute-time RMS for  $(\omega_p, \omega_s, v_b, s)$ , omitting  $\dot{s}$ . Its combined value is the square root of the mean of the four squared component RMS values, using the scales above. Unlike the formal norm, it is an unweighted average over the compact reference sample times. The original candidate trace has a 2 ms reporting spacing plus event endpoints and is linearly interpolated onto that reference trace. Shared timestamps retain the outgoing state. This compact diagnostic can therefore interpolate across a reset and includes timing and reporting-grid effects. It is retained as a gross-motion description, not used for formal hybrid acceptance or to infer a convergence order. The plotted shift traces instead retain separate native segments.

Figure D.2 retains this exploratory comparison and identifies the selected loose setting. Of 4,681 completed dense runs, 4,072 have the exact reference signature and 609 do not; 264 integrations fail. The latter completed group has a median gross RMS of  $1.2248411 \times 10^{-4}$  and a 90th percentile of  $3.493 \times 10^{-4}$ . Sorting that group by gross RMS selects the middle row at  $\text{rtol} = 0.012684971009663653$ ,  $\text{atol} = 1.2684971009663654 \times 10^{-5}$  and  $\Delta t_{\text{max}} = 0.025041423884469123$  s. Its trajectory was not retained in the original dense archive, so those settings were replayed with the installed frozen release and current study decoder. The replay reproduces 12 transitions and 486 native solver points, with gross RMS  $1.2249757 \times 10^{-4}$  rather than the archived  $1.2248411 \times 10^{-4}$ . These are identified as separate executions; the replay supplies the displayed history, not a claimed recovery of the original event timestamps.

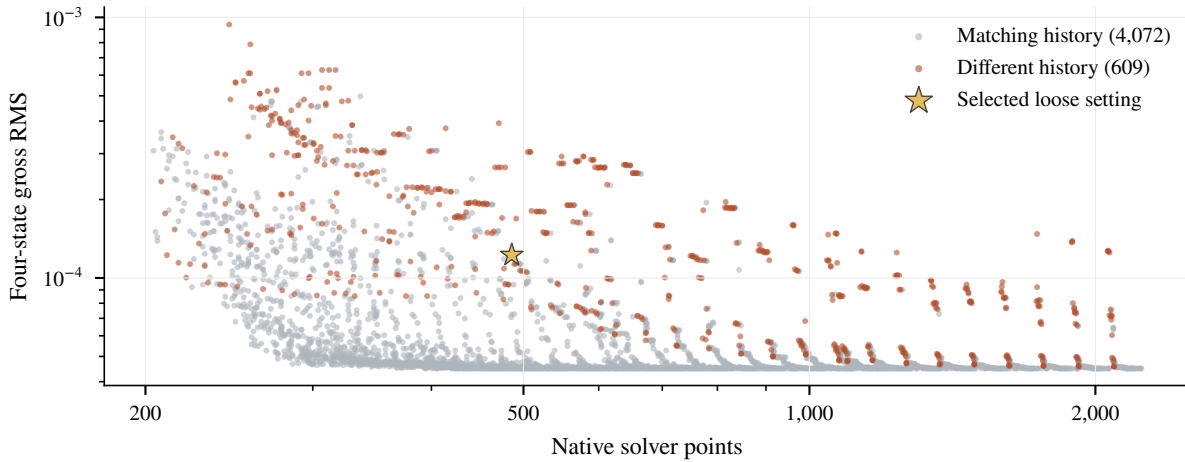


Figure D.2: Supplementary exploratory population and selection of the loose setting. The four-state compact gross-motion RMS is plotted against retained native solver points, with groups distinguished by their complete transition sequence. The star is the archived median among completed different-sequence calculations; its later replay supplies the resolved motion in Figure 4.2. Failed integrations have no trajectory norm and are excluded. This diagnostic omits shift speed and is not the formal five-state acceptance measure.

| Excursion | Reference [ $\mu$ s] | Intermediate [ $\mu$ s] | Loose replay [ $\mu$ s] |
|-----------|----------------------|-------------------------|-------------------------|
| 1         | 1878.269             | 1877.716                | 1412.204                |
| 2         | 331.672              | 331.407                 | 192.146                 |
| 3         | 60.418               | 60.292                  | 7.219                   |
| 4         | 11.051               | 11.000                  | —                       |
| 5         | 2.023                | 1.983                   | —                       |

Table D.14: Successive primary-separated intervals after first engagement and before low-ratio seating. Dashes mean that no further excursion occurs, not an interval of zero duration. The replay also retains a zero-duration engaged segment at its final engagement/seating timestamp; neither event is discarded.

The seating marks in [Figure 4.2](#) occur at 66.012 ms (reference), 66.004 ms (intermediate) and 64.830 ms (loose replay). The first separated excursions reach sampled depths of 49.016  $\mu$ m, 48.980  $\mu$ m and 22.326  $\mu$ m, respectively. The markers use the first outgoing low-ratio support event, not an inference from the apparent flattening of the position trace. The progressively shorter returns in [Table D.14](#) terminate under the fixed numerical seating rule. They describe the retained implementation and are not offered as independently resolved physical chatter. Their true durations are retained in the segmentwise motion curves, including intervals narrower than a display pixel; the full contact ledger remains in the retained evidence.

The small gross RMS is not a uniform dimensional bound. In the replay’s compact absolute-time comparison, the sampled maximum differences are 0.12045 rad/s in primary speed, 0.07551 rad/s in secondary speed, 0.21239 m/s in belt speed and 0.13422 mm in shift position. These maxima include the differing event times and the reporting convention just described. Native solver-point count likewise has a limited role: it sums retained adaptive points over segments, including repeated boundary timestamps, and is not a count of right-hand-side evaluations. In the archived formal execution, the intermediate run retains 1,156 points and takes 7.257 s for hybrid integration, compared with 4,451 points and 25.665 s for the tight reference. The original machine is not identified, so these are within-execution comparisons, not a portable speed claim or an optimal-setting result.

The maintained `solver-convergence/run.py` provides an archive-import path, the selected-history replay and a checked `--plot-only` path. Registered archive and member hashes identify the retained formal and dense tables; both-side caches permit direct recomputation of the intermediate-run metrics. The other per-point caches were not retained, so their tabulated guards are rechecked without claiming a fresh integration or native-history audit for every row. Exact replay inputs, numerical output hashes and the original-versus-replayed distinction are recorded in [the study provenance note](#).

## D.4 Closure Conditioning and Root Structure

The study separates the linear response at prescribed traction from the nonlinear traction selection required by sticking. Its fixed-traction condition number is  $\kappa_2(\mathbf{A}_s)$ , where  $\mathbf{A}_s$  is the row/column-equilibrated eight-variable matrix in [Equations \(B.2.1\)](#) and [\(B.2.3\)](#). The retained pulley torques are implementation unknowns; eliminating them recovers the six-variable mechanical formulation. The sticking-root Jacobian is a different object: the derivative of the frozen dimensional relative-acceleration residuals with respect to the two tractions. It is reconstructed by central differences at

the reported root, using  $h = 10^{-5}$ . The optimizer’s working Jacobian is not substituted for this derivative, because continuation can update that matrix from previous states. Across all 15 roots, the relative spread in  $\kappa_2(\mathbf{J}_R)$  over  $10^{-6} \leq h \leq 3 \times 10^{-4}$  is at most  $2.01 \times 10^{-6}$ .

#### D.4.1 Operating-State Maps and Starting-Guess Census

The retained full run uses the configuration in [closure-conditioning/study.json](#). Four states come from the reference launch; eleven are constructed stick–stick cases from the shared verification library. The low-ratio seat, mid-shift, upper-stop and **free\_shift\_opening\_70** states receive traction-plane maps. The last is a free opening state at 70% engaged travel and  $-20$  mm/s. The low-ratio seat already has both the largest actual-root mechanical condition number and the largest actual-root Jacobian condition number, so the worst-state selection adds no fifth map.

| Quantity                                       | Value                                      |
|------------------------------------------------|--------------------------------------------|
| Reference launch duration / reporting interval | 10 s / 2 ms                                |
| Reference launch rtol / atol / maximum step    | $10^{-4}$ / $10^{-7}$ / 10 ms              |
| Static / kinetic traction magnitudes           | $\mu_s = 0.65$ / $\mu_k = 0.55$            |
| Static-box starting grid                       | $7 \times 7$ at each of 15 states          |
| Expanded-box starting grid                     | $9 \times 9$ at each of four mapped states |
| Root-clustering distance                       | $10^{-6}$ in $(\lambda_p, \lambda_s)$      |

Table D.15: Retained full-run settings. Starting grids include their endpoints; the expanded census is separate from the 735 static-box attempts.

| Domain          | Grid per state   | Definition and scope                                                         |
|-----------------|------------------|------------------------------------------------------------------------------|
| Static capacity | $241 \times 241$ | $[-0.65, 0.65]^2$ , four states. Contact and support checks remain separate. |
| Expanded        | $321 \times 321$ | $[-2.6, 2.6]^2$ , four states; mathematical continuation.                    |
| Broad           | $501 \times 501$ | $[-10, 10]^2$ , mid-shift only; structural diagnostic.                       |

Table D.16: Retained traction domains. The expanded half-width is  $\max(2.5, 4\mu_s) = 2.6$ . Map maxima are sampled values, not bounds between grid points or condition numbers at the accepted roots.

All 644,488 sampled systems in the static and expanded maps have numerical rank eight. This statement does not extend to the broad diagnostic: 76,746 of its 251,001 points are numerically rank deficient. The static-map maxima at the low-ratio seat, mid-shift, upper stop and opening state are, respectively,  $1.3260 \times 10^4$ ,  $3.6957 \times 10^4$ ,  $1.0540 \times 10^6$  and  $8.2612 \times 10^3$ . The upper-stop maximum occurs at  $(\lambda_p, \lambda_s) = (-0.216667, -0.254583)$ ; negative integrated normal force, tension and local wrap-normal loading make that trial inadmissible independently of the friction limits.

Every retained map includes signed normal resultants, minimum belt tension, minimum local wrap-normal loading, an actuator-contact margin and the support reaction. Its original mask includes the one-sided secondary-helix guard. For the two maps in [Figure 4.4](#), the same mask also applies to the bilateral reference: every rejected grid point independently fails a belt-loading or support condition, while every accepted point passes both original actuator-contact checks. Removing the secondary flank inequality therefore changes neither plotted mask and does not discard a primary-contact failure. There are 35,817 admissible points out of 58,081 at the upper stop, and 51,198 at mid-shift. This equivalence is checked for these two maps only; the historical masks of the other retained maps keep their original meaning.

The focused two-root field in [Figure 4.5](#) uses a separate  $121 \times 161$  contact audit under the bilateral reference. The primary-contact guard remains active. All 19,481 recalculated residual pairs agree exactly with the retained field; 19,005 grid points pass the contact checks. The dimensional residual norm is plotted on a logarithmic colour scale, with the two signed components supplying their separate zero contours. Cells touching a rejected point are excluded from these contours, so an interpolated sign change across a contact-invalid region cannot appear as a sticking solution. Root markers come from the corrected solutions, not from interpolation of the colour field.

| Frozen state          | Accepted | Not accepted | $\kappa_2(A_s)$    | $\kappa_2(J_R)$ |
|-----------------------|----------|--------------|--------------------|-----------------|
| low_ratio_seat        | 49       | 0            | $1.04 \times 10^4$ | 280             |
| mid_shift             | 42       | 7            | $9.44 \times 10^3$ | 22.8            |
| late_shift            | 42       | 7            | $8.02 \times 10^3$ | 28.0            |
| upper_stop            | 35       | 14           | $8.56 \times 10^3$ | 93.8            |
| stick_mid_forward     | 49       | 0            | $1.57 \times 10^3$ | 1.97            |
| stick_mid_reverse     | 49       | 0            | $1.57 \times 10^3$ | 1.97            |
| stick_low_forward     | 42       | 7            | $5.90 \times 10^3$ | 21.8            |
| stick_high_forward    | 46       | 3            | $2.52 \times 10^3$ | 4.68            |
| stick_mid_stressed    | 45       | 4            | $1.82 \times 10^3$ | 6.62            |
| free_shift_closing_30 | 42       | 7            | $3.76 \times 10^3$ | 22.0            |
| free_shift_closing_50 | 49       | 0            | $1.62 \times 10^3$ | 2.39            |
| free_shift_closing_70 | 42       | 7            | $3.17 \times 10^3$ | 18.5            |
| free_shift_opening_30 | 42       | 7            | $3.83 \times 10^3$ | 20.3            |
| free_shift_opening_50 | 49       | 0            | $1.52 \times 10^3$ | 1.66            |
| free_shift_opening_70 | 43       | 6            | $3.62 \times 10^3$ | 13.6            |
| Total                 | 666      | 69           |                    |                 |

Table D.17: All 735 static-box starts and condition numbers at the accepted operating roots. Every state’s accepted solutions form one cluster. Forward/reverse and closing/opening cases remain distinct.

Numerical acceptance requires optimizer success and the release’s sticking-compatibility checks. It does not itself certify every contact inequality. Here the accepted solutions all lie within the static limits and coincide, within the clustering tolerance, with the mechanically accepted operating roots. The expanded census accepts 66, 27, 37 and 33 of 81 starts at the low-ratio seat, mid-shift, upper stop and opening state, respectively. Thus 163 of 324 expanded starts are accepted, again in one cluster per state, and none supplies an additional root. Unsuccessful starts remain in both censuses. These finite searches neither exhaust traction space nor establish global uniqueness.

All 69 rejected static-box searches returned optimizer success but failed sticking compatibility. Their residual norms range from 13.31 to 424.36 m/s<sup>2</sup>, compared with at most  $3.10 \times 10^{-11}$  m/s<sup>2</sup> for the accepted searches. In 65 of them, the secondary traction reaches the numerical search bound  $|\lambda_s| = 1.95$ ; the other four terminate inside that wider search interval. All 69 also exceed the physical secondary static-friction limit. These are unsuccessful searches for a compatibility root, rather than additional roots rejected only on physical grounds. The retained rows do not include the optimizer’s individual termination messages. At the 15 accepted operating roots, the mechanical condition number ranges from  $1.52 \times 10^3$  to  $1.04 \times 10^4$ , indicating substantially less numerical sensitivity than the rejected upper-stop static-map maximum.

### D.4.2 Selected Folded Branches

The two-contact example in [Figure 4.5](#) fixes position at 95% of engaged travel and shift speed at  $-40$  mm/s. Primary speed is 3000 rpm; secondary speed is 3286.525553 rpm and belt speed is 22.72382860 m/s, chosen to match both represented surfaces. The attached primary and secondary bench inertias are 0.03 and 0.05 kg m<sup>2</sup>, respectively. They are not the nominal engine and vehicle boundaries. At the selected slice, the primary boundary torque is  $-81.14811271$  N m and the secondary boundary torque is  $+189.96446922$  N m: with both shafts rotating positively, the secondary boundary supplies power and the primary boundary absorbs it.

| Quantity                                       | Root 1          | Root 2          |
|------------------------------------------------|-----------------|-----------------|
| Primary traction $\lambda_p$                   | $-0.2676104653$ | $-0.3003678175$ |
| Secondary traction $\lambda_s$                 | $+0.3340981242$ | $+0.5521247953$ |
| Primary normal resultant [N]                   | 3077.56         | 3122.29         |
| Secondary normal resultant [N]                 | 2400.85         | 1668.60         |
| Primary acceleration [rad/s <sup>2</sup> ]     | $-575.334$      | $-346.401$      |
| Secondary acceleration [rad/s <sup>2</sup> ]   | $+2919.426$     | $+2736.666$     |
| Belt acceleration [m/s <sup>2</sup> ]          | $-72.4980$      | $-55.9388$      |
| Shift acceleration [m/s <sup>2</sup> ]         | $+142.544$      | $+128.944$      |
| Minimum belt tension [N]                       | 174.100         | 170.668         |
| Minimum primary local normal loading [N/rad]   | 65.7586         | 44.8540         |
| Minimum secondary local normal loading [N/rad] | 20.8365         | 0.242602        |
| $\kappa_2(A_s)$                                | 6305.92         | 4364.80         |
| $\kappa_2(J_R)$                                | 13.8870         | 27.5920         |

Table D.18: Retained frozen evaluation of the two traction pairs under installed CINDER 1.1.2, at identical state and boundary torques. Both satisfy the friction and mechanical-contact inequalities. The signed accelerations show that the two roots define different local responses.

The smaller static margin over both roots and interfaces is  $\mu_s - |\lambda_j| = 0.0978752$ ; it is not a normalized percentage. Both fixed-traction matrices have rank eight. The checked two-component sticking-residual norms are below  $5.5 \times 10^{-11}$  m/s<sup>2</sup>. Each root’s relative  $\kappa_2(\mathbf{J}_R)$  spread over the same step range is below  $1.57 \times 10^{-6}$ .

The fixed-primary-torque branch is corrected from the retained inverse-grid curve by solving both sticking residuals for  $\lambda_p$  and secondary boundary torque at each prescribed  $\lambda_s$ . The displayed open segment has 184 points over  $0.30 \leq \lambda_s \leq 0.59$ , including both roots and the refined turn. Direct CINDER evaluation checks each point; no endpoint is joined to the other. Its local maximum is 197.3901853 N m at  $(\lambda_p, \lambda_s) = (-0.2856750, 0.4134494)$ , 7.4257161 N m above the selected loading. The turn is mechanically admissible and the fixed-traction matrix remains rank eight throughout this segment. The root Jacobian becomes nearly singular there, as expected when the traction response to loading turns; this is a numerical local-extremum calculation, not an exact singularity certificate. The dotted part of the curve has negative secondary local wrap-normal loading and is excluded from the admissible branch.

The mixed example fixes the same travel fraction and opening speed but uses 2500 rpm primary speed, 2736.737483 rpm secondary speed and 18.93652384 m/s belt speed. It retains the same bench inertias. Boundary torques are  $-260$  N m primary and  $+59.34019609$  N m secondary. With  $\lambda_p = -\mu_k = -0.55$ , the secondary sticking roots are 0.4342879568 and 0.5458610486. Their scalar residual slopes  $\partial R_s / \partial \lambda_s$  are approximately  $+54.6254$  and  $-28.0426$  m/s<sup>2</sup>, respectively; a two-contact Jacobian condition number is not used for this scalar problem. Both roots have zero relative speed

at the selected instant. Primary relative accelerations of  $+364.835$  and  $+379.770$  m/s<sup>2</sup> have the required sign for onset of the negative-traction slip branch. The secondary residual magnitudes are below  $1.8 \times 10^{-10}$  m/s<sup>2</sup>. Both roots pass the contact checks, although the minimum secondary local normal loading falls from 2.60109 to 0.00616308 N/rad. The targeted search has 62 states and 15 candidate rows, but those rows represent one distinct state/branch fold geometry under different fixed primary torques. They are not 15 independent fold discoveries or a frequency estimate.

### D.4.3 Retained Evidence and Reproduction

The earlier recap used `artifacts(10).zip`: 25 starts at each of 15 states, and  $81^2$ ,  $101^2$  and  $121^2$  maps. Its 375 attempts and upper-stop static-map maximum  $3.38405 \times 10^5$  therefore do not describe the full protocol above. The later retained `artifacts(20260912-044618).zip` supplies the full census and maps; no launch, broad map or multistart sweep was rerun for this finalization. The selected folded branches come from separately registered targeted archives. The two-contact curve's archived interpolated maximum, 197.4374 N m, is replaced by the locally corrected value above, with both versions retained in the evidence.

The canonical `run.py` provides a checked `--import-retained` path for the three original archives and a `--plot-only` path for the delivered evidence. The corrected roots and branch were recovered from the earlier finalization package and verified against their registered hashes. This revision reuses those calculations, all full maps and both starting-guess censuses. New mechanical evaluations are limited to the focused contact grid above; `--audit-contours` reproduces that check without a transient or multistart search. Both execution records preserve exact input snapshots and output hashes under installed `cinder-cvt==1.1.2`. The newer live simulator is not imported.

Original archive/member hashes, the recovered package identity, numerical checks and exact reproduction commands are recorded in [the study provenance note](#). The retained archives do not contain a complete original machine and dependency lock, so reproducing the plots and the selected frozen checks does not establish bitwise regeneration of the historical full search.

## D.5 Independent Ballew Benchmark

Exact benchmark inputs supporting [Section 4.3](#) are defined in `study.json`. The reconstruction choices that translate Ballew's distributed-belt case into the quantities required by CINDER are recorded in `RECONSTRUCTION.md`. The benchmark assembly and shaft boundaries are constructed in `case.py`, with the source and reconstructed constants collected in `constants.py`. The comparison is model-to-model; the reference traces are Ballew's simulated vehicle-acceleration results rather than experimental measurements.

### D.5.1 Source Data and Reconstruction

The reconstruction retains Ballew's published pulley geometry, belt mass, shaft inertias, vehicle data, initial shaft speeds, applied torque, secondary clamp, and controller gains where they are reported. Quantities required by CINDER but not specified by the source are introduced explicitly rather than fitted to the digitized trajectories.

| Source-native quantity                 | Value                   |
|----------------------------------------|-------------------------|
| Sheave half-angle                      | 15°                     |
| Pulley centre distance                 | 0.2421 m                |
| Published primary radius limits        | 0.0159 m to 0.0838 m    |
| Published secondary radius limits      | 0.0159 m to 0.0838 m    |
| One-dimensional belt length            | 0.8636 m                |
| Belt mass                              | 1.000 kg                |
| Primary pulley + engine inertia        | 0.008 kg m <sup>2</sup> |
| Secondary pulley inertia               | 0.002 kg m <sup>2</sup> |
| Secondary pulley + ATV inertia         | 1.275 kg m <sup>2</sup> |
| Static / kinetic friction coefficients | 0.55/0.40               |
| Final-drive reduction                  | 8.93                    |
| ATV mass                               | 226 kg                  |
| Tire radius                            | 0.317 m                 |
| Rolling-resistance coefficient         | 0.048                   |
| Aerodynamic drag coefficient           | 1.0                     |
| Frontal area                           | 1.39 m <sup>2</sup>     |
| Primary applied torque                 | 18 N m                  |
| Secondary closing force                | 2000 N                  |
| Initial primary / secondary speed      | 2500/1136 rpm           |
| Simulated interval                     | 5 s                     |

Table D.19: Source-native quantities retained in the Ballew reconstruction.

Ballew’s transient belt is one-dimensional, whereas CINDER distinguishes the outer belt geometry from the cord-line effective path. The reconstruction identifies Ballew’s published 0.8636 m belt length with the CINDER effective path. The smooth section taken from Ballew’s Figure 39 is treated as an equivalent load-carrying core with height 7.62 mm, outer width 29.972 mm, inner width 26.670 mm, and cord depth 1.270 mm. The corresponding CINDER outer length is 0.871 579 645 m. An effective density of 5316.543 kg/m<sup>3</sup> then preserves the published 1.000 kg belt mass exactly. This density is a mass-preserving reconstruction parameter, not a measured material density.

The published minimum radii are search limits rather than a simultaneously compatible fixed-length endpoint. At the reconstructed low-ratio end, the secondary effective radius is fixed at its published maximum, 0.0838 m, and belt-length closure gives the compatible primary effective radius 0.033 647 778 m. The always-engaged Ballew case uses zero deadzone width, and the resulting CINDER shift range is 0.026 876 495 m. Solving within that geometry for the exact published shaft-speed ratio gives

$$\begin{aligned}
n_p(0) &= 2500 \text{ rpm}, & n_s(0) &= 1136 \text{ rpm}, \\
s(0) &= 0.001\,754\,497 \text{ m}, & \dot{s}(0) &= 0, \\
r_{p,\text{eff}}(0) &= 0.036\,921\,714 \text{ m}, & r_{s,\text{eff}}(0) &= 0.081\,253\,771 \text{ m}, \\
v_b(0) &= 9.666\,082 \text{ m/s}.
\end{aligned} \tag{D.5.1}$$

The two models use different normal-force conventions. Preserving the same gross tangential capacity gives

$$\lambda = \frac{\mu \cos \beta}{2}, \tag{D.5.2}$$

and therefore  $\lambda_s = 0.2656296022$  and  $|\lambda_k| = 0.1931851653$  for  $\beta = 15^\circ$ . This is a force-normalization translation rather than a friction fit.

The source traces are digitized from Ballew’s Figures 41 and 45. Their raw WebPlotDigitizer exports, project files, prepared CSVs, and the source thesis are stored under [reference/](#); file hashes are recorded in [manifest.json](#). Preparation adds explicit headers, preserves the two native Figure 41 time grids, averages exact duplicate Figure 45 timestamps, and extends the first visible Figure 45 force back to  $t = 0$  by zero-order hold. No smoothing, filtering, curve fitting, or resampling is applied.

### D.5.2 Shaft Boundaries

Ballew publishes a combined primary-pulley/engine inertia, so the full  $0.008 \text{ kg m}^2$  is assigned to the primary shaft boundary and is not duplicated inside the CVT. On the secondary side, CINDER owns the published  $0.002 \text{ kg m}^2$  output-pulley inertia. Vehicle translation reflected through the fixed 8.93 reduction contributes  $0.284\,789\,55 \text{ kg m}^2$ ; the remaining  $0.988\,210\,45 \text{ kg m}^2$  is carried as direct secondary-boundary inertia so that the published  $1.275 \text{ kg m}^2$  output-pulley–ATV total is preserved.

| Quantity                                             | CINDER<br>boundary                   | Value                           |
|------------------------------------------------------|--------------------------------------|---------------------------------|
| Primary applied torque $\tau_{B,p}$                  | Constant                             | 18 N m                          |
| Primary referred inertia $I_{B,p}$                   | torque–inertia<br>Constant           | $0.008 \text{ kg m}^2$          |
| Final-drive reduction $G$                            | torque–inertia<br>Locked final drive | 8.93                            |
| Tire radius $r_w$                                    | Locked final drive                   | 0.317 m                         |
| ATV mass $m_v$                                       | Locked final drive                   | 226 kg                          |
| Wheel rotational inertia $I_w$                       | Locked final drive                   | $0 \text{ kg m}^2$              |
| Rolling-resistance coefficient $C_{rr}$              | Road load                            | 0.048                           |
| Aerodynamic drag coefficient $C_d$                   | Road load                            | 1.0                             |
| Frontal area $A_f$                                   | Road load                            | $1.39 \text{ m}^2$              |
| Road grade $\gamma$                                  | Constant grade                       | 0                               |
| Air density $\rho$                                   | Reconstruction<br>choice             | $1.225 \text{ kg/m}^3$          |
| Gravity $g$                                          | Reconstruction<br>choice             | $9.806\,65 \text{ m/s}^2$       |
| Rolling-speed regularization $v_\epsilon$            | Reconstruction<br>choice             | $0.01 \text{ m/s}$              |
| Reflected ATV translation $m_v(r_w/G)^2$             | Locked final drive                   | $0.284\,789\,55 \text{ kg m}^2$ |
| Direct secondary-boundary inertia $I_{s,\text{dir}}$ | Locked final drive                   | $0.988\,210\,45 \text{ kg m}^2$ |
| Total secondary boundary inertia $I_{B,s}$           | Locked final drive                   | $1.273 \text{ kg m}^2$          |

Table D.20: Shaft-boundary quantities used by the independent Ballew benchmark.

### D.5.3 Actuation and Host State

The benchmark-specific axial-force laws are defined in [Appendix C.3.2](#). Both protocols retain the same reconstructed geometry, inertias, friction translation, secondary clamp, and vehicle boundary; only the primary clamping law differs.

| Protocol     | Primary actuator                                                 | Secondary actuator            | Host state                                           |
|--------------|------------------------------------------------------------------|-------------------------------|------------------------------------------------------|
| Force replay | Prepared Figure 45 force history; piecewise-linear interpolation | Constant 2000 N closing force | Secondary-shaft angle                                |
| Closed loop  | PI force law of <a href="#">Equation (C.3.7)</a>                 | Constant 2000 N closing force | Secondary-shaft angle and controller integral $\eta$ |

Table D.21: Actuator and host configuration of the two Ballew benchmark protocols.

For the reconstructed closed-loop protocol, the controller uses  $e = n_p - 2500$  rpm,  $K_{ff} = 1.2$ ,  $K_p = 5$ , and  $K_i = 75$ . The feed-forward term is interpreted as  $K_{ff}F_{s,act} = 2400$  N, with  $\eta(0) = 0$ . No saturation or anti-windup law is added because neither is specified by the source.

#### D.5.4 Numerical Settings

| Protocol     | rtol      | atol      | Max. step | Report step | Duration |
|--------------|-----------|-----------|-----------|-------------|----------|
| Force replay | $10^{-7}$ | $10^{-9}$ | 1 ms      | 5 ms        | 5 s      |
| Closed loop  | $10^{-7}$ | $10^{-9}$ | 1 ms      | 0.2 ms      | 5 s      |

Table D.22: Numerical settings used by the two Ballew benchmark protocols.

Both protocols use LSODA, permit up to 2000 hybrid transitions, and retain dense output. The separate closed-loop refinement check is defined in [NUMERICAL\\_STABILITY.md](#).

| Refinement case               | rtol               | atol                | Max. step |
|-------------------------------|--------------------|---------------------|-----------|
| Nominal                       | $10^{-7}$          | $10^{-9}$           | 1.00 ms   |
| Nominal, refined step         | $10^{-7}$          | $10^{-9}$           | 0.50 ms   |
| Nominal, further refined step | $10^{-7}$          | $10^{-9}$           | 0.25 ms   |
| Tighter tolerances            | $3 \times 10^{-8}$ | $3 \times 10^{-10}$ | 0.50 ms   |

Table D.23: Closed-loop numerical-refinement cases defined for the Ballew benchmark.

All four cases complete the 5 s interval. The reference errors remain 110.32 rpm in primary speed, 32.580 rpm in secondary speed and 1196.25 N in primary clamp at the displayed precision. To compare the actual trajectories, the solver’s dense solution is evaluated at common 0.2 ms times within each continuous segment. Values are never interpolated across an event; a time exactly on an event uses the outgoing state. [Table D.24](#) reports differences from the nominal 1 ms run, rather than differences from Ballew’s reference.

| Case             | Primary speed<br>RMS [rpm] | Primary clamp<br>RMS [N] | Shift position<br>max. [mm] | Matched event<br>max. [ $\mu$ s] |
|------------------|----------------------------|--------------------------|-----------------------------|----------------------------------|
| Nominal, 0.50 ms | 0.00942                    | 0.0471                   | 0.00384                     | 0.513                            |
| Nominal, 0.25 ms | 0.01296                    | 0.0649                   | 0.00516                     | 0.688                            |
| Tighter, 0.50 ms | 0.03203                    | 0.1604                   | 0.01258                     | 1.673                            |

Table D.24: Closed-loop refinement differences from the nominal 1 ms trajectory. Speed, clamp and position are compared at the same times over 0 s to 5 s; event times are matched by the ordered event records after excluding slip-direction updates, as described below. The complete state and event comparisons are retained in the publication evidence audit.

The raw event totals are 1735, 1735, 1722 and 1421, in the order of [Table D.23](#). They include 320, 320, 307 and 6 zero-crossing updates of kinetic-slip direction. Excluding only those updates leaves the same ordered 1415 event reasons and incoming/outgoing sticking, sliding and support labels in all four runs. These include 207 arrivals at the low-ratio seat. Their largest matched timing difference is 1.673  $\mu$ s. The large oscillations and repeated seating therefore persist while the direction-update count is sensitive to the integration settings.

Shift velocity requires care at those arrivals because it changes instantaneously. At the common sample  $t = 0.827$  s, the nominal run is already seated while the 0.25 ms and tighter-tolerance runs are still arriving. The resulting same-time velocity difference is about 6.41 m/s; it compares opposite sides of the capture. Among common-grid samples with matching contact and support labels, the largest shift-speed difference across the refinements is 0.002 62 m/s. Event times and continuous quantities are therefore reported separately, with both event sides retained in the evidence.

The slip-loss curves integrate  $|(\tau_j/r_{j,\text{eff}})v_{\text{rel},j}|$  separately on each continuous segment. Reintegrating the retained power samples reproduces the reported primary and secondary totals, 12.3008 kJ and 7.4050 kJ. Retaining every second report sample, while keeping both segment endpoints, changes their sum by 7.98 J, or 0.041%. This sampling check supports the 19.7 kJ scale used in the main text; it is not a complete energy-balance or efficiency calculation.

### D.5.5 Reproduction and Retained Evidence

The benchmark and refinement runs use `cinder-cvt==1.1.2`, at the simulator commit identified in [Section 4.1](#). The maintained study’s [README.md](#) gives the environment setup and complete commands. From the repository root, the canonical runner [run.py](#) accepts `--with-refinement` to run both protocols, the three additional refinements and the evidence audit. Its `--plot-only` option reproduces the final figures from retained inputs without repeating the simulations.

The exact executed source files, solver settings, reference hashes and output hashes are recorded for every run. The [publication\\_inputs/](#) directory retains compact plot and refinement arrays, selected without resampling. Their [checksum manifest](#) identifies every retained file. The full state and event exports accompany the manuscript delivery. The publication plotter draws each continuous segment separately, including its endpoints, and applies no smoothing. The audit recomputes the comparison errors, checks the retained event sides and evaluates the refinement quantities reported here.

## D.6 Primary-Actuator Dynamics

The comparisons in [Section 4.4.1](#) use the reference assembly and boundaries of [Appendices C.1](#) and [C.2](#), with the primary reduction in [Appendix C.3.1](#). The full and quasi-static models have the same initial total primary shaft-axis inertia,  $0.00491861 \text{ kg m}^2$ , excluding the engine boundary. The reduction keeps this value constant and retains the translating mass; the full model retains the position-dependent flyweight inertia and pivot motion. The secondary remains dynamic and uses the reference bilateral helix topology. These are comparisons within CINDER 1.1.2.

The baseline starts at 1800 rpm with all other CVT velocities and the shift coordinate zero, and runs for 10 s on level ground. The controlled screen starts from naturally reached states near 20%, 50% and 80% of engaged travel. At each position, the six torque increments  $\pm 5$ ,  $\pm 10$  and  $\pm 20 \text{ N.m}$  are combined with the four rise times 5, 20, 100 and 250 ms. Each increment follows  $A(3u^2 - 2u^3)$ , where  $u = (t - 0.05 \text{ s})/T$  during the ramp, and is then held for 0.3 s. It is added to the unchanged engine torque boundary. The 72 recovered screen runs completed without a post-onset contact or support transition. The plotted screen peaks use 2 ms reporting; they are sampled maxima, not continuous-time error bounds.

The selected  $A = -20 \text{ N.m}$ ,  $T = 5 \text{ ms}$  case is reintegrated from the archived physical states with their original free, stick-stick modes. These states were reached separately by the two models. Each receives both the torque ramp and an unforced control, giving

$$\delta y_{\text{QS}} - \delta y_{\text{full}}, \quad \delta y = y_{\text{forced}} - y_{\text{control}}. \quad (\text{D.6.1})$$

Both continuations remain free and sticking, so the comparison over 0.05 s to 0.355 s requires no interpolation across events. The secondary-angle host is rebased to zero; under this time-programmed level-road boundary it changes distance metadata only.

Exact state vectors, torque inputs and numerical settings are in [primary\\_publication.json](#). The individual screen rows are retained in [primary\\_screen.csv](#). The full source and output hashes, masks, force checks and unrounded metrics are recorded in [primary\\_publication\\_audit.json](#).

| Setting or measured quantity                      | Nominal            | Tighter            |
|---------------------------------------------------|--------------------|--------------------|
| Relative tolerance                                | $1 \times 10^{-4}$ | $1 \times 10^{-5}$ |
| Absolute tolerance                                | $1 \times 10^{-7}$ | $1 \times 10^{-8}$ |
| Baseline maximum step [ms]                        | 10                 | 1                  |
| Selected-ramp maximum step [ms]                   | 0.625              | 0.625              |
| First engagement, full [ms]                       | 61.96531           | 61.97135           |
| First engagement, quasi-static [ms]               | 60.03489           | 60.04125           |
| Approach speed, full [mm/s]                       | 565.138            | 565.163            |
| Approach speed, quasi-static [mm/s]               | 672.780            | 673.331            |
| Maximum paired shift difference [ $\mu\text{m}$ ] | 2.52668            | 2.52652            |
| Maximum paired primary-speed difference [rpm]     | 0.670406           | 0.670415           |

Table D.25: Focused LSODA refinement for the primary comparisons. Baseline values refer to the first engagement; paired differences refer to the selected torque ramp. Published traces use the tighter settings.

Baseline output is sampled every 1 ms, with additional 0.025 ms samples between 40 ms and 70 ms; every continuous segment retains its exact endpoints. The selected ramp is sampled every 0.1 ms.

This resolves its peak correction as  $-1.83099\text{ N}$ , against  $3592.58\text{ N}$  centrifugal force. Its minimum centrifugal denominator after ramp onset exceeds  $2995\text{ N}$ . The baseline’s largest engaged force correction is  $75.4878\text{ N}$ ; its largest force fraction is  $5.34288\%$ , at a different instant. Over  $0.1\text{ s}$  to  $10\text{ s}$ , the largest sampled fraction is  $0.00147132\%$ .

The first-engagement comparison is stable at the scales reported in the main text. The tiny subsequent seating sequence is less uniform: the quasi-static baseline records 14 transitions at nominal settings and 16 at tighter settings; the full model records 16 at both. The figure therefore shows the approach and first capture, and no invariant rebound count is claimed. The subsequent smooth torque-ramp comparisons contain no events. Across the selected response-difference traces, tightening the integration tolerances changes shift by at most  $0.0044\mu\text{m}$  and primary speed by  $0.00031\text{ rpm}$ .

The shaft-inertia difference is kept separate from the axial force. Written as an additional torque relative to constant initial inertia, it is

$$\Delta\tau_{p,\text{inertia}} = -[J_{f,p}(s) - J_{f,p}(0)]\dot{\omega}_p - \frac{dJ_{f,p}}{ds}\dot{s}\omega_p. \quad (\text{D.6.2})$$

It ranges from  $-1.369\text{ N m}$  to  $1.321\text{ N m}$  over engaged baseline motion, and from  $-0.2344\text{ N m}$  to  $-0.000114\text{ N m}$  after  $0.1\text{ s}$ . The selected ramp’s post-onset range is  $-0.1672\text{ N m}$  to  $-0.00830\text{ N m}$ . The integrated response differences include this coupling as well as the pivot-force reduction.

The sampled free-primary force balance closes to within  $4.1 \times 10^{-12}\text{ N}$  across the new runs. At the selected peak, the torque ramp lowers the actuator force by  $11.79\text{ N}$  and the belt’s opening reaction by  $15.23\text{ N}$  relative to the full model’s unforced control, accounting for the additional closing acceleration. Both integrated normal loads remain positive, and the sampled stick–stick tractions remain inside the static limits.

The maintained entry point `run.py` accepts `--plot-only` to regenerate the two figures from verified compact outputs, and `--primary-publication` to repeat these baseline and selected-ramp refinements before plotting. The latter preserves the study’s other outputs. Recovered screen provenance and the relationship to the earlier actuator archive are documented with the study.

## D.7 Secondary-Helix Dynamics

### D.7.1 Inputs and comparisons

The comparisons in [Section 4.4.2](#) use CINDER 1.1.2, with the reference boundaries and the secondary reduction of [Appendix C.3.1](#). The primary remains dynamic. The full and reduced secondary models retain the same total rotational inertia for rigid shaft motion; the reduction removes the movable member’s relative helix dynamics by assigning its rotational inertia to the fixed shaft. Both use the zero-clearance bilateral slot.

The continuous comparison replaces only the secondary helix profile and movable rotational inertia with the selected component estimates. The underlying hardware references, dimensional assumptions and annular-face plus hub calculation are recorded in [ESTIMATION\\_METHOD.md](#) and its accompanying `inputs/PARAMETER_PROVENANCE.csv`. The listed  $31^\circ$  angle is sourced; the movable mass distribution and  $45\text{ mm}$  to  $60\text{ mm}$  effective helix radii are engineering estimates. [Table D.26](#) retains the low and high comparisons. The additional  $28^\circ$  local-angle sensitivity gives a high-estimate reflected axial inertia of 3.72 times the reference; it is not a reconstruction of an entire progressive profile.

| Quantity                                        | Low      | Nominal  | High     |
|-------------------------------------------------|----------|----------|----------|
| Movable inertia [kg m <sup>2</sup> ]            | 0.005556 | 0.012875 | 0.020469 |
| Helix radius [mm]                               | 60       | 50       | 45       |
| Local reflected inertia [kg]                    | 4.275    | 14.265   | 27.998   |
| Peak force correction [%]                       | 3.147    | 7.870    | 13.604   |
| Maximum shift-response difference [mm]          | 0.1088   | 0.2742   | 0.4535   |
| Maximum primary-speed response difference [rpm] | 15.43    | 40.36    | 68.49    |

Table D.26: Recovered 31° component-estimate sensitivity under the same added secondary torque. The response subtracts each model’s own unforced continuation. These are engineering scenarios, not confidence bounds. The nominal case is also replayed and refined for the published figures.

The commercial screen contains secondary torque disturbances at selected travel positions and rise times. The recorded selection rule seeks an uninterrupted response with a reaction correction between 10% and 20%, while retaining a quasi-static reaction away from zero. No eligible case reaches that band. The selected `S_s50_m120_r25ms` is the recorded largest eligible continuous case; its attained nominal correction is 7.87%. All low, nominal and high loaded/control continuations remain free and stick–stick in the recovered records. Fresh local-wrap and refinement checks here concern the selected nominal comparison.

The added torque follows  $A(3u^2 - 2u^3)$  during its ramp and is then held. For the continuous case,  $A = -120$  N m, the ramp starts at 0.05 s, lasts 25 ms, and is followed for 0.35 s. Each model uses its own archived, naturally conditioned state near 50% active travel. For the reversal, the primary boundary blends from its engine torque to  $-28$  N m, beginning at 0.03 s and lasting 50 ms; the common end time is 0.43 s. It uses 270° torsional preload. The stock comparison instead adds  $-120$  N m to the secondary over 2 ms, beginning at 0.03 s and ending at 0.382 s, with 300° torsional preload. Both paired cases start near 30% active travel. The stock case is therefore not a preload-only comparison with the reversal.

The paired cases use identical continuous coordinates and the supplied stick–stick mode in both models. This preserves the intended sticking restart instead of classifying a small inherited speed mismatch as fresh sliding. The largest initial mismatch is 0.0471 mm/s in the reversal and 0.006 46 mm/s in the stock case, below the 1 mm/s sticking audit tolerance used in [Section 4.2.1](#). Both initial traction pairs are within their static limits, and both wraps have positive local loading. Reclassifying the archived coordinates instead produces a very brief sliding sequence with negative local secondary loading; those initialization states are not used in the published pairs. The exact five-coordinate vectors and archived source hashes are in [secondary\\_initial\\_conditions.json](#). The accumulated secondary-shaft angle is reset to zero. It is a passive reporting coordinate for these time-programmed boundaries and does not change the applied loads.

## D.7.2 Extraction and numerical checks

The signed helix correction is evaluated using the executed assembly’s movable inertia. This is necessary for the commercial transplant: the old generic trajectory diagnostics used the reference inertia, whereas the separate component trace and the actual actuator force used the correct value. The present extraction verifies the helix force plus axial spring against the delivered secondary-actuator force to within  $5 \times 10^{-13}$  N in the nominal refined comparison. The 109.1 N maximum difference between its loaded-minus-control force responses is an actuator closing-force difference, not an integrated belt-normal force.

| Setting or measured quantity                      | Nominal            | Tighter            |
|---------------------------------------------------|--------------------|--------------------|
| Relative tolerance                                | $1 \times 10^{-4}$ | $1 \times 10^{-5}$ |
| Absolute tolerance                                | $1 \times 10^{-7}$ | $1 \times 10^{-8}$ |
| Continuous-case maximum step [ms]                 | 1                  | 1                  |
| Reversal maximum step [ms]                        | 0.5                | 0.25               |
| Stock-case maximum step [ms]                      | 0.1667             | 0.1667             |
| Continuous shift-response difference [mm]         | 0.274157           | 0.274148           |
| Continuous primary-speed difference [rpm]         | 40.36482           | 40.36479           |
| Reversal: earlier quasi-static limit arrival [ms] | 4.66493            | 4.66513            |
| Stock maximum shift difference [mm]               | 0.0514228          | 0.0514230          |

Table D.27: Focused LSODA refinement. The continuous case uses each model’s own unforced control; the other pairs compare motion from common initial coordinates. The reversal timing concerns arrival at the traction limit, not a validated post-slip continuation.

Published traces use the tighter settings. Continuous-case output is sampled every 0.5 ms; the paired cases use 0.1 ms, with the exact endpoints of every continuous segment retained. Both sides of a contact or support event are kept separately. No force curve is interpolated across a jump. Tightening the integration settings changes the continuous shift-response difference trace by at most  $2.02 \times 10^{-5}$  mm and its primary-speed difference by 0.000 882 rpm. The stock difference trace changes by at most  $3.76 \times 10^{-6}$  mm. These changes are small compared with the reported response differences.

Local contact is checked from the recovered tension field. With belt linear density  $q$ , the normal loading per unit wrap angle is

$$\frac{dN_j}{d\phi} = \frac{T_j(\phi) - q(v_b^2 - r_{j,\text{cm}}\ddot{r}_{j,\text{cm}})}{\sin \beta}. \quad (\text{D.7.1})$$

The frozen wrap solution is monotone between its endpoint tensions, so the minimum follows from the smaller endpoint value. This tests the entire spatial profile at each examined time, rather than only its resultant. The temporal audit includes sampled interiors and both native event sides; it is not a continuous-time proof. For the nominal commercial loaded/control runs, the smallest refined primary and secondary values are 228.8 N/rad and 224.5 N/rad. In the refined stock pair they are 9.50 N/rad and 20.34 N/rad. Tension is also positive. Deadzone states have no primary wrap and are excluded from this engaged contact check.

### D.7.3 Reversal and the local contact limit

In [Figure 4.10\(c\)](#), the plotted demand is  $|\lambda_p|/\mu_s$ , with  $\mu_s = 0.65$ . The primary traction is negative throughout this late window; both incoming events reach  $\lambda_p = -0.65$ . The normalization shows the remaining static capacity, not a change in the sign convention. The displayed curves stop before the first failed local-contact state.

The full and quasi-static reversal models reach the primary traction limit at 0.423 610 23 s and 0.418 945 10 s, respectively. Before the earlier limit, their largest sampled shift separation on the common clock is only 0.0417 mm. The full model’s incoming event state has 7.481 96 kg reflected helix inertia and 2.184 91 kg other direct shift inertia. The 77.4% share concerns this direct coefficient, before eliminating the coupled shaft and contact equations.

The force changes immediately with the contact regime. On the two sides of the full model's event, the helix reaction is 1.627 58 N m and  $-0.805\,46$  N m; the quasi-static diagnostic at those same full-model states is 1.425 06 N m and 2.001 91 N m. The outgoing shift-acceleration contribution is  $-3.020\,86$  N m, while the shaft contribution is 0.213 48 N m. The negative net reaction requests the opposite slot flank, which the bilateral constraint can supply. There is no resolved continuous zero-reaction state between the two event sides.

The belt imposes a separate restriction. Secondary normal resultant remains positive, changing from 1051.25 N to 716.33 N, but the outgoing primary-wrap minimum is  $-0.076\,03$  N/rad. The quasi-static model's outgoing primary minimum is  $-0.732\,01$  N/rad. Both violate the  $1 \times 10^{-6}$  N/rad numerical allowance used for negative local loading; tightening the integration does not remove either violation. The formal continuation to 0.43 s gives a maximum paired shift separation of 2.027 mm, close to the archived 2.026 mm result. Because both trajectories pass through locally inadmissible states, this number is retained as an audit result and is not used to establish a physical post-slip performance difference. A later return to positive loading cannot repair the intervening trajectory. Nor does positive integrated normal force establish local compressive contact everywhere. The main comparison therefore ends at the incoming event states. [Figure D.3\(c,d\)](#) retains the discontinuous force solution and identifies the local violation.

The maintained entry point [run.py](#) accepts `--plot-only --unit secondary` to reproduce these figures from the hashed compact outputs, and `--secondary-publication` to repeat the selected continuations and audits before plotting. The complete settings, event identities, masks, local minima and source/output hashes are in [secondary\\_publication\\_audit.json](#). Archive comparisons and the exploratory checks are preserved separately. No live simulator source is imported by these release-pinned runs.

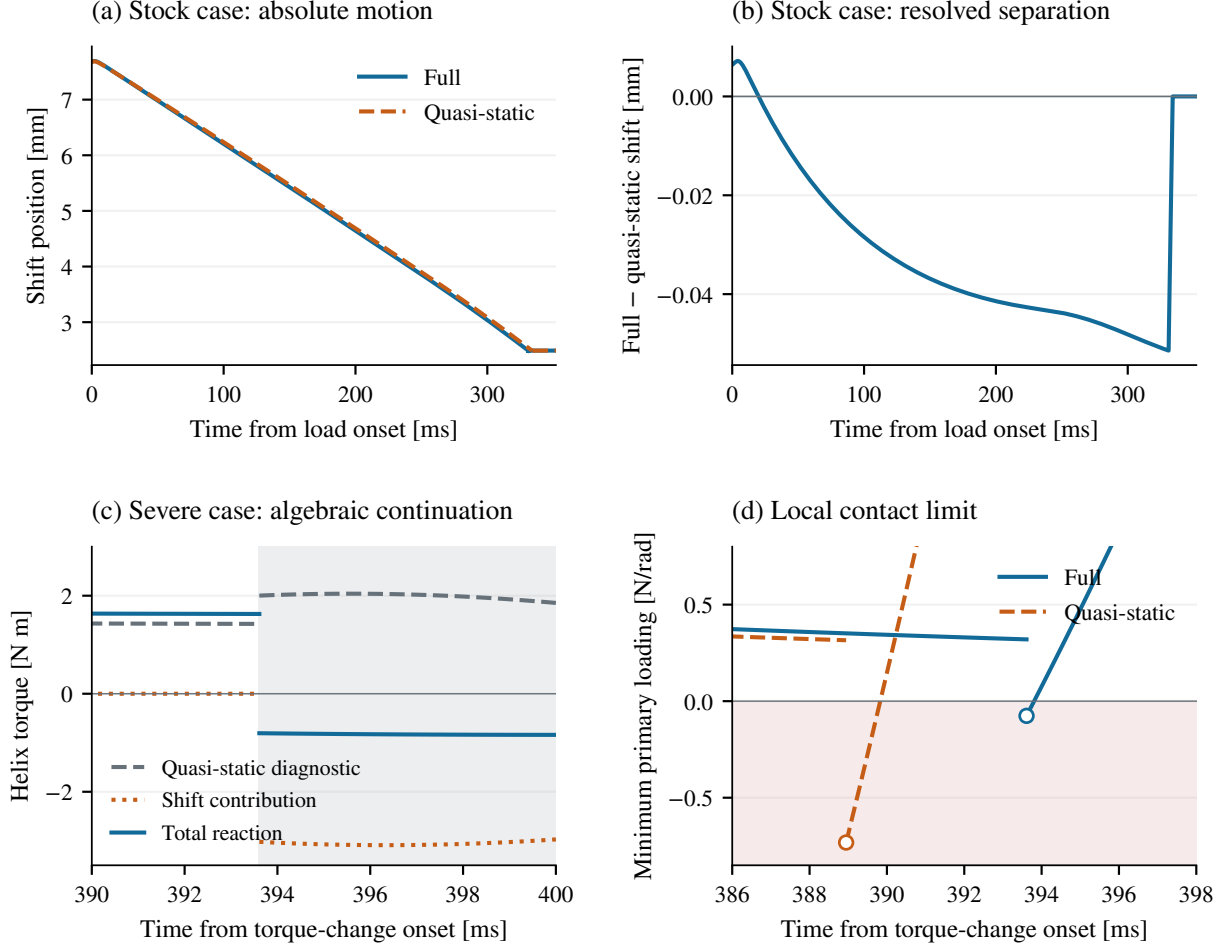


Figure D.3: Supporting comparisons and the reversal limit. (a,b) Stock-case motion and its small, resolved shift separation. (c) The full reversal’s signed reaction, with the region after the local-contact failure shaded. The quasi-static diagnostic is evaluated on the full trajectory; it is not the separately integrated reduced model. (d) Local primary-wrap minima for the two formal continuations. Hollow points denote the exact outgoing states; the region below zero requires tensile contact. Curves on different event sides are not joined. The post-event traces in (c,d) diagnose the failed contact assumption and are not admissible continuations.

For the stock contrast, at 1.8 ms after load onset the shaft and shift contributions are approximately 0.444 N m and  $-0.891$  N m. Their net contribution is approximately  $-0.448$  N m, against 40.43 N m quasi-static reaction. The later non-sticking sequence follows low-ratio seating and re-engagement; it is not solely a static-traction failure. These observations explain why a fast applied load and nonzero individual terms need not produce a large shift difference.

The relative-sheave contribution in [Figure 4.10\(a\)](#) combines the recorded shift-acceleration and geometric-curvature contributions, giving the full  $-I_{s,M}\ddot{\theta}_s$  term. Its signed sum with  $-I_{s,M}\dot{\omega}_s$  is the reaction correction. The first 8 ms are shown to resolve the imposed 2 ms ramp; the shift difference in panel (b) uses the complete 352 ms post-onset interval. The two panels therefore resolve different timescales of the same response.

## D.8 Common-Course Operation and Tuning

Exact inputs supporting [Section 4.5](#) are defined by `study.json`. The selected road and hardware cases are frozen in `course.json` and `competitors.json`; their selected-input hashes are recorded in `selection.lock.json`. The maintained selection revision is `section4p5-final-v3-r26b7-m170`.

All selected cases use CINDER 1.1.2, the reference CVT of [Appendix C.1](#), the bilateral secondary-helix topology of [Appendix C.1.1](#), the full-throttle primary boundary of [Appendix C.2.2](#), and the locked-final-drive vehicle boundary of [Appendix C.2.3](#). They begin from the common state in [Equation \(C.1.1\)](#). The hardware variants therefore change the CVT itself without changing the engine, vehicle, belt/contact model, initial state, or numerical method.

### D.8.1 Distance-Indexed Road

Road position is signed travelled distance along the road surface. Constant grades are joined by the  $C^2$  quintic transition

$$S(u) = u^3 (10 - 15u + 6u^2), \quad 0 \leq u \leq 1, \quad (\text{D.8.1})$$

so a ramp from  $\gamma_0$  to  $\gamma_1$  over length  $L$  is  $\gamma(x) = \gamma_0 + (\gamma_1 - \gamma_0)S((x - x_0)/L)$ . The cyclic region uses a 12 m-wavelength sinusoidal grade with amplitude  $32^\circ$ , multiplied by the same smooth envelope over the first and last 6 m. The complete selected road is listed in [Table D.28](#).

| Start [m] | End [m] | Road feature                                                          |
|-----------|---------|-----------------------------------------------------------------------|
| 0         | 120     | Level launch and opening upshift                                      |
| 120       | 132     | Smooth $0 \rightarrow 38^\circ$ entry                                 |
| 132       | 212     | Constant $38^\circ$ climb                                             |
| 212       | 224     | Smooth $38^\circ \rightarrow 0$ exit                                  |
| 224       | 264     | Level recovery                                                        |
| 264       | 336     | Six 12 m cycles at $\pm 32^\circ$ , zero mean, with 6 m end envelopes |
| 336       | 348     | Level separation                                                      |
| 348       | 356     | Smooth $0 \rightarrow 18^\circ$ entry                                 |
| 356       | 596     | Constant $18^\circ$ climb                                             |
| 596       | 604     | Smooth $18^\circ \rightarrow 0$ exit                                  |
| 604       | 624     | Level recovery                                                        |
| 624       | 636     | Smooth $0 \rightarrow -22^\circ$ descent entry                        |
| 636       | 696     | Constant $-22^\circ$ descent                                          |
| 696       | 708     | Smooth $-22^\circ \rightarrow 0$ exit                                 |
| 708       | 732     | Final level segment                                                   |

Table D.28: Distance-indexed road used by every selected common-course case.

The road implementation is defined in `course.py`. The signed distance is road arc length, so the elevation derivative is  $\sin \gamma$ , not  $\tan \gamma$ . Full throttle is retained throughout; the course changes only the road grade supplied to the secondary vehicle boundary.

### D.8.2 Selected Hardware Configurations

The reference flyweight has a fixed arm/body mass of 13.646 g and a replaceable tip mass of 250 g per flyweight. Mass-scale cases change the tip mass and its associated first and second mass moments consistently while leaving the arm/body mass unchanged. Spring-rate case P300 triples the primary

spring stiffness while choosing the installed compression that preserves the reference spring force at the low-ratio engagement geometry. The scalar actuator quantities resolved by `tunes.py` are listed in Table D.29.

| Case     | Tip [g] | $k_p$ [kN/m] | $c_{p,0}$ [mm] | $c_{s,0}$ [mm] | $\theta_{s,\text{pre}}$ [deg] | Helix [deg] |
|----------|---------|--------------|----------------|----------------|-------------------------------|-------------|
| R00      | 250.0   | 12.784       | 94.090         | 110.0          | 300                           | 20          |
| W85      | 212.5   | 12.784       | 94.090         | 110.0          | 300                           | 20          |
| P300     | 250.0   | 38.352       | 29.704         | 110.0          | 300                           | 20          |
| RC10     | 250.0   | 12.784       | 94.090         | 110.0          | 300                           | 20          |
| R26B7    | 250.0   | 12.784       | 94.090         | 110.0          | 300                           | 20          |
| RC40L    | 250.0   | 12.784       | 94.090         | 110.0          | 300                           | 20          |
| H28      | 250.0   | 12.784       | 94.090         | 110.0          | 300                           | 28          |
| U55      | 137.5   | 12.784       | 94.090         | 110.0          | 300                           | 20          |
| D01      | 325.0   | 12.784       | 75.272         | 88.0           | 240                           | 28          |
| D02      | 162.5   | 12.784       | 108.204        | 110.0          | 300                           | 20          |
| D02_M    | 250.0   | 12.784       | 108.204        | 110.0          | 300                           | 20          |
| D02_M170 | 425.0   | 12.784       | 108.204        | 110.0          | 300                           | 20          |
| D02_P    | 162.5   | 12.784       | 94.090         | 110.0          | 300                           | 20          |

Table D.29: Resolved scalar actuator quantities for the thirteen selected common-course configurations.  $c_{p,0}$  and  $c_{s,0}$  are installed axial spring compressions. The helix angle is measured from the circumferential direction. Unlisted actuator quantities retain the R00 reference values.

The three selected ramp cases retain the same fixed-pivot hardware and flyweight mass properties as R00. Only the contacted ramp profile is changed. The baseline profile consists of a 5 mm linear 35° segment, a 3 mm  $C^3$  blend, and a 30 mm circular segment from 35° to 20°. The selected modifications are

| Case  | Unchanged prefix [mm] | $C^3$ blend [mm] | Arc-tail end angle |
|-------|-----------------------|------------------|--------------------|
| RC10  | 10                    | 5                | 10°                |
| R26B7 | 10                    | 7                | 26°                |
| RC40L | 18                    | 5                | 40°                |

Table D.30: Selected primary-ramp modifications used on the common course. The prefix and blend distances are physical coordinates along the ramp, not movable-sheave travel.

All other selected cases retain the baseline ramp. In particular, D02\_M restores the reference tip mass while retaining D02’s 115% primary preload, D02\_P restores the reference primary preload while retaining D02’s 65% tip mass, and D02\_M170 uses 170% tip mass while retaining the 115% primary preload. No quasi-static actuator replacement, force scaling, or fitted speed curve is introduced in these course cases.

### D.8.3 Numerical and Stopping Controls

The selected common-course study runs thirteen cases on the 732 m road and two supporting cases, R00 and U55, on a separate 800 m flat road. All fifteen trajectories use the same tight numerical settings.

| Setting                             | Value              |
|-------------------------------------|--------------------|
| Integrator                          | LSODA              |
| Relative tolerance                  | $3 \times 10^{-5}$ |
| Absolute tolerance                  | $3 \times 10^{-8}$ |
| Maximum integration step            | 5 ms               |
| Diagnostic/report interval          | 5 ms               |
| Maximum simulated time              | 180 s              |
| Maximum total hybrid transitions    | 5000               |
| Checkpoint interval                 | 2 s                |
| Maximum transitions per checkpoint  | 600                |
| Event-time tolerance                | $10^{-10}$ s       |
| Per-case wall timeout               | 1800 s             |
| Progress monitor begins after       | 20 s               |
| No-progress review window           | 15 s               |
| Minimum progress over review window | 0.25 m             |
| Rollback stop speed                 | -0.25 m/s          |

Table D.31: Numerical and execution controls of the selected common-course study.

A progress or rollback stop is recorded as a physical observed outcome of the specified case rather than as a numerical or model-domain failure. Conversely, integration, construction, or model-domain errors remain distinct stopping classes. The runner does not restart a sector from a prescribed state: each common-course trajectory is continuous from the shared initial condition until completion or its recorded stopping condition.

The separate 800 m flat experiment uses the same numerical settings and contains only R00 and U55. It is a supporting test of whether U55's incomplete opening shift on the common course is merely caused by the first hill arriving at 120 m; it is not appended to the common road and does not alter the selected course.

#### D.8.4 Settling and Archived Case Records

Interior settling checks use a 5 s window advanced in 0.25 s increments. A candidate window must remain at least 0.25 mm from either shift stop, maintain forward speed above 0.1 m/s, and remain on constant grade. The principal variation limits are listed in [Table D.32](#); the complete criteria are defined in [settling\\_criteria.json](#).

| Settling criterion                     | Limit                               |
|----------------------------------------|-------------------------------------|
| Maximum $ \dot{s} $                    | 0.05 mm/s                           |
| Maximum vehicle acceleration magnitude | 0.01 m/s <sup>2</sup>               |
| Maximum primary RPM rate magnitude     | 2 rpm/s                             |
| Maximum secondary RPM rate magnitude   | 2 rpm/s                             |
| Maximum belt acceleration magnitude    | 0.01 m/s <sup>2</sup>               |
| Maximum shift acceleration magnitude   | $5 \times 10^{-4}$ m/s <sup>2</sup> |
| Shift range within window              | 0.1 mm                              |
| Vehicle-speed range within window      | 0.05 m/s                            |
| Primary-speed range within window      | 10 rpm                              |
| Secondary-speed range within window    | 10 rpm                              |
| Belt-speed range within window         | 0.05 m/s                            |

Table D.32: Principal finite-window settling criteria used by the common-course analysis. Passing a window establishes only the stated finite-window condition.

The final study output is self-describing. The top-level `suite.json` records the selected study identity, numerical controls, and environment. Each case stores a complete `resolved_case.json`, `model_identity.json`, event history, sampled diagnostics, and stopping summary. The common-course report also stores the resolved physical tune table and the analytic road sectors. The intended appendix-facing inventory is summarized in [APPENDIX\\_HANDOFF.md](#). These records, rather than the human-readable case labels, define the exact simulation inputs used by the study.